Approximate Syzygies of Smoothed Polynomials and Applications

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Abstract

Given a set of n-variate homogenous degree-d polynomials $p_1(\mathbf{x}), \ldots, p_m(\mathbf{x}) \in \mathbb{R}[\mathbf{x}]$, a syzygy is defined as a tuple of homogenous polynomials $(h_1(\mathbf{x}), \ldots, h_m(\mathbf{x}))$ satisfying $\sum_{i=1}^{m} p_i(\mathbf{x})h_i(\mathbf{x}) = 0$. An approximate syzygy satisfies $\sum_i p_i(\mathbf{x})h_i(\mathbf{x}) \approx 0$, where the error is measured in a natural coefficient norm of the polynomial. Much like condition numbers in linear systems, understanding approximate syzygies is central to stability questions in computational algebra. Approximate syzygies also play a key role in algorithms for learning latent variable models and arithmetic circuits in the smoothed analysis setting. A long line of work in algebraic geometry focuses on the question of characterizing the structure of exact syzygies for generic polynomials; however they do not provide any quantitative guarantees for approximate syzygies.

We develop a systematic theory of approximate syzygies in the *smoothed analysis* setting. Our generators are powers $p_i(\mathbf{x}) = a_i(\mathbf{x})^D$ of independently ρ-smoothed degree-d polynomials $a_i(\mathbf{x})$, including ordinary smoothed polynomials when $D = 1$. For constant d, D, any fixed $\varepsilon > 0$ with $m \leq n^{dD-\varepsilon}$, we prove that, with exponentially small failure probability in the smoothed analysis setting, every η-approximate syzygy $h_1, \ldots, h_m$ is at most $\eta \cdot \mathrm{poly}(n, \rho^{-1})$ close to the well-understood space of *Koszul syzygies*. In particular, m can be larger than n and is only a $n^{-\varepsilon}$ factor away from the best possible. Our results imply new bounds on the condition number of Macaulay matrices which are important in numerical computational algebra. Furthermore, we resolve the open question of polynomial time smoothed analysis guarantees for power sum decompositions, and prove robust uniqueness of symmetric third order tensors of super-linear rank. Our proofs combine ideas from algebraic geometry and high-dimensional probability to establish uniform robustness in the presence of a random space of exact cancellations.

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1 Introduction

Solving a system of linear equations of the form $M\mathbf{z} = c$ is a fundamental primitive in mathematics and computing. The kernel of the coefficient matrix M characterizes the ambiguity for exact solutions. When there are errors in M or c due to noise, numerical precision or misspecification, we require a quantitative understanding of approximate solutions to $M\mathbf{z} \approx c$. The stability of the solution space is controlled by the smallest non-zero singular value of M, which captures how a small residual (error in c) translates to proximity to the exact solution space.[1] While worst-case matrices can be arbitrarily ill-conditioned, i.e., the smallest non-zero singular value being very close to 0, the elegant framework of *smoothed analysis* provides an explanation for the stability of typical linear systems in practice, by proving that a small random perturbation to M guarantees inverse-polynomial least singular value bounds with high probability [ST04, SST06].

In this work, we study the analogous question for systems whose coefficients and unknowns are multivariate polynomials. These systems are foundational primitives in computational algebra. Given homogeneous polynomials $p_1(\mathbf{x}), \ldots, p_m(\mathbf{x}) \in \mathbb{R}[\mathbf{x}]^{=d}$ of degree d over n variables $\mathbf{x} = (x_1, \ldots, x_n)$, a tuple of homogeneous polynomials $(h_1(\mathbf{x}), \ldots, h_m(\mathbf{x}))$ of degree $t - d$ constitutes a *syzygy* if they satisfy

$$p_1(\mathbf{x})h_1(\mathbf{x}) + \cdots + p_m(\mathbf{x})h_m(\mathbf{x}) = 0. \tag{1}$$

The polynomials $p_1, \ldots, p_m$ are called the *generator* polynomials, while $\mathsf{Syz}_{\mathbf{p}}^{(t)}$ corresponds to the space of solution tuples $(h_1, \ldots, h_m)$ of degree $t-d$ satisfying (1). In this work we study *approximate syzygies:* our goal is to characterize tuples of polynomials $(h_1(\mathbf{x}), \ldots, h_m(\mathbf{x}))$ satisfying

$$\left\| \sum_{i=1}^{m} p_i(\mathbf{x})h_i(\mathbf{x}) \right\| \leq \eta \left(\sum_{i=1}^{m} \|h_i\|^2 \right)^{1/2}. \tag{2}$$

Here we use the Bombieri-Weyl norm $\|h\|$ for a polynomial $h(\mathbf{x})$, or equivalently, the Frobenius norm of the canonical symmetric tensor that represents $h(\mathbf{x})$. We will be interested in the setting where the degrees d, t are constant, and in approximate syzygies that are robust up to η being inverse polynomial in n. The main question is whether every approximate syzygy must be quantitatively close to the space $\mathsf{Syz}_{\mathbf{p}}^{(t)}$ imposed by the exact algebraic structure. As in numerical linear algebra, such a characterization also controls the stability of solving the corresponding inhomogeneous equations of the form $\sum_i p_i(\mathbf{x})h_i(\mathbf{x}) = q(\mathbf{x})$ with the solutions understood up to the syzygies $\mathsf{Syz}_{\mathbf{p}}^{(t)}$.

Quantitative bounds on approximate syzygies provide stability guarantees for several problems in computational algebra and machine learning:

- **Numerical Computational Algebra:** Given a polynomial $q(\mathbf{x})$ in the *ideal* generated by $p_1, \ldots, p_m$, finding multipliers satisfying $\sum_i p_i h_i = q$ is the ideal membership problem, which is key in Nullstellensatz certificates and Gröbner basis reductions. In terms of the coefficient matrix, this linear system is specified using the *Macaulay matrix* (which we will describe later). Quantitative bounds on approximate syzygies control how close an approximate representation is to an exact solution up to the ambiguity in the syzygy [BDD14]. Approximate syzygies have also been used algorithmically to compute approximate greatest common divisors of multivariate polynomials [Lic19].

[1] The least singular value $\sigma_{\min}$ or the condition number $\kappa = \sigma_{\max}(M)/\sigma_{\min}(M)$ also controls how errors in the coefficient matrix M translate to errors in solution; see [Bha97].

- **Power-sum decompositions:** Given a polynomial $f(\mathbf{x}) \approx_\eta \sum_{i=1}^m (a_i(\mathbf{x}))^D$, where the $a_i(\mathbf{x})$ are degree-d polynomials and the approximation error η can be a sufficiently small inverse-polynomial in n, the goal is to design an algorithm that estimates each of the components $a_i(\mathbf{x})$. This is a general class of decomposition problems that arises in learning arithmetic circuits, and moment methods for learning latent-variable models. The problem is NP-hard in the worst-case even for tensor decomposition, where the $a_i(\mathbf{x})$ are linear forms [Hås90, HL13]. Algorithms based on algebraic circuit lower bounds exploit algebraic relations between polynomial powers and their derivatives [GKS20, BHKX22, CGK+24]. Smoothed polynomial time guarantees have been open except in some special settings, and require quantitative bounds on approximate syzygies.
- **Robust uniqueness of symmetric tensor decompositions:** The famous uniqueness results of tensor decomposition show that a symmetric $n \times n \times n$ tensor $T = \sum_i^m \ell_i^{\otimes 3}$ where $\ell_i \in \mathbb{R}^n$ has a unique decomposition (up to relabeling terms) of rank at most m under some non-degeneracy conditions [Har72, Kru77, CO12]; such uniqueness is false for matrices, but uniqueness results are known for "non-degenerate" tensors of order 3 and above. The work of [COV17] shows uniqueness up to rank $\sim n^2/6$ when the components ℓ_i are *generic*.[2] Robust uniqueness shows that any decomposition incurring an error η is component-wise close up to $\mathrm{poly}(\eta, n)$ error. Robust uniqueness is crucial for applications to efficient learning algorithms where we only have polynomial samples, yet existing robust uniqueness results are only known for linear rank $m = O(n)$ [BCV14]. Generic uniqueness for exact decompositions relies on powerful statements from algebraic geometry involving syzygies of the form $\sum_i \ell_i(\mathbf{x})^2 h_i(\mathbf{x}) = 0$ and corresponds to the rank of the tangent map represented by the Terracini matrix [Ott13]. Quantitative bounds on approximate syzygies are necessary for obtaining robust uniqueness results for tensor decompositions with rank $m = \omega(n)$, and other questions on robust unique decompositions [BV18].

Linear systems with scalar coefficients h are characterized by linear dependencies of the generators $p_i(\mathbf{x})$. However, polynomial systems inherently possess non-trivial algebraic structure as soon as the degree of the $h_i(\mathbf{x})$ (unknowns) is larger. Most notably, when the total degree $t = 2d$, any pair of polynomials satisfies the trivial commutativity relation $p_i(\mathbf{x})p_j(\mathbf{x}) - p_j(\mathbf{x})p_i(\mathbf{x}) = 0$. More generally for $t \geq 2d$, we have the class of *Koszul syzygies* given by any skew-symmetric matrix of polynomials $\Omega_{ij}(\mathbf{x}) = -\Omega_{ji}(\mathbf{x})$ of degree $t - 2d$, by choosing $h_i(\mathbf{x}) = \sum_{j=1}^m \Omega_{ij}(\mathbf{x})p_j(\mathbf{x})$. Furthermore, we may have other non-trivial syzygies if the polynomials $p_i(\mathbf{x})$ share a common factor, say $p_1(\mathbf{x}) = g(\mathbf{x})q_1(\mathbf{x})$ and $p_2(\mathbf{x}) = g(\mathbf{x})q_2(\mathbf{x})$ with $\deg(g) \geq 1$. For pairs of polynomials ($m = 2$), quantitative co-primeness of pairs of polynomials is characterized in [BL98] using the condition number of the Sylvester resultant matrix. With more generators $m > 2$, the algebraic structure of the syzygies for an arbitrary set of polynomials $p_1, \ldots, p_m$ could be complicated. Strong characterization results on the structure of the syzygies exist in algebraic geometry for generic polynomials $p_1, \ldots, p_m$. For these reasons we will also focus on the generic or smoothed setting of the problem.

Smoothed setting. Motivated by the above settings, we study approximate syzygies when the polynomials $p_1, \ldots, p_m$ are of the form $p_i(\mathbf{x}) = a_i(\mathbf{x})^{D_i}$ for a generic or "smoothed" polynomial $a_i(\mathbf{x})$ of degree d_i. Let $\rho > 0$ be the smoothing parameter (think of $\rho = 1/\mathrm{poly}(n)$). Starting from arbitrary homogeneous polynomials $\bar{a}_1, \ldots, \bar{a}_m$, we independently perturb their coefficients to

[2] A property holds generically, if it is satisfied for all but an exceptional set that is Zariski closed (a zero measure set).

obtain ρ-smoothed polynomials

$$a_i(\mathbf{x}) = \bar{a}_i(\mathbf{x}) + g_i(\mathbf{x}),$$

where each $g_i(\mathbf{x})$ is a random polynomial where each coefficient of $g_i(\mathbf{x})$ is an independent Gaussian $\mathcal{N}(0, \rho^2)$ with mean 0 and variance ρ^2, with a scaling that is associated with the Bombieri-Weyl norm (see Section 3 for a formal definition). We consider:

- *Smoothed polynomials:* the generators are $p_i(\mathbf{x}) = a_i(\mathbf{x})$. This is the special case when $D_i = 1$.
- *Powers of smoothed polynomials:* the generators are $p_i(\mathbf{x}) = a_i(\mathbf{x})^{D_i}$, where the D_i are fixed positive integers. Here the perturbation is applied to the underlying low-degree polynomial $a_i(\mathbf{x})$ and preserves the algebraic structure relevant to decomposition problems.

In our setting, the degrees and powers d_i, D_i are all fixed or constants, while n and m grow. Our goal is to obtain bounds polynomial in n, m and the inverse perturbation scale ρ^{-1}. As standard in smoothed analysis [BCMV14], we require the failure probability to be exponentially small in n or powers of n. The main question we address in this work is:

> **Question:** *When can we establish that every η-approximate syzygy in (2) is $\mathrm{poly}(\eta, n, \rho)$ close to an exact syzygy with high probability, even when the number of generators m is much larger than n?*

Establishing inverse-polynomial robustness bounds for approximate syzygies requires overcoming two distinct sets of challenges — one rooted in algebraic geometry, and the other in high-dimensional probability.

Challenge 1: Algebraic Geometry and Structural Syzygies. Systems with non-scalar multipliers h_i inherently possess non-trivial algebraic structure. Classic results from algebraic geometry show that when the number of generators $m \leq n$, then syzygies of powers of generic polynomials are characterized by Koszul syzygies for all degrees t (since generic homogeneous polynomials form a regular sequence) [Eis95]. When $m > n$, non-Koszul syzygies necessarily occur in sufficiently higher degree, potentially growing with n.

In our regime of interest with $t = O(1)$, strong algebraic geometry results exclude non-Koszul syzygies. For example, Hochster and Laksov considered the case of linear multipliers h_i, and proved that for generic homogeneous degree-d polynomials, the only syzygies are Koszul syzygies, for $m \leq c_d n^d$ generators for a sharp constant $c_d > 0$ [HL87]. More generally, for the same degree $d \geq 2$ of the $\{a_i : i \in [m]\}$, and power $D \geq 1$ and sufficiently large n, there are no nonzero syzygies for generic polynomial powers of total degree $t < 2dD$ when $m \leq c_{d,D}\binom{n+dD-1}{dD}$, for a specific constant $c_{d,D} > 0$ [Nen17]; see also [BDL26]. More broadly, Fröberg's conjectures imply the precise range of m, through conjectured predictions about Hilbert series of ideals generated by generic forms [Frö85]. The conjecture of Nicklasson [Nic17] conjectures similar bounds for powers of generic forms of degree $d \geq 2$; see also [FH94] for powers of linear forms. Our goal corresponds to quantitative versions of these statements with inverse polynomial robustness, but for a smaller range of m that incurs multiplicative losses in terms of the predicted maximum range of m (e.g., $m \leq O(n^{Dd-\varepsilon})$ for constant t, d, D).

Challenge 2: High-Dimensional Probability, Anti-Concentration and the Macaulay matrix. Even when equipped with an algebraic characterization of exact syzygies, establishing quantitative robustness in the smoothed setting represents a different probabilistic challenge. In the appropriate coefficient basis, the multiplication map $\mathsf{M}_{\mathbf{p}}^{(t)} : (h_i : i \in [m]) \mapsto \sum_i p_i h_i$ is represented

by a *Macaulay matrix*, whose smallest nonzero singular value controls quantitative robustness. Even in the simplest setting when h_i are scalars, this reduces to approximate linear dependencies among the generators p_i. Establishing inverse-polynomial robustness in the smoothed model corresponds to least singular value bounds for random matrices with highly dependent entries — the matrix entries are dependent polynomials of the underlying perturbations [BCMV14, MSS16, ADM+18, BCPV19, BEG+24]. This requires substantial *anti-concentration* machinery even for powers of smoothed linear forms, and for powers of smoothed polynomials [BCPV19, BEG+24]. Polynomial multipliers introduce additional dependencies: the same random coefficients of each generator are reused across many columns of the Macaulay matrix. When Koszul relations are present, they also create an unavoidable high-dimensional kernel $\mathcal{K}_{\mathbf{p}}^{(t)}$ that varies with the perturbed generators.

Our goal is to build a quantitative theory of approximate syzygies that connects the algebraic structure of exact syzygies, with stability in the smoothed analysis setting. Developing this theory requires a rich interplay of ideas from the two distinct areas of algebraic geometry and high-dimensional probability.

1.1 Main Result: Approximate Syzygies

Our main result proves that for powers of smoothed polynomials, the only syzygies are the "trivial" Koszul syzygies, in the following robust sense.

Informal Theorem 1.1 (Theorem 4.1, informal)**.** *Let $a_1, \dots, a_m$ be ρ-perturbed homogeneous polynomials of degrees $d_1, \dots, d_m$ in $\mathbf{x} = (x_1, \dots, x_n)$, with centers $\bar{a}_1, \dots, \bar{a}_m$ of norm at most 1. Let $b_1, \dots, b_m$ be polynomials such that $b_i = a_i^{D_i}$, for integers $D_i \geq 1$. Let $d_{\min} = \min_i d_i$, and let m, n satisfy $\sum_{i\in[m]} n^{-d_i D_i} \leq n^{-\varepsilon}$ for a small, fixed $\varepsilon > 0$. Then for large enough n and a constant t, with probability $1 - \exp(-\Omega(n^{d_{\min}}))$ we have*

$$\left\| \sum_{i\in[m]} b_i(\mathbf{x}) h_i(\mathbf{x}) \right\| \geq \frac{\|\mathbf{h}\|}{\mathrm{poly}(n, 1/\rho)}, \tag{3}$$

simultaneously for all $\mathbf{h} \perp \mathcal{K}_{\mathbf{b}}^{(t)}$.
In particular, for $d_i = d$, $D_i = D$, the above holds true whenever $m \leq n^{dD-\varepsilon}$. Moreover when $t \leq 2dD - 1$, $\mathcal{K}_{\mathbf{b}}^{(t)} = \{0\}$, so (3) holds for all $\mathbf{h}$.

In the above theorem, recall that $\mathcal{K}_{\mathbf{b}}^{(t)}$ is the set of Koszul syzygies, defined as[3]

$$\mathcal{K}_{\mathbf{b}}^{(t)} = \left\{ \big(h_1(\mathbf{x}), \dots, h_m(\mathbf{x})\big) : h_i(\mathbf{x}) = \sum_{j\in[m]} \Omega_{ij}(\mathbf{x}) b_j(\mathbf{x}), \Omega_{ij}(\mathbf{x}) = -\Omega_{ji}(\mathbf{x}) \in \mathbb{R}[\mathbf{x}]^{=t-d_iD_i-d_jD_j} \right\}$$

We remark that in the above theorem, the condition on $\|\bar{a}_i\| \leq 1$ is just a convenient scaling. The same conclusions hold when the norms of $\bar{a}_i$ are polynomially bounded. The theorem is stated with different degrees d_i, D_i, but with the same total degree t. This is also without loss of generality, since the theorem can be applied for each $t' \leq t$ to get a similar conclusion.

[3] By convention, we assume that $\mathbb{R}[\mathbf{x}]^{=0} = \mathbb{R}$ and $\mathbb{R}[\mathbf{x}]^{=d} = \{0\}$ for all $d < 0$. This allows for $\mathcal{K}_{\mathbf{b}}^{(t)}$ to be empty for certain values of t. For example, if b_1, b_2 are quadratics, then $\mathcal{K}_{\mathbf{b}}^{(3)} = \{0\}$, whereas $\mathcal{K}_{\mathbf{b}}^{(4)} = \{(h_1, h_2) : h_1 = \alpha b_2, h_2 = -\alpha b_1, \alpha \in \mathbb{R}\}$

Our result is interesting even in the setting where $\deg(a_i) = d, \deg(b_i) = dD$ for each $i \in [m]$. Theorem 1.1 shows that the approximate syzygies of $b_1, \ldots, b_m$ are close to the unavoidable Koszul syzygies. In other words, if $\left\|\sum_{i\in[m]} b_i(\mathbf{x})h_i(\mathbf{x})\right\| \leq \eta$, then

$$\mathsf{dist}(\mathbf{h}, \mathcal{K}^{(t)}_{\mathbf{b}}) \leq \eta \cdot \mathrm{poly}(n, \rho^{-1}).$$

To illustrate the result, consider the setting of squares of smoothed quadratics i.e., $d = 2, D = 2$, with $t = 8$. The generators are $b_i = a_i^2$, where a_i are smoothed quadratics, and the multipliers have degree $t - dD = 4$. In this setting the Koszul syzygy corresponds to the basic commutativity relations given by $h_i(\mathbf{x}) = b_j(\mathbf{x}), h_j(\mathbf{x}) = -b_i(\mathbf{x})$. The above theorem shows that these are the only syzygies. Moreover, every approximate syzygy also needs to lie close to one of these Koszul syzygies with high probability (over the random perturbations in a_i).

The result holds when the number of generators is $m \leq n^{dD-\varepsilon}$, which is near-optimal. For $m > \binom{n+dD-1}{dD}$, the polynomials are linearly dependent, and we cannot hope for such a claim even for $t = \deg b_i$. Hence, the bound $m \lesssim n^{dD}$ is the optimal number of generators for which a result like this could hold. We remark that our bounds remain the same for any choices of d', D' whose product is equal to dD; this includes the choice of $d' = dD, D' = 1$. Hence in the spirit of Nicklasson's conjecture [Nic17], our result shows for approximate syzygies the Dth powers of smoothed degree-d polynomials essentially behave like smoothed polynomials of degree Dd up to the loss of $n^{-\varepsilon}$ in m.

Theorem 1.1 can equivalently be viewed as a singular value lower bound for the (transposed) *Macaulay matrix*, a basic object in computational algebra [CLO05]. For homogeneous polynomials $\mathbf{p} = (p_1, \ldots, p_m)$, and an integer t, define the multiplication map $\mathsf{M}^{(t)}_{\mathbf{p}}$ as

$$\mathsf{M}^{(t)}_{\mathbf{p}} : \bigoplus_{i=1}^{m} \mathbb{R}[\mathbf{x}]^{=t-\deg p_i} \to \mathbb{R}[\mathbf{x}]^{=t}, \qquad \mathbf{h} \mapsto \sum_{i=1}^{m} p_i h_i.$$

The Macaulay matrix is the matrix representation of the map $\mathsf{M}^{(t)}_{\mathbf{p}}$, in the orthogonal bases corresponding to the Bombieri inner product (see Section 2.1). The image of the map $\mathsf{M}^{(t)}_{\mathbf{p}}$ corresponds to the degree-t part of the ideal generated by the polynomials $p_1, \ldots, p_m$, and the kernel of $\mathsf{M}^{(t)}_{\mathbf{p}}$ corresponds to the degree-t syzygy space.

Theorem 1.1 shows that, when the polynomials $b_1, \ldots, b_m$ are powers of smoothed polynomials, the kernel of $\mathsf{M}^{(t)}_{\mathbf{b}}$ consists of only the Koszul syzygies. Furthermore, it shows a least singular value lower bound on $\mathsf{M}^{(t)}_{\mathbf{b}}$, restricted to the orthogonal complement of the Koszul syzygy space. Namely, under the conditions of Theorem 1.1, we have:

$$\ker \mathsf{M}^{(t)}_{\mathbf{b}} = \mathcal{K}^{(t)}_{\mathbf{b}}, \text{ and } \sigma_{\min}\left(\mathsf{M}^{(t)}_{\mathbf{b}}\Big|_{(\mathcal{K}^{(t)}_{\mathbf{b}})^{\perp}}\right) \geq \frac{1}{\mathrm{poly}(n, 1/\rho)}.$$

Viewed in this way, our result adds to the literature for least singular value lower bounds in smoothed analysis [BCPV19, BEG+24, BCMV14, ADM+18]. Furthermore, this singular value lower bound for the Macaulay matrix directly gives us robust guarantees for the ideal membership problem, in the setting where the ideal is generated by a collection of smoothed polynomials. See Section 7 for more details.

Next, we present some more applications of our main result.

1.2 Power Sum Decomposition: Smoothed Polynomial Time Guarantees

We consider the algorithmic problem of decomposing "power-sum" polynomials. Our object of interest is the *power-sum* polynomial:

$$p(\mathbf{x}) = \sum_{i=1}^{m} a_i(\mathbf{x})^{3D},$$

where the underlying $a_i(\mathbf{x})$ are homogeneous degree-d polynomials, and $d, D \geq 1$ are given integers.

The goal is to design a polynomial time algorithm that recovers the base polynomials $\{a_i(\mathbf{x}) : i \in [m]\}$ from $p(\mathbf{x})$. In fact, we would like to be able to recover estimates of the $a_i(\mathbf{x})$, even when we are only given access to $\widehat{p}(\mathbf{x})$ which is an estimate of $p(\mathbf{x})$. This is motivated by applications in learning theory and high-dimensional statistics, where $p(\mathbf{x})$ corresponds to a moment of a distribution, which we can estimate up to inverse polynomial error (with polynomial samples), but we cannot compute exactly. In particular, the problem of estimating the parameters of a mixture of non-spherical zero-mean Gaussians reduces to the power-sum decomposition problem with $d = 2$ using the degree-$6D$ moments of the distribution.

For this to be tractable, we require a non-degeneracy condition on the $a_i(\mathbf{x})$. The work of Bafna, Hsieh, Kothari and Xu designs polynomial time algorithms in the case where the $a_i(\mathbf{x})$ are random i.e., the coefficients are random mean-0 Gaussians [BHKX22]. In that setting they are able to establish that the algorithm is robust to noise in $\widehat{p}(\mathbf{x})$ of magnitude $\mathrm{poly}(1/n)$. They leave the question of obtaining polynomial time guarantees for constant d, D in the more general *smoothed analysis* setting as an open problem.

Our main result resolves this open question by obtaining polynomial time smoothed analysis guarantees. Precisely, we assume that the $a_i(\mathbf{x})$ are ρ-smoothed, for some parameter $\rho \in (0, 1]$. That is, for each $i \in [m]$:

$$a_i(\mathbf{x}) = \bar{a}_i(\mathbf{x}) + \zeta_i(\mathbf{x}),$$

where $\bar{a}_i(\mathbf{x})$ are arbitrary polynomials of norm at most 1, and the $\zeta_i(\mathbf{x})$ have coefficients drawn independently from $\mathcal{N}(0, \rho^2)$. Under this smoothed analysis setting we are able to design an algorithm that can recover estimates for $a_i(\mathbf{x})$ from an estimate of $p(\mathbf{x})$, that is robust to noise in $\widehat{p}(x)$ of magnitude $\mathrm{poly}(1/n, \rho)$.

Theorem 1.2 (Power-sum decomposition (Same as Theorem 5.1))**.** *For constant degrees d, D, and $m = O(n^{2dD/(3d-2)})$, suppose $a_1(\mathbf{x}), \ldots, a_m(\mathbf{x}) \in \mathbb{R}[\mathbf{x}]^{=d}$ are ρ-smoothed homogeneous degree-d polynomials. For any $\varepsilon > 0$, given as input an estimate $\widehat{p}(\mathbf{x})$ of $p(\mathbf{x}) = \sum_{i=1}^{m} a_i(\mathbf{x})^{3D} \in \mathbb{R}[\mathbf{x}]^{=3dD}$ such that $\|\widehat{p}(\mathbf{x}) - p(\mathbf{x})\| < \varepsilon$, with high probability we can recover in polynomial time estimates $\{\widetilde{a}_i(\mathbf{x}) : i \in [m]\}$ (up to relabeling terms) such that $\|\widetilde{a}_i - a_i\| \leq \mathrm{poly}(\varepsilon, n, 1/\rho)$ for all $i \in [m]$.*

We follow the same algorithm as in [BHKX22], who design and analyze the algorithm for the setting where $a_i(\mathbf{x})$ are random. A key step in their algorithm and analysis is to robustly estimate the span of the $a_i(\mathbf{x})^D$. Our robust syzygy claim is the main technical ingredient that allows us to analyze this step and establish a smoothed analysis guarantee for this algorithm and problem. This is discussed in more detail in the technical overview Section 2.2.

1.3 Robust Uniqueness of Tensor Decompositions

The next application of our approximate syzygy bounds is to the surprising fact of *uniqueness* for tensor decompositions of order 3. Given a symmetric tensor T represented as a polynomial $T(\mathbf{x})$,

the Waring rank or symmetric rank is defined as the smallest integer m for which we can write $T(\mathbf{x}) = \sum_{i=1}^{m} \ell_i(\mathbf{x})^3$, where each ℓ_i is a linear form (in the variables $\mathbf{x}$). Given a homogeneous cubic polynomial $T(\mathbf{x})$ and a decomposition $\mathbf{a} = (a_1, a_2, \ldots, a_m)$ such that $T(\mathbf{x}) = \sum_{i=1}^{m} a_i(\mathbf{x})^3$, we say that $\mathbf{a}$ is the unique rank-m decomposition of T if for all other decompositions $\widehat{\mathbf{a}} = (\widehat{a}_1, \ldots, \widehat{a}_m)$, we have that $\sum_i a_i(\mathbf{x})^3 = \sum_i \widehat{a}_i(\mathbf{x})^3$ implies that there is a permutation (relabeling) of the indices such that $a_i(\mathbf{x}) = \widehat{a}_i(\mathbf{x})$ for all $i \in [m]$.

In applications to learning and numerical analysis, tensors are usually estimated from data, and are thus known only up to a small error. In many of these settings (e.g., in tensors estimated via empirical moments; see [AGH+14]), the error can be driven down, at the expense of sample complexity. In particular, using N data samples, we can only estimate tensor entries up to an error $\approx N^{-1/2}$. Thus to conclude approximate *identifiability* of the components,[4] we need a more robust notion of uniqueness. We say a decomposition $\mathbf{a} = (a_1, \ldots, a_m)$ of a tensor T is *unique with polynomial robustness* if the following holds: for any tuple of linear functions $\widehat{\mathbf{a}} = (\widehat{a}_1, \ldots, \widehat{a}_m)$ which have polynomially bounded norms, and with $\left\| \sum_{i=1}^{m} a_i(\mathbf{x})^3 - \sum_{i=1}^{m} \widehat{a}_i(\mathbf{x})^3 \right\| \leq \varepsilon$, there is a permutation (relabeling) of the indices $[m]$ such that $\|a_i(\mathbf{x}) - \widehat{a}_i(\mathbf{x})\| \leq \varepsilon \cdot \mathrm{poly}(n, 1/\rho)$.

Unlike matrices, tensors of small rank *typically* satisfy the unique decomposition property. The classic result of Kruskal [Kru77] showed that for *generic* tensors of rank $m < \lfloor 3n/2 \rfloor$, the decomposition is unique. The work of [BCV14] extended this result by showing that under a smoothed analysis model, tensors of rank $m < \lfloor 3n/2 \rfloor$ satisfy the robust uniqueness property (defined above), with polynomial robustness. This was shown to have applications to machine learning, specifically to the recoverability of parameters in latent variable models.

It is natural to ask if the bound $m < \lfloor 3n/2 \rfloor$ can be improved. Following the work of Strassen [Str83], Chiantini, Ottaviani, and Vannieuwenhoven [CO12, COV17] showed that uniqueness of decomposition extends to a much larger range of m. Using arguments from algebraic geometry, specifically Terracini's lemma about tangent spaces of secant varieties, they showed that the unique decomposition property holds as long as $m < n^2/6$. However, their method of proof does not yield a robust analog.

This raises the following natural question (see e.g., the open problems and discussion in [BCV14, Vij20]): *For what range of m do smoothed rank-m tensors satisfy robust uniqueness?*

The following theorem shows robust uniqueness of smoothed tensors of superlinear rank, even up to $n^{3/2}$.

Theorem 1.3 (Robust uniqueness of tensor decompositions (Same as Theorem 6.3))**.** *Suppose $m = O(n^{3/2})$ and $\rho \in (0, 1]$, and let $\mathbf{a} = (a_1(\mathbf{x}), \ldots, a_m(\mathbf{x}))$ be an m-tuple of ρ-perturbed linear homogeneous polynomials. For any $\varepsilon > 0$, there exist constants c, C_0 such that with probability at least $1 - \exp(-cn)$, the decomposition $\mathbf{a}$ is robustly unique with polynomial robustness.*

This result significantly improves on previously known results of this nature [BCV14], from $m < \lfloor 3n/2 \rfloor$ to $m < O(n^{3/2})$. Our robust uniqueness theorem uses our quantitative bounds on approximate syzygies as a key ingredient. In fact, Theorem 1.1 applied with squares of linear forms can be used to give condition number bounds for the Terracini map, and prove a robust local uniqueness result up to $m \leq n^{2-\varepsilon}$. The final theorem provides a robust global uniqueness result by combining it with a global compression argument inspired by a result of Kane [Kan17]. For more details, refer to the proof overview in Section 2.3.

[4] Identifiability is an information theoretical notion that says that given T, we can uniquely recover the decomposition $\mathbf{a}$. The notion does not require an efficient algorithm for recovery.

It is natural to ask if polynomial robustness can be improved all the way to $m = O(n^{2-\varepsilon})$ rank. Indeed, when prompted with our work, ChatGPT Astra was able to obtain such a result, using a much more intricate version of the compression scheme we present below.[5] However, the two proofs have the same overall structure, and more relevantly for us, the same use of the approximate syzygy theorem, so we present only the proof until rank $n^{3/2}$.

1.4 Comparison to Related work

In addition to the related works we mention in the introduction, the problems we study in this work have connections to many existing topics in algebraic geometry, high-dimensional probability, arithmetic circuits and theoretical machine learning. The study of approximate syzygies is of interest in numerical algebraic geometry. For example, the works of [BDD14] and [Lic19] relate stability of computational algebraic geometry problems to condition number bounds for the corresponding Macaulay matrix. In particular, ideal membership is a fundamental problem in computational algebra [CLO15] and a central motivation for Gröbner basis methods, including Buchberger's algorithm [Buc65] and Faugère's F4 and F5 [Fau99, Fau02]. Our work provides the necessary condition number bounds in the smoothed analysis framework.

To the best of our knowledge, we are the first to study the characterization of approximate syzygies in the smoothed analysis setting, and their applications. The two most closely related works are the work of Bafna, Hsieh, Kothari and Xu [BHKX22], and the previous work of [BEG+24]. We give more detailed comparisons below.

Comparison with [BHKX22] Building on [GKS20], [BHKX22] develop an algorithm for power-sum decomposition of generic polynomials, with inverse-polynomial noise tolerance in the *random* setting, where the underlying polynomials have independent Gaussian coefficients. Their algorithm follows the "desymmetrize+tensor decompose" method outlined in [GHK15]. A crucial step in their algorithm involves analyzing the syzygy space of powers of random polynomials. In particular, for decomposing a sum of $3D$-th powers of degree-d polynomials, this step involves analyzing the degree-$(3d-2)D$ syzygy space of $(a_1^D, a_2^D, \ldots, a_m^D, g)$ where $a_1, \ldots, a_m$ are random degree-d homogeneous polynomials, and g is a random degree-$2(d-1)D$ homogeneous polynomial. They show that, for a particular range of parameters, the syzygy space consists of only the Koszul syzygies, and also provide robustness guarantees by showing singular value lower bounds for relevant matrices. For the setting $D \leq 2$ e.g., squares of degree-d polynomials ($D = 2$), they handle $m \leq n^{Dd}/\operatorname{poly}\log(n)$, which is better than our bound by the n^{ε} factor. For general constant D (e.g., cubes of polynomials), they handle $m \leq n^{dD/2}/\operatorname{poly}\log(n)$ which is worse than our $m \leq n^{dD-\varepsilon}$ bound. To prove a lower bound on the least singular value of a random matrix M, they show concentration (upper) bounds on the largest singular value of the deviation matrix $M - \mathbb{E}[M]$, and use the fact that $\sigma_{\min}(M) \geq \sigma_{\min}(\mathbb{E}[M]) - \sigma_{\max}(M - \mathbb{E}[M])$. To bound $\sigma_{\max}(M - \mathbb{E}[M])$, they use techniques from the theory of graph matrices [AMP16, MRX20, JPR+22].

The main technical distinction between our works is that our techniques work in the more general smoothed analysis setting. While the concentration techniques work in the random setting, they seem challenging to extend to the more general smoothed setting. Our techniques on the other hand are based more on anticoncentration. Our result applies when the polynomials $b_1, \ldots, b_m$ are given by $b_i = a_i^{D_i}$, where each a_i is a *smoothed* polynomial of degree d_i; both d_i and D_i may depend on $i \in [m]$. This can be viewed as a generalization of the result of [BHKX22] to the smoothed

[5] In the final version, we will outline details of this improvement in an appendix.

setting, and also handles much more general cases where the underlying polynomials have different degrees and are raised to different powers.

Comparison with [BEG+24]. The work [BEG+24] develops new techniques for showing least singular value lower bounds for random matrices arising in smoothed analysis applications. One of the applications of their techniques is to a specific setting of the power-sum decomposition algorithm of [BHKX22] in the smoothed setting, for cubes of quadratics i.e., $d = 2, D = 3$. One of the steps in the argument involved a degree-4 syzygy claim for smoothed quadratics, which turned out to be incorrect. Theorem 4.1 provides, as a special case, the degree-4 syzygies result for smoothed quadratics, but with a loss of $n^{-\varepsilon}$ in m. We also provide a full smoothed analysis of the power-sum decomposition problem. The analysis of our algorithm still uses the bounds on the least singular values of linear operators applied to Kronecker products of smoothed matrices, or lifts of smoothed subspaces proven in [BEG+24]. See Section 5 for further details.

2 Technical Overview

Now we give an overview of each of our proofs. For exposition, we will consider special cases of our main results and introduce some informal definitions.

2.1 Approximate Syzygies of Powers of Smoothed Polynomials

Here, we present an outline of our proof for the setting of degree-4 syzygies of squares of linear forms. Let $a_1, \ldots, a_m$ be arbitrary linear forms, each independently perturbed by a small Gaussian vector. Let $b_i = a_i^2$ for each $i \in [m]$, and we use boldface to represent vectors of polynomials, e.g., $\mathbf{b} = (b_1, \ldots, b_m)$.

Our main result concerns the kernel of the operator $\mathsf{M}_{\mathbf{b}}^{(4)}$, the *multiplication map* of $\mathbf{b}$ at degree 4:

$$\mathsf{M}_{\mathbf{b}}^{(4)} : \bigoplus_{i\in[m]} \mathbb{R}[\mathbf{x}]^{=2} \to \mathbb{R}[\mathbf{x}]^{=4}, \qquad \mathsf{M}_{\mathbf{b}}^{(4)}(h_1, \ldots, h_m) = \sum_{i\in[m]} b_i(\mathbf{x})h_i(\mathbf{x}).$$

For a concrete example, take $n = m = 2$ and write $a_i(y, z) = u_i y + v_i z$. In ordinary monomial bases[6], its matrix representation is the transposed Macaulay matrix:

$$\mathsf{M}_{\mathbf{b}}^{(4)} = \begin{array}{c} \\ y^4 \\ y^3z \\ y^2z^2 \\ yz^3 \\ z^4 \end{array} \begin{array}{c} \overbrace{\begin{array}{ccc} y^2 & yz & z^2 \end{array}}^{h_1} \quad \overbrace{\begin{array}{ccc} y^2 & yz & z^2 \end{array}}^{h_2} \\ \left[\begin{array}{ccc|ccc} u_1^2 & 0 & 0 & u_2^2 & 0 & 0 \\ 2u_1v_1 & u_1^2 & 0 & 2u_2v_2 & u_2^2 & 0 \\ v_1^2 & 2u_1v_1 & u_1^2 & v_2^2 & 2u_2v_2 & u_2^2 \\ 0 & v_1^2 & 2u_1v_1 & 0 & v_2^2 & 2u_2v_2 \\ 0 & 0 & v_1^2 & 0 & 0 & v_2^2 \end{array}\right] \end{array}$$

The kernel of this linear map is the degree-4 syzygy space of $\mathbf{b}$. Since both the generators b_i's and coefficients h_i's are quadratic, the space of Koszul syzygies consists of (the span) of relations of the form $h_1 = -b_2$ and $h_2 = b_1$, with other h_i set to 0. In the example above, we have $\mathsf{M}_{\mathbf{b}}^{(4)}\mathbf{h} = 0$ if and only if $\mathbf{h}$ is a multiple of $(-u_2^2, -2u_2v_2, -v_2^2, u_1^2, 2u_1v_1, v_1^2)^\top$.

[6] In our proofs, we will instead work with the Bombieri-normalized monomial basis, which replaces yz with $\sqrt{2}yz$. Here we use ordinary monomial bases for exposition.

For $m \le n^{2-\varepsilon}$ with fixed $\varepsilon > 0$, our main theorem shows that, with high probability, these are the *only* degree-4 syzygies, in a robust sense:

$$\ker \mathsf{M}_{\mathbf{b}}^{(4)} \;=\; \mathcal{K}_{\mathbf{b}}^{(4)}, \qquad \text{and for } \mathbf{h} \perp \mathcal{K}_{\mathbf{b}}^{(4)}, \quad \left\|\mathsf{M}_{\mathbf{b}}^{(4)}\mathbf{h}\right\| \;\ge\; n^{-C}\rho^{C}\|\mathbf{h}\|.$$

The challenge. Compared to prior analysis of robust non-singularity of smoothed matrices [BCPV19, BEG+24], this matrix poses several new and significant challenges. The columns are highly interdependent: each of the first three columns exposes the randomness in the other two columns in that same block. Moreover, across the blocks, the algebraic relations produce a known null space $\mathcal{K}_{\mathbf{b}}^{(4)}$, which itself depends on the *same* random variables u_i, v_i. We therefore cannot simply restrict the input to the orthogonal complement of this space. These intra- and inter-block dependencies also make it hard to apply a standard leave-one-out argument, which usually asserts that certain columns generate sufficient magnitude from the randomness, even when projecting out the span of the other columns.

To work around these difficulties, we make use of the polynomial structure and apply a more carefully designed "project out" operation.

Variable separation and $\mathcal{S}^{\perp}$-gradation. We partition the variables as follows. Let $\mathcal{S} \subseteq \mathbb{R}^n$ be a coordinate subspace, so that $\mathbb{R}^n = \mathcal{S} \oplus \mathcal{S}^{\perp}$ corresponds to a partition of the variables $\mathbf{x} = (x_1, \ldots, x_n)$ into $\mathbf{x} = (\mathbf{y}, \mathbf{z})$. For a homogeneous polynomial $p \in \mathbb{R}[\mathbf{x}]^{=d}$, we use

$$p[r] \;\in\; \mathbb{R}[\mathbf{y}]^{=d-r} \cdot \mathbb{R}[\mathbf{z}]^{=r}$$

to denote its terms with degree exactly $r \in \{0, \ldots, d\}$ on $\mathbf{z}$ variables. .

The blocks $p[r]$ capture the normal Taylor layers of p along $\mathcal{S}$ (i.e., differentiate with respect to directions in $\mathcal{S}^{\perp}$, but evaluated at points in $\mathcal{S}$). Their use in our inductive argument is in the spirit of the differential Horace method of Alexander and Hirschowitz [AH95, BO08]. While the usual argument considers hyperplanes, leaving only one normal direction ($\dim(\mathcal{S}^{\perp}) = 1$), here we choose both $\dim(\mathcal{S}), \dim(\mathcal{S}^{\perp}) = \Omega(n)$, so the normal block exposes enough fresh randomness for our quantitative anti-concentration arguments, while at the same time allowing us to carry out the induction.

The starting point of our argument is the observation that an anti-concentration result is much easier if in $\sum_i b_i(\mathbf{x})h_i(\mathbf{x})$, the polynomials b_i and coefficients h_i are supported only on disjoint sets of variables $\mathbf{y}, \mathbf{z}$. In that case,

$$\sum_{i=1}^{m} b_i(\mathbf{0}, \mathbf{z})h_i(\mathbf{y}, \mathbf{0}) \;=\; \sum_{i=1}^{m} b_i[2]\, h_i[0] \;\approx\; 0.$$

Since the product of homogeneous polynomials on different variables is essentially a tensor product[7], we now have to argue about the robust linear independence of the $b_i[2]$'s (which are squares of perturbed linear forms in $\mathbf{z}$ i.e. $a_i[1]$'s). We can hence use existing least singular value bounds for smoothed matrices [BCPV19] and directly get that $\|\sum_i b_i[2]\, h_i[0]\| \gtrsim \|h_i[0]\|$, where $\gtrsim$ hides a $\mathrm{poly}(n/\rho)$ factor.

Much of the technical work in our proof lies in reducing the problem on shared variables $\mathbf{x}$ to a variant of the setting above. We now present how we do this reduction. Let $f = \sum_i b_i h_i$. Observe

[7] There is a unitary isomorphism that maps the polynomial products to their tensor products after a normalization.

that $f[2]$, the part of f with $\mathbf{z}$-degree 2, includes the part $\sum_i b_i[2]\, h_i[0]$. However, it is not the only set of terms present, as there are 3 ways one can obtain a polynomial with $\mathbf{z}$-degree of 2:

$$f[2] \;=\; \sum_{i=1}^{m} \Big(b_i[2]\, h_i[0] \;+\; b_i[1]\, h_i[1] \;+\; b_i[0]\, h_i[2] \Big).$$

Let $a_i(\mathbf{x}) = u_i(\mathbf{y}) + v_i(\mathbf{z})$, so that $b_i[2] = v_i^2$. We want to isolate the contribution of $v_i^2 \cdot h_i[0]$ in the expression for $f[2]$: if we can argue that the contribution from this term is never canceled by the other terms whenever $\mathbf{h}$ is far from $\mathcal{K}_{\mathbf{b}}^{(4)}$, then we are done.

However, while $\sum_{i=1}^m v_i^2\, h_i[0]$ has anti-concentration properties, it is *impossible* that $\|f[2]\| \gtrsim \|\mathbf{h}[0]\|$ for all $\mathbf{h}[0]$ simultaneously, due to the existence of Koszul syzygies. For example, when $h_1 = -b_2$ and $h_2 = b_1$, the first and third terms in $f[2]$ will simply cancel. So although $\sum_i v_i^2\, h_i[0]$ might have large magnitude for all $\mathbf{h}$ with high probability, it does not lead to the same result for $f[2]$.

The key to our proof lies in identifying a workaround to this problem: instead of trying to argue that $\|f[2]\| \gtrsim \|\mathbf{h}[0]\|\ \forall \mathbf{h}$, we instead identify a suitable subspace $\mathcal{J} \subseteq \bigoplus_{i=1}^m \mathbb{R}[\mathbf{y}]^{=2}$ and consider the projection of $\mathbf{h}$ orthogonal to $\mathcal{J}$, which we denote by $\mathbf{g} := \Pi_{\mathcal{J}}^{\perp}\, \mathbf{h}[0]$. We would like $\mathbf{g}$ to possess the following two desirable properties:

1. With high probability, $\|f\| \gtrsim \|\mathbf{g}\|$ for all $\mathbf{h}$.
2. When $\|f\| \approx 0$, $\mathbf{g}$ has large magnitude whenever $\mathbf{h}$ is far from $\mathcal{K}_{\mathbf{b}}^{(4)}$.

Indeed, our proof would be complete if we identify such a $\mathcal{J}$.

We show that $\mathcal{J}$ can be taken to be the following space:

$$\mathcal{J} \;=\; \bigoplus_{i=1}^{m} \mathcal{J}_i, \quad \text{where} \quad \mathcal{J}_i \;=\; \operatorname{span}\{u_i \cdot \ell_i(\mathbf{y}) : \ell_i \in \mathbb{R}[\mathbf{y}]^{=1}\} \;+\; \operatorname{span}\big\{u_j^2 : j \in [m],\ j \neq i\big\},$$

We first show why $\mathcal{J}_i$ is a natural subspace to project out from $h_i[0]$ when trying to isolate the contribution of $v_i^2\, h_i[0]$ from the output block $f[2]$.

Isolating a single $v_i^2 \cdot g_i$. Since u_i is linear in $\mathbf{y}$ and v_i is linear in $\mathbf{z}$, for a fixed i we can expand $f[2]$ explicitly:

$$f[2] \;=\; v_i^2 \cdot h_i[0] \;+\; \underbrace{\sum_{j \in [m], j \neq i} v_j^2 \cdot h_j[0] \;+\; \sum_{j=1}^{m} 2u_j v_j \cdot h_j[1] \;+\; \sum_{j=1}^{m} u_j^2 \cdot h_j[2]}_{\text{nuisance}}. \tag{4}$$

Now, note that a lot of the "nuisance" terms in the above expression have parts contained in $\mathcal{J}_i$. In particular, the term $u_i v_i \cdot h_i[1]$ and the term $\sum_{j=1}^m u_j^2 \cdot h_j[2]$ lie in $\mathcal{J}_i \otimes \mathbb{R}[\mathbf{z}]^{=2}$.

Hence, if we project out $\mathcal{J}_i \otimes \mathbb{R}[\mathbf{z}]^{=2}$ from equation 4, we get

$$(\Pi_{\mathcal{J}_i}^{\perp} \otimes I)(f[2]) \;=\; v_i^2 \cdot g_i \;+\; \underbrace{(\Pi_{\mathcal{J}_i}^{\perp} \otimes I)\Big(\sum_{j \in [m], j \neq i} v_j^2 \cdot h_j[0] \;+\; \sum_{j \in [m], j \neq i} 2u_j v_j \cdot h_j[1] \Big)}_{\text{remaining nuisance}}. \tag{5}$$

Finally, note that the subspace $\mathcal{J}$ is defined in a way that it is independent of v_i. This allows us to condition on $\mathcal{J}_i$ and the remaining nuisance terms in equation (5), while still retaining the "fresh

randomness" in v_i. This allows us to bound $f[2]$ in terms of $\|\mathbf{g}\|$ (up to polynomial terms). We next describe this step formally.

Now we outline the proof for the two "desirable properties" of $\mathbf{g}$. We will assume $\|f\| \approx 0$ for convenience.

A substitution for proving $\|f\| \gtrsim \|\mathbf{g}\|$. As outlined above, after projection, the $f[2]$ block produces Equation (5), and we now need to ensure the remaining nuisance terms do not cancel $v_i^2 \cdot g_i$.

Assume that $\mathbf{h}$ is an approximate syzygy so that $\|f\| \approx 0$. The first two blocks of f give:

$$0 \approx f[0] = \sum_{i=1}^{m} u_i^2 \cdot h_i[0], \tag{6}$$

$$0 \approx f[1] = \sum_{i=1}^{m} 2u_i v_i \cdot h_i[0] + \sum_{i=1}^{m} u_i^2 \cdot h_i[1]. \tag{7}$$

By the induction hypothesis, we may assume that the degree-3 syzygies of squares of smoothed linear forms consist of only the Koszul syzygies, which in this case correspond to empty syzygies. To make use of this fact, we differentiate both sides of Equation (6) to reduce the total output degree by 1. Using the inductive hypothesis and Euler's identity for homogeneous polynomials, we can "reconstruct" $\mathbf{h}[0]$ up to some terms in the Koszul syzygy i.e. we can write $\mathbf{h}[0] \approx G_0(\mathbf{g}) + \mathbf{k}[0]$, where G_0 is a linear function with $\mathrm{poly}(n/\rho)$-bounded norm, and $\mathbf{k}[0] \in \mathcal{K}^{(4)}_{\mathbf{b}[0]}[0]$ is in the Koszul syzygy space of $\mathbf{b}[0]$, on the $\mathbf{y}$-polynomials:

$$\mathcal{K}^{(4)}_{\mathbf{b}[0]} \;:=\; \Big\{ \Omega\, \mathbf{b}[0] \;:\; \Omega_{ij} = -\Omega_{ji} \in \mathbb{R}[\mathbf{y}]^{=0} \Big\}.$$

Intuitively, $\|f\| \approx 0$ greatly constrains the behavior of $\mathbf{h}$ (hence reconstruction is possible), but this constraint is only modulo the Koszul syzygy. To see why $\mathbf{k}[0]$ is necessary, note that when $\mathbf{h} \in \mathcal{K}^{(t)}_{\mathbf{b}}$, we have $\mathbf{g} = 0$, so $\mathbf{h}[0]$ itself cannot be written as a linear map on $\mathbf{g}$.

Now, we move to the equation corresponding to $\mathbf{z}$-degree 1. We have $f[1] \approx 0$, and we can substitute the $h_i[0]$ obtained above into Equation (7) to get

$$\sum_{i=1}^{m} u_i^2 \cdot h_i[1] \;\approx\; -\sum_{i=1}^{m} 2u_i v_i \cdot \big(\mathsf{G}_0(\mathbf{g})_i + k_i[0]\big).$$

Again, by the inductive hypothesis, the mapping $\mathbf{h}[1] \mapsto \sum_i u_i^2 \cdot h_i[1]$, denoted by $\mathsf{M}^{(3)}_{\mathbf{b}[0]}$ is (robustly) injective. We can hence apply the inverse operator of $\mathsf{M}^{(3)}_{\mathbf{b}[0]}$ to both sides and obtain:

$$\mathbf{h}[1] \;\approx\; \mathsf{G}_1(\mathbf{g}) \;+\; \mathbf{k}[1],$$

where G_1 is a linear function with $\mathrm{poly}(n/\rho)$-bounded norm, and $\mathbf{k}[1]$ is a block in the same $\mathbf{k}$.

Finally, we can substitute *both* $\mathbf{h}[0]$ and $\mathbf{h}[1]$ into Equation (5). The term $\mathbf{k}$ vanishes from the projection $\Pi^{\perp}_{\mathcal{J}_i} \otimes I$ since it is in the Koszul syzygy, and we therefore get:

$$v_i^2 \cdot g_i \;+\; \mathsf{C}_i(\mathbf{g}) \;\approx_f\; 0, \tag{8}$$

where C_i is a linear function with $\mathrm{poly}(n/\rho)$-bounded norm, which may depend on all v_j's for $j \neq i$.

We then show this implies $\|g_i\| \approx 0$ using a technical lemma, which we call successive shrinking of candidate space (Proposition 4.8). Intuitively, we expose randomness in v_i sequentially by another partitioning of the variables. We maintain a candidate space for "bad $\mathbf{g}$'s", which iteratively gets smaller each time we expose fresh randomness. The lemma applies more broadly than prior anticoncentration results (e.g., [BCPV19]), to cases where there are different linear corrections $\mathsf{C}_i(\mathbf{g})$ for each i, with C_i depending on all other v_j's.

Another substitution for $\|\mathbf{g}\| \gtrsim \mathsf{dist}(\mathbf{h}, \mathcal{K}^{(4)}_{\mathbf{b}})$. The final step is to show that $\|\mathbf{g}\|$ faithfully represents the distance of $\mathbf{h}$ to the prescribed Koszul syzygies. Using the previous substitution, we already can express $\mathbf{h}[0] - \mathbf{k}[0]$ and $\mathbf{h}[1] - \mathbf{k}[1]$ as bounded linear functions of $\mathbf{g}$, modulo some Koszul syzygy term $\mathbf{k}$. It now suffices to show that $\mathbf{h}[2]$ is also bounded in terms of $\mathbf{g}$ modulo a Koszul term.

To see this, we use the equation for $f[2]$ Equation (4) and substitute the expressions for $\mathbf{h}[0]$ and $\mathbf{h}[1]$ obtained above. Since $\mathbf{k}$ is a Koszul syzygy, $\sum_i b_i k_i$ vanishes, and we have

$$\left(\sum_{i=1}^{m} b_i(h_i - k_i)\right)[2] = f[2]$$
$$= \sum_{i=1}^{m} \Big(v_i^2 \cdot (h_i[0] - k_i[0]) + 2u_i v_i \cdot (h_i[1] - k_i[1]) + u_i^2 \cdot (h_i[2] - k_i[2])\Big).$$

Substituting the previous expressions and rearranging, we have:

$$\sum_{i=1}^{m} u_i^2 \cdot \big(h_i[2] - k_i[2]\big) \;\approx_f\; -\sum_{i=1}^{m} \Big(2u_i v_i \cdot \mathsf{G}_1(\mathbf{g})_i + v_i^2 \cdot \mathsf{G}_0(\mathbf{g})_i\Big).$$

Again, by inductive hypothesis, the mapping $(c_1, \ldots, c_m) \mapsto \sum_i u_i^2 c_i$ on scalar coefficients is robustly injective. We can hence invert this multiplication map on both sides, and obtain

$$\mathbf{h}[2] \;\approx_f\; \mathsf{G}_2(\mathbf{g}) \;+\; \mathbf{k}[2],$$

where G_2 is a linear function with $\mathrm{poly}(n/\rho)$-bounded norm, and $\mathbf{k}[2]$ is a block in the same $\mathbf{k}$. Combining the three bounded linear maps for the three blocks of $\mathbf{h} - \mathbf{k}$, we get

$$\mathsf{dist}\big(\mathbf{h}, \mathcal{K}^{(4)}_{\mathbf{b}}\big) \;\le\; \|\mathbf{h} - \mathbf{k}\| \;\le\; \mathrm{poly}(n/\rho)\big(\|f\| + \|\mathbf{g}\|\big),$$

as desired.

2.2 Power-Sum Decompositions: Polynomial Time Smoothed Analysis Guarantees

Recall that in power-sum decomposition, we are given a polynomial of the form

$$p(\mathbf{x}) = \sum_{i=1}^{m} a_i(\mathbf{x})^{3D},$$

and our goal is to recover the underlying base polynomials $a_i(\mathbf{x}) \in \mathbb{R}[\mathbf{x}]^{=d}$.

We analyze the algorithm of [BHKX22]. They present and analyze the algorithm for random instances, i.e. where the base polynomials $a_i(\mathbf{x})$ have i.i.d. mean-zero coefficients. In this setting

they establish that the algorithm is tolerant to noise of inverse polynomial magnitude. They conjecture that a similar bound should hold in the smoothed analysis setting. We establish a robust guarantee for this algorithm on smoothed instances using our robust analysis of syzygies.

We can identify $p(\mathbf{x})$ with the tensor

$$\sum_{i=1}^{m} a_i(\mathbf{x})^{\otimes 3D}.$$

In fact, if this was the representation that we had $p(\mathbf{x})$ in, then we could use existing algorithms for robust tensor decomposition to recover the factors $a_i(\mathbf{x})$ from the tensor $\sum_{i=1}^{m} a_i(\mathbf{x})^{\otimes 3D}$.

However, since $p(\mathbf{x})$ is a polynomial over commuting variables, at first, we only have access to the entries of $\sum_{i=1}^{m} a_i(\mathbf{x})^{\otimes 3D}$ up to symmetry. That is, we are given an estimate for

$$\mathsf{Sym}_{d\to 3dD} \sum_{i=1}^{m} a_i(\mathbf{x})^{\otimes 3D}.$$

The algorithm of [BHKX22] proceeds by computing the *desymmetrized* underlying tensor, from the symmetrized polynomial form, using the power sum structure. Then, once the desymmetrized form is recovered, they use robust tensor decomposition to recover the factors.

A key observation is that the symmetrization operator is actually invertible when restricted to the span of the components $a_i(\mathbf{x})$. Thus, if we can recover the span of the $a_i(\mathbf{x})$, we can invert the symmetrization, and apply tensor decomposition.

To estimate the span the algorithm takes partial derivatives of $p(\mathbf{x})$. A $2D$th order partial derivative of $p(\mathbf{x})$ looks like

$$\sum_{i=1}^{m} a_i(\mathbf{x})^D h_i(\mathbf{x}), \quad \text{for some degree-}q\text{ polynomials } h_i(\mathbf{x}),$$

where $q = 2D(d-1)$. Thus, the derivatives live in the span of multiples of the $a_i(\mathbf{x})^D$ terms that we are hoping to recover. We would like to remove the multipliers, but a priori, we do not have much control over the multiplier polynomials $h_i(\mathbf{x})$. To get around this, we restrict the derivatives to a subset $\mathbf{y} \subseteq \mathbf{x}$ of the variables. Writing $b_i(\mathbf{y})$ for $a_i(\mathbf{x})$ restricted to $\mathbf{y}$, we can show that the restricted partials robustly span the degree-$(dD+q)$ *ideal* generated by the $b_i(\mathbf{y})^D$.

To establish this, it helps to view the statement through the lens of syzygies. Let $H_{\mathbf{y}}$ be the matrix of partial derivatives. We want to establish a least singular value lower bound for $H_{\mathbf{y}}$. We factor $H_{\mathbf{y}} = M_{\mathbf{b}^D}^{(dD+q)} R$, where $M_{\mathbf{b}^D}^{(dD+q)} : (\mathbb{R}[\mathbf{y}]^{=q})^{\times m} \to \mathbb{R}[\mathbf{y}]^{=dD+q}$ is the *Macaulay matrix* that takes a collection of multiplier polynomials $h_1(\mathbf{y}), \ldots, h_m(\mathbf{y}) \in \mathbb{R}[\mathbf{y}]^{=q}$ to the corresponding member of the ideal $\sum_{i=1}^{m} b_i(\mathbf{y})^D h_i(\mathbf{y})$. This allows us to bound the least singular values of $M_{\mathbf{b}^D}^{(dD+q)}$ and R separately to achieve the desired bound. That is, it lets us separately argue that we are covering the space of possible multipliers $h_i(\mathbf{y})$, and that those multipliers cover the ideal. The least singular value bound of $M_{\mathbf{b}^D}^{(dD+q)}$ is exactly our robust syzygy statement.

Once we have $H_{\mathbf{y}}$ that spans the ideal generated by the $b_i(\mathbf{y})^D$, we would like to extract the span of $b_i(\mathbf{y})^D$ from the ideal, which spans all degree-$(dD+q)$ multiples. We do this by sampling a random degree-q polynomial $h^\star(\mathbf{y})$, and intersecting the ideal with the span of multiples of $h^\star(\mathbf{y})$. Our syzygy theorem allows us to argue that the only vectors that are close to this intersection

are of the form $b'(\mathbf{y})h^\star(\mathbf{y})$ for $b'(\mathbf{y}) \in \mathrm{span}\{b_i(\mathbf{y})^D\}$. This is by the robust characterization of the Koszul syzygies. Once we have this, we can divide by $h^\star(\mathbf{y})$ to recover the span of $b_i(\mathbf{y})^D$.

The main novel components of our analysis for the smoothed setting are facilitated by the robust syzygy theorem. It is particularly useful in analyzing the partial derivatives of these structured polynomials, which could be applicable in many other settings of learning arithmetic circuits.

2.3 Robust Uniqueness of Tensor Decompositions

We now provide a rough overview of the proof of uniqueness for smoothed symmetric decompositions of tensors up to rank $m \leq O(n^{3/2})$. To recall, a decomposition is given by an m-tuple $\mathbf{a} = (a_1, a_2, \ldots, a_m)$ of linear forms a_i in the variables $\mathbf{x}$. For such a tuple, define

$$\Phi(\mathbf{a}) = \sum_{i=1}^{m} a_i^3,$$

where the RHS is viewed as an element of $\mathbb{R}[\mathbf{x}]^{=3}$. We wish to prove that for a smoothed $\mathbf{a}$, if we have an alternate decomposition $\mathbf{b}$, $\|\Phi(\mathbf{b}) - \Phi(\mathbf{a})\|$ being small implies that $\mathbf{b}$ is close to $\mathbf{a}$, up to a permutation. (See the definition of $d_{\mathrm{perm}}(\mathbf{a}, \mathbf{b})$ in Section 6 for the formal definition.)

The first step in the proof is to show a "local" uniqueness. For this, we look at the Jacobian of the map Φ above, denoted $\mathsf{D}\Phi(\mathbf{a})$. This has a simple form: it takes as input a tuple of linear forms $\mathbf{h} = (h_1, h_2, \ldots, h_m)$ and outputs

$$\mathsf{D}\Phi(\mathbf{a})[\mathbf{h}] = 3\sum_{i=1}^{m} a_i^2 h_i.$$

By definition, it is the direction of $\Phi(\mathbf{a}+\delta\mathbf{h}) - \Phi(\mathbf{a})$ as $\delta \to 0$. Now we invoke our result on smoothed syzygies, Theorem 1.1. For these parameters, it implies that there are no syzygies $\mathbf{h}$. Thus, we have that the map Φ is locally unique. Indeed, in the non-robust setting, [CO12] use (a genericity variant of) this argument and appeal to Terracini's lemma to obtain generic uniqueness. In our setting, we need a new argument to rule out alternate decompositions $\mathbf{b}$ that are "non-local", or far from $\mathbf{a}$, but still have $\Phi(\mathbf{a}) - \Phi(\mathbf{b})$ being small.

The key quantity for this turns out to be the following. Let $N_d = \dim \mathbb{R}[\mathbf{x}]^{=d} = \binom{n+d-1}{d}$ denote the dimension of the degree-d polynomial space, let $\mathcal{U} \subset \mathbb{R}^{N_2}$ denote the linear span of $\{a_i^2\}_{i=1}^m$, and let $\mathcal{V}$ denote the span of $\{b_i^2\}_{i=1}^m$. We define s to be (informally) the number of distinct directions of $\mathcal{V}$ that are "far" from $\mathcal{U}$. The proof splits into two cases, based on whether $s \leq c_* n$ for a small constant c_* or not. In the former case, a clever linear algebraic argument shows that if $\|\Phi(\mathbf{a}) - \Phi(\mathbf{b})\|$ is small enough, then in fact, $d_{\mathrm{perm}}(\mathbf{a}, \mathbf{b})$ must be small, which lets us reduce to the local case. (See Section 6.3.) The more challenging case turns out to be $s > c_* n$. Here, the key is to consider the Jacobian of the *joint map*:

$$\Psi(\mathbf{a}, \mathbf{b}) := \Phi(\mathbf{a}) - \Phi(\mathbf{b}).$$

This is a map with $2mn$ inputs, mapping into the space $\mathbb{R}^{N_3}$. We can argue that this map has at least $mn + ns/10$ singular values that are sufficiently large. Intuitively, this is because the Jacobian of the joint map behaves as a direct sum of the maps of $\Phi(\mathbf{a})$ and $\Phi(\mathbf{b})$. We have mn high singular value directions from the former map, because of a syzygy statement for smoothed $\mathbf{a}$; now by the assumption on $\mathcal{V}$ having many directions orthogonal to $\mathcal{U}$, we obtain $\approx ns$ new directions from $\Phi(\mathbf{b})$. This is formalized in Section 6.4. Once we show that Ψ has many large singular directions, we can use a geometric volume-counting idea very close to one considered in [Kan17]. The idea is to

bound a *joint* measure on $(\mathbf{a}, \mathbf{b})$, specifically showing that in this case, the measure of all $(\mathbf{a}, \mathbf{b})$ with $\|\Psi(\mathbf{a}, \mathbf{b})\| < \eta$ drops proportionally to $\eta^{mn+sn/10}$. (This can be viewed as an anti-concentration statement.) Once this is bounded, we can bound the probability of the event "$\mathbf{a}$ has an alternate decomposition" as follows: for any such $\mathbf{a}$, we show that there is a Euclidean ball of sufficiently large volume around $\mathbf{b}$ that should all be included in the joint volume computation. Dividing the two volumes thus bounds the desired probability.

The steps above, unfortunately, only yield the conclusion of Theorem 6.3 when $\delta(\mathbf{a}, \mathbf{b}) < \exp(-n)$. To improve the bound to polynomial, the main idea is an argument that we term *variable compression*. In effect, it argues that when $s > c_* n$ and $\delta(\mathbf{a}, \mathbf{b})$ is a small enough inverse polynomial, then the components of $\mathbf{b}$ can actually be defined using far fewer than $p = mn$ "variables" or coordinates. Roughly speaking, one can show that there exists a subset of the p variables, such that all the coordinates of $\mathbf{b}$ can be approximated by *rational functions* of the chosen coordinates and $\mathbf{a}$. The number of selected coordinates is $O(ns + q)$, where q corresponds to the low-singular value directions of Ψ. Each of these steps —identifying the coordinates to "retain", expressing the remaining coordinates as rational functions, bounding the number and error— turns out to require novel ideas. We present them in Sections 6.5, 6.6, 6.7. Finally, we can use a stronger version of the volume-counting argument (using ellipsoids instead of balls as described above) and thus obtain Theorem 6.3. The final step, along with a crucial error reduction step, is presented in Section 6.8.

3 Preliminaries and Notation

We now introduce our notation, and some preliminaries on polynomials. We denote vectors and polynomials by lower-case letters, matrices/higher-order tensors and linear operators by capital letters, and subspaces by calligraphic letters.

Polynomials and Tensors. We denote the vector space of homogenous, degree-d polynomials over the n variables $\mathbf{x} = (x_1, \ldots, x_n)$ as $\mathbb{R}[\mathbf{x}]^{=d}$. We adopt the convention that $\mathbb{R}[\mathbf{x}]^{=d} = \{0\}$ whenever $d < 0$, and $\mathbb{R}[\mathbf{x}]^{=0} = \mathbb{R}$. A basis for $\mathbb{R}[\mathbf{x}]^{=d}$ is given by the set of all possible monomials in $\mathbf{x}$ variables, which we can index by the set of multi-indices i.e. tuples $\alpha = (\alpha_1, \ldots, \alpha_n) \in \mathbb{Z}^n_{\geq 0}$ such that $\alpha_1 + \alpha_2 + \cdots + \alpha_n = d$. The monomial corresponding to a tuple α is given by $\mathbf{x}^\alpha = x_1^{\alpha_1} x_2^{\alpha_2} \cdots x_n^{\alpha_n}$.

We denote the set of all symmetric tensors of order d over $\mathbb{R}^n$ as $\mathsf{Sym}^d(\mathbb{R}^n)$. Given a polynomial $p(\mathbf{x}) \in \mathbb{R}[\mathbf{x}]^{=d}$, we can define a unique symmetric tensor P such that

$$p(\mathbf{x}) = \left\langle P, \mathbf{x}^{\otimes d} \right\rangle \tag{9}$$

This allows us to identify the space $\mathbb{R}[\mathbf{x}]^{=d}$ with $\mathsf{Sym}^d(\mathbb{R}^n)$ in a natural way.

We equip $\mathbb{R}[\mathbf{x}]^{=d}$ with the Bombieri inner product, defined by $\langle \mathbf{x}^\alpha, \mathbf{x}^\alpha \rangle = \alpha!/d!$ for a multi-index α, with monomials corresponding to distinct multi-indices are all orthogonal to each other. This inner product makes the above described identification of $\mathbb{R}[\mathbf{x}]^{=d}$ with $\mathsf{Sym}^d(\mathbb{R}^n)$ an isomorphism. Formally, given polynomials $p(\mathbf{x}), q(\mathbf{x})$ with corresponding symmetric tensor representations P, Q (as described by equation (9)), we have

$$\langle p(\mathbf{x}), q(\mathbf{x}) \rangle = \langle P, Q \rangle$$

where we use the standard inner product for tensors. We present some useful properties of the Bombieri inner product in Proposition A.1.

We will frequently consider vectors of homogeneous polynomials that are not necessarily of the same degree; we denote such a vector by $\mathbf{p} = (p_1, \ldots, p_m)$ and equip it with the product Bombieri norm, $\|\mathbf{p}\|^2 = \sum_{i=1}^m \|p_i\|^2$. We sometimes use the tuple and vector interchangeably, regarding an ordered tuple of polynomials as a column vector. In particular, we will use notations such as $\mathbf{b}^\top \mathbf{h} = \sum_i b_i h_i$, and polynomial matrices act on such a vector by $(\Omega \mathbf{b})_i = \sum_j \Omega_{ij} b_j$. Finally, the norms induce the following notion of distance to a set:

$$\mathsf{dist}(\mathbf{p}, E) := \inf_{\mathbf{p}' \in E} \|\mathbf{p} - \mathbf{p}'\|.$$

Smoothed Vectors and Polynomials. We say that a random vector $\widetilde{v} \in \mathbb{R}^n$ is ρ-perturbed if $\widetilde{v} = \bar{v} + \rho\zeta, \zeta \sim \mathcal{N}(0, I_n)$. We similarly define ρ-perturbed matrices and tensors as matrices (resp. tensors) where each entry is perturbed independently by a Gaussian random variable with variance ρ^2. We define a ρ-perturbed polynomial to be a random polynomial given by

$$\widetilde{p}(\mathbf{x}) = \sum_{\alpha : |\alpha| = d} \widetilde{p}_\alpha \sqrt{\frac{d!}{\alpha!}} \mathbf{x}^\alpha$$

where $\widetilde{p}_\alpha$ are independent ρ-perturbed coefficients in $\mathbb{R}$. Equivalently, the coefficient of each Bombieri-normalized monomial term is ρ-perturbed.

Ideals and Syzygies. Given polynomials $p_1, \ldots, p_m$ of degrees $d_1, \ldots, d_m$, we define the degree-t ideal of $\mathbf{p} = (p_1, \ldots, p_m)$ as the span of polynomial multiples of $p_1, \ldots, p_m$ of degree-t:

$$\mathcal{I}_{\mathbf{p}}^{(t)} = \operatorname{span}\left\{ p_i(\mathbf{x}) h_i(\mathbf{x}) : h_i \in \mathbb{R}[\mathbf{x}]^{=t-d_i}, i \in [m] \right\}$$

We also use the notation $\langle p_1, \ldots, p_m \rangle^{=t}$ to denote $\mathcal{I}_{\mathbf{p}}^{(t)}$. We can equivalently define the ideal at degree-t as the image of the Macaulay matrix $\mathsf{M}_{\mathbf{p}}^{(t)}$ defined above .

Next, we formally define syzygies, our objects of study.

Definition 3.1 (Syzygy Space)**.** Let $p_1, \ldots, p_m$ be homogeneous polynomials. We define the *degree-t syzygy space* of $\mathbf{p} = (p_1, \ldots, p_m)$ as

$$\mathsf{Syz}_{\mathbf{p}}^{(t)} = \left\{ (h_1(\mathbf{x}), \ldots, h_m(\mathbf{x})) \in \bigoplus_{i \in [m]} \mathbb{R}[\mathbf{x}]^{=t-\deg p_i} : \sum_{i \in [m]} p_i(\mathbf{x}) h_i(\mathbf{x}) = 0 \right\}.$$

Equivalently, for the vector $\mathbf{p} = (p_1, \ldots, p_m)$, define the linear map $\mathsf{M}_{\mathbf{p}}^{(t)} : \bigoplus_{i \in [m]} \mathbb{R}[\mathbf{x}]^{=t-\deg p_i} \to \mathbb{R}[\mathbf{x}]^{=t}$ by

$$\mathsf{M}_{\mathbf{p}}^{(t)}(h_1, \ldots, h_m) = \sum_{i \in [m]} p_i(\mathbf{x}) h_i(\mathbf{x}).$$

The degree-t syzygy space is the kernel of this linear map:

$$\mathsf{Syz}_{\mathbf{p}}^{(t)} = \ker \mathsf{M}_{\mathbf{p}}^{(t)}.$$

Note that t in the definition above denotes the output degree — it is the degree of the product $h_i(\mathbf{x}) p_i(\mathbf{x})$ for each $i \in [m]$. Also note that we allow $t < \deg p_i$ in the definition; in that case, by

our convention, $h_i = 0$. When it is clear what the polynomials $p_1, \ldots, p_m$ are, we omit them and write $\mathsf{Syz}^{(t)}$ for $\mathsf{Syz}_{\mathbf{p}}^{(t)}$.

As outlined in the introduction, for large enough output degrees t, there is always a set of "trivial" relations that lie in the syzygy space; these are the *Koszul syzygies*[8].

Definition 3.2 (Koszul Syzygies)**.** For homogeneous polynomials $p_1, \ldots, p_m$ of degrees $d_1, \ldots, d_m$, their *degree-t Koszul syzygies* are defined as

$$\mathcal{K}_{\mathbf{p}}^{(t)} = \left\{ \big(h_1(\mathbf{x}), \ldots, h_m(\mathbf{x})\big) : h_i(\mathbf{x}) = \sum_{j \in [m]} \Omega_{ij}(\mathbf{x}) p_j(\mathbf{x}), \Omega_{ij}(\mathbf{x}) = -\Omega_{ji}(\mathbf{x}) \in \mathbb{R}[\mathbf{x}]^{=t-d_i-d_j} \right\}.$$

In the above definition, h_i has degree $t - d_i$. Once again, we write $\mathcal{K}^{(t)}$ in place of $\mathcal{K}_{\mathbf{p}}^{(t)}$ when the polynomials are clear from context. If $t - \deg p_i - \deg p_j < 0$, we have $\Omega_{ij}(\mathbf{x}) = 0$ (the zero polynomial), and the space of Koszul syzygies $\mathcal{K}_{\mathbf{p}}^{(t)}$ could be $\{0\}$ for certain choices of t. For example, when all p_i's are quadratics, $\mathcal{K}_{\mathbf{p}}^{(3)} = \{0\}$. In this case, $\mathcal{K}_{\mathbf{p}}^{(4)}$ consists of skew-symmetric linear combinations of the p_i polynomials.

It is easy to see that $\mathcal{K}_{\mathbf{p}}^{(t)} \subseteq \mathsf{Syz}_{\mathbf{p}}^{(t)}$, and in general, the converse need not hold. Our main result shows that, when the polynomials $p_1, \ldots, p_m$ are (powers of) smoothed polynomials, then the syzygy space $\mathsf{Syz}_{\mathbf{b}}^{(t)}$ consists of only the Koszul syzygies.

Gradations of Polynomial Spaces. Throughout this paper, we will frequently partition the variables $\mathbf{x}$ into two disjoint sets of variables $(\mathbf{y}, \mathbf{z})$. This corresponds to a decomposition $\mathbb{R}^n = \mathcal{S} \oplus \mathcal{S}^\perp$ where the space $\mathcal{S}$ (resp. $\mathcal{S}^\perp$) corresponds to the variables $\mathbf{y}$ (resp. $\mathbf{z}$). This also allows us to decompose the space $\mathbb{R}[\mathbf{x}]^{=d}$ into $d+1$ orthogonal subspaces:

$$\mathbb{R}[\mathbf{x}]^{=d} \;=\; \bigoplus_{r=0}^{d} \mathbb{R}[\mathbf{z}]^{=r} \cdot \mathbb{R}[\mathbf{y}]^{=d-r}$$

We refer to this decomposition of $\mathbb{R}[\mathbf{x}]^{=d}$ as a *gradation*. By grouping the monomials based on their $\mathcal{S}^\perp$-degree, each homogeneous polynomial $f \in \mathbb{R}[\mathbf{x}]^{=d}$ can be uniquely decomposed into $d+1$ homogeneous polynomials according to the above decomposition of $\mathbb{R}[\mathbf{x}]^{=d}$:

$$f = f[0] + f[1] + \ldots + f[d],$$

where each $f[r] \in \mathbb{R}[\mathbf{z}]^{=r} \cdot \mathbb{R}[\mathbf{y}]^{=d-r}$ is the sum of all terms in f with $\mathbf{z}$-degree r and $\mathbf{y}$-degree $d-r$, for $r \in \{0, 1, \ldots, d\}$. Note that here, we consider an $\mathcal{S}^\perp$ gradation (or a $\mathbf{z}$-gradation), so that $f[r]$ denotes the part of f with $\mathbf{z}$-degree r. We could equivalently consider a $\mathbf{y}$-gradation, with the indices reversed.

As a concrete example, for $f(x_1, x_2) = (x_1 + x_2)^2$, taking $\mathcal{S} = \mathrm{span}\{e_1\}$ and $\mathcal{S}^\perp = \mathrm{span}\{e_2\}$, this gradation is just:

$$f = \underbrace{x_1^2}_{f[0]} + \underbrace{2x_1x_2}_{f[1]} + \underbrace{x_2^2}_{f[2]}.$$

Since $\mathcal{S}$ and $\mathcal{S}^\perp$ correspond to disjoint variables, the space $\mathbb{R}[\mathbf{y}]^{=d-r} \cdot \mathbb{R}[\mathbf{z}]^{=r}$, which contains $f[r]$, can be identified with the *tensor product* of the $\mathcal{S}$-polynomial and $\mathcal{S}^\perp$-polynomial spaces. Formally,

[8] See [Eis95, Chapter 17] for more background.

there is an isomorphism $\varphi : \mathbb{R}[\mathbf{y}]^{=d-r} \cdot \mathbb{R}[\mathbf{z}]^{=r} \to \mathbb{R}[\mathbf{y}]^{=d-r} \otimes \mathbb{R}[\mathbf{z}]^{=r}$, defined on products $h(\mathbf{y}) \cdot p(\mathbf{z})$ (and extended linearly to the whole space) as

$$\varphi\big(h(\mathbf{y}) \cdot p(\mathbf{z})\big) = \binom{d}{r}^{-1/2} h(\mathbf{y}) \otimes p(\mathbf{z}),$$

for any $h \in \mathbb{R}[\mathbf{y}]^{=d-r}$ and $p \in \mathbb{R}[\mathbf{z}]^{=r}$. The normalizing constant $\binom{d}{r}^{-1/2}$ ensures the Bombieri norm of $f[r]$ is the same as the tensor-product Bombieri norm of $\varphi(f[r])$, and hence φ is an isometry.

4 Syzygies of Smoothed Polynomial Powers

4.1 Setup and Result

In this section, we state and prove our main result (Theorem 4.1).

Theorem 4.1. *Let $m \in \mathbb{N}$, $\rho \in (0,1], \varepsilon > 0$ and $T \in \mathbb{N}$ be fixed, and let $a_1, \dots, a_m$ be independently ρ-perturbed n-variate homogeneous polynomials with centers $\bar{a}_1, \dots, \bar{a}_m$ of norm at most 1. Let $d_i = \deg a_i \in [d_{\max}]$, and let $D_1, D_2, \dots, D_m \in [D_{\max}]$. Define $b_i = a_i^{D_i}$ for each $i \in [m]$.*

Then, there is a constant C depending only on $\varepsilon, d_{\max}, D_{\max}$, and T, such that, for all sufficiently large n satisfying $\sum_{i=1}^m n^{-d_i D_i} \le n^{-\varepsilon}$, we have, simultaneously for all $t \in [T]$,

$$\mathsf{dist}\big(\mathbf{h}, \mathcal{K}_{\mathbf{b}}^{(t)}\big) \le n^C \rho^{-C} \left\| \sum_{i=1}^m b_i(\mathbf{x}) h_i(\mathbf{x}) \right\| \qquad \text{for all} \quad \mathbf{h} \in \bigoplus_{i=1}^m \mathbb{R}[\mathbf{x}]^{=t-d_i D_i},$$

with probability $1 - \exp\big(-\Omega(n^{\min_i d_i})\big)$. In particular, if $d_i = d$ and $D_i = D$ for all i, then this holds whenever $m \le n^{dD-\varepsilon}$.

We now describe some additional definitions we need in our proof, and provide a roadmap. To keep track of the notation and definitions, we refer to the Table 1.

Additional definitions. Let $f(\mathbf{x}) = \sum_{i \in [m]} b_i(\mathbf{x}) h_i(\mathbf{x}) \in \mathbb{R}[\mathbf{x}]^{=t}$ be the output. We want to prove that

$$\mathsf{dist}(\mathbf{h}, \mathcal{K}_{\mathbf{b}}^{(t)}) \le \mathrm{poly}(n, 1/\rho) \|f\|.$$

Our proof applies induction on the output degree t. In the rest of this section, we will work with a fixed output degree $t \in [T]$. We note that the base case $t = 0$ holds trivially, since $\deg b_i > 0$ forces all h_i's to be 0. Throughout the proof, we will treat $\varepsilon, T, d_{\max}, D_{\max}$ as constants.

We partition the variables $\mathbf{x} = (\mathbf{y}, \mathbf{z})$ with the corresponding coordinate decomposition $\mathbb{R}^n = \mathcal{S} \oplus \mathcal{S}^\perp$, and consider $\mathbf{z}$-gradation defined by the partition. We also write $\mathcal{K}_{\mathbf{b}}^{(t)}[0] := \{\mathbf{k}[0] : \mathbf{k} \in \mathcal{K}_{\mathbf{b}}^{(t)}\}$. Equivalently, this is the Koszul syzygy space of $\mathbf{b}[0]$ over $\mathbb{R}[\mathbf{y}]$.[9]

For $i \in [m]$, we define $\mathcal{J}_i \subseteq \mathbb{R}[y]^{=t-d_i D_i}$ as the following degree-$t - d_i D_i$ ideal:

$$\mathcal{J}_i := \big\langle b_1[0], \dots, b_{i-1}[0], a_i[0], b_{i+1}[0], \dots, b_m[0] \big\rangle_{t-d_i D_i}, \quad \text{and} \quad \mathcal{J} := \bigoplus_{i=1}^m \mathcal{J}_i.$$

Note that $\mathcal{K}_{\mathbf{b}}^{(t)}[0] \subseteq \mathcal{J}$. Let $\mathbf{g} = (g_1, \dots, g_m)$ be defined by $g_i = \Pi_{\mathcal{J}_i}^{\perp} h_i[0]$.

[9] We note that this is the element-wise projection of $\mathcal{K}_{\mathbf{b}}^{(t)}$ onto the pure-$\mathbf{y}$ polynomial space, *not its intersection* with the pure-$\mathbf{y}$ space, and in fact $\mathcal{K}_{\mathbf{b}}^{(t)}[0] \not\subseteq \mathcal{K}_{\mathbf{b}}^{(t)}$ in general.

Notation	Description
n, m	Number of variables and number of generators.
$\mathbf{x}, \mathbf{y}, \mathbf{z}$	Ambient variables $\mathbf{x} = (x_1, \dots, x_n) = (\mathbf{y}, \mathbf{z})$; $\mathbf{y}$ and $\mathbf{z}$ are the variables on $\mathcal{S}$ and $\mathcal{S}^\perp$.
$\mathbb{R}[\mathbf{x}]^{=d}$	Homogeneous degree-d polynomials in the indicated variables.
t, T	Output degree t and its fixed upper bound T.
$a_i, \bar{a}_i$	Independently perturbed root polynomials and their deterministic centers.
d_i, D_i	Root degree $d_i \le d_{\max}$ and power exponent $D_i \le D_{\max}$.
ρ, ε	Gaussian perturbation scale $\rho \in (0, 1]$ and slack $\varepsilon > 0$ in $\sum_i n^{-d_i D_i} \le n^{-\varepsilon}$.
$\mathbf{b} = (b_1, \dots, b_m)$	Generators $b_i = a_i^{D_i}$, of degree $d_i D_i$.
$\mathbf{p} = (p_1, \dots, p_m)$	General homogeneous generator family in the deterministic lemmas.
$\mathbf{h} = (h_1, \dots, h_m)$	Coefficient vectors with $h_i \in \mathbb{R}[\mathbf{x}]^{=t-\deg p_i}$.
f	Multiplication output: $f = \mathbf{b}^\top \mathbf{h}$.
$f[r], \mathbf{h}[r]$	Terms of $\mathbf{z}$-degree r in a polynomial or vector of polynomials.
$\mathsf{M}_{\mathbf{p}}^{(t)}$	Multiplication maps of output degree t, using the generators p_i.
$\mathsf{Syz}_{\mathbf{p}}^{(t)}, \mathcal{K}_{\mathbf{p}}^{(t)}$	Degree-t syzygy space $\ker \mathsf{M}_{\mathbf{p}}^{(t)}$ and its subspace of Koszul syzygies.
$\mathsf{W}_{\mathbf{p}}^{(s)}[0], \tau$	Pseudo-inverse of $\mathsf{M}_{\mathbf{p}}^{(s)}[0]$; $\tau = \max_{0 \le s < t} \|\mathsf{W}_{\mathbf{p}}^{(s)}[0]\|_{\mathrm{op}}$.
$\mathcal{J}_i, \mathcal{J}$	Ideal piece $\mathcal{J}_i = \langle a_i[0],\, b_j[0]\ (j \ne i) \rangle_{t-d_i D_i}$ and product $\mathcal{J} = \bigoplus_i \mathcal{J}_i$.
$\Pi_E, \Pi_E^\perp$	Orthogonal projections onto a subspace E and its orthogonal complement.
$\mathbf{g} = (g_1, \dots, g_m)$	The important part of coefficients, $g_i = \Pi_{\mathcal{J}_i}^\perp h_i[0]$.
r, R	Layer indices during substitution, $0 \le r \le R \le t$.
$\mathsf{F}_r, \mathsf{H}_r$	Linear maps expressing $\mathbf{h}'[r]$ in terms of $f[0], \dots, f[r]$ and a prescribed pure-$\mathbf{y}$ part $\mathbf{h}_0$.

Table 1: Notation for the syzygy argument in Section 4.

Multiplication maps and pseudoinverses. For positive-degree homogeneous generators $\mathbf{p} = (p_1, \dots, p_m)$ and $s \ge 0$, let $\mathsf{M}_{\mathbf{p}}^{(s)}[0]$ be the multiplication map of the $\mathbf{y}$-parts of p_i's restricted to $\mathbf{y}$:

$$\mathsf{M}_{\mathbf{p}}^{(s)}[0] : \bigoplus_i \mathbb{R}[\mathbf{y}]^{=s-\deg p_i} \to \mathbb{R}[\mathbf{y}]^{=s}, \qquad \mathsf{M}_{\mathbf{p}}^{(s)}[0](\mathbf{u}) = \sum_{i=1}^{m} p_i[0] u_i.$$

Let $\mathsf{W}_{\mathbf{p}}^{(s)}[0]$ be the Moore-Penrose pseudoinverse of $\mathsf{M}_{\mathbf{p}}^{(s)}[0]$, so that

$$\mathsf{W}_{\mathbf{p}}^{(s)}[0] : \mathbb{R}[\mathbf{y}]^{=s} \to \left(\ker \mathsf{M}_{\mathbf{p}}^{(s)}[0]\right)^\perp.$$

For this generator family, we define $\tau := \max_{0 \le s < t} \|\mathsf{W}_{\mathbf{p}}^{(s)}[0]\|_{\mathrm{op}}$.

Proof outline. As described in the technical overview, our proof consists of the following steps.

- *Reconstructing* $\mathbf{h}[0]$ *and Substitution.* First, we express the non-Koszul part of $\mathbf{h}[0]$ as a bounded linear function of $\mathbf{g}$, and $f[0]$, as in Lemma 4.2. For this, we use differentiation and Euler's identity.

  Next, we express $\mathbf{h}[r]$ for $r > 0$ as bounded linear functions of $\mathbf{h}[0]$, and the output blocks $f[0], \dots, f[r]$, upto the Koszul syzygy terms (see Lemma 4.3). We use this and the above step to show that $\mathsf{dist}(\mathbf{h}, \mathcal{K}_{\mathbf{b}}^{(t)}) \le \mathrm{poly}(n, 1/\rho)\big(\|f\| + \|\mathbf{g}\|\big)$.

- *Isolating fresh randomness.* In Section 4.4, we modify the generators and use substitution and projection to isolate $v_i^{D_i} g_i$, where v_i is the pure-$\mathbf{z}$ part of the Gaussian perturbation of a_i.

The remaining terms form a linear correction in $\mathbf{g}$, and are independent of v_i. The product of $v_i^{D_i}$ and g_i, which are on disjoint variables, produces a tensor product structure, that allows us to argue (in the next step) that $\|\mathbf{g}\| \le \mathrm{poly}(n, 1/\rho)\|f\|$.

- *Successive shrinking.* We further partition $\mathcal{S}^\perp$ into disjoint coordinate blocks $\mathcal{V}^{(\ell)}$'s, and use subspace anticoncentration to successively shrink the bad candidate space for $\mathbf{g}$ (Section 4.5). This bounds $\|\mathbf{g}\|$ in terms of $\|f\|$.
- *Triangular Elimination.* Finally, substituting this expression for $\|\mathbf{g}\|$, we get that $\mathsf{dist}(\mathbf{h}, \mathcal{K}_{\mathbf{b}}^{(t)}) \le \mathrm{poly}(n, 1/\rho)\|f\|$, which we wanted to show.

We prove the required lemmas in the subsequent sections, and finally put everything together in Section (Section 4.6).

4.2 Reconstructing $\mathbf{h}[0]$.

In this step, we look at the equation corresponding to $\mathbf{z}$-degree 0:

$$f[0] = \sum_{i\in[m]} b_i[0]h_i[0].$$

As described, we bound the distance of the $\mathbf{y}$ part $\mathbf{h}[0]$ from the Koszul syzygy $\mathcal{K}_{\mathbf{b}}^{(t)}[0]$ in terms of $f[0]$, and $\mathbf{g}$. In particular, we prove the following lemma.

Lemma 4.2. *In the setting of Theorem 4.1, let $\mathcal{J}, \mathbf{g}, f$ be as defined above. Assume that $\ker \mathsf{M}_{\mathbf{b}}^{(t-1)}[0] = \mathcal{K}_{\mathbf{b}}^{(t-1)}[0]$. Then, for all $\mathbf{h} \in \bigoplus_{i\in[m]} \mathbb{R}[\mathbf{x}]^{=t-d_iD_i}$, we have*

$$\mathbf{h}[0] \;=\; \mathbf{k}_0 + \mathsf{L}_1(\mathbf{g}) + \mathsf{L}_2(f[0])$$

where $\mathbf{k}_0 \in \mathcal{K}_{\mathbf{b}}^{(t)}[0]$, and $\mathsf{L}_1, \mathsf{L}_2$ are linear maps depending only on $\mathbf{b}[0]$ that satisfy

$$\|\mathsf{L}_1\|_{\mathrm{op}} \le 1 + \tau\|\mathbf{b}[0]\| \;\text{ and }\; \|\mathsf{L}_2\|_{\mathrm{op}} \le \tau,$$

where $\tau = \max_{0\le s<t} \|\mathsf{W}_{\mathbf{b}}^{(s)}[0]\|_{\mathrm{op}}$.

Proof. For this proof, since every polynomial is over $\mathbf{y}$ we will drop the [0] symbol and use $\mathbf{h}, f, \mathbf{a}, \mathbf{b}$ to denote $\mathbf{h}[0], f[0], \mathbf{a}[0], \mathbf{b}[0]$ respectively. All multiplication maps (e.g., $\mathsf{M}_{\mathbf{b}}^{(t)}$), their pseudoinverses (e.g., $\mathsf{W}_{\mathbf{b}}^{(t)}$), and Koszul syzygy spaces (e.g., $\mathcal{K}_{\mathbf{b}}^{(t)}$) in this proof are also taken over the $\mathbf{y}$ variables.

We decompose $\mathbf{h}$ into its projection onto $\mathcal{J}$ and its complement $\mathcal{J}^\perp = \bigoplus_{i=1}^m \mathcal{J}_i^\perp$:

$$\mathbf{h} = \Pi_{\mathcal{J}}\mathbf{h} \;+\; \Pi_{\mathcal{J}}^\perp\mathbf{h}.$$

Let $\mathbf{g} = \Pi_{\mathcal{J}}^\perp\mathbf{h}$. Since $\mathcal{J}_i$ is the ideal of degree $t - d_iD_i$ generated by $\{a_i, b_j : j \ne i\}$, we can write h_i as

$$h_i(\mathbf{y}) = \sum_{\substack{j=1\\ j\ne i}}^{m} \Gamma_{ij}(\mathbf{y})b_j(\mathbf{y}) + \Gamma_{ii}(\mathbf{y})a_i(\mathbf{y}) + g_i(\mathbf{y}) \tag{10}$$

where $\Gamma_{ij}(\mathbf{y}) \in \mathbb{R}[\mathbf{y}]^{=t-d_iD_i-d_jD_j}$ for all $j \ne i$, and $\Gamma_{ii}(\mathbf{y}) \in \mathbb{R}[\mathbf{y}]^{=t-d_iD_i-d_i}$.

Let $f(\mathbf{y}) = \sum_{i\in[m]} b_i(\mathbf{y})h_i(\mathbf{y})$. Then

$$f(\mathbf{y}) = \sum_{\substack{i,j=1\\ i\neq j}}^{m} \Gamma_{ij}(\mathbf{y})b_i(\mathbf{y})b_j(\mathbf{y}) + \sum_{i=1}^{m} \Gamma_{ii}(\mathbf{y})b_i(\mathbf{y})a_i(\mathbf{y}) + \sum_{i=1}^{m} g_i(\mathbf{y})b_i(\mathbf{y}) \tag{11}$$

We want to use the fact that the degree-$(t-1)$ syzygy space consists of only the Koszul syzygies. Hence, we want an equation of a smaller output degree. To get this, we differentiate equation 11 w.r.t y_ℓ, for each $\ell \in [\dim \mathcal{S}]$, to get

$$\frac{\partial f}{\partial y_\ell} = \sum_{\substack{i,j=1\\ i\neq j}}^{m} \left(\frac{\partial \Gamma_{ij}}{\partial y_\ell} b_i b_j + (\Gamma_{ij} + \Gamma_{ji}) b_i \frac{\partial b_j}{\partial y_\ell} \right) + \sum_{i=1}^{m} \left(\frac{\partial \Gamma_{ii}}{\partial y_\ell} b_i a_i + (D_i+1)\Gamma_{ii} b_i \frac{\partial a_i}{\partial y_\ell} + \frac{\partial}{\partial y_\ell}(g_i(\mathbf{y})b_i(\mathbf{y})) \right)$$

where we use the fact that $b_i = a_i^{D_i}$, to differentiate b_i and simplify. Rearranging the above, we get

$$r_\ell(\mathbf{y}) = \sum_{i=1}^{m} b_i(\mathbf{y}) u_{\ell i}(\mathbf{y}) \tag{12}$$

where $u_{\ell i}(\mathbf{y})$ is defined as

$$u_{\ell i}(\mathbf{y}) = \sum_{\substack{j=1\\ j\neq i}}^{m} \left(\frac{\partial \Gamma_{ij}}{\partial y_\ell} b_j + (\Gamma_{ij} + \Gamma_{ji}) \frac{\partial b_j}{\partial y_\ell} \right) + \frac{\partial \Gamma_{ii}}{\partial y_\ell} a_i + (D_i+1)\Gamma_{ii} \frac{\partial a_i}{\partial y_\ell}.$$

and $r_\ell(\mathbf{y})$ is defined as

$$r_\ell(\mathbf{y}) = \frac{\partial f(\mathbf{y})}{\partial y_\ell} - \frac{\partial}{\partial y_\ell}\left(\mathsf{M}_{\mathbf{b}}^{(t)}(\mathbf{g}) \right).$$

Let $\mathsf{M}_{\mathbf{b}}^{(t-1)}$ be the operator such that $\mathsf{M}_{\mathbf{b}}^{(t-1)} : (v_1, \ldots, v_m) \mapsto \sum_{i\in[m]} b_i(\mathbf{y}) v_i(\mathbf{y})$. We can write equation 12 as

$$\mathsf{M}_{\mathbf{b}}^{(t-1)}(\mathbf{u}_\ell) = r_\ell(\mathbf{y}).$$

The null space of $\mathsf{M}_{\mathbf{b}}^{(t-1)}$ consists of only the degree-$(t-1)$ Koszul syzygies, $\mathcal{K}_{\mathbf{b}}^{(t-1)}$. So, if $\mathsf{W}_{\mathbf{b}}^{(t-1)}$ denotes the pseudoinverse of $\mathsf{M}_{\mathbf{b}}^{(t-1)}$, there exists $\mathbf{u}'_\ell \in \mathcal{K}_{\mathbf{b}}^{(t-1)}$

$$\mathbf{u}_\ell = \mathsf{W}_{\mathbf{b}}^{(t-1)}(r_\ell(\mathbf{y})) + \mathbf{u}'_\ell \tag{13}$$

Note that $\mathsf{W}_{\mathbf{b}}^{(t-1)}(r_\ell(\mathbf{y})) \in \bigoplus_{i\in[m]} \mathbb{R}[\mathbf{y}]^{=t-1-d_iD_i}$. Similarly, $u'_{\ell i} \in \mathbb{R}[\mathbf{y}]^{=t-1-d_iD_i}$ for each $i \in [m]$.

We now apply Euler's identity to recover our original polynomials. For each $i \in [m]$,

$$\begin{aligned}
\sum_{\ell=1}^{\dim \mathcal{S}} y_\ell u_{\ell i}(\mathbf{y}) &= \sum_{\substack{j=1\\ j\neq i}}^{m} \left(\sum_{\ell=1}^{\dim \mathcal{S}} y_\ell \frac{\partial \Gamma_{ij}}{\partial y_\ell} b_j + (\Gamma_{ij} + \Gamma_{ji}) \sum_{\ell=1}^{\dim \mathcal{S}} y_\ell \frac{\partial b_j}{\partial y_\ell} \right) \\
&\quad + \sum_{\ell=1}^{\dim \mathcal{S}} y_\ell \frac{\partial \Gamma_{ii}}{\partial y_\ell} a_i + (D_i+1)\Gamma_{ii} \sum_{\ell=1}^{\dim \mathcal{S}} y_\ell \frac{\partial a_i}{\partial y_\ell} \\
&= \sum_{\substack{j=1\\ j\neq i}}^{m} ((t - d_i D_i - d_j D_j)\Gamma_{ij} b_j + d_j D_j (\Gamma_{ij} + \Gamma_{ji}) b_j) \\
&\quad + (t - d_i D_i - d_i)\Gamma_{ii} a_i + d_i (D_i + 1)\Gamma_{ii} a_i \\
&= t \left(\sum_{\substack{j=1\\ j\neq i}}^{m} \Gamma_{ij} b_j + \Gamma_{ii} a_i \right) + \sum_{\substack{j=1\\ j\neq i}}^{m} (d_j D_j \Gamma_{ji} - d_i D_i \Gamma_{ij}) b_j
\end{aligned}$$

In the above expression, the second term is a skew-symmetric linear combination of b_j's, and so the tuple consisting of these terms lies in the Koszul syzygy $\mathcal{K}_{\mathbf{b}}^{(t)}$, and the first expression is just $t(h_i - g_i)$. From equation 13, we also get, for each $i \in [m]$,

$$\sum_{\ell=1}^{\dim \mathcal{S}} y_\ell u_{\ell i}(\mathbf{y}) = \sum_{\ell=1}^{\dim \mathcal{S}} y_\ell (\mathsf{W}_{\mathbf{b}}^{(t-1)}(r_\ell))_i + \sum_{\ell=1}^{\dim \mathcal{S}} y_\ell u'_{\ell i}(\mathbf{y})$$

Since $\mathbf{u}'_\ell \in \mathcal{K}_{\mathbf{b}}^{(t-1)}$, it can be seen that $\left(\sum_{\ell=1}^{\dim \mathcal{S}} y_\ell u'_{\ell i}(\mathbf{y})\right)_{i\in[m]} \in \mathcal{K}_{\mathbf{b}}^{(t)}$. Substituting in the above equation, we get that, there exists a vector $\mathbf{k}_0 \in \mathcal{K}_{\mathbf{b}}^{(t)}$ such that, for each $i \in [m]$,

$$h_i(\mathbf{y}) - g_i(\mathbf{y}) = \frac{1}{t} \sum_{\ell=1}^{\dim \mathcal{S}} y_\ell (\mathsf{W}_{\mathbf{b}}^{(t-1)}(r_\ell))_i + (\mathbf{k}_0)_i(\mathbf{y}). \tag{14}$$

Finally, we substitute the definition of r_ℓ into equation 14, we obtain

$$\mathbf{h}(\mathbf{y}) = \mathbf{k}_0(\mathbf{y}) + \mathbf{g}(\mathbf{y}) - \frac{1}{t} \sum_{\ell=1}^{\dim \mathcal{S}} y_\ell \mathsf{W}_{\mathbf{b}}^{(t-1)} \left(\frac{\partial}{\partial y_\ell} (\mathsf{M}_{\mathbf{b}}^{(t)}(\mathbf{g})) \right) + \frac{1}{t} \sum_{\ell=1}^{\dim \mathcal{S}} y_\ell \mathsf{W}_{\mathbf{b}}^{(t-1)} \left(\frac{\partial f}{\partial y_\ell} \right)$$

Now we define the two linear operators to be

$$\mathsf{L}_2(u) := \frac{1}{t} \sum_{\ell=1}^{\dim \mathcal{S}} y_\ell \, \mathsf{W}_{\mathbf{b}}^{(t-1)} \left(\frac{\partial u}{\partial y_\ell} \right), \qquad \mathsf{L}_1 := I - \mathsf{L}_2 \mathsf{M}_{\mathbf{b}}^{(t)}.$$

Then $\mathbf{h} = \mathbf{k}_0 + \mathsf{L}_1(\mathbf{g}) + \mathsf{L}_2(f)$, and both linear maps depend only on $\mathbf{b}$. Thus, we have written $\mathbf{h}$ in the desired form. To bound the operator norms, we use Proposition A.1:

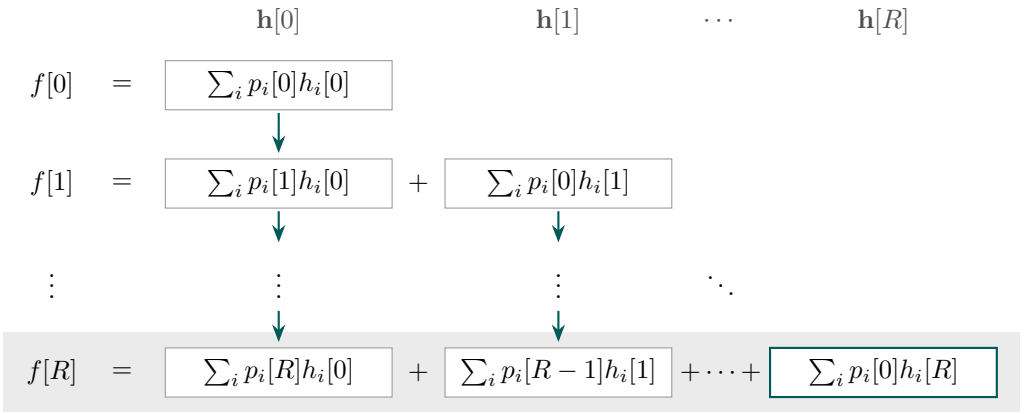


Figure 1: Substitution structure of the output equations $f[r] = \sum_{s=0}^{r} \sum_i p_i[r-s]h_i[s]$, shown for rows $r = 0, 1, \ldots, R$. Each column collects the terms involving the same input component $\mathbf{h}[s]$; vertical arrows indicate that this component $\mathbf{h}[s]$ is substituted into the later output equations in which it reappears. The shaded row is the current row R, and the highlighted diagonal term is the only one involving the new component $\mathbf{h}[R]$.

$$\begin{aligned}\frac{1}{t}\left\|\sum_{\ell=1}^{\dim \mathcal{S}} y_\ell \mathsf{W}_{\mathbf{b}}^{(t-1)}\left(\frac{\partial f}{\partial y_\ell}\right)\right\| &\le \frac{1}{t}\left(\sum_{\ell=1}^{\dim \mathcal{S}}\left\|\mathsf{W}_{\mathbf{b}}^{(t-1)}\left(\frac{\partial f}{\partial y_\ell}\right)\right\|^2\right)^{1/2} \\ &\le \frac{1}{t}\left\|\mathsf{W}_{\mathbf{b}}^{(t-1)}\right\|_{\mathrm{op}}\left(\sum_{\ell=1}^{\dim \mathcal{S}}\left\|\frac{\partial f}{\partial y_\ell}\right\|^2\right)^{1/2} \\ &= \left\|\mathsf{W}_{\mathbf{b}}^{(t-1)}\right\|_{\mathrm{op}} \cdot \|f\|.\end{aligned}$$

Similarly, we get that

$$\begin{aligned}\frac{1}{t}\left\|\sum_{\ell=1}^{\dim \mathcal{S}} y_\ell \mathsf{W}_{\mathbf{b}}^{(t-1)}\left(\frac{\partial}{\partial y_\ell}(\mathsf{M}_{\mathbf{b}}^{(t)}(\mathbf{g}))\right)\right\| &\le \left\|\mathsf{M}_{\mathbf{b}}^{(t)}\right\|_{\mathrm{op}}\left\|\mathsf{W}_{\mathbf{b}}^{(t-1)}\right\|_{\mathrm{op}} \cdot \|\mathbf{g}\| \\ &\le \left\|\mathsf{W}_{\mathbf{b}}^{(t-1)}\right\|_{\mathrm{op}} \cdot \|\mathbf{b}\| \cdot \|\mathbf{g}\|\end{aligned}$$

Using the fact that $\|\mathsf{W}_{\mathbf{b}}^{(t-1)}\|_{\mathrm{op}} \le \tau$, we get the required result. □

4.3 A Substitution Lemma

In this subsection we show the following substitution lemma, which will also be useful in Section 4.4. Intuitively, this propagates the previous bounded linear operator result for $\mathbf{h}[0]$ to the entire $\mathbf{h}$, modulo a Koszul syzygy $\mathbf{k}$, done by a substitution through a gradation on $\mathbf{z}$-degree as illustrated in Figure 1.

We write $f[0:r] := (f[0], \ldots, f[r])$ and use the same notation for vectors of polynomials, with $[r]$ applied coordinatewise.

Lemma 4.3 (Substitution)**.** *Let $\mathbf{p} = (p_1, \ldots, p_m)$ be positive-degree homogeneous generators satisfying $\ker \mathsf{M}_{\mathbf{p}}^{(s)}[0] = \mathcal{K}_{\mathbf{p}}^{(s)}[0]$ for every $0 \le s < t$, and let $\tau = \max_{0 \le s < t} \|\mathsf{W}_{\mathbf{p}}^{(s)}[0]\|_{\mathrm{op}}$. Let R be the final*

substitution row with $0 \le R \le t$, and let $\mathbf{h} \in \bigoplus_i \mathbb{R}[\mathbf{x}]^{=t-\deg p_i}$ be a coefficient vector with $f = \mathbf{p}^\top \mathbf{h}$. For every target pure-$\mathbf{y}$ part $\mathbf{h}_0 \in \mathbf{h}[0] + \mathcal{K}_{\mathbf{p}}^{(t)}[0]$, there exists some $\mathbf{k} \in \mathcal{K}_{\mathbf{p}}^{(t)}$, such that:

1. *$\mathbf{k}[0] = \mathbf{h}[0] - \mathbf{h}_0$, and*
2. *For each $r = 0, \dots, R$, there are linear maps F_r and H_r, depending only on $\mathbf{p}[0:r]$, such that*

$$\mathbf{h}[r] \;=\; \mathbf{k}[r] \;+\; \mathsf{F}_r(f[0:r]) \;+\; \mathsf{H}_r(\mathbf{h}_0),$$

and their operator norms are bounded by:

$$\|\mathsf{F}_r\|_{\mathrm{op}}, \|\mathsf{H}_r\|_{\mathrm{op}} \le 1 + 2^{T/2} \cdot \tau \cdot \max\Big\{\big(1 + 2^{T/2}\tau\|\mathbf{p}\|\big)^{R-1},\ \|\mathbf{p}\|\big(1 + 2^{T/2}\tau\|\mathbf{p}\|\big)^{R-1}\Big\}.$$

Proof. We construct $\mathbf{h}'$ with $\mathbf{h}'[0] = \mathbf{h}_0$ and $\mathbf{h} - \mathbf{h}' \in \mathcal{K}_{\mathbf{p}}^{(t)}$, such that $\mathbf{h}'[r] = \mathsf{F}_r(f[0:r]) + \mathsf{H}_r(\mathbf{h}_0)$ for every $0 \le r \le R$. Taking $\mathbf{k} = \mathbf{h} - \mathbf{h}'$ then gives the claimed conclusion. We apply induction on the maximum row number R. When $R = 0$, we can modify $\mathbf{h}$ into $\mathbf{h}'$ so that its pure-$\mathbf{y}$ part equals the given $\mathbf{h}_0$. To do this, take a matrix of homogeneous $\mathbf{y}$-polynomials Ω, such that

$$\mathbf{h}_0 - \mathbf{h}[0] = \Omega\, \mathbf{p}[0], \qquad \Omega_{ij} = -\Omega_{ji} \in \mathbb{R}[\mathbf{y}]^{=t-\deg p_i - \deg p_j}, \text{ for all } i, j \in [m].$$

We take $\mathbf{h}' = \mathbf{h} + \Omega\mathbf{p}$. This satisfies the desired properties because $\mathbf{h}'[0] = \mathbf{h}[0] + \Omega\mathbf{p}[0] = \mathbf{h}_0$, and the difference $\mathbf{h} - \mathbf{h}'$ is in the Koszul syzygy space:

$$\mathbf{h} - \mathbf{h}' \;=\; -\Omega\,\mathbf{p} \;\in\; \mathcal{K}_{\mathbf{p}}^{(t)}.$$

Taking $\mathsf{F}_0 = 0$ and $\mathsf{H}_0 = I$, this proves the base case.

We now assume this claim holds for all $r < R$, and show it for row R. Then we have some $\mathbf{h}'$ as described above, but the desired properties hold only up to row $r \le R - 1$. Since $\mathbf{h}' - \mathbf{h} \in \mathcal{K}_{\mathbf{p}}^{(t)}$, we have $\mathbf{p}^\top \mathbf{h}' = f$, and we will consider the Rth row in Figure 1 with coefficient $\mathbf{h}'$ in place of $\mathbf{h}$. Here, the output block $f[R]$ is written as:

$$f[R] \;=\; \sum_{r=0}^{R} \mathbf{p}[R-r]^\top \mathbf{h}'[r],$$

which we reorganize into

$$\mathbf{p}[0]^\top \mathbf{h}'[R] \;=\; f[R] - \sum_{r=0}^{R-1} \mathbf{p}[R-r]^\top \mathbf{h}'[r]. \tag{15}$$

The left-hand side can be viewed as a multiplication operator applied to $\mathbf{h}'[R]$. We invert this operator (up to the kernel) to get an expression for $\mathbf{h}'[R]$ in terms of linear function in $f, \mathbf{h}'[0:R-1]$ on the right.

Formally, equation 15 can be written as

$$\widehat{\mathsf{M}}_{\mathbf{p}}^{(t-R)}(\mathbf{h}'[R]) = f[R] - \sum_{r=0}^{R-1} \mathbf{p}[R-r]^\top \mathbf{h}'[r].$$

where $\widehat{\mathsf{M}}_{\mathbf{p}}^{(t-R)}(\mathbf{h}'[R])$ is defined as

$$\widehat{\mathsf{M}}_{\mathbf{p}}^{(t-R)} : \bigoplus_{i=1}^{m} \mathbb{R}[\mathbf{y}]^{=t-\deg p_i-R} \cdot \mathbb{R}[\mathbf{z}]^{=R} \;\to\; \mathbb{R}[\mathbf{y}]^{=t-R} \cdot \mathbb{R}[\mathbf{z}]^{=R}, (u_1, \ldots, u_m) \mapsto \sum_{i=1}^{m} p_i[0]u_i.$$

Define $\widehat{\mathsf{W}}_{\mathbf{p}}^{(t-R)}$ to be the following map:

$$\widehat{\mathsf{W}}_{\mathbf{p}}^{(t-R)}\left(\sum_{|\alpha|=R} \mathbf{z}^{\alpha} q_{\alpha}(\mathbf{y})\right) := \sum_{|\alpha|=R} \mathbf{z}^{\alpha} \mathsf{W}_{\mathbf{p}}^{(t-R)}[0]\big(q_{\alpha}(\mathbf{y})\big),$$

Then, note that

$$\widehat{\mathsf{M}}_{\mathbf{p}}^{(t-R)} \widehat{\mathsf{W}}_{\mathbf{p}}^{(t-R)} u = u$$

for any u in the image of $\widehat{\mathsf{M}}_{\mathbf{p}}^{(t-R)}$. Thus, $\widehat{\mathsf{W}}_{\mathbf{p}}^{(t-R)}$ acts as an extended pseudo-inverse.

We now show that the operator norm of this linear map is not too much larger than that of $\mathsf{W}_{\mathbf{p}}^{(t-R)}[0]$. Note that for every homogeneous polynomial $q(\mathbf{y})$ of degree $t-R$, we always have $\|\mathbf{z}^{\alpha} q(\mathbf{y})\| = \binom{t}{R}^{-1/2} \|\mathbf{z}^{\alpha}\| \|q(\mathbf{y})\|$; since different monomials are orthogonal, each input to $\widehat{\mathsf{W}}_{\mathbf{p}}^{(t-R)}$ has norm:

$$\left\| \sum_{|\alpha|=R} \mathbf{z}^{\alpha} q_{\alpha}(\mathbf{y}) \right\|^2 \;=\; \binom{t}{R}^{-1} \sum_{|\alpha|=R} \|\mathbf{z}^{\alpha}\|^2 \|q_{\alpha}\|^2.$$

Moreover, using the fact that multiplication is contractive under Bombieri norm (see Proposition A.1, the output is bounded by

$$\begin{aligned}
\left\| \widehat{\mathsf{W}}_{\mathbf{p}}^{(t-R)} \left(\sum_{|\alpha|=R} \mathbf{z}^{\alpha} q_{\alpha} \right) \right\|^2 &= \sum_{\alpha} \left\| \mathbf{z}^{\alpha} \mathsf{W}_{\mathbf{p}}^{(t-R)}[0](q_{\alpha}) \right\|^2 \\
&\le \sum_{\alpha} \|\mathbf{z}^{\alpha}\|^2 \left\| \mathsf{W}_{\mathbf{p}}^{(t-R)}[0](q_{\alpha}) \right\|^2 \\
&\le \left\| \mathsf{W}_{\mathbf{p}}^{(t-R)}[0] \right\|_{\mathrm{op}}^2 \cdot \sum_{\alpha} \|\mathbf{z}^{\alpha}\|^2 \|q_{\alpha}\|^2.
\end{aligned}$$

and combining the two gives

$$\left\| \widehat{\mathsf{W}}_{\mathbf{p}}^{(t-R)} \right\|_{\mathrm{op}} \le \binom{t}{R}^{1/2} \left\| \mathsf{W}_{\mathbf{p}}^{(t-R)}[0] \right\|_{\mathrm{op}} \le 2^{T/2} \left\| \mathsf{W}_{\mathbf{p}}^{(t-R)}[0] \right\|_{\mathrm{op}}.$$

We now apply the extended pseudo-inverse $\widehat{\mathsf{W}}_{\mathbf{p}}^{(t-R)}$ to both sides of Equation (15). This recovers $\mathbf{h}'[R]$ from $\mathbf{p}[0]^{\top}\mathbf{h}'[R]$ only upto the kernel of $\widehat{\mathsf{M}}_{\mathbf{p}}^{(t-R)}$. If we write out the terms in $\mathbf{h}'[R]$ corresponding to each $\mathbf{z}$-monomial, we get:

$$\mathbf{h}'[R] = \sum_{|\alpha|=R} \mathbf{z}^{\alpha} \mathbf{u}_{\alpha}(\mathbf{y}), \quad \mathbf{u}_{\alpha} \in \bigoplus_{i=1}^{m} \mathbb{R}[\mathbf{y}]^{=t-\deg p_i-R}.$$

Then we have that

$$\widehat{\mathsf{W}}_{\mathbf{p}}^{(t-R)}\big(\mathbf{p}[0]^{\top}\mathbf{h}'[R]\big) \;=\; \sum_{|\alpha|=R} \mathbf{z}^{\alpha} \cdot \Pi^{\perp}_{\mathcal{K}_{\mathbf{p}}^{(t-R)}[0]} \mathbf{u}_{\alpha},$$

which is not equal to $\mathbf{h}'[R]$ in general, due to the projection operator acting on $\mathbf{u}_\alpha$'s.

However, this mismatch is not an issue for our purpose, as we only want to find *some* $\mathbf{h}'$ satisfying the desired properties up to row R. Now we show that, from the current $\mathbf{h}'$ (which satisfies the properties up to row $R-1$), we can construct some $\mathbf{h}''$ such that:

1. For $0 \le r < R$, $\mathbf{h}'[r] = \mathbf{h}''[r]$, so the right-hand side of Equation (15) remains the same if we replace $\mathbf{h}'$ with $\mathbf{h}''$.
2. $\mathbf{h}''[R] = \widehat{\mathsf{W}}_{\mathbf{p}}^{(t-R)}\big(\mathbf{p}[0]^\top \mathbf{h}'[R]\big)$, so its Rth row is exactly recovered by pseudo-inversion; and
3. $\mathbf{h}''$ also does not alter the final output:

$$\mathbf{p}^\top \mathbf{h} \;=\; \mathbf{p}^\top \mathbf{h}' \;=\; \mathbf{p}^\top \mathbf{h}'' \;=\; f.$$

And as a final step, we can simply take this $\mathbf{h}''$ as our $\mathbf{h}'$, and apply induction.

To show such a $\mathbf{h}''$ exists, we explicitly define it as follows. For each multi-index α, the projection of $\mathbf{u}_\alpha$ onto the Koszul syzygy space can be expressed using a skew-symmetric matrix of polynomials Ω_α:

$$\Pi_{\mathcal{K}_{\mathbf{p}}^{(t-R)}[0]} \mathbf{u}_\alpha \;=\; \Omega_\alpha \mathbf{p}[0], \qquad \Omega_\alpha^\top = -\Omega_\alpha,$$

where each $(\Omega_\alpha)_{ij}$ has degree $t - R - \deg p_i - \deg p_j$ over $\mathbf{y}$. To remove this part from $\mathbf{h}'[R]$ without affecting the previous rows, we can take:

$$\mathbf{h}'' = \mathbf{h}' \;-\; \sum_{|\alpha|=R} \Omega_\alpha \mathbf{z}^\alpha \mathbf{p}.$$

Now we verify that $\mathbf{h}''$ has the three properties claimed above. Each term $\mathbf{z}^\alpha \Omega_\alpha \mathbf{p}$ that we subtracted is a multiple of some $\mathbf{z}^\alpha$, so the previous rows with $\mathbf{z}$-degree less than R are unaffected. By design, each $\mathbf{y}$-coefficient of every $\mathbf{z}^\alpha$ in $\mathbf{h}''[R]$ is perpendicular to $\mathcal{K}_{\mathbf{p}}^{(t-R)}[0]$, so $\mathbf{h}''[R]$ is recovered by the pseudo-inversion applied to Equation (15). Finally, $\mathbf{h}''$ does not alter the output f, since each subtracted term $\Omega_\alpha \mathbf{z}^\alpha \mathbf{p}$ is in the Koszul syzygy:

$$(\mathbf{z}^\alpha \Omega_\alpha)_{ij} = -(\mathbf{z}^\alpha \Omega_\alpha)_{ji}, \text{ where } \mathbf{z}^\alpha \text{ is a scalar,}$$

hence $(\mathbf{z}^\alpha \Omega_\alpha)\mathbf{p} \in \mathcal{K}_{\mathbf{p}}^{(t)}$.

Linear maps $\mathsf{F}_R, \mathsf{H}_R$. Now it suffices to show that the corresponding linear maps $\mathsf{F}_R, \mathsf{H}_R$ have the stated properties. We can write $\mathbf{h}''[R]$ explicitly as:

$$\mathbf{h}''[R] = \widehat{\mathsf{W}}_{\mathbf{p}}^{(t-R)} \left(f[R] - \sum_{r=0}^{R-1} \mathbf{p}[R-r]^\top \mathbf{h}'[r] \right),$$

and by inductive hypothesis we have, for $r < R$,

$$\mathbf{h}'[r] = \mathsf{F}_r(f[0:r]) + \mathsf{H}_r(\mathbf{h}_0).$$

Plugging in these F_r and H_r's, we have:

$$\mathbf{h}''[R] = \widehat{\mathsf{W}}_{\mathbf{p}}^{(t-R)} \left(f[R] - \sum_{r=0}^{R-1} \mathbf{p}[R-r]^\top \big(\mathsf{F}_r(f[0:r]) + \mathsf{H}_r(\mathbf{h}_0)\big) \right).$$

We can define $\mathsf{F}_R, \mathsf{H}_R$ explicitly:

$$\mathsf{F}_R(f[0:R]) := \widehat{\mathsf{W}}_{\mathbf{p}}^{(t-R)} \left(f[R] - \sum_{r=0}^{R-1} \mathbf{p}[R-r]^\top \mathsf{F}_r(f[0:r]) \right),$$

and

$$\mathsf{H}_R(\mathbf{h}_0) := -\widehat{\mathsf{W}}_{\mathbf{p}}^{(t-R)} \left(\sum_{r=0}^{R-1} \mathbf{p}[R-r]^\top \mathsf{H}_r(\mathbf{h}_0) \right).$$

We have shown in prior calculations that $\|\widehat{\mathsf{W}}_{\mathbf{p}}^{(t-R)}\|_{\mathrm{op}} \le 2^{T/2}\tau$. Moreover, by Proposition A.1, polynomial multiplication is contractive under Bombieri norm. We can therefore bound the outputs of the linear maps by:

$$\begin{aligned}
\big\|\mathsf{F}_R(f[0:R])\big\| &\le 2^{T/2}\tau \left(\big\|f[R]\big\| + \|\mathbf{p}\| \sum_{r=0}^{R-1} \big\|\mathsf{F}_r\big\|_{\mathrm{op}} \big\|f[0:r]\big\| \right) \\
&\le 2^{T/2}\tau \left(1 + \|\mathbf{p}\| \sum_{r=0}^{R-1} \big\|\mathsf{F}_r\big\|_{\mathrm{op}} \right) \cdot \big\|f[0:R]\big\| \\
&\le 2^{T/2}\tau\big(1 + R\|\mathbf{p}\|\big)\Big(1 + \max_{0\le r<R} \|\mathsf{F}_r\|_{\mathrm{op}}\Big) \cdot \big\|f[0:R]\big\|,
\end{aligned}$$

and

$$\begin{aligned}
\big\|\mathsf{H}_R(\mathbf{h}_0)\big\| &\le 2^{T/2}\tau\|\mathbf{p}\| \left(\sum_{r=0}^{R-1} \|\mathsf{H}_r\|_{\mathrm{op}} \right) \|\mathbf{h}_0\| \\
&\le 2^{T/2}\tau R\|\mathbf{p}\| \max_{0\le r<R} \|\mathsf{H}_r\|_{\mathrm{op}} \cdot \|\mathbf{h}_0\|
\end{aligned}$$

From the base case $\mathbf{h}'[0] = \mathbf{h}_0$, we have $\mathsf{F}_0 = 0$ and $\mathsf{H}_0 = I$. We can hence recursively bound the operator norms by inductively showing:

$$\begin{aligned}
&\|\mathsf{F}_1\|_{\mathrm{op}} \le 2^{T/2}\tau, && \|\mathsf{H}_1\|_{\mathrm{op}} \le 2^{T/2}\tau\|\mathbf{p}\| \\
&\|\mathsf{F}_2\|_{\mathrm{op}} \le 2^{T/2}\tau\big(1 + 2^{T/2}\tau\|\mathbf{p}\|\big), && \|\mathsf{H}_2\|_{\mathrm{op}} \le 2^{T/2}\tau\|\mathbf{p}\|\big(1 + 2^{T/2}\tau\|\mathbf{p}\|\big) \\
&\qquad \cdots \\
&\|\mathsf{F}_R\|_{\mathrm{op}} \le 2^{T/2}\tau\big(1 + 2^{T/2}\tau\|\mathbf{p}\|\big)^{R-1}, && \|\mathsf{H}_R\|_{\mathrm{op}} \le 2^{T/2}\tau\|\mathbf{p}\|\big(1 + 2^{T/2}\tau\|\mathbf{p}\|\big)^{R-1}
\end{aligned}$$

This gives us the required result. □

4.4 Reduction to Linear Maps

Now we will apply the substitution lemma again, this time to isolate $v_i^{D_i} \cdot g_i$, where v_i is the Gaussian perturbation on $\mathbf{z}$-variables.

Let $\mathcal{V} \subseteq \mathcal{S}^\perp$ be a coordinate subset, and in this subsection, we consider all polynomials to be *restricted* to the coordinates $\mathcal{S} \oplus \mathcal{V}$. In this subsection, $\mathbf{y}$ and $\mathbf{z}$ denote the coordinates in $\mathcal{S}$ and $\mathcal{V}$, respectively. For exposition, we will drop the restriction and write $p\big|_{\mathcal{S}\oplus\mathcal{V}}$ simply as p. The notation $p[r]$ will now denote the terms in p with $\mathcal{V}$-degree r.

Let v_i be the Gaussian perturbation restricted on the $\mathcal{V}$-variables, and for each smoothed polynomial $a_i \in \mathbb{R}[\mathbf{y}, \mathbf{z}]^{=d_i}$, we can isolate v_i:

$$a_i \;=\; c_i + v_i,$$

where c_i contains all terms containing an $\mathcal{S}$-variable (including the Gaussian perturbations), as well as the pure-$\mathcal{V}$ terms of the fixed center $\bar{a}_i$. Note that v_i is independent of c_i and all other a_j, where $i \neq j \in [m]$. In most of this subsection, we will fix an $i \in [m]$ and show an analysis that applies across all i.

Setting up the substitution. Denote the output by $f = \sum_{i=1}^m b_i h_i$. On a high level, for each $i \in [m]$, we would like to analyze the magnitude of $v_i^{D_i} g_i$, and show that this term has sufficient magnitude even after projecting out a subspace containing other nuisance terms. Since $v_i^{D_i} g_i$ has $\mathcal{V}$-degree exactly $d_i D_i$, we will apply a substitution up to row $R = d_i D_i$ in order to isolate this term. We consider $R \leq t$ below; when $R > t$, we have $h_i = g_i = 0$ and there is nothing to prove.

First, we consider the difference $f - v_i^{D_i} h_i$, which is a part of the nuisance terms that we'd like to project out:

$$f - v_i^{D_i} h_i = \sum_{j \in [m], j \neq i} b_j h_j \;+\; (b_i - v_i^{D_i}) h_i.$$

Because both b_i and $v_i^{D_i}$ are D_ith powers, we have:

$$b_i - v_i^{D_i} \;=\; (a_i - v_i) \underbrace{(a_i^{D_i - 1} + a_i^{D_i - 2} v_i + \ldots + v_i^{D_i - 1})}_{=:q}.$$

We can hence write the difference as:

$$f - v_i^{D_i} h_i = \sum_{j \in [m], j \neq i} b_j h_j \;+\; (a_i - v_i)(q h_i),$$

and we can view this again as some $f - v_i^{D_i} h_i = \sum_j \widetilde{b}_j \widetilde{h}_j$, where the modified generators $\{\widetilde{b}_j\}$ and coefficients $\{\widetilde{h}_j\}$ are defined as:

$$\widetilde{b}_j = \begin{cases} b_j & \text{if } j \neq i, \\ a_i - v_i & \text{if } j = i; \end{cases} \quad \text{, and} \quad \widetilde{h}_j = \begin{cases} h_j & \text{if } j \neq i, \\ q h_j & \text{if } j = i. \end{cases}$$

We will then apply Lemma 4.3 with $\mathbf{p} = \widetilde{\mathbf{b}} = (\widetilde{b}_1, \ldots, \widetilde{b}_m)$, up to row $R = d_i D_i$.

Projecting out a nuisance space. Consider the pure-$\mathcal{S}$ terms in our new $\widetilde{\mathbf{b}}$ and $\widetilde{\mathbf{h}}$. Since v_i does not have any $\mathcal{S}$-variable, and that $q[0] = a_i[0]^{D_i - 1}$, we have:

$$\widetilde{b}_j[0] = \begin{cases} b_j[0] & \text{if } j \neq i, \\ a_i[0] & \text{if } j = i; \end{cases} \quad \text{, and} \quad \widetilde{h}_j[0] = \begin{cases} h_j[0] & \text{if } j \neq i, \\ a_i[0]^{D_i - 1} h_j[0] & \text{if } j = i. \end{cases}$$

Moreover, $(f - v_i^{D_i} h_i)[r] = f[r]$ for all $r < R$.

By Lemma 4.3, for all $0 \leq r < R$, we have some alternative coefficient $\widetilde{\mathbf{h}}'$ and linear maps $\mathsf{F}_r, \mathsf{H}_r$ such that $\widetilde{\mathbf{h}} - \widetilde{\mathbf{h}}' \in \mathcal{K}^{(t)}_{\widetilde{\mathbf{b}}}$, $\widetilde{\mathbf{h}}'[0] = \widetilde{\mathbf{h}}[0]$, and

$$\widetilde{\mathbf{h}}'[r] = \mathsf{F}_r(f[0:r]) + \mathsf{H}_r(\widetilde{\mathbf{h}}[0]),$$

Now we consider row R. Note $(v_i^{D_i} h_i)[R] = v_i^{D_i} h_i[0]$, since $v_i^{D_i}$ already has $\mathcal{V}$-degree R. Plugging in earlier linear maps $\mathsf{F}_r, \mathsf{H}_r$, we have:

$$f[R] - v_i^{D_i} h_i[0] \;=\; \widetilde{\mathbf{b}}[0]^\top \widetilde{\mathbf{h}}'[R] \;+\; \sum_{r=0}^{R-1} \widetilde{\mathbf{b}}[R-r]^\top \overbrace{\left(\mathsf{F}_r(f[0:r]) + \mathsf{H}_r(\widetilde{\mathbf{h}}[0])\right)}^{\widetilde{\mathbf{h}}'[r]}.$$

On both sides of the equation, we will apply projection operator $\Pi^{\perp}_{\mathcal{J}_i}$ from the $\mathcal{S}$-polynomial coefficient of each $\mathcal{V}$-monomial. Recall that $\mathcal{J}_i$ is generated by $a_i[0]$ and $b_j[0]$ for $j \neq i$. By design, this projection annihilates $\widetilde{\mathbf{b}}[0]^\top \widetilde{\mathbf{h}}'[R]$. Moreover, $h_i[0]$ becomes g_i under this projection.

Re-organizing the terms, this yields:

$$v_i^{D_i} g_i + \Pi^{\perp}_{\mathcal{J}_i}\left(\sum_{r=0}^{R-1} \widetilde{\mathbf{b}}[R-r]^\top \mathsf{H}_r(\widetilde{\mathbf{h}}[0])\right) \;=\; \Pi^{\perp}_{\mathcal{J}_i}\left(f[R] - \sum_{r=0}^{R-1} \widetilde{\mathbf{b}}[R-r]^\top \mathsf{F}_r(f[0:r])\right). \tag{16}$$

Now, the right-hand side is a bounded linear function of f, and the left-hand side is our desired $v_i^{D_i} g_i$ plus the image of $\widetilde{\mathbf{h}}[0]$ under a linear operator independent of v_i.

We still have to consider all $i \in [m]$ to ensure that at least *one* of the g_i's are non-negligible. Formally, we summarize the arguments in this subsection in the following lemma:

Lemma 4.4 (Reduction to linear correction)**.** *For each $i \in [m]$, suppose that both $\mathbf{b}$ and $\widetilde{\mathbf{b}}$ satisfy the kernel condition in Lemma 4.3, and let $\tau = \max_{\mathbf{p}\in\{\mathbf{b},\widetilde{\mathbf{b}}\}} \max_{0\le s<t} \|\mathsf{W}_{\mathbf{p}}^{(s)}[0]\|_{\mathrm{op}}$. Then there exists a linear map $\mathsf{C}_i : \bigoplus_{j=1}^m \mathcal{J}_j^{\perp} \to \mathbb{R}[\mathbf{y},\mathbf{z}]^{=t}$, independent of v_i, such that for every $\mathbf{h} \in \bigoplus_{j=1}^m \mathbb{R}[\mathbf{y},\mathbf{z}]^{=t-d_jD_j}$, we have*

$$\left\| v_i^{D_i} g_i + \mathsf{C}_i(\mathbf{g}) \right\| \le \pi_T(\tau, \|\widetilde{\mathbf{b}}\|)\|f\|, \qquad \|\mathsf{C}_i\|_{\mathrm{op}} \le \pi_T(\tau, \|\widetilde{\mathbf{b}}\|).$$

Here, π_T is a polynomial that depends only on T and is the same across all choices of $i \in [m]$.

Proof. If $R = d_iD_i > t$, then $g_i = 0$ and we can take $\mathsf{C}_i = 0$.

Otherwise, use the maps $\mathsf{F}_r, \mathsf{H}_r$ constructed above, and let π_T be a polynomial upper bound on their operator norms from Lemma 4.3. Its degree and coefficients depend only on T, and we may enlarge it as needed, as long as the final choice of π_T still only depends on T. For $\mathbf{u} \in \bigoplus_{j=1}^m \mathbb{R}[\mathbf{y}]^{=t-d_jD_j}$, let $\widetilde{u}_j = u_j$ for $j \neq i$ and $\widetilde{u}_i = a_i[0]^{D_i-1} u_i$ (the same modification that takes $\mathbf{h}$ to $\widetilde{\mathbf{h}}$), and define the following linear map:

$$\mathsf{A}_i(\mathbf{u}) \;:=\; \Pi^{\perp}_{\mathcal{J}_i} \sum_{r=0}^{R-1} \widetilde{\mathbf{b}}[R-r]^\top \mathsf{H}_r(\widetilde{\mathbf{u}}).$$

The modified generators $\widetilde{\mathbf{b}}$ are independent of v_i, and the maps H_r depend only on $\widetilde{\mathbf{b}}[0:r]$. Both the map $\mathbf{u} \mapsto \widetilde{\mathbf{u}}$ and $\Pi^{\perp}_{\mathcal{J}_i}$ depend only on $\widetilde{\mathbf{b}}[0]$. We therefore conclude that A_i is independent of v_i.

Since $\widetilde{b}_i[0] = a_i[0]$ and $D_i \le R \le T$, by contractivity of polynomial multiplication, we have

$$\|\widetilde{\mathbf{u}}\| \le (1 + \|\widetilde{\mathbf{b}}\|)^T \|\mathbf{u}\|.$$

Applying the bounds on H_r, and since projection is contractive, we have

$$\|\mathsf{A}_i\|_{\mathrm{op}} \le R\|\widetilde{\mathbf{b}}\|\pi_T(\tau, \|\widetilde{\mathbf{b}}\|)(1 + \|\widetilde{\mathbf{b}}\|)^T.$$

For $\mathbf{u} = \mathbf{h}[0]$, we have $\widetilde{\mathbf{u}} = \widetilde{\mathbf{h}}[0]$, so Equation (16) gives the following bound:

$$\left\| v_i^{D_i} g_i + \mathsf{A}_i(\mathbf{h}[0]) \right\| = \left\| \Pi_{\mathcal{J}_i}^{\perp} \left(f[R] - \sum_{r=0}^{R-1} \widetilde{\mathbf{b}}[R-r]^{\top} \mathsf{F}_r(f[0:r]) \right) \right\| \\ \leq \left(1 + R\|\widetilde{\mathbf{b}}\| \pi_T(\tau, \|\widetilde{\mathbf{b}}\|)\right) \cdot \|f\|.$$

Since $R \leq T$, we can enlarge π_T so that $\|\mathsf{A}_i\|_{\mathrm{op}} \leq \pi_T(\tau, \|\widetilde{\mathbf{b}}\|)$ and the last bound is at most $\pi_T(\tau, \|\widetilde{\mathbf{b}}\|) \cdot \|f\|$.

To express the correction in terms of $\mathbf{g}$, we apply Lemma 4.2 to obtain

$$\mathbf{h}[0] = \mathbf{k}_0 + \mathsf{L}_1(\mathbf{g}) + \mathsf{L}_2(f[0]), \quad \text{where} \quad \mathbf{k}_0 \in \mathcal{K}_{\mathbf{b}}^{(t)}[0].$$

The maps $\mathsf{L}_1, \mathsf{L}_2$ depend only on $\mathbf{b}[0]$, and satisfy $\|\mathsf{L}_1\|_{\mathrm{op}} \leq 1 + \tau\|\mathbf{b}[0]\|$ and $\|\mathsf{L}_2\|_{\mathrm{op}} \leq \tau$. Every $\mathbf{k}_0 \in \mathcal{K}_{\mathbf{b}}^{(t)}[0]$ has a lift $\mathbf{k} \in \mathcal{K}_{\mathbf{b}}^{(t)}$ with $\mathbf{k}[0] = \mathbf{k}_0$, obtained by replacing each $b_j[0]$ with b_j in a Koszul representation, as in proof of Lemma 4.3.

Consider plugging in $\mathbf{k}$ for $\mathbf{h}$ in the bound above. Then, f vanishes since $\mathbf{k} \in \mathcal{K}_{\mathbf{b}}^{(t)}$, and $\mathbf{g}$ vanishes since $\mathbf{k}[0] = \mathbf{k}_0 \in \mathcal{K}_{\mathbf{b}}^{(t)}[0] \subseteq \bigoplus_j \mathcal{J}_j$. Hence, $\mathsf{A}_i(\mathbf{k}_0) = 0$. In other words, replacing $\mathbf{h}[0]$ with $\mathsf{L}_1(\mathbf{g}) + \mathsf{L}_2(f[0])$ does not change the correction $\mathsf{A}_i(\mathbf{h}[0])$.

Define $\mathsf{C}_i = \mathsf{A}_i \circ \mathsf{L}_1$ on $\bigoplus_j \mathcal{J}_j^{\perp}$. It is linear and independent of v_i, and

$$\left\| v_i^{D_i} g_i + \mathsf{C}_i(\mathbf{g}) \right\| = \left\| v_i^{D_i} g_i + \mathsf{A}_i(\mathbf{h}[0]) - \mathsf{A}_i \mathsf{L}_2(f[0]) \right\| \\ \leq (1+\tau)\pi_T(\tau, \|\widetilde{\mathbf{b}}\|) \cdot \|f\|,$$

and it also satisfies

$$\|\mathsf{C}_i\|_{\mathrm{op}} \leq \pi_T(\tau, \|\widetilde{\mathbf{b}}\|) \cdot \left(1 + \tau\|\mathbf{b}[0]\|\right).$$

Since we have $\|\mathbf{b}[0]\| \leq \|\widetilde{\mathbf{b}}\| + \|\widetilde{\mathbf{b}}\|^{D_i}$, where $D_i \leq T$, we can once again enlarge π_T and obtain the desired bounds with the same polynomial for every i. □

4.5 Successive Shrinkage of the Candidate Solution Space

In this subsection, we show a technical lemma that lower bounds the left-hand side of Lemma 4.4 with high probability using $\|g_i\|$, which, in turn, will show that $\|g_i\| \approx 0$ whenever $\|f\|$ is small.

On a high level, we iteratively use fresh randomness to obtain strong restrictions on candidate solutions $\mathbf{g}$ where the desired claim is violated; eventually, we will conclude that the violating $\mathbf{g}$'s must be in the trivial subspace $\{0\}$. We state this result as a general *successive shrinkage lemma*, which can then be applied to the outcome of Lemma 4.4.

Let $\mathcal{W}_1, \ldots, \mathcal{W}_m$ be fixed Euclidean spaces, and let $\mathcal{W} := \bigoplus_i \mathcal{W}_i$ be the space of coefficients. The candidate space $\mathcal{G} \subseteq \mathcal{W}$ is a subspace consists of possible tuples $\mathbf{g} = (g_1, \ldots, g_m)$, and has dimension at most N_G. In our case, we have $\mathcal{W}_i = \mathcal{J}_i^{\perp}$, and $\mathcal{G} = \mathcal{W}$. For large enough n, its dimension is at most

$$N_G \;\leq\; \sum_{i=1}^{m} \dim \mathbb{R}[\mathbf{y}]^{=t-d_iD_i} \;\leq\; n^{t-\varepsilon}.$$

From the previous subsection, for $\mathbf{y} \in \mathcal{S}$ and $\mathbf{z} \in \mathcal{V} \subseteq \mathcal{S}^{\perp}$, we have, whenever $\|f\|$ is small,

$$\zeta_i(\mathbf{z}) g_i(\mathbf{y}) \;+\; \mathsf{C}_i(\mathbf{g}) \;\approx\; 0, \qquad i \in [m].$$

Here, the function ζ_i only depends on the "fresh randomness" v_i on the $\mathcal{V}$-variables. The linear correction C_i may depend on the other ζ_j $(j \neq i)$, but ζ_i is independent of C_i.

To show that this leads to $\|g_i\| \approx 0$, the challenge is to deal with the fact that the candidate $\mathbf{g}$ can depend on all of $\zeta, \mathsf{C}_1, \ldots, \mathsf{C}_m$. Indeed, if the linear map $\mathsf{C}_i(\mathbf{g})$ did not exist, then we can use the randomness in ζ_i, to argue that with high probability $\zeta_i \otimes g_i$ is significantly far from the zero polynomial when g_i has non-negligible length. However, the "fixed" term $\mathsf{C}_i(\mathbf{g})$ can cancel the first term, since $\mathbf{g}$ can be chosen based on the randomness in ζ. We will instead use the fact that the correction term is restricted to the subspace $\mathsf{C}_i(\mathcal{G})$.

Crucially, since C_i is a linear map, $\mathsf{C}_i(\mathcal{G})$ is restricted to a fixed subspace of dimension at most N_G. From the tensor product structure, we show that with high probability, it cannot cancel out $\zeta_i \otimes g_i$ unless g_i lies close to a subspace $\mathcal{W}_i' \subset \mathcal{W}_i$ that has dimension $O(n^{-d_i D_i} N_G)$. Combining this restriction across different i's, the "fresh randomness" from each ζ_i allows us to bring down the dimension of the bad candidate space $\mathcal{G}$ by a factor of $O(n^{-\varepsilon})$.

We will then partition $\mathcal{S}^\perp$ into disjoint coordinate subspaces, $\mathcal{S}^\perp = \mathcal{V}^{(1)} \oplus \ldots \oplus \mathcal{V}^{(L)}$, each with dimension $\approx n/L$ for some $L \approx T/\varepsilon$, where T is the maximum output degree in our main theorem, viewed as a constant. For each $\ell \in [L]$, the variables in $\mathcal{V}^{(\ell)}$ provide fresh randomness unexposed by previous analysis, so the candidate space $\mathcal{G}$ will successively shrink $\Theta(1/\varepsilon)$ times, each time by an $O(n^{-\varepsilon})$ factor. Eventually, we have $\dim \mathcal{G} = 0$ and conclude our lemma.

The successive contraction argument seems to be potentially of more general interest in dealing with settings where there is a "non-random shift" that is given by a linear map of the test vector or candidate $\mathbf{g}$.

Before we describe the general lemma, we describe the main anti-concentration property we assume for our random variables.

Definition 4.5 (Subspace anticoncentrated r.v.)**.** Suppose $\gamma, \eta_0 > 0$. A random variable $\zeta \in \mathbb{R}^N$ is (θ, η_0, γ)-subspace anticoncentrated if and only if for any fixed subspace $\mathcal{E} \subset \mathbb{R}^N$ of dimension at least θN, we have

$$\mathop{\mathbb{P}}_{\zeta}[\|\Pi_{\mathcal{E}} \zeta\|_2 \leq \eta_0] \;\leq\; \exp(-\gamma), \tag{17}$$

where $\Pi_{\mathcal{E}}$ is the projector onto the subspace $\mathcal{E}$. This implies the same bound for every $\eta \leq \eta_0$.

If $\zeta \in \mathbb{R}^N$ is a ρ-smoothed vector in $\mathbb{R}^N$, then it is subspace anticoncentrated with $\eta_0 = \Omega(\rho\sqrt{N})$ and $\gamma = \Omega(N)$ for any constant $\theta > 0$. For tensor powers of smoothed vectors, the following result of Bhaskara, Chen, Perreault and Vijayaraghavan [BCPV19] gives a vector-valued variant of the Carbery-Wright inequality that provides the stronger anticoncentration property we need with $\gamma = \Omega(N^{1/D}), \eta_0 = c_D'(\rho/\sqrt{n})^D$, for any fixed θ. Here, $N = \binom{n+D-1}{D}$ is the dimension of the symmetric tensor space.

Theorem 4.6 (Vector-valued Carbery-Wright inequality (Theorem 4.1 of [BCPV19]))**.** *For any $D \in \mathbb{N}_{\geq 1}$, there exist constants $c_D, c_D' > 0$ with the following property. For any $\theta \in (0,1)$, let $\mathcal{E}$ be a fixed subspace of $\mathsf{Sym}^D(\mathbb{R}^n)$ of dimension at least $\theta\binom{n+D-1}{D}$, and let $\bar{a} \in \mathbb{R}^n$ be any fixed vector. We have*

$$\mathop{\mathbb{P}}_{a \sim \mathcal{N}(\bar{a}, \rho^2 I)} \left[\left\| \Pi_{\mathcal{E}} a^{\otimes D} \right\|_2 \leq c_D' \frac{\rho^D}{n^{D/2}} \right] \leq \exp(-c_D \theta n). \tag{18}$$

Main lemmas. We first formalize one step in the candidate shrinking process (Lemma 4.7), then use it to prove the whole result Proposition 4.8, and finally prove the one-step lemma at the end.

We will state the results in their general form in terms of tensor products, which also applies to products between polynomials on different variables.

Lemma 4.7 (One-step shrinking)**.** *Fix a candidate space $\mathcal{G} \subseteq \mathcal{W}$ of dimension $\dim(\mathcal{G}) \le N_G$, and suppose $\zeta = (\zeta_1, \dots, \zeta_m)$ with each $\zeta_i \in \mathcal{U}_i$ drawn independently from a $(\frac{1}{4}, \eta_0, \gamma)$-subspace anticoncentrated distribution. Suppose $\dim(\mathcal{U}_i) \ge 4$ for all $i \in [m]$. Consider any set of m linear maps $\mathsf{C}_i : \mathcal{G} \to \mathcal{U}_i \otimes \mathcal{W}_i$, where C_i can depend on ζ_j for $j \ne i$, but ζ_i is independent of C_i. Then with probability at least $1 - m\exp(-\gamma)$ over the randomness in ζ, there exists a subspace $\mathcal{G}' \subseteq \mathcal{G}$ with*

$$\dim(\mathcal{G}') \;\le\; 2\sum_{i=1}^{m} \frac{N_G}{\dim(\mathcal{U}_i)}, \quad \textit{and every approximate solution } \mathbf{g} \in \mathcal{G} \textit{ satisfies}$$

$$\forall i \in [m],\ \left\|\zeta_i \otimes g_i + \mathsf{C}_i(\mathbf{g})\right\| \;\le\; \eta \quad\Longrightarrow\quad \operatorname{dist}(\mathbf{g}, \mathcal{G}') \;\le\; \frac{\eta\sqrt{\sum_i \dim(\mathcal{U}_i)^2}}{\eta_0}. \tag{19}$$

Consider the setting when $t = 4$ and $d_i = 1, D_i = 2$ for all $i \in [m]$. Initially $\dim(\mathcal{G}) = N_G \le mn^2$, while $\dim(\mathcal{U}_i) \sim n^2$. After one application of the above lemma, the dimension *shrinks* to at most $\dim(\mathcal{G}') \le O(m/n^2)N_G \le O(n^{-\varepsilon})N_G$, for our choice of $m \le n^{2-\varepsilon}$ in this setting.

We can then successively apply the above lemma L times using fresh randomness from coordinate blocks $\mathcal{V}^{(1)}, \dots, \mathcal{V}^{(L)} \subseteq \mathcal{S}^{\perp}$ with corresponding blocks of variables $\mathbf{z}^{(1)}, \dots, \mathbf{z}^{(L)}$. To apply this to our main theorem, we will later take $L = \lfloor T/\varepsilon \rfloor$ to ensure $\mathcal{G}_L = \{0\}$.

In each intermediate step $\ell \in [L]$, we expose the randomness in $\zeta^{(\ell)} = \big(\zeta_i^{(\ell)}(\mathbf{z}^{(\ell)}) : i \in [m]\big)$. We also have a candidate space $\mathcal{G}_{\ell-1} \subseteq \mathcal{G}$ before step ℓ with $\mathcal{G}_0 = \mathcal{G}$, and the linear map $\mathsf{C}_i^{(\ell)}(\mathbf{g})$ can depend on $\zeta^{(1)}, \dots, \zeta^{(\ell-1)}$ and $(\zeta_j^{(\ell)} : j \in [m] \setminus \{i\})$. To control the accumulation of errors, we also need an operator norm bound $\beta \ge 1$ on the maps $\mathsf{C}_i^{(\ell)}$ and a norm bound on the random inputs. We have the following proposition.

Proposition 4.8 (Successive shrinking)**.** *Fix a candidate space $\mathcal{G}_0 \subseteq \mathcal{W} = \bigoplus_{i=1}^m \mathcal{W}_i$ with dimension at most N_G. Suppose for each $\ell \in [L]$, conditioned on the earlier blocks, for each $i \in [m]$, $\zeta_i^{(\ell)} \in \mathcal{U}_i^{(\ell)}$ is drawn independently from an $(\frac{1}{4}, \eta_0, \gamma)$-subspace anticoncentrated distribution, and linear maps $\mathsf{C}_i^{(\ell)} : \mathcal{G}_0 \to \mathcal{U}_i^{(\ell)} \otimes \mathcal{W}_i$, where $\mathsf{C}_i^{(\ell)}$ depends only on $\{\zeta_j^{(\ell')} : \ell' < \ell,\ \vee\ (\ell' = \ell\ \wedge\ j \ne i)\}$. Let $n_{\ell i} := \dim \mathcal{U}_i^{(\ell)} \ge 4$, and suppose also that*

$$Q_L = \max_{\ell \in [L]} \frac{\sqrt{\sum_i n_{\ell i}^2}}{\eta_0}, \quad N_G \prod_{\ell=1}^{L} \left(2\sum_{i=1}^{m} \frac{1}{n_{\ell i}}\right) < 1, \ \textit{and } \forall \ell \in [L], i \in [m], \quad \|\zeta_i^{(\ell)}\| + \|\mathsf{C}_i^{(\ell)}\| \le \beta. \tag{20}$$

Then with probability at least $1 - mL\exp(-\gamma)$, simultaneously for every $\mathbf{g} \in \mathcal{G}_0$:

$$\forall \ell \le L\ \forall i \in [m], \left\|\zeta_i^{(\ell)} \otimes g_i + \mathsf{C}_i^{(\ell)}(\mathbf{g})\right\| \;\le\; \eta \quad\Longrightarrow\quad \forall i \in [m],\ \ \|g_i\| \;\le\; \eta \cdot (1 + \beta Q_L)^L\,. \tag{21}$$

In our application, we will take $\gamma = \Omega(n^{\min_i d_i})$ as a common lower bound on the anticoncentration exponents of all inputs. We will use a polynomial upper bound on $1/\eta_0$ and β. If the random variables $\zeta^{(\ell)}$ are not bounded by β almost surely, the failure probability for that event can be incorporated into the high probability bound in the proposition. Moreover, our parameters are chosen so that $2\sum_{i=1}^m \frac{1}{n_{\ell i}} = O(n^{-\varepsilon})$, hence choosing $L = O_T(1/\varepsilon)$ and η to be a sufficiently small inverse polynomial suffices. We now prove the proposition using Lemma 4.7.

Proof of Proposition 4.8. At round ℓ, define $\mathcal{G}_\ell \subseteq \mathcal{G}_{\ell-1}$ using the construction of $\mathcal{G}'$ provided by the one-step shrinking in Lemma 4.7. The current candidate depends only on earlier blocks

$\mathcal{V}^{(1)}, \ldots, \mathcal{V}^{(\ell-1)}$, so it is fixed when the new block $\mathcal{V}^{(\ell)}$ (with corresponding variables $\mathbf{z}^{(\ell)}$) is exposed. Hence the random variable $\zeta^{(\ell)}$ is independent after conditioning on previous steps. Now we apply Lemma 4.7 with candidate space $\mathcal{G}_{\ell-1}$ and dimension bound $\dim \mathcal{G}_{\ell-1}$. A union bound over rounds gives

$$\dim(\mathcal{G}_\ell) \le N_G \prod_{\ell'=1}^{\ell} \left(2 \sum_{i=1}^{m} \frac{1}{n_{\ell' i}} \right) \quad \text{for every } \ell$$

outside an event of probability at most $mLe^{-\gamma}$. The dimension condition $N_G \prod_{\ell=1}^{L} \left(2 \sum_{i=1}^{m} \frac{1}{n_{\ell i}} \right) < 1$ implies $\mathcal{G}_L = \{0\}$. We condition on the successful event (involving the subspace anticoncentration of random variables $\zeta^{(\ell)}_{\ell \in [L]}$) for the rest of the proof.

Fix any $\mathbf{g} \in \mathcal{G}_0$ satisfying the residual bounds in Proposition 4.8. Let $\delta_\ell := \mathsf{dist}(\mathbf{g}, \mathcal{G}_\ell)$ and $\mathbf{g}' = \Pi_{\mathcal{G}_{\ell-1}} \mathbf{g}$. The new residual for $\mathbf{g}'$ is at most $\eta + \beta \delta_{\ell-1}$, since for each $i \in [m]$,

$$\begin{aligned} \|\zeta_i^{(\ell)} \otimes g_i' + \mathsf{C}_i^{(\ell)}(\mathbf{g}')\| &\le \eta + \left(\|\zeta_i^{(\ell)}\| + \|\mathsf{C}_i^{(\ell)}\|_{\mathrm{op}} \right) \|\mathbf{g}' - \mathbf{g}\| \\ &\le \eta + \beta \delta_{\ell-1}. \end{aligned}$$

Using the distance bound for $\mathbf{g}'$ from $\mathcal{G}' = \mathcal{G}_\ell$ in Lemma 4.7 and the triangle inequality gives

$$\delta_\ell \le \delta_{\ell-1} + \mathsf{dist}(\mathbf{g}', \mathcal{G}_\ell) \le \delta_{\ell-1} + Q_L(\eta + \beta \delta_{\ell-1}) \le (1 + Q_L \beta) \delta_{\ell-1} + \eta Q_L.$$

Since $\delta_0 = 0$ and $\mathcal{G}_L = \{0\}$,

$$\|\mathbf{g}\| = \delta_L \le Q_L \eta \sum_{j=0}^{L-1} (1 + Q_L \beta)^j = \eta \cdot \frac{(1 + Q_L \beta)^L - 1}{\beta} \le \eta \cdot (1 + Q_L \beta)^L.$$

This calculation holds for all $\mathbf{g}$. This concludes the proof. □

We now proceed to the proof of Lemma 4.7. We will use the following claim about a robust version of a simple dimension sum lemma. Let $\mathcal{U}, \mathcal{W}$ be subspaces and consider a subspace $\mathcal{X}$ of the product space $\mathcal{U} \otimes \mathcal{W}$. Then, there exists a subspace $\mathcal{U}_{\mathrm{null}} \subseteq \mathcal{U}$ of dimension at most $\dim(\mathcal{X})/q$, which contains every $u \in \mathcal{U}$ for which there is a subspace $\mathcal{W}_u \subseteq \mathcal{W}$ of dimension at least q satisfying $u \otimes \mathcal{W}_u \subseteq \mathcal{X}$. The following lemma proves a robust version of it (a related claim of this flavor is present in [BCMV14]). The subspace $\mathcal{U}_{\mathrm{null}}$ will identify the subspace with which we will apply the *subspace anticoncentration* property of the r.v. ζ_i to prove Lemma 4.7.

For a linear map M and a parameter $\lambda \ge 0$, let $\nu_\lambda(M)$ denote the number of small singular values at most λ.

Lemma 4.9 (Robust dimension and direct sum)**.** *Let $\mathcal{U}, \mathcal{W}$ be Euclidean spaces and $\mathcal{X} \subseteq \mathcal{U} \otimes \mathcal{W}$ have dimension at most N_X. Put $M_u w = \Pi_{\mathcal{X}^\perp}(u \otimes w)$ for $w \in \mathcal{W}$. For any integer $q \ge 1$ with $\lfloor N_X/q \rfloor \le \dim(\mathcal{U}) - 1$ and any $\lambda \ge 0$, there is a fixed subspace $\mathcal{U}_{null} \subseteq \mathcal{U}$ of dimension at most $\lfloor N_X/q \rfloor$ such that*

$$\{u : \nu_\lambda(M_u) \ge q\} \subseteq \left\{ u : \mathsf{dist}(u, \mathcal{U}_{null}) \le \lambda \left(\frac{N_X}{q} + 1 \right) \right\}. \tag{22}$$

When $\lambda = 0$ this reduces to the statement of the existence of a proper subspace $\mathcal{U}_{\mathrm{null}}$ of dimension at most $\dim(\mathcal{X})/q$ that contains all the $u \in \mathcal{U}$ with large intersection $\dim\left((u \otimes \mathcal{W}) \cap \mathcal{X}\right) \ge q$.

Proof. Let $r = \lfloor N_X/q \rfloor + 1$ and let $\mathcal{B} = \{u : \nu_\lambda(M_u) \geq q\}$; if this set is empty, the result is immediate. For any $u_1, \ldots, u_r \in \mathcal{B}$, let V_j have q orthonormal columns spanning small right singular directions of M_{u_j}, so $\|M_{u_j} V_j\| \leq \lambda$. Set

$$U = [u_1 \ \cdots \ u_r], \qquad F = [u_1 \otimes V_1 \ \cdots \ u_r \otimes V_r].$$

It is easy to see that $\sigma_{rq}(F) \geq \sigma_r(U)$. Each block of $\Pi_{\mathcal{X}^\perp} F$ has operator norm at most λ, so $\|\Pi_{\mathcal{X}^\perp} F\| \leq \sqrt{r}\,\lambda$. Since $\Pi_{\mathcal{X}} F$ has rank at most $N_X < rq$, the singular-value bound for a rank-deficient approximation gives

$$\sigma_r(U) \leq \sigma_{rq}(F) \leq \|F - \Pi_{\mathcal{X}} F\| \leq \sqrt{r}\,\lambda. \tag{23}$$

For $r = 1$, take $\mathcal{U}_{\text{null}} = \{0\}$. If $\lambda = 0$, every r vectors in $\mathcal{B}$ are dependent, so take $\mathcal{U}_{\text{null}} = \operatorname{span} \mathcal{B}$.

For the remaining case, first restrict to $\mathcal{B}_s = \{u \in \mathcal{B} : \|u\| \leq s\}$, which is closed and bounded by continuity of singular values. If $\dim \operatorname{span} \mathcal{B}_s < r$, use that span. Otherwise choose $u_1, \ldots, u_{r-1} \in \mathcal{B}_s$ maximizing their $(r-1)$-dimensional volume, and let $\mathcal{U}_{\text{null}}$ be their span.

We now want to prove that every $u \in \mathcal{B}_s$ has $\mathsf{dist}(u, \mathcal{U}_{\text{null}}) \leq r\lambda$. For $u \in \mathcal{B}_s \setminus \mathcal{U}_{\text{null}}$, append $u_r = u$ and put $\delta = \mathsf{dist}(u, \mathcal{U}_{\text{null}})$. Every base obtained by deleting one column of $U = [u_1 \ \cdots \ u_r]$ has volume at most the chosen base. The base-times-height formula gives that the leave-one-out-distance

$$\min_{j \in [r]} \mathsf{dist}\big(u_j, \operatorname{span}\{u_k : k \neq j\}\big) \geq \delta.$$

From the standard relation of $\sigma_r(U)$ to leave-one-out-distance, we have

$$\mathsf{dist}(u, \mathcal{U}_{\text{null}}) = \delta \leq \min_{j \in [r]} \mathsf{dist}\big(u_j, \operatorname{span}\{u_k : k \neq j\}\big) \leq \sqrt{r}\sigma_r(U) \leq r\lambda.$$

Finally, take a convergent subsequence of the resulting orthogonal projectors as $s \to \infty$. Its limit is an orthogonal projector of rank at most $r - 1$, and the distance bound passes to the limit for every $u \in \mathcal{B}$. □

We now prove the one-step shrinkage lemma.

Proof of Lemma 4.7. We will use Lemma 4.9 and anticoncentration of ζ_i to restrict each coordinate to a subspace $\mathcal{W}_i'$ of dimension $O(\dim(\mathcal{G})/n_i)$, and then construct a smaller candidate space $\mathcal{G}' \subseteq \mathcal{G}$.

Write $N_X = \dim \mathcal{G} \leq N_G$ and $n_i = \dim \mathcal{U}_i$. Project each correction onto $\mathcal{U}_i \otimes \mathcal{W}_i$ and call the resulting map $\overline{\mathsf{C}}_i$. This only decreases residual norms. Fix i and condition on all inputs ζ_j, $j \neq i$. Then $\mathcal{X}_i := \overline{\mathsf{C}}_i(\mathcal{G})$ is fixed, has dimension at most N_X, and is independent of the fresh input ζ_i. Put

$$M_{i,u} w = \Pi_{\mathcal{X}_i^\perp}(u \otimes w), \qquad \lambda_i = \frac{\eta_0}{n_i}, \qquad q_i = \left\lfloor \frac{2N_X}{n_i} \right\rfloor + 1.$$

Lemma 4.9 places all u with $\nu_{\lambda_i}(M_{i,u}) \geq q_i$ within distance $(N_X/q_i + 1)\lambda_i \leq \eta_0$ of a fixed subspace $\mathcal{U}_{\text{null},i} \subseteq \mathcal{U}_i$ of dimension $\lfloor N_X/q_i \rfloor < n_i/2$. The orthogonal complement of $\mathcal{U}_{\text{null},i}$ has dimension at least $n_i/4$, so subspace anticoncentration of ζ_i gives

$$\mathop{\mathbb{P}}_{\zeta_i}\left[\nu_{\lambda_i}(M_{i,\zeta_i}) \geq q_i \,|\, (\zeta_j : j \neq i)\right] \leq \mathop{\mathbb{P}}_{\zeta_i}\left[\|\Pi_{\mathcal{U}_{\text{null},i}^\perp} \zeta_i\| \leq \eta_0 \,\middle|\, (\zeta_j : j \neq i)\right] \leq e^{-\gamma}.$$

Taking a union bound over $i \in [m]$, with probability at least $1 - me^{-\gamma}$ every M_{i,ζ_i} has at most $q_i - 1$ singular values at most λ_i.

On this event, let $\mathcal{W}_i' \subseteq \mathcal{W}_i$ be spanned by the right singular directions of M_{i,ζ_i} with singular values at most λ_i. Thus $\dim \mathcal{W}_i' \le q_i - 1$. Outside these directions, the map expands lengths by a factor greater than λ_i, so

$$\operatorname{dist}(g_i, \mathcal{W}_i') \le \frac{\|M_{i,\zeta_i} g_i\|}{\lambda_i} \le \frac{\|\zeta_i \otimes g_i + \mathsf{C}_i(\mathbf{g})\|}{\lambda_i} \le \frac{\eta}{\lambda_i}.$$

The second inequality holds because projection onto $\mathcal{X}_i^{\perp}$ kills the projected correction. In particular, an equation with residual at most η confines g_i to within η/λ_i of $\mathcal{W}_i'$.

Combine these coordinate subspaces and project into the current candidate:

$$\mathcal{W}' = \bigoplus_i \mathcal{W}_i', \qquad \mathcal{G}' = \Pi_{\mathcal{G}}(\mathcal{W}') \subseteq \mathcal{G},$$

$$\text{and } \dim \mathcal{G}' \le \sum_i \dim \mathcal{W}_i' \le \sum_i (q_i - 1) \le 2N_X \sum_i \frac{1}{n_i} \le 2 \sum_i \frac{N_G}{n_i}.$$

For $\mathbf{g} \in \mathcal{G}$, the vector $\Pi_{\mathcal{G}} \Pi_{\mathcal{W}'} \mathbf{g}$ belongs to $\mathcal{G}'$. The residuals being bounded by η give

$$\mathsf{dist}(\mathbf{g}, \mathcal{G}') \le \|\mathbf{g} - \Pi_{\mathcal{W}'} \mathbf{g}\| = \left(\sum_i \operatorname{dist}(g_i, \mathcal{W}_i')^2 \right)^{1/2} \le \eta \left(\sum_i \lambda_i^{-2} \right)^{1/2} = \frac{\eta \sqrt{\sum_i n_i^2}}{\eta_0}.$$

These subspaces $\mathcal{W}'$ and $\mathcal{G}'$ are chosen based on ζ and the realized maps alone, so the same event works simultaneously for every $\mathbf{g} \in \mathcal{G}$ and every $\eta \ge 0$. □

4.6 Putting things together

Now we put everything together and prove our main theorem by induction.

Proof for Theorem 4.1. We apply induction on the degree t. The base case $t = 0$ is trivial because all $\mathbf{h}$ must be zero. We will work with the regime where ε is small, and in particular we assume $\varepsilon \le T$ where the possible coefficient space for $\mathbf{h}$ is non-trivial. Now we assume Theorem 4.1 holds for all output degrees $s < t$, and we show the same holds for output degree t.

Set $L = \lfloor T/\varepsilon \rfloor$, and partition variables $\mathbf{x}$ by:

$$\mathbb{R}^n \;=\; \mathcal{S} \oplus \mathcal{V}^{(1)} \oplus \ldots \oplus \mathcal{V}^{(L)},$$

where $\dim \mathcal{S} = \lceil n/2 \rceil$, and $\mathcal{S}^{\perp}$ is partitioned uniformly so $\dim \mathcal{V}^{(\ell)}$ has dimension roughly n/L. We will only use the fact that these are all $\Theta(n)$ assuming ε and T are fixed. We will also consider $d_{\max}$ and $D_{\max}$ to be fixed constants.

The inductive hypothesis. Our first two steps use the multiplicative map $\mathsf{M}_{\mathbf{b}}^{(s)}[0]$ on $\mathbf{b}[0]$, and in Lemma 4.4, we also replace $b_i[0]$ with $a_i[0]$ when processing each i. We can still use induction by checking that $\mathbf{b}[0]$ satisfies the conditions for Theorem 4.1. Indeed, the polynomials $a_i[0]$ are independently ρ-perturbed, with centers $\bar{a}_i[0]$ of norm at most 1.

The final condition to check is the $\sum_{i \in [m]} n^{-d_i D_i} \le n^{-\varepsilon}$ requirement. For this we separate two cases. For the reconstruction and substitution steps, we use generators $\mathbf{b}[0]$, and in Section 4.4 we use the alternative generators $\widetilde{\mathbf{b}}^{(i)}[0] = (a_i[0], b_j[0] : j \ne i)$ for each i.

- For $\mathbf{b}[0]$, we have

$$\sum_i (\dim \mathcal{S})^{-d_i D_i} \le \sum_i 2^{d_i D_i} n^{-d_i D_i} \le 2^{d_{\max} D_{\max}} n^{-\varepsilon} = O(n^{-\varepsilon}).$$

- For $\widetilde{\mathbf{b}}^{(i)}$, the ith generator is in degree d_i instead of $d_i D_i$. We thus have

$$\sum_{j \neq i} (\dim \mathcal{S})^{-d_j D_j} + (\dim \mathcal{S})^{-d_i} \le O(n^{-\varepsilon}) + 2n^{-1} = O(n^{-\varepsilon} + n^{-1}).$$

For sufficiently large n, both bounds are at most $(\dim \mathcal{S})^{-\varepsilon'}$, where $\varepsilon' = \min\{\varepsilon/2, 1/2\}$. We can hence apply the induction hypothesis with this smaller slack parameter instead, and we can still treat it as a constant that only depends on the original ε.

Formally, with probability at least $1 - \exp(-\Omega(n^{\min_i d_i}))$, simultaneously for all $\mathbf{p} \in \{\mathbf{b}\} \cup \{\widetilde{\mathbf{b}}^{(i)} : i \in [m]\}$, we now have

$$\ker \mathsf{M}_{\mathbf{p}}^{(s)}[0] = \mathcal{K}_{\mathbf{p}}^{(s)}[0] \quad (0 \le s < t), \qquad \max_{0 \le s < t} \|\mathsf{W}_{\mathbf{p}}^{(s)}[0]\|_{\mathrm{op}} \le (n/\rho)^C,$$

where C depends only on $\varepsilon, T, d_{\max}, D_{\max}$. For the rest of the proof, let τ be the maximum of these pseudoinverse norms over all these generator families and all $0 \le s < t$, so that $\tau \le (n/\rho)^C$.

Reducing h to linear map in g. Now we show that

$$\mathsf{dist}(\mathbf{h}, \mathcal{K}_{\mathbf{b}}^{(t)}) \ \le n^C \rho^{-C} \big(\|f\| + \|\mathbf{g}\|\big).$$

For now, consider gradation to be over the entire $\mathcal{S}^\perp$. Using Lemma 4.2, we can express the coefficient block $\mathbf{h}_0$ on the zeroth level, which is equivalent to $\mathbf{h}[0]$ modulo a Koszul syzygy on $\mathbf{y}$:

$$\mathbf{h}_0 := \mathsf{L}_1(\mathbf{g}) + \mathsf{L}_2(f[0]) \in \mathbf{h}[0] + \mathcal{K}_{\mathbf{b}}^{(t)}[0],$$

and by the operator norm bounds on $\mathsf{L}_1, \mathsf{L}_2$, we have

$$\|\mathbf{h}_0\| \le \big(1 + \tau \|\mathbf{b}[0]\|\big)\|\mathbf{g}\| + \tau \|f[0]\|.$$

Now we apply Lemma 4.3 with $\mathbf{p} = \mathbf{b}$ up to row $R = t$, and with the target zeroth level block being this $\mathbf{h}_0$. This gives a single $\mathbf{h}'$ with $\mathbf{h} - \mathbf{h}' \in \mathcal{K}_{\mathbf{b}}^{(t)}$, such that for each $r = 0, \dots, t$,

$$\begin{aligned} \mathbf{h}'[r] &= \mathsf{F}_r(f[0:r]) + \mathsf{H}_r(\mathbf{h}_0) \\ &= \mathsf{F}_r(f[0:r]) + \mathsf{H}_r \mathsf{L}_2(f[0]) + \mathsf{H}_r \mathsf{L}_1(\mathbf{g}). \end{aligned}$$

Note that each row is linear in $f[0:r]$ and $\mathbf{g}$. By enlarging π_T, we can absorb the bounds on $\mathsf{L}_1, \mathsf{L}_2$ into π_T, and obtain that

$$\|\mathbf{h}'[r]\| \le \pi_T(\tau, \|\mathbf{b}\|) \cdot (\|f\| + \|\mathbf{g}\|).$$

And since the different graded blocks are orthogonal, using the bounds on F_r and H_r, we have:

$$\begin{aligned} \mathsf{dist}\big(\mathbf{h}, \mathcal{K}_{\mathbf{b}}^{(t)}\big) \ &\le \ \|\mathbf{h}'\| \\ &\le \ \sqrt{t+1}\,\pi_T(\tau, \|\mathbf{b}\|)(\|f\| + \|\mathbf{g}\|) \\ &\le \ n^C \rho^{-C}(\|f\| + \|\mathbf{g}\|), \end{aligned}$$

by enlarging C if necessary, while having it still being a constant. Here, the last inequality is from a standard high probability bound for the norm of the Gaussian random vectors (see Lemma A.2). This lemma bounds the size of each Gaussian perturbation to $\bar{a}_i$, and since $m \le n^{d_{\max} D_{\max}}$, we can apply a union bound over all i's. Hence, on an event with probability $1 - \exp(-\Omega(n^{\min_i d_i}))$, we have, simultaneously all i,

$$\|a_i - \bar{a}_i\| \;\le\; 2\rho\sqrt{\binom{n+d_i-1}{d_i}}.$$

Recall that $\rho \le 1$ and the centers $\bar{a}_i$ all have norm at most 1. Since polynomial multiplication is contractive (Proposition A.1), on this event we have $\|\mathbf{b}\| \le \sqrt{m}\max_i \|a_i\|^{D_i} \le n^{O(1)}$. This proves the desired claim for this step.

Isolating every $v_i^{D_i} \cdot g_i$. Now we fix one coordinate block from $\mathcal{S}^\perp$. Let it be $\mathcal{V}^{(1)}$ WLOG. Now we consider every function to be on $\mathcal{S} \oplus \mathcal{V}^{(1)}$ and use $[r]$ to denote $\mathcal{V}^{(1)}$ gradation instead. This restriction has a few effects, none of which affects our arguments later on:

1. The output norm $\|f\|$ and generator norm $\|\mathbf{b}\|$ can only decrease.
2. The pure-$\mathcal{S}$ terms (hence the inductive hypothesis) remain the same, including $\mathbf{b}[0], \mathbf{a}[0], \mathcal{J}, \mathbf{g}$.
3. Every ρ-perturbed a_i remains ρ-perturbed for the same ρ.

Recall that for each i, we define a separate $\widetilde{\mathbf{b}}^{(i)}$, written simply as $\widetilde{\mathbf{b}}$ for short below, by

$$\widetilde{b}_i = a_i - v_i \quad \text{and} \quad \widetilde{h}_i = h_i \cdot \frac{a_i^{D_i} - v_i^{D_i}}{a_i - v_i},$$

and for all $j \ne i$ we have $\widetilde{b}_j = b_j$ and $\widetilde{h}_j = h_j$.

Note that v_i is purely on $\mathbf{z}$, so $\widetilde{b}_i[0] = a_i[0]$ and $\widetilde{b}_j[0] = b_j[0]$ for $j \ne i$. Hence, both $\mathbf{b}$ and $\widetilde{\mathbf{b}}$ satisfy conditions verified in inductive hypothesis, and in particular their pseudoinverses on the pure-$\mathbf{y}$ parts are bounded by the same τ.

Applying Lemma 4.4, for each $i \in [m]$, we obtain a linear map C_i, independent of v_i, such that

$$\left\| v_i^{D_i} g_i + \mathsf{C}_i(\mathbf{g}) \right\| \le \pi_T(\tau, \|\widetilde{\mathbf{b}}\|)\|f\|, \qquad \|\mathsf{C}_i\|_{\mathrm{op}} \le \pi_T(\tau, \|\widetilde{\mathbf{b}}\|).$$

Successive shrinking of bad candidates $\mathbf{g}$. The same argument above applies to every $\ell \in [L]$, with $v_i^{(\ell)}$ and $\mathsf{C}_i^{(\ell)}$ being the corresponding random polynomials and linear maps.

Now we apply Proposition 4.8 with the following parameters:

- The initial bad candidate space is $\mathcal{G}_0 = \bigoplus_i \mathcal{J}_i^\perp$;
- The space of random variables is $\mathcal{U}_i^{(\ell)} = \mathbb{R}[\mathbf{z}^{(\ell)}]^{=d_i D_i}$, and
- The random variables themselves are $\zeta_i^{(\ell)} = (v_i^{(\ell)})^{D_i}$.

By the projection onto $\mathcal{J}_i^\perp$ in its construction, the image of $\mathsf{C}_i^{(\ell)}$ lies in $\mathbb{R}[\mathbf{z}^{(\ell)}]^{=d_i D_i} \cdot \mathcal{J}_i^\perp$, which we can identify as the tensor product space as required in the lemma, again up to a $\Theta(1)$ factor. Note that polynomial products on disjoint variables can be identified with tensor products, up to a degree-dependent normalizing term which, when T is fixed, is universally upper- and lower-

bounded by constants (Section 3). We can hence absorb such a constant in the prior polynomial bound, and for each $\ell \in [L]$ and $i \in [m]$, reduce the problem to:

$$\left\| \zeta_i^{(\ell)} \otimes g_i + \mathsf{C}_i^{(\ell)}(\mathbf{g}) \right\| \le (n/\rho)^C \|f\|, \qquad \forall \ell, i.$$

Now we check the other stated conditions in Proposition 4.8. First, we would want a norm bound $\|\zeta_i^{(\ell)}\| + \|\mathsf{C}_i^{(\ell)}\|_{\mathrm{op}} \le \beta$. Since $\zeta_i^{(\ell)} = (v_i^{(\ell)})^{D_i}$ is just a power of a Gaussian random polynomial, we can again use Lemma A.2 and contractivity of multiplication to obtain $\|\zeta_i^{(\ell)}\| \le \|v_i^{(\ell)}\|^{D_i} \le n^{O(1)}$ on the same Gaussian norm event as in the previous step. Meanwhile, by Lemma 4.4, we have $\|\mathsf{C}_i^{(\ell)}\|_{\mathrm{op}} \le \pi_T(\tau, \|\widetilde{\mathbf{b}}\|)$. On this event, the modified generators can also be bounded by:

$$\|\widetilde{\mathbf{b}}\| \le \|\mathbf{b}\| + \|a_i\| + \|v_i^{(\ell)}\| \le n^{O(1)}, \qquad \forall \ell, i.$$

Combining these bounds with $\tau \le (n/\rho)^C$ from the inductive hypothesis, we have $\|\zeta_i^{(\ell)}\| + \|\mathsf{C}_i^{(\ell)}\|_{\mathrm{op}} \le \beta \le (n/\rho)^C$ simultaneously for all ℓ, i.

Now we check the dependencies on randomness. Condition on all Gaussian coefficients except those defining the inputs $(v_i^{(\ell)})_{\ell,i}$. Then $\mathcal{G}_0$ is fixed by the pure-$\mathcal{S}$ parts of $\mathbf{a}$ and $\mathbf{b}$, and each $\mathsf{C}_i^{(\ell)}$ depends only on the other inputs $v_j^{(\ell)}$, $j \ne i$, within its block. As described, we will expose these underlying Gaussian inputs one block ℓ at a time. The proof of Proposition 4.8 extends to this setting: in each round we condition on the other underlying inputs $v_j^{(\ell)}$, which fixes $\mathsf{C}_i^{(\ell)}$ without changing the distribution of $\zeta_i^{(\ell)} = (v_i^{(\ell)})^{D_i}$.

To check anticoncentration, we view each $v_i^{(\ell)}$ as a Gaussian vector in $\mathbb{R}[\mathbf{z}^{(\ell)}]^{=d_i}$. Under the symmetric-tensor identification, the multiplication map which sends the tensor power $(v_i^{(\ell)})^{\otimes D_i} \in \mathsf{Sym}^{D_i}(\mathbb{R}[\mathbf{z}^{(\ell)}]^{=d_i})$ to the polynomial power $(v_i^{(\ell)})^{D_i} \in \mathbb{R}[\mathbf{z}^{(\ell)}]^{=d_i D_i}$ is the orthogonal projection onto the fully symmetric order-$d_i D_i$ tensors, and the latter space has a constant fraction of the former's dimension.

Thus, we can apply Theorem 4.6 and obtain the required $(1/4, \eta_0, \gamma)$-subspace anticoncentration with parameters $\eta_0^{-1} \le (n/\rho)^C$ and $\gamma = \Omega(n^{\min_i d_i})$. Let $n_{\ell i} = \dim \mathcal{U}_i^{(\ell)}$. Then we have $n_{\ell i} = \binom{\dim \mathcal{V}^{(\ell)} + d_i D_i - 1}{d_i D_i} = \Theta(n^{d_i D_i})$, and for sufficiently large n we have the following bounds:

$$N_G \le n^{t-\varepsilon}, \qquad 2\sum_i n_{\ell i}^{-1} \le C_0 n^{-\varepsilon}, \qquad Q_L \le (n/\rho)^C.$$

For the first bound, we use $N_G = \dim \mathcal{G}_0$ where $\mathcal{G}_0 = \bigoplus_i \mathcal{J}_i^{\perp}$. Hence,

$$\begin{aligned} N_G &\le \sum_{i : d_i D_i \le t} \dim \mathbb{R}[\mathbf{y}]^{=t-d_i D_i} \le \sum_{i : d_i D_i \le t} n^{t - d_i D_i} \\ &\le n^t \sum_i n^{-d_i D_i} \le n^{t-\varepsilon}. \end{aligned}$$

For the second bound, $n_{\ell i} = \Theta(n^{d_i D_i})$ gives, uniformly for all ℓ,

$$2\sum_i n_{\ell i}^{-1} \;\le\; C_0 \sum_i n^{-d_i D_i} \;\le\; C_0 n^{-\varepsilon},$$

where C_0 depends only on the fixed parameters. Finally, by the definition of Q_L in Proposition 4.8,

$$Q_L = \max_{\ell \in [L]} \frac{\sqrt{\sum_i n_{\ell i}^2}}{\eta_0} \;\le\; \frac{\sqrt{m}\max_{\ell,i} n_{\ell i}}{\eta_0} \;\le\; (n/\rho)^C,$$

using $m \le n^{d_{\max} D_{\max}}$, $\max_{\ell,i} n_{\ell i} = O(n^{d_{\max} D_{\max}})$, and the choice of $\eta_0^{-1} \le (n/\rho)^C$ above, after enlarging C if necessary.

Since $L = \lfloor T/\varepsilon \rfloor$ and $t \le T < (L+1)\varepsilon$, for sufficiently large n the bad candidate space eventually shrinks to the trivial space:

$$N_G \prod_{\ell=1}^{L} \left(2 \sum_i n_{\ell i}^{-1} \right) \le C_0^L n^{t-(L+1)\varepsilon} < 1.$$

The successful event in Proposition 4.8 is uniform in the residual tolerance η, so we can take $\eta = (n/\rho)^C \|f\|$. It follows that

$$\|\mathbf{g}\| \le \sqrt{m}\,\eta(1+\beta Q_L)^L \le (n/\rho)^C \|f\|,$$

where C is again larger than before, but still a constant, since L is constant. Combining this with the previous distance bound, we have, simultaneously for all $\mathbf{h}$,

$$\mathsf{dist}(\mathbf{h}, \mathcal{K}_{\mathbf{b}}^{(t)}) \le (n/\rho)^C \|f\|$$

as desired.

Failure probability Finally, we apply a union bound over all failure probabilities. There are three types of failures: induction hypothesis on all output degrees $s < t$, the Gaussian norm event on $\mathbf{b}$ and $\widetilde{\mathbf{b}}$, and the failure in the shrinking lemma Proposition 4.8.

For induction hypothesis, by assumption the probability of failure on all lower degrees $s < t$ is $\exp(-\Omega(n^{d_{\min}}))$.

For the Gaussian norm event, we note that the *same* event bounding $\|\mathbf{b}\|$ can also bound the norm for every modified generator $\widetilde{\mathbf{b}}$, for all $i \in [m]$ and $\ell \in [L]$. The failure probability here, as established before, is also $\exp(-\Omega(n^{d_{\min}}))$.

Finally, Proposition 4.8 states the failure probability as $mL\exp(-\gamma)$, where $m = \mathrm{poly}(n)$, L is a constant, and $\gamma = \Omega(n^{d_{\min}})$ as stated in the previous step.

Since our main theorem is simultaneously over $t = 0, \ldots, T$, and that the induction hypothesis is only on the $\mathbf{y}$-variables, we have to apply a final union bound over all induction steps $t \in [T]$. We hence conclude that

$$\mathbb{P}[\exists t \in [T] : \text{the degree-}t \text{ conclusion fails}] \le \exp\big(-\Omega(n^{d_{\min}})\big).$$

And the proof is finished since all $n^C \rho^{-C}$ terms arising in calculation have C depending only on $\varepsilon, T, d_{\max}, D_{\max}$.

□

5 Application: Polynomial-time Power Sum Decomposition

5.1 Algorithm and Overview

We analyze the algorithm of [BHKX22] for power-sum decomposition. [BHKX22] present this algorithm and analyze it in the *random* case. They conjecture that it is robust in the smoothed analysis setting as well.

The analysis presented here is self-contained, and we review the algorithm. We make some minor modifications for ease of analysis. The first step is stated differently than it is in [BHKX22], we provide some exposition as to why this is doing essentially the same thing.

Our object of interest is the *power-sum* polynomial:

$$p(\mathbf{x}) = \sum_{i=1}^{m} a_i(\mathbf{x})^{3D},$$

where the underlying $a_i(\mathbf{x})$ are homogeneous degree-d polynomials, and $D \geq 1$ is a given integer. Our goal is to recover the base polynomials $a_i(\mathbf{x})$ from $p(\mathbf{x})$.

In fact, we would like to be able to recover estimates of the $a_i(\mathbf{x})$, even when we are only given access to $\widehat{p}(\mathbf{x})$ which is an estimate of $p(\mathbf{x})$. This is motivated by applications in learning theory and high-dimensional statistics, where $p(x)$ corresponds to a moment of a distribution, which we can estimate, but we cannot compute exactly.

For this to be tractable, we require a non-degeneracy condition on the $a_i(\mathbf{x})$. [BHKX22] analyze this algorithm in the case where the $a_i(\mathbf{x})$ have random mean-0 coefficients. In that setting they are able to establish that the algorithm is robust to noise in $\widehat{p}(\mathbf{x})$ of magnitude $\mathrm{poly}(1/n)$.

We consider the more general setting of *smoothed analysis*. Precisely, we assume that the $a_i(\mathbf{x})$ are ρ-smoothed, for some parameter $\rho \in (0, 1]$. That is, for each $i \in [m]$:

$$a_i(\mathbf{x}) = \bar{a}_i(\mathbf{x}) + \zeta_i(\mathbf{x}),$$

where $\bar{a}_i(\mathbf{x})$ are arbitrary polynomials of norm at most 1, and the $\zeta_i(\mathbf{x})$ have coefficients drawn independently from $\mathcal{N}(0, \rho^2)$.

Under this smoothed analysis condition we are able to design an algorithm that can recover estimates for $a_i(\mathbf{x})$ from an estimate of $p(\mathbf{x})$, that is robust to noise in $\widehat{p}(x)$ of magnitude $\mathrm{poly}(1/n, \rho)$. The algorithm is summarized in Algorithm 1. The analysis shows that the operations performed by this algorithm are well-conditioned when the $a_i(\mathbf{x})$ are ρ-perturbed.

We can identify $p(\mathbf{x})$ with the tensor

$$\sum_{i=1}^{m} a_i(\mathbf{x})^{\otimes 3D}.$$

In fact, if this was the representation that we had $p(\mathbf{x})$ in, then we could use existing algorithms for robust tensor decomposition to recover the factors $a_i(\mathbf{x})$ from the tensor $\sum_{i=1}^{m} a_i(\mathbf{x})^{\otimes 3D}$.

However, since $p(\mathbf{x})$ is a polynomial over commuting variables, at first, we only have access to the entries of $\sum_{i=1}^{m} a_i(\mathbf{x})^{\otimes 3D}$ up to symmetry. That is, we are given an estimate for

$$\mathsf{Sym}_{d \to 3dD} \sum_{i=1}^{m} a_i(\mathbf{x})^{\otimes 3D}.$$

Algorithm 1 Polynomial-time Power-sum decomposition

Input: d, D, $\widehat{p}(\mathbf{x})$ such that $\|\widehat{p}(\mathbf{x}) - p(\mathbf{x})\| \leq \varepsilon$ for ε sufficiently small inverse polynomial of n and smoothing parameter ρ, for $p(x) = \sum_{i=1}^{n} a_i(\mathbf{x})^{3D}$, for $a_i(\mathbf{x})$ homogeneous degree-d polynomials with norm < 1, smoothed by Gaussian noise of magnitude ρ, with $m = O(n^D)$

Output: estimates $\widehat{a}_i(\mathbf{x})$ for $i \in [m]$ with $\|\widehat{a}_i(\mathbf{x})\| \leq \mathrm{poly}(\varepsilon, n, 1/\rho)$.

For $\mathrm{poly}(n)$ randomly chosen subsets $\mathbf{y}$ of the variables $\mathbf{x}$:

1. Estimate $\widehat{B}_{\mathbf{y}} \approx \mathrm{span}\{b_i(\mathbf{y})^D : i \in [m]\}$, where $b_i(\mathbf{y}) = a_i(\mathbf{y}, 0)$
   - Use span of $\widehat{p}(x)$ terms with $\mathbf{z}$-degree exactly $2D$ to estimate span of the degree-$dD + 2D(d-1)$ ideal of $b_i(\mathbf{y})^D$s
   - Draw a random degree $2D(d-1)$ polynomial $h^\star(\mathbf{y})$, and intersect the span of the ideal with multiples of $h^\star(\mathbf{y})$ to estimate the span of $b_i(\mathbf{y})^D h^\star(\mathbf{y})$. Divide by $h^\star(\mathbf{y})$ to extract span of $b_i(\mathbf{y})^D$.
2. Desymmetrize $\widehat{p}(\mathbf{y})$ over $\widehat{B}_{\mathbf{y}}$ to get $\widehat{T}_{\mathbf{y}} \approx \sum_{i=1}^{m} (b_i(\mathbf{y})^D)^{\otimes 3}$

Aggregate $\widehat{T}_{\mathbf{y}}$s to get a total tensor $\widehat{T}_{\mathbf{x}}$.

3. Use Jennrich's algorithm for tensor decomposition on $\widehat{T}_{\mathbf{x}}$ to recover powered components $a_i(\mathbf{x})^D$. Take polynomial roots to recover base components $a_i(\mathbf{x})$.

The algorithm of [BHKX22] proceeds by computing the *desymmetrized* underlying tensor, from the symmetrized polynomial form, using the power sum structure. Then, once the desymmetrized form is recovered, they use robust tensor decomposition to recover the factors.

In general, it is not possible to desymmetrize a tensor. Given a symmetric tensor T_{sym} there are infinitely many asymmetric tensors such that $\mathsf{Sym}_{d \to 3dD} T_{\mathrm{asym}} = T_{\mathrm{sym}}$.

The key observation is that the symmetrization operator $\mathsf{Sym}_{dD \to 3dD}$ is actually invertible when *restricted* to the span of the $a_i(\mathbf{x})$. This is because the $a_i(\mathbf{x})$ are perturbed, and so their span looks like a random low-dimensional subspace, that is unlikely to intersect the kernel of the symmetrization operator $\mathsf{Sym}_{d \to 3dD}$. To perform this inversion algorithmically, we must also estimate the span of the $a_i(\mathbf{x})$.

It turns out that it suffices to partially desymmetrize the tensor into the form

$$T_{\mathbf{x}} = \sum_{i=1}^{m} (a_i(\mathbf{x})^D)^{\otimes 3}.$$

Then we can use order-3 tensor decomposition to recover the components, which are now given by the Dth powers of the $a_i(\mathbf{x})$.

Thus the algorithm is 3 main steps.

Step 1: Estimating span of components. The first step is to estimate $\mathrm{span}\{a_i(\mathbf{x})^D : i \in [m]\}$, so that we can use it to desymmetrize in step 2. [BHKX22] do this by taking order $2D$ partial derivatives of $p(\mathbf{x})$. Each of these partial derivatives results in a term that looks like

$$\sum_{i=1}^{m} a_i(\mathbf{x})^D q_i(\mathbf{x}), \quad \text{for some degree-} 2D(d-1) \text{ polynomials } q_i(\mathbf{x}).$$

That is, each of the partial derivatives lives in the *ideal* generated by the $a_i(\mathbf{x})^D$. A priori, we do not have much control over the $q_i(\mathbf{x})$ multipliers. [BHKX22] observe that if they *restrict* the partial derivatives to a smaller (random) subset of variables $\mathbf{y} \subseteq \mathbf{x}$, then the restricted partials will span the entire restricted ideal. That is, letting $b_i(\mathbf{y}) = a_i(\mathbf{y}, 0)$ be the restricted polynomials, they recover the space

$$\mathcal{I}_{\mathbf{b}^D} = \operatorname{span}\left\{b_i(\mathbf{y})^D h_i(\mathbf{y}) : i \in [m], h_i(\mathbf{y}) \in \mathbb{R}[\mathbf{y}]^{=2D(d-1)}\right\}.$$

Then, they are able to extract the span of the $b_i(\mathbf{y})^D$ from the ideal by sampling a random $h^\star(\mathbf{y}) \in \mathbb{R}[\mathbf{y}]^{=2D(d-1)}$, and taking all multiples of $h^\star(\mathbf{y})$ in the ideal $\mathcal{I}_{\mathbf{b}^D}$. Since $b_i(\mathbf{y})$ and $h^\star(\mathbf{y})$ are random, they have no nontrivial factors in common, and this gives us exactly the space

$$\operatorname{span}\left\{b_i(\mathbf{y})^D h^\star(\mathbf{y}) : i \in [m]\right\}.$$

Dividing by $h^\star(\mathbf{y})$ gives us the span of the $b_i(\mathbf{y})^D$.

The analysis requires two key parts. The first is showing that the restricted partial derivatives span the entire ideal $\mathcal{I}_{\mathbf{b}^D}$. The second is proving quantitatively that everything in $\mathcal{I}_{\mathbf{b}^D}$ that is *close* to a multiple of $h^\star(\mathbf{y})$ must look like $b(\mathbf{y})h^\star(\mathbf{y})$ for some $b(\mathbf{y}) \in \operatorname{span}\{b_i(\mathbf{y})^D\}$. For the first part, [BHKX22] do a clever but intricate analysis that works out what the structure of the $q_i(\mathbf{y})$ multipliers are for each of the partial derivatives, and argue that they are linearly independent. This is particularly complicated because the $q_i(\mathbf{y})$ are dependent on the $a_i(\mathbf{y})$. To handle the second part [BHKX22] prove an analogous claim to our main syzygy theorem (Theorem 4.1) for the case of random polynomials.

We rephrase this algorithm and analysis in terms of restrictions of $p(\mathbf{x})$. We begin by choosing an (arbitrary) subset of variables $\mathbf{y} \subseteq \mathbf{x}$ to restrict to. Let $\mathbf{z}$ be the rest of the variables not in $\mathbf{y}$. We look at $p(\mathbf{y}, \mathbf{z})[2D]$, which is $p(\mathbf{y}, \mathbf{z})$ restricted to the terms with z-degree exactly $2D$. Then we think of this as a matrix $H_{\mathbf{y}} : \mathbb{R}[\mathbf{z}]^{=2D} \to \mathbb{R}[\mathbf{y}]^{=dD+2D(d-1)}$, where the α, β entry of $H_{\mathbf{y}}$ is the coefficient of $\mathbf{y}^\alpha, \mathbf{z}^\beta$ in $p(\mathbf{y}, \mathbf{z})[2D]$.

This change from partial derivatives to restrictions of $p(\mathbf{x})$ is simply one of perspective. The columns of $H_{\mathbf{y}}$ are indeed partial derivatives of $p(\mathbf{x})$ with respect to the $\mathbf{z}$ variables, then restricted to the $\mathbf{y}$ variables. The only difference from [BHKX22] is that we do not consider partial derivatives that include the $\mathbf{y}$ variables, which is just for convenience and does not make much difference.

In a departure from the analysis of [BHKX22], we prove that $H_{\mathbf{y}}$ indeed spans the entire ideal $\mathcal{I}_{\mathbf{b}^D}$ using a reduction to our main syzygy theorem, Theorem 4.1. This allows us to streamline the analysis, as syzygies are a central object in this problem. This analysis is given in Lemma 5.2.

To recover the span of the $b_i(\mathbf{y})^D$ from $\mathcal{I}_{\mathbf{b}^D}$, we are again able to use our main syzygy theorem. This analysis is given in Lemma 5.3. Thus the syzygy theorem for smoothed polynomials (rather than random) allows us to analyze both parts of the span-finding subroutine.

Step 2: Desymmetrization. Once we have the span of $b_i(\mathbf{y})^D$ we can desymmetrize $p(\mathbf{y})$ to recover the form

$$T_{\mathbf{y}} = \sum_{i=1}^{m} (b_i(\mathbf{y})^D)^{\otimes 3}.$$

We do this by observing that the original components

$$b_i(\mathbf{y})^{3D} \in \operatorname{Im}(\mathsf{Sym}_{dD \to 3dD} B^{\circledast 3})$$

where B is the space of $b_i(\mathbf{y})^D$ and $\circledast$ is the symmetric lift. Thus, we can recover $T_{\mathbf{y}}$ so long as we can estimate the (pseudo)inverse of $\mathsf{Sym}_{dD\to 3dD}B^{\circledast 3}$. [BEG+24] gives us the tools to analyze matrices of this form, and allows us to get the robust analysis. This analysis can be found in Lemma 5.4. Finally, once we have $T_{\mathbf{y}}$ estimated for many choices of $\mathbf{y}$, we can aggregate to put together the full tensor $T_{\mathbf{x}}$. This is in Lemma 5.6.

Step 3: Tensor decomposition. The final step proceeds by applying Jennrich's algorithm for tensor decomposition to recover the $a_i(\mathbf{x})^D$ from $T_{\mathbf{x}}$. This follows from the robust analysis of [BCMV14]. Finally, we can take the polynomial root to recover $a_i(\mathbf{x})$. This can be found in Lemma 5.7.

This allows us to establish the following theorem.

Theorem 5.1 (Polynomial-time Power-sum decomposition)**.** *For constant degrees $d \geq 2, D \geq 1$, let $m < n^{2dD/(3d-2)-\iota}$ for some fixed $\iota > 0$. Let $a_1(\mathbf{x}), \ldots, a_m(\mathbf{x}) \in \mathbb{R}[\mathbf{x}]^{=d}$ be homogeneous degree-d polynomials that are independently ρ-perturbed for a parameter $\rho \in (0, 1]$. That is, $a_i(\mathbf{x}) = \bar{a}_i(\mathbf{x}) + \zeta_i(\mathbf{x})$, where $\bar{a}_i(\mathbf{x})$ is arbitrary of norm at most 1, and $\zeta_i(\mathbf{x})$ has independent $\mathcal{N}(0, \rho^2)$ coefficients. Let $p(x)$ be the* power-sum *polynomial*

$$p(x) = \sum_{i=1}^{m} a_i(\mathbf{x})^{3D} \in \mathbb{R}[\mathbf{x}]^{=3dD}.$$

Let $\widehat{p}(\mathbf{x})$ be an estimate of $p(\mathbf{x})$ such that

$$\|\widehat{p}(x) - p(x)\| < \varepsilon.$$

Then, for $\varepsilon < \mathrm{poly}(1/n, \rho)$ sufficiently small, with probability $1 - \mathrm{poly}(1/n)$, in polynomial time, we can estimate a matrix $\widehat{A}_{\mathrm{base}}$ (up to signs when D is even), such that

$$\|\widehat{A}_{\mathrm{base}} - A_{\mathrm{base}} M_\pi\| < \mathrm{poly}(\varepsilon, n, 1/\rho),$$

where $A_{\mathrm{base}} : \mathbb{R}^m \to \mathbb{R}[\mathbf{x}]^{=d}$ is the matrix with column i given by $a_i(\mathbf{x})$, and $M_\pi : \mathbb{R}^m \to \mathbb{R}^m$ is a permutation of the columns of A_{base}.

Proof. This is the final result of Lemmas 5.6 and 5.7. □

The supporting lemmas are proven in order of the algorithm in the following subsections.

5.2 Step 1: Estimating span of components

Lemma 5.2 ($H_{\mathbf{y}}$ robustly spans y-restricted ideal)**.** *Let $d \geq 2, D \geq 1$ be constants. (That is, n^{dD} is $\mathrm{poly}(n)$.) Let n be sufficiently large with respect to d, D, ι. Take $m < n^{2dD/(3d-2)-\iota}$ for some fixed $\iota > 0$, and $|\mathbf{y}| = c_2 n^{2/(3d-2)}$ for c_2 an appropriately chosen constant depending on d, D. Let $p(\mathbf{x}) = \sum_{i=1}^m a_i(\mathbf{x})^{3D}$, for $a_i(\mathbf{x})$ each an independently ρ-perturbed homogeneous degree-d polynomial with center of norm at most 1, where $\rho \in (0, 1]$. Let $\mathbf{z}$ be the variables $\mathbf{x} \setminus \mathbf{y}$.*

Let $p(\mathbf{x})[2D]$ be the restriction of p to terms with $\mathbf{z}$ degree exactly $2D$. Define the $(\mathbf{y}, \mathbf{z})$-flattening of $p(\mathbf{y}, \mathbf{z})[2D]$ as the matrix $H_{\mathbf{y}} : \mathbb{R}[\mathbf{z}]^{2D} \to \mathbb{R}[\mathbf{y}]^{dD+q}$ for $q = 2D(d-1)$, where the α, β entry of $H_{\mathbf{y}}$ is the coefficient of the $\mathbf{y}^\alpha \mathbf{z}^\beta$ term in $p(\mathbf{x})[2D]$. Let $b_i(\mathbf{y}) = a_i(\mathbf{y}, 0)$ be the restriction of a_i to the $\mathbf{y}$ variables. Then,

$$\mathrm{Im}(H_{\mathbf{y}}) = \mathcal{I}_{\mathbf{b}^D},$$

where $\mathcal{I}_{\mathbf{b}^D}$ is the space spanned by multiples of the $b_i(\mathbf{y})^D$s. Furthermore, we have that

$$\sigma_{\dim(\mathcal{I}_{\mathbf{b}^D})}(H_{\mathbf{y}}) \geq \mathrm{poly}(1/n, \rho),$$

with probability $1 - \exp(-\mathrm{poly}(n))$.

Proof. Let $p(\mathbf{x}) = p(\mathbf{y}, \mathbf{z})$ be the power sum polynomial

$$p(\mathbf{x}) = \sum_{i=1}^{m} a_i(\mathbf{x})^{3D}.$$

Identifying the span. For an arbitrary homogeneous polynomial $p'(\mathbf{x}) = p'(\mathbf{y}, \mathbf{z}) \in \mathbb{R}[\mathbf{x}]^{=d'}$, for a restriction to z-degree r, we will define the $(\mathbf{y}, \mathbf{z})$-*flattening* of $p'(\mathbf{y}, \mathbf{z})[r]$ to be a matrix $M : \mathbb{R}[\mathbf{z}]^{=r} \to \mathbb{R}[\mathbf{y}]^{=d'-r}$, where the entries of M are the corresponding coefficients of $p'(\mathbf{y}, \mathbf{z})[r]$:

$$p'(\mathbf{y}, \mathbf{z})[r] = \sum_{\alpha,\beta} M_{\alpha,\beta} e_{\mathbf{y}^\alpha} e_{\mathbf{z}^\beta},$$

for $e_{\mathbf{y}^\alpha}, e_{\mathbf{z}^\beta}$ the basis vectors of $\mathbb{R}[y]^{=d'-r}$ and $\mathbb{R}[\mathbf{z}]^{=r}$ respectively.

Let $H_{\mathbf{y}} : \mathbb{R}[\mathbf{z}]^{=2D} \to \mathbb{R}[\mathbf{y}]^{=dD+2D(d-1)}$ be the $(\mathbf{y}, \mathbf{z})$-flattening of $p(\mathbf{y}, \mathbf{z})[2D]$.

Define $b_i(\mathbf{y}) = a_i(\mathbf{y}, \mathbf{z})[0]$ as a_i restricted to the subset of variables $\mathbf{y}$. We would like to prove that the columns of $H_{\mathbf{y}}$ robustly span the degree-$(dD + q)$ ideal $\mathcal{I}_{\mathbf{b}^D}$ generated by the $b_i(\mathbf{y})^D$. That is, for all choices of m degree q polynomials $h_i(\mathbf{y})$,

$$\sum_{i=1}^{m} b_i(\mathbf{y})^D h_i(\mathbf{y}) \in \mathrm{Ran}(H_{\mathbf{y}}),$$

and the least singular value of $H_{\mathbf{y}}$ is polynomially lower-bounded.

We have that $p(\mathbf{y}, \mathbf{z})[2D] = \sum_{i=1}^{m} a_i(\mathbf{y}, \mathbf{z})^{3D}[2D]$. Let $\Delta_i(\mathbf{y}, \mathbf{z}) = a_i(\mathbf{y}, \mathbf{z}) - a_i(\mathbf{y}, \mathbf{z})[0]$. We can expand

$$\begin{aligned} a_i(\mathbf{y}, \mathbf{z})^{3D}[2D] &= (b_i(\mathbf{y}) + \Delta_i(\mathbf{y}, \mathbf{z}))^{3D}[2D] \\ &= \sum_{k=0}^{3D} \binom{3D}{k} b_i(\mathbf{y})^{3D-k} (\Delta_i(\mathbf{y}, \mathbf{z})^k[2D]) \\ &= b_i(\mathbf{y})^D \sum_{k=0}^{2D} \binom{3D}{k} b_i(\mathbf{y})^{2D-k} (\Delta_i(\mathbf{y}, \mathbf{z})^k[2D]), \end{aligned}$$

where the last line uses the observation that for terms with $k > 2D$, since every term in $\Delta_i(\mathbf{y}, \mathbf{z})$ has z-degree at least 1, all terms in $\Delta_i(\mathbf{y}, \mathbf{z})^k$ will have z-degree greater than $2D$, and the restriction will vanish.

Thus, $a_i(\mathbf{y}, \mathbf{z})^{3D}[2D]$ restricted to multiples of any given degree-$2D$ z monomial is a multiple of $b_i(\mathbf{y})^D$, and the range of $H_{\mathbf{y}}$ is contained in $\mathcal{I}_{\mathbf{b}^D}$. Define

$$q_i(\mathbf{y}, \mathbf{z}) := \sum_{k=0}^{2D} \binom{3D}{k} b_i(\mathbf{y})^{2D-k} \Delta_i(\mathbf{y}, \mathbf{z})^k[2D], \tag{24}$$

the polynomial coefficient associated with $b_i(\mathbf{y})^D$. We have that the column of $H_{\mathbf{y}}$ corresponding to monomial $\mathbf{z}^\beta$ is given by

$$\sum_{i=1}^{m} b_i(\mathbf{y})^D q_{i,\beta}(\mathbf{y}),$$

where we define $q_{i,\beta}(\mathbf{y})$ as $q_i(\mathbf{y}, \mathbf{z})$, restricted to terms that are multiples of $\mathbf{z}^\beta$, divided by $\mathbf{z}^\beta$.

Factoring $H_{\mathbf{y}}$. This motivates factoring $H_{\mathbf{y}}$ into two parts:

$$H_{\mathbf{y}} = \mathsf{M}_{\mathbf{b}^D}^{(dD+q)} R.$$

The first part, $\mathsf{M}_{\mathbf{b}^D}^{(dD+q)} : (\mathbb{R}[\mathbf{y}]^{=q})^m \to \mathbb{R}[\mathbf{y}]^{=dD+q}$ is the *Macaulay matrix*, or multiplication map, that maps a collection of polynomial coefficients to the resulting member of the ideal $\mathcal{I}_{\mathbf{b}^D}$:

$$h_1(\mathbf{y}), \ldots, h_m(\mathbf{y}) \in \mathbb{R}[\mathbf{y}]^{=q} \;\longmapsto\; \sum_{i=1}^{m} b_i(\mathbf{y})^D h_i(\mathbf{y}) \in \mathbb{R}[\mathbf{y}]^{=dD+q}.$$

The second part, $R : \mathbb{R}[\mathbf{z}]^{=2D} \to (\mathbb{R}[\mathbf{y}]^{=q})^m$, implements the map from $\mathbf{z}$ monomials to the polynomial coefficients that they implement for each i:

$$\mathbf{z}^\beta \in \mathbb{R}[\mathbf{z}]^{=2D} \;\longmapsto\; q_{1,\beta}(\mathbf{y}), \ldots, q_{m,\beta}(\mathbf{y}) \in \mathbb{R}[\mathbf{y}]^{=q}.$$

To establish a least singular value bound for $H_{\mathbf{y}}$, it suffices to bound both $\mathsf{M}_{\mathbf{b}^D}^{(dD+q)}$ and R. Since the $b_i(\mathbf{y})$ are independently perturbed by Gaussian noise, our main syzygy theorem Theorem 4.1, establishes that

$$\sigma_{\dim(\mathcal{I}_{\mathbf{b}^D})}\left(\mathsf{M}_{\mathbf{b}^D}^{(dD+q)}\right) > \mathrm{poly}(1/n, \rho), \tag{25}$$

with probability $1 - \exp(-\mathrm{poly}(n))$, since our choice of $m < n^{2dD/(3d-2)-\iota}$ for some $\iota > 0$, and $|\mathbf{y}| = c_2 n^{2/(3d-2)}$ for c_2 a constant, ensures that $m < |\mathbf{y}|^{dD-\iota}$. Thus, it only remains to argue that R is well-conditioned.

Bounding R. R consists of vertically stacked blocks $R_i : \mathbb{R}[\mathbf{z}]^{=2D} \to \mathbb{R}[\mathbf{y}]^{=q}$,

$$R = \begin{bmatrix} R_1 \\ \hline R_2 \\ \hline \cdots \\ \hline R_m \end{bmatrix},$$

where R_i maps $\mathbf{z}^\beta$ to $q_{i,\beta}(\mathbf{y})$. We want to show that the rows of R are robustly linearly independent, equivalently, that the columns of $R^\top$ are linearly independent. We do this via a block leave-one-out argument.

For each i, we would like to establish that the least singular value of $R_i^\top$ is sufficiently large, even after projecting out the column spans of all $R_j^\top$, $j \neq i$. $R_i^\top$ is the sum of many terms that correspond to the different, dependent, terms in $q_i(\mathbf{y}, \mathbf{z})$. We begin by identifying a term that is sufficiently independent from the rest of $q_i(\mathbf{y}, \mathbf{z})$ that we can use to establish the least singular value bound.

$$\begin{aligned} q_i(\mathbf{y}, \mathbf{z}) &= \sum_{k=0}^{2D} \binom{3D}{k} b_i(\mathbf{y})^{2D-k} \left(a_i(\mathbf{y}, \mathbf{z})[1] + \cdots + a_i(\mathbf{y}, \mathbf{z})[d]\right)^k [2D] \\ &= \sum_{k=0}^{2D} \binom{3D}{k} b_i(\mathbf{y})^{2D-k} \sum_{k'=0}^{k} \binom{k}{k'} \left(\left(a_i(\mathbf{y}, \mathbf{z})[1]\right)^{k-k'} \left(a_i(\mathbf{y}, \mathbf{z})[2] + \cdots + a_i(\mathbf{y}, \mathbf{z})[d]\right)^{k'}\right) [2D] \end{aligned}$$

Of the terms in the above expression, when $k = 2D, k' = 0$, we get $\binom{3D}{2D} a_{i,(1)}^{2D}(\mathbf{y},\mathbf{z})[2D]$. For every other choice of k, k', we have that $k - k' < 2D$, and therefore $(a_i(\mathbf{y},\mathbf{z})[1])^{k-k'}[2D] = 0$. Thus all of the other surviving terms are multiples of at least one of the $a_i(\mathbf{y},\mathbf{z})[r]$ for $r \geq 2$. We write

$$q_i(\mathbf{y},\mathbf{z}) = \binom{3D}{2D}(a_i(\mathbf{y},\mathbf{z})[1])^{2D}[2D] + k_i(\mathbf{y},\mathbf{z}), \qquad k_i(\mathbf{y},\mathbf{z}) \in \bigoplus_{r=2}^{d} a_i(\mathbf{y},\mathbf{z})[r] \cdot \mathbb{R}[\mathbf{y},\mathbf{z}]^{=2dD-d}, \quad (26)$$

where we use the $\oplus$ notation to denote the span of the subspaces spanned by multiples of each of the polynomials $a_i(\mathbf{y},\mathbf{z})[r]$.

Let $Z_i : \mathbb{R}[\mathbf{z}]^{=1} \to \mathbb{R}[\mathbf{y}]^{=d-1}$ be the the $(\mathbf{y},\mathbf{z})$-flattening of $a_i(\mathbf{y},\mathbf{z})[1]$. This lets us write the power $(a_i(\mathbf{y},\mathbf{z})[1])^{2D}[2D]$ in a similar flattened form $\mathsf{Sym}_{2D(d-1)}\ Z_i^{\circledast 2D}$, where $\circledast$ denotes the symmetric product,[10] and $\mathsf{Sym}_q : (\mathbb{R}[\mathbf{y}]^{=d-1})^{\circledast 2D} \to \mathbb{R}[\mathbf{y}]^{=q}$, is the operation that reformats the symmetric product into a multiplied polynomial of degree q.[11][12]

Let $K_i : \mathbb{R}[\mathbf{z}]^{=2D} \to \mathbb{R}[\mathbf{y}]^{=q}$ be the $(\mathbf{y},\mathbf{z})$-flattening of $k_i(\mathbf{y},\mathbf{z})$. Then we can write

$$R_i = \binom{3D}{2D}\mathsf{Sym}_{2D(d-1)}\ Z_i^{\circledast 2D} + K_i.$$

We show that the leading Z_i term has large least singular value, even when projected orthogonal to all of the R_j for $j \neq i$, and K_i. Let $U_{i,(r)} : \mathbb{R}[\mathbf{z}]^{=r} \to \mathbb{R}[\mathbf{y}]^{=d-r}$ be the $(\mathbf{y},\mathbf{z})$-flattening of $a_i(\mathbf{y},\mathbf{z})[r]$. By (26) we have that the rows of K_i are contained in the space of polynomial multiples of rows of the $U_{i,(r)}$:

$$\mathrm{Im}(K_i^\top) \subseteq \bigoplus_{r=2}^{\min\{d,2D\}} \mathrm{Im}(U_{i,(r)}^\top) \cdot \mathbb{R}[\mathbf{z}]^{=2D-r} := \mathcal{U}_i, \quad (27)$$

where we use $\mathrm{Im}(U_{i,(r)}^\top) \cdot \mathbb{R}[\mathbf{z}]^{=2D-r}$ to denote the span of multiples of polynomials in $\mathrm{Im}(U_{i,(r)}^\top)$. We have that

$$\begin{aligned}
\dim(\mathcal{U}_i) &\leq \sum_{r=2}^{\min\{d,2D\}} \dim(\mathrm{Im}(U_{i,(r)}^\top) \cdot \mathbb{R}[\mathbf{z}]^{=2D-r}) \\
&\leq \sum_{r=2}^{\min\{d,2D\}} \dim(\mathrm{Im}(U_{i,(r)}^\top)) \cdot \dim(\mathbb{R}[\mathbf{z}]^{=2D-r}) \\
&\leq \sum_{r=2}^{\min\{d,2D\}} \binom{|\mathbf{y}| + d - r - 1}{d-r}\binom{|\mathbf{z}| + 2D - r - 1}{2D-r}
\end{aligned}$$

[10] We use $\circledast$ to denote the symmetric product. That is, for a matrix $M : \mathbb{R}^m \to \mathbb{R}^n$, we define $M^{\circledast \ell} : \mathbb{R}^{\binom{m+\ell-1}{\ell}} \to \mathbb{R}^{n^\ell}$. The columns of $M^{\circledast \ell}$ are indexed by multisets of size ℓ over $[m]$. A multiset $I = \{i_1, \ldots, i_\ell\}$ corresponds to a column $\mathsf{Sym}_\ell(M_{i_1} \otimes \cdots \otimes M_{i_\ell})$, up to a constant factor depending only on ℓ, where Sym_ℓ takes the average over all of the symmetric terms: $\mathsf{Sym}_\ell(M_{i_1} \otimes \cdots \otimes M_{i_\ell}) = \frac{1}{\ell!}\sum_{\pi \in S_\ell} M_{i_{\pi(1)}} \otimes \cdots \otimes M_{i_{\pi(\ell)}}$. We can also see it as $M^{\circledast \ell} = M^{\otimes \ell}\mathsf{Sym}_\ell^\top$, where $\otimes$ is the usual Kronecker product, and Sym_ℓ sends order ℓ tensors over $\mathbb{R}^m$ to ℓ-way symmetric tensors.

[11] Interpreted as a tensor operation, Sym_q takes a tensor that is $2D$-way symmetric and $(d-1)$-way symmetric and symmetrizes it into a fully $q = 2D(d-1)$-way symmetric tensor.

[12] While the rows of $Z_i^{\circledast 2D}$ also correspond to symmetric lifts, we do not need an extra symmetrization operator since we started with monomials of degree 1, so their lift is already fully symmetric.

Define the space

$$\mathcal{N}_i = \mathcal{R}_i \oplus \mathcal{U}_i, \qquad \text{where } \mathcal{R}_i = \bigoplus_{j \neq i} \mathrm{Im}(R_j^\top).$$

We have that

$$\dim(\mathcal{N}_i) \leq (m-1)\binom{|\mathbf{y}|+q-1}{q} + \dim(\mathcal{U}_i). \tag{28}$$

Recall that $m < n^{2dD/(3d-2)-\iota}$ for some $\iota > 0$, and $|\mathbf{y}| = c_2 n^{2/(3d-2)}$ for c_2 an appropriately chosen constant depending on d, D. We can bound the dimension of $\mathcal{U}_i$

$$\begin{aligned}
\dim(\mathcal{U}_i) &\leq \sum_{r=2}^{\min\{d,2D\}} \binom{|\mathbf{y}|+d-r-1}{d-r}\binom{|\mathbf{z}|+2D-r-1}{2D-r} \\
&\leq \min\{d,2D\} \cdot \binom{|\mathbf{y}|+d-2-1}{d-2}\binom{|\mathbf{z}|+2D-2-1}{2D-2} \\
&\leq \min\{d,2D\}\,(|\mathbf{y}|+d-2)^{d-2}\binom{|\mathbf{z}|+2D-1}{2D}\frac{2D(2D-1)}{(|\mathbf{z}|+2D-1)(|\mathbf{z}|+2D-2)} \\
&\leq \min\{d,2D\}(2c_2)^{d-2}n^{2(d-2)/(3d-2)}\binom{|\mathbf{z}|+2D-1}{2D}\frac{8D(2D-1)}{n^2} \\
&\leq \min\{d,2D\}8D(2D-1)(2c_2)^{d-2}n^{-4(d-1)/(3d-2)}\binom{|\mathbf{z}|+2D-1}{2D}.
\end{aligned}$$

We can bound the first term of $\dim(\mathcal{N}_i)$ from (28)

$$\begin{aligned}
&\leq (m-1)\binom{|\mathbf{y}|+q-1}{q} \\
&\leq n^{2dD/(3d-2)-\iota} \cdot (2|\mathbf{y}|)^{2D(d-1)} \\
&\leq n^{2dD/(3d-2)-\iota} \cdot 2^{2D(d-1)}(c_2 n^{2/(3d-2)})^{2D(d-1)} \\
&\leq 2^{2D(d-1)} c_2^{2D(d-1)} n^{2D-\iota}
\end{aligned}$$

Taking n sufficiently large in terms of d, D, ι, c_2, we get that

$$\dim(\mathcal{N}_i) \leq \frac{1}{2}\binom{|\mathbf{z}|+2D-1}{2D} = \frac{1}{2}\dim(\mathbb{R}[\mathbf{z}]^{=2D}).$$

Finally, this allows us to complete our leave-one-out bound for $R^\top$. Let $\sigma_{\min}$ denote the least singular value indexed by the number of columns, which is $\binom{|\mathbf{y}|+q-1}{q}$ for $R_i^\top$. We have

$$\begin{aligned}
\sigma_{\min}(\Pi_{\mathcal{R}_i}^{\perp} R_i^\top) &\geq \sigma_{\min}(\Pi_{\mathcal{N}_i}^{\perp} R_i^\top) && \text{since } \mathcal{R}_i \subseteq \mathcal{N}_i \\
&= \sigma_{\min}\left(\binom{3D}{2D}\Pi_{\mathcal{N}_i}^{\perp}((Z_i^\top)^{\circledast 2D}\mathsf{Sym}_{2D(d-1)}^\top + K_i^\top)\right) \\
&= \sigma_{\min}\left(\binom{3D}{2D}\Pi_{\mathcal{N}_i}^{\perp}(Z_i^\top)^{\circledast 2D}\mathsf{Sym}_{2D(d-1)}^\top\right) && \text{by (27)} \\
&\geq \sigma_{\min}\left(\binom{3D}{2D}\Pi_{\mathcal{N}_i}^{\perp}(Z_i^\top)^{\circledast 2D}\right) && \text{since } \mathsf{Sym}_{2D(d-1)}^\top \text{ is an isometry} \\
&\geq \Omega_D\left(\frac{\rho^{2D}}{n^{2D}}\right) && \text{with prob. } 1-\exp(-\Omega_{D,c_2}(n)),
\end{aligned}$$

by Theorem 5.1 in [BEG+24], which says that for a fixed projection onto a large enough space (here, $\mathcal{N}_i^\perp$), the projection of a symmetric lift of a smoothed matrix will be well-conditioned.

This tells us that each block $R_i^\top$ contributes sufficient linear independence. Taking a union bound over the m blocks, we have that with probability $1 - m\exp(-\Omega_{D,c_2}(n)) \geq 1 - \mathrm{poly}(n)\exp(-\Omega(n))$, we have

$$\sigma_{\min}\left(R^\top\right) \geq \frac{1}{\sqrt{m}}\Omega_D\left(\frac{\rho^{2D}}{n^{2D}}\right) \geq \mathrm{poly}(1/n, \rho).$$

Final bound on $H_{\mathbf{y}}$. Combining with our least singular value bound on $\mathsf{M}_{\mathbf{b}^D}^{(dD+q)}$ from (25), and conditioning on the above bound for R, we have

$$\begin{aligned}\sigma_{\dim(\mathcal{I}_{\mathbf{b}^D})}(H_{\mathbf{y}}) &\geq \sigma_{\dim(\mathcal{I}_{\mathbf{b}^D})}\left(\mathsf{M}_{\mathbf{b}^D}^{(dD+q)}\right)\sigma_{\min}(R)\\ &\geq \mathrm{poly}(1/n, \rho).\end{aligned}$$

□

Lemma 5.3 (Robustly recovering multiples of $b_i(\mathbf{y})^D$)**.** *Let $H_{\mathbf{y}}$ be as defined in Lemma 5.2. Let δ be the least singular value of $H_{\mathbf{y}}$,*

$$\delta = \sigma_{\dim(\mathcal{I}_{\mathbf{b}^D})}(H_{\mathbf{y}}),$$

where $\delta > \mathrm{poly}(1/n, \rho)$ by Lemma 5.2. Let $\widehat{H}_{\mathbf{y}}$ be such that $\|H_{\mathbf{y}} - \widehat{H}_{\mathbf{y}}\| \leq \varepsilon$, for ε sufficiently smaller than δ (by $\mathrm{poly}(1/n, \rho)$ factors).

Then, we can recover a matrix $\widehat{B}_{\mathbf{y}}$ such that

$$\forall v \in \mathrm{Im}(\widehat{B}_{\mathbf{y}}),\ \|v\| = 1, \qquad \|\Pi_{\mathbf{b}^D}^\perp v\| < \varepsilon \cdot \mathrm{poly}(n, 1/\rho),$$

where $\Pi_{\mathbf{b}^D}^\perp$ is the projection orthogonal to $\mathrm{span}\{b_i(\mathbf{y})^D\}$, with probability $1 - \exp(-\mathrm{poly}(n))$.

Proof. Let $h^\star(\mathbf{y}) \in \mathbb{R}[\mathbf{y}]^{=q}$ be a polynomial drawn by the algorithm, with independent $\mathcal{N}(0, \rho^2)$ coefficients. Let $\mathsf{M}_{h^\star}^{(dD+q)} : \mathbb{R}[\mathbf{y}]^{=dD} \to \mathbb{R}[\mathbf{y}]^{=dD+q}$ be the multiplication map (Macaulay matrix) of degree dD polynomials by $h^\star$. Let

$$B_{\mathbf{y}} := \ker\left(\Pi_{H_{\mathbf{y}}}^\perp M_{h^\star}^{(dD+q)}\right).$$

Note that $M_{h^\star}^{(dD+q)} B_{\mathbf{y}}$ spans the intersection of $\mathrm{Im}(H_{\mathbf{y}})$ and $\mathrm{Im}(\mathsf{M}_{h^\star}^{(dD+q)})$. Analogously, define the estimate

$$\widehat{B}_{\mathbf{y}} := \ker\left(\Pi_{\widehat{H}_{\mathbf{y}}}^\perp M_{h^\star}^{(dD+q)}\right),$$

which is computed using $\widehat{H}_{\mathbf{y}}$ instead of $H_{\mathbf{y}}$, and taking ker to be the *approximate* kernel given by the bottom m right singular space of $\Pi_{\widehat{H}_{\mathbf{y}}}^\perp M_{h^\star}^{(dD+q)}$. Note that $M_{h^\star}^{(dD+q)}$ is still exact because we generate it ourselves.

Write $\widehat{H}_{\mathbf{y}} = H_{\mathbf{y}} + E$ for $\|E\| < \varepsilon$. We have

$$\begin{aligned}\widehat{H}_{\mathbf{y}}\widehat{H}_{\mathbf{y}}^\top &= (H_{\mathbf{y}} + E)(H_{\mathbf{y}} + E)^\top\\ \|\widehat{H}_{\mathbf{y}}\widehat{H}_{\mathbf{y}}^\top - H_{\mathbf{y}}H_{\mathbf{y}}^\top\| &\leq \varepsilon \cdot \mathrm{poly}(n),\end{aligned}$$

since $\|H_{\mathbf{y}}\| < \mathrm{poly}(n)$.

Let $r = \dim(\mathcal{I}_{\mathbf{b}^D})$. We have that the rth singular value of $H_{\mathbf{y}}H_{\mathbf{y}}^\top$ is at least δ^2. By the above bound, we have that the least rth singular value of $\widehat{H}_{\mathbf{y}}\widehat{H}_{\mathbf{y}}^\top$ is at least $\delta^2 - \varepsilon \cdot \mathrm{poly}(n)$. We assume that ε is chosen as a sufficiently small (inverse polynomial of n) that $\delta^2 - \varepsilon \cdot \mathrm{poly}(n) \geq \delta^2/2$.

Let $\Pi_{H_{\mathbf{y}}}$ be the projector onto the image of $H_{\mathbf{y}}H_{\mathbf{y}}^\top$, and $\Pi^\perp_{H_{\mathbf{y}}}$ be the projector orthogonal to the image of $H_{\mathbf{y}}H_{\mathbf{y}}^\top$. Define $\Pi_{\widehat{H}_{\mathbf{y}}}$ as the projector onto the top r-singular space of $\widehat{H}_{\mathbf{y}}$ and $\Pi^\perp_{\widehat{H}_{\mathbf{y}}}$ as the orthogonal projector. We have

$$\begin{aligned}
\Pi^\perp_{H_{\mathbf{y}}} - \Pi^\perp_{\widehat{H}_{\mathbf{y}}} &= (\Pi^\perp_{H_{\mathbf{y}}} - \Pi^\perp_{\widehat{H}_{\mathbf{y}}})(\Pi_{\widehat{H}_{\mathbf{y}}} + \Pi^\perp_{\widehat{H}_{\mathbf{y}}}) \\
&= \Pi^\perp_{H_{\mathbf{y}}}\Pi_{\widehat{H}_{\mathbf{y}}} + \Pi^\perp_{H_{\mathbf{y}}}\Pi^\perp_{\widehat{H}_{\mathbf{y}}} - \Pi^\perp_{\widehat{H}_{\mathbf{y}}}\Pi_{\widehat{H}_{\mathbf{y}}} - \Pi^\perp_{\widehat{H}_{\mathbf{y}}}\Pi^\perp_{\widehat{H}_{\mathbf{y}}} \\
&= \Pi^\perp_{H_{\mathbf{y}}}\Pi_{\widehat{H}_{\mathbf{y}}} + (\Pi^\perp_{H_{\mathbf{y}}} - I)\Pi^\perp_{\widehat{H}_{\mathbf{y}}} \\
&= \Pi^\perp_{H_{\mathbf{y}}}\Pi_{\widehat{H}_{\mathbf{y}}} - \Pi_{H_{\mathbf{y}}}\Pi^\perp_{\widehat{H}_{\mathbf{y}}}.
\end{aligned}$$

By the Davis-Kahan theorem (see e.g. Theorem VII.3.1 in [Bha97]) we have

$$\|\Pi^\perp_{H_{\mathbf{y}}}\Pi_{\widehat{H}_{\mathbf{y}}}\| \leq \frac{2}{\delta^2}\|H_{\mathbf{y}}H_{\mathbf{y}}^\top - \widehat{H}_{\mathbf{y}}\widehat{H}_{\mathbf{y}}^\top\| \leq \frac{\varepsilon \cdot \mathrm{poly}(n)}{\delta^2}, \qquad \|\Pi_{H_{\mathbf{y}}}\Pi^\perp_{\widehat{H}_{\mathbf{y}}}\| \leq \frac{2}{\delta^2}\|H_{\mathbf{y}}H_{\mathbf{y}}^\top - \widehat{H}_{\mathbf{y}}\widehat{H}_{\mathbf{y}}^\top\| \leq \frac{\varepsilon \cdot \mathrm{poly}(n)}{\delta^2}.$$

Thus,

$$\|\Pi^\perp_{H_{\mathbf{y}}} - \Pi^\perp_{\widehat{H}_{\mathbf{y}}}\| \leq \|\Pi^\perp_{H_{\mathbf{y}}}\Pi_{\widehat{H}_{\mathbf{y}}}\| + \|\Pi_{H_{\mathbf{y}}}\Pi^\perp_{\widehat{H}_{\mathbf{y}}}\| \leq \varepsilon \cdot \mathrm{poly}(n, 1/\delta).$$

Since $\|M_{h^\star}^{(dD+q)}\| \leq \mathrm{poly}(n)$ we have

$$\left\|\Pi^\perp_{H_{\mathbf{y}}} M_{h^\star}^{(dD+q)} - \Pi^\perp_{\widehat{H}_{\mathbf{y}}} M_{h^\star}^{(dD+q)}\right\| < \varepsilon \cdot \mathrm{poly}(n, 1/\delta).$$

Since $\mathrm{Im}(H_{\mathbf{y}}) = \mathcal{I}_{\mathbf{b}^D}$, by Theorem 4.1, we have that with probability $1 - \exp(-\mathrm{poly}(n))$,

$$\begin{aligned}
&\Pi^\perp_{H_{\mathbf{y}}} M_{h^\star}^{(dD+q)} b = 0 && \text{for } b \in \mathrm{span}\{b_i(\mathbf{y})^D\} \\
&\left\|\Pi^\perp_{H_{\mathbf{y}}} M_{h^\star}^{(dD+q)} x\right\| \geq \mathrm{poly}(1/n, \rho)\|x\| && \text{for } x \perp \mathrm{span}\{b_i(\mathbf{y})^D\}.
\end{aligned}$$

Take $\widehat{B}_{\mathbf{y}}$ to be a well-conditioned basis spanning the bottom m singular space of $\Pi^\perp_{\widehat{H}_{\mathbf{y}}} M_{h^\star}^{(dD+q)}$. By a similar Davis-Kahan argument, we have that for

$$\forall v \in \mathrm{Im}(\widehat{B}_{\mathbf{y}}),\ \|v\| = 1, \qquad \|\Pi^\perp_{\mathbf{b}^D} v\| < \varepsilon \cdot \mathrm{poly}(n, 1/\rho),$$

where $\Pi^\perp_{\mathbf{b}^D}$ is the projection orthogonal to $\mathrm{span}\{b_i(\mathbf{y})^D\}$.

□

5.3 Step 2: Desymmetrization

Lemma 5.4 (One block desymmetrization)**.** *Let $B_{\mathbf{y}}, \widehat{B}_{\mathbf{y}}$ be as defined in Lemma 5.3, $\|B_{\mathbf{y}} - \widehat{B}_{\mathbf{y}}\| \leq \varepsilon$ for ε sufficiently small in terms of $\mathrm{poly}(1/n, \rho)$. Let $p(\mathbf{y})$ be the power-sum polynomial $p(x)$ restricted to variables y, where $|\mathbf{y}| > \Omega(n^{2/(3d-2)})$, and let $\widehat{p}(y)$ be an estimate of $p(y)$ with $\|\widehat{p}(y) - p(y)\| < \varepsilon$.*

Define $T_{\mathbf{y}} := \sum_{i=1}^m (b_i(y)^D)^{\otimes 3} \in (\mathbb{R}[y]^{=dD})^{\otimes 3}$. Then, we can recover $\widehat{T}_{\mathbf{y}} \in (\mathbb{R}[y]^{=dD})^{\otimes 3}$ such that

$$\|T_{\mathbf{y}} - \widehat{T}_{\mathbf{y}}\| < \varepsilon \cdot \mathrm{poly}(n, 1/\rho).$$

Proof. Let $\mathsf{Sym}_{dD\to 3dD} : (\mathbb{R}[\mathbf{y}]^{=dD})^{\circledast 3} \to \mathbb{R}[\mathbf{y}]^{=3dD}$ be the symmetrization operator that takes an order 3 symmetric tensor on degree-dD polynomials to a degree-$3dD$ polynomial. This can be seen as reformatting the multiplication operation. Let $p(\mathbf{y})$ be $p(\mathbf{x})$ restricted to the variables $\mathbf{y}$. Let

$$T_{\mathbf{y}} := B_{\mathbf{y}}^{\circledast 3}(\mathsf{Sym}_{dD\to 3dD} B_{\mathbf{y}}^{\circledast 3})^{\dagger} p(\mathbf{y}).$$

Here $T_{\mathbf{y}}$ and $p(\mathbf{y})$ are both vectors from the perspective of the linear algebra.

We define the estimate $\widehat{T}_{\mathbf{y}}$ using the estimates $\widehat{B}_{\mathbf{y}}$ and $\widehat{p}(\mathbf{y})$ in place of the true values.

$$\widehat{T}_{\mathbf{y}} := \widehat{B}_{\mathbf{y}}^{\circledast 3} \left(\mathsf{Sym}_{dD\to 3dD} \widehat{B}_{\mathbf{y}}^{\circledast 3}\right)^{\dagger} \widehat{p}(\mathbf{y})$$

Conditioning of $\mathsf{Sym}_{dD\to 3dD} B_{\mathbf{y}}^{\circledast 3}$. Let $B^{\star} : \mathbb{R}^m \to \mathbb{R}[\mathbf{y}]^{=dD}$, where column i of $B^{\star}$ is given by $B_i^{\star} = b_i(\mathbf{y})^D$.

$$B_{\mathbf{y}} = B^{\star} K,$$

By assumption we have that $\sigma_{\max}(B), \sigma_{\max}(B^{\star}) < \mathrm{poly}(n)$. By Lemma 5.3 we have that $\sigma_{\min}(B) > \mathrm{poly}(1/n, \rho)$. Let $B_{\mathrm{base}} : \mathbb{R}^m \to \mathbb{R}[\mathbf{y}]^{=d}$ be the matrix with each column i containing $b_i(\mathbf{y})$. Note that B_{base} is ρ-perturbed. We have that

$$\mathrm{Im}(B) = \mathrm{Im}(\mathsf{Sym}_{d\to dD} B_{\mathrm{base}}^{\odot D}),$$

since the left hand side is exactly spanned by the polynomials $b_i(\mathbf{y})^D$. Thus, there exists a transformation $K : \mathbb{R}^m \to \mathbb{R}^m$ such that

$$B_{\mathbf{y}} = \mathsf{Sym}_{d\to dD} B_{\mathrm{base}}^{\odot D} K, \quad \sigma_{\max}(K) < \mathrm{poly}(n, 1/\rho),\ \sigma_{\min}(K) > \mathrm{poly}(1/n, \rho).$$

This gives us that

$$\begin{aligned}
\mathsf{Sym}_{dD\to 3dD} B_{\mathbf{y}}^{\circledast 3} &= \mathsf{Sym}_{dD\to 3dD} \left(\mathsf{Sym}_{d\to dD} B_{\mathrm{base}}^{\odot D} K\right)^{\circledast 3} \\
&= \mathsf{Sym}_{dD\to 3dD} \mathsf{Sym}_{d\to dD}^{\otimes 3} (B_{\mathrm{base}}^{\odot D})^{\otimes 3} K^{\circledast 3} \\
&= \mathsf{Sym}_{3dD} (B_{\mathrm{base}}^{\odot D})^{\otimes 3} K^{\circledast 3}.
\end{aligned}$$

Note that the columns of $K^{\circledast 3}$ are 3-way symmetric by definition. Therefore, the above is equal to

$$\begin{aligned}
\mathsf{Sym}_{dD\to 3dD} B_{\mathbf{y}}^{\circledast 3} &= \mathsf{Sym}_{3dD} (B_{\mathrm{base}}^{\odot D})^{\otimes 3} \mathsf{Sym}_3^{\top} \mathsf{Sym}_3 K^{\circledast 3} \\
&= \mathsf{Sym}_{3dD} (B_{\mathrm{base}}^{\odot D})^{\circledast 3} \mathsf{Sym}_3 K^{\circledast 3}.
\end{aligned}$$

By Lemma 5.5 we have that

$$\sigma_{\binom{m+3-1}{3}} \left(\mathsf{Sym}_{3dD} (B_{\mathrm{base}}^{\odot D})^{\circledast 3}\right) > \mathrm{poly}(1/n, \rho).$$

For the second part, we have that

$$\begin{aligned}
\sigma_{\binom{m+3-1}{3}} \left(\mathsf{Sym}_3 K^{\circledast 3}\right) &= \sigma_{\binom{m+3-1}{3}} \left(K^{\circledast 3}\right) && \text{since } \mathsf{Sym}_3 \text{ isometric on } \mathrm{Im}(K^{\circledast 3}) \\
&\geq \sigma_{m^3} \left(K^{\otimes 3}\right) \\
&\geq \sigma_m(K)^3 \\
&\geq \mathrm{poly}(1/n, \rho).
\end{aligned}$$

Thus, we have that

$$\begin{aligned}
\sigma_{\binom{m+3-1}{3}} \left(\mathsf{Sym}_{dD\to 3dD} B_{\mathbf{y}}^{\circledast 3}\right) &\geq \sigma_{\binom{m+3-1}{3}} \left(\mathsf{Sym}_{3dD} (B_{\mathrm{base}}^{\odot D})^{\circledast 3}\right) \sigma_{\binom{m+3-1}{3}} \left(\mathsf{Sym}_3 K^{\circledast 3}\right) \\
&\geq \mathrm{poly}(1/n, \rho).
\end{aligned}$$

Perturbation bound for $(\mathsf{Sym}_{dD\to 3dD}B_{\mathbf{y}}^{\circledast 3})^\dagger$. Since $\|B_{\mathbf{y}} - \widehat{B}_{\mathbf{y}}\| < \varepsilon \cdot \mathrm{poly}(n, 1/\rho)$, and $\|B_{\mathbf{y}}\| < \mathrm{poly}(n), \|\mathsf{Sym}_{dD\to 3dD}\| < \mathrm{poly}(n)$, we have that

$$\left\|\mathsf{Sym}_{dD\to 3dD}B_{\mathbf{y}}^{\circledast 3} - \mathsf{Sym}_{dD\to 3dD}\widehat{B}_{\mathbf{y}}^{\circledast 3}\right\| \le \varepsilon \cdot \mathrm{poly}(n, 1/\rho).$$

Taking ε to be a sufficiently small, we have that

$$\sigma_{\binom{m+3-1}{3}}\left(\mathsf{Sym}_{dD\to 3dD}\widehat{B}_{\mathbf{y}}^{\circledast 3}\right) \ge \sigma_{\binom{m+3-1}{3}}\left(\mathsf{Sym}_{dD\to 3dD}B_{\mathbf{y}}^{\circledast 3}\right) - \varepsilon \cdot \mathrm{poly}(n, 1/\rho) \ge \mathrm{poly}(1/n, \rho).$$

Since both $\mathsf{Sym}_{dD\to 3dD}B_{\mathbf{y}}^{\circledast 3}$ and $\mathsf{Sym}_{dD\to 3dD}\widehat{B}_{\mathbf{y}}^{\circledast 3}$ have rank exactly $\binom{m+3-1}{3}$ (the number of columns), we have that

$$\left\|(\mathsf{Sym}_{dD\to 3dD}B_{\mathbf{y}}^{\circledast 3})^\dagger\right\| \le \mathrm{poly}(n, 1/\rho),$$
$$\left\|(\mathsf{Sym}_{dD\to 3dD}\widehat{B}_{\mathbf{y}}^{\circledast 3})^\dagger\right\| \le \mathrm{poly}(n, 1/\rho).$$

Using the perturbation bound of [Wed73] (Theorem 4.1) for pseudoinverses, we have that

$$\begin{aligned}
&\left\|(\mathsf{Sym}_{dD\to 3dD}B_{\mathbf{y}}^{\circledast 3})^\dagger - (\mathsf{Sym}_{dD\to 3dD}\widehat{B}_{\mathbf{y}}^{\circledast 3})^\dagger\right\| \\
\le\ &\left\|(\mathsf{Sym}_{dD\to 3dD}B_{\mathbf{y}}^{\circledast 3})^\dagger\right\| \left\|(\mathsf{Sym}_{dD\to 3dD}\widehat{B}_{\mathbf{y}}^{\circledast 3})^\dagger\right\| \left\|\mathsf{Sym}_{dD\to 3dD}B_{\mathbf{y}}^{\circledast 3} - \mathsf{Sym}_{dD\to 3dD}\widehat{B}_{\mathbf{y}}^{\circledast 3}\right\| \\
\le\ &\varepsilon \cdot \mathrm{poly}(n, 1/\rho).
\end{aligned}$$

Final bound. Finally, since we have that

$$\begin{aligned}
\|p(\mathbf{y}) - \widehat{p}(\mathbf{y})\| &\le \varepsilon \\
\left\|B_{\mathbf{y}}^{\circledast 3} - \widehat{B}_{\mathbf{y}}^{\circledast 3}\right\| &\le \varepsilon \cdot \mathrm{poly}(n, 1/\rho), \\
\left\|(\mathsf{Sym}_{dD\to 3dD}B_{\mathbf{y}}^{\circledast 3})^\dagger - (\mathsf{Sym}_{dD\to 3dD}\widehat{B}_{\mathbf{y}}^{\circledast 3})^\dagger\right\| &\le \varepsilon \cdot \mathrm{poly}(n, 1/\rho),
\end{aligned}$$

and all of these objects have norm bounded by $\mathrm{poly}(n, 1/\rho)$, we have that

$$\begin{aligned}
\left\|T_{\mathbf{y}} - \widehat{T}_{\mathbf{y}}\right\| &= \left\|B_{\mathbf{y}}^{\circledast 3}(\mathsf{Sym}_{dD\to 3dD}B_{\mathbf{y}}^{\circledast 3})^\dagger p(\mathbf{y}) - \widehat{B}_{\mathbf{y}}^{\circledast 3}\left(\mathsf{Sym}_{dD\to 3dD}\widehat{B}_{\mathbf{y}}^{\circledast 3}\right)^\dagger \widehat{p}(\mathbf{y})\right\| \\
&\le \varepsilon \cdot \mathrm{poly}(n, 1/\rho).
\end{aligned}$$

□

Lemma 5.5 (Symmetric Lift of Khatri-Raos)**.** *Let $d, D \in \mathbb{N}$. Let $A_1, \ldots, A_m$ be independently ρ-perturbed d-th order symmetric tensors over $\mathbb{R}^n$, and let A be the matrix with columns $A_1, \ldots, A_m$. Let $B = A^{\odot D}$, and $\Phi : \mathsf{Sym}^{3dD}(\mathbb{R}^n) \to \mathbb{R}^R$ be a linear operator of rank δn^{3dD}, for a small $\delta \in (0, 1)$, with $\sigma_{\delta n^{3dD}}(\Phi) \ge 1$. Then, we have that*

$$\sigma_{\binom{m+2}{3}}(\Phi\mathsf{Sym}_{3dD}B^{\circledast 3}) \ge \Omega_{d,D}\left(\frac{\rho^{3D}}{m^{3/2}n^{3dD/2}}\right)$$

with probability at least $1 - \exp(-\Omega_{d,D,\delta}(n^d))$, whenever $m \le cn^{dD}$, for a small enough constant c depending only on d, D, δ.

Proof. Let $M = \Phi \mathsf{Sym}_{3dD}(B^{\circledast 3})$. Then, for $1 \le i \le j \le k \le m$, we can write the (i,j,k)-th column of M as

$$M_{ijk} = \Phi \cdot \mathsf{Sym}_{3dD}(A_i^{\otimes D} \otimes A_j^{\otimes D} \otimes A_k^{\otimes D}).$$

We will use a leave one out bound. Let $\mathcal{M}_{-(i,j,k)}$ be the space spanned by the columns of M, except the (i,j,k)-th column. Using the leave one out lemma , we have

$$\sigma_{\binom{m+2}{3}}(M) \ge \frac{1}{\sqrt{\binom{m+2}{3}}} \min_{1 \le i \le j \le k \le m} \left\| \Pi^{\perp}_{\mathcal{M}_{-(i,j,k)}} M_{ijk} \right\|.$$

We bound $\left\| \Pi^{\perp}_{\mathcal{M}_{-(i,j,k)}} M_{ijk} \right\|$ by successively projecting out the columns - we first project out a space that contains all the columns that do not involve any of i,j,k in each mode, and then we argue about the remaining block.

Let $I = \{i,j,k\}$, and define the subspace given by

$$\mathcal{S}_I = \Phi \cdot \mathsf{Sym}_{3dD}\left(\mathrm{span}\{A_\ell^{\otimes D} : \ell \notin I\} \otimes (\mathbb{R}^n)^{\otimes dD} \otimes (\mathbb{R}^n)^{\otimes dD} \right).$$

Note that $\mathcal{S}_I$ contains all columns of M that do not involve any of the indices in I, in each of the modes. Further, $\dim \mathcal{S}_I \le (m - |I|) n^{2dD} \le \frac{1}{2}\delta n^{3dD} = \frac{1}{2}\mathsf{rank}\Phi$, if we pick $m \le cn^{dD}$ for small enough c. Let $\Psi_I = \Pi^{\perp}_{\mathcal{S}_I}\Phi$; from the earlier argument, we have $\mathsf{rank}(\Psi_I) \ge \mathsf{rank}\Phi - \dim \mathcal{S}_I \ge \frac{1}{2}\delta n^{3dD}$. We condition on all columns $A_\ell, \ell \notin I$; thus, Ψ_I is a fixed linear operator of rank at least $\frac{1}{2}\delta n^{3dD}$, in fact we have $\sigma_{1/2\delta n^{3dD}}(\Psi_I) \ge 1$.

Projecting out these columns from M, we are left with only the columns that involve only the indices in I in each of the modes i.e. the surviving columns form the matrix

$$C_I = \Psi_I \mathsf{Sym}_{3dD}\big(A_\ell^{\otimes D} : \ell \in I\big)^{\circledast 3}$$

Out of these surviving columns, we want to show that the column given by $\Psi_I \mathsf{Sym}_{3dD}(A_i^{\otimes D} \otimes A_j^{\otimes D} \otimes A_k^{\otimes D})$ is far away from the span of the other columns. We do this by showing that the matrix C_I has a large least singular value. To do this, note that the matrix C_I is a submatrix of a much larger matrix, whose least singular value we can bound. Consider the matrix defined by

$$P_I = \Psi_I \mathsf{Sym}_{3dD}\big(A_\ell : \ell \in I\big)^{\circledast 3D}$$

By Theorem 5.1 in [BEG+24], we have that $\sigma_{min}(P_I) \ge \Omega_{d,D}\left(\frac{\rho^{3D}}{n^{3dD/2}}\right)$ with probability at least $1 - \exp(-\Omega(n^d))$. Since C_I is a submatrix of P_I, this gives us that the distance of $\Psi_I \mathsf{Sym}_{3dD}(A_i^{\otimes D} \otimes A_j^{\otimes D} \otimes A_k^{\otimes D})$ from the other columns of C_I is at least $\Omega_{d,D}\left(\frac{\rho^{3D}}{n^{3dD/2}}\right)$. Hence, we have that

$$\left\| \Pi^{\perp}_{\mathcal{M}_{-(i,j,k)}} M_{ijk} \right\| \ge \Omega_{d,D}\left(\frac{\rho^{3D}}{n^{3dD/2}}\right)$$

with probability at least $1 - \exp(-\Omega(n^d))$. Finally, we take a union bound over all the $\binom{m+2}{3}$ choices of i,j,k to get that

$$\sigma_{\binom{m+2}{3}}(M) \ge \Omega_{d,D}\left(\frac{\rho^{3D}}{m^{3/2} n^{3dD/2}}\right)$$

with probability $1 - \exp(-\Omega(n^d))$. □

Lemma 5.6 (Full desymmetrization by tiling $\mathbf{y}$)**.** *Let d, D be constants. Let $\widehat{p}(\mathbf{x})$ be an estimated polynomial such that $\|p(\mathbf{x}) - \widehat{p}(\mathbf{x})\| < \varepsilon$, for $p(x) = \sum_{i=1}^{m} a_i(\mathbf{x})^{3D}$, and $m < n^{2dD/(3d-2)-\iota}$ for some $\iota > 0$. Let $T_{\mathbf{x}}, \widehat{T}_{\mathbf{x}}$ be defined as in Lemma 5.4. Then, we can estimate a $\widehat{T}_{\mathbf{x}}$ such that*

$$T_{\mathbf{x}} := \sum_{i=1}^{m} (a_i(\mathbf{x})^D)^{\otimes 3}$$
$$\|\widehat{T}_{\mathbf{x}} - T_{\mathbf{x}}\| \leq \varepsilon \cdot \mathrm{poly}(n, 1/\rho), \text{ with probability } 1 - \mathrm{poly}(1/n).$$

Proof. Let $S_1, \ldots, S_N$ be independently randomly selected subsets of $\mathbf{x}$, each of size $|\mathbf{y}| = c_2 n^{2/(3d-2)}$, for $N = \Omega(n^{3dD} \log n)$ for some constant c_2.

For any given multiset of indices $I \subseteq [n]$ of $|I| = 3dD$, we have that the probability that a single draw S captures all of I is at least

$$\begin{aligned}
\frac{\binom{n-3dD}{|\mathbf{y}|-3dD}}{\binom{n}{|\mathbf{y}|}} &= \frac{(n-3dD)!}{(|\mathbf{y}|-3dD)!(n-|\mathbf{y}|)!} \cdot \frac{|\mathbf{y}|!(n-|\mathbf{y}|)!}{n!} \\
&= \frac{|\mathbf{y}|(|\mathbf{y}|-1)\ldots(|\mathbf{y}|-3dD+1)}{n(n-1)\ldots(n-3dD+1)} \\
&\geq \left(\frac{|\mathbf{y}|-3dD+1}{n-3dD+1}\right)^{3dD} \\
&\geq \frac{1}{n^{3dD}}.
\end{aligned}$$

Thus, the probability that one of the N draws captures all of I is at least

$$\begin{aligned}
1 - \left(1 - \frac{1}{n^{3dD}}\right)^N &\geq 1 - e^{-N/n^{3dD}} \\
&\geq 1 - \mathrm{poly}(1/n).
\end{aligned}$$

Conditioning on the above, and applying Lemmas 5.2, 5.3, and 5.4 with $\mathbf{y}$ restricted to each S_j, we have that with probability $1 - \mathrm{poly}(1/n)$, we can estimate every entry of

$$T_{\mathbf{x}} = \sum_{i=1}^{m} (a_i(\mathbf{x})^D)^{\otimes 3},$$

upto additive error $\varepsilon \cdot \mathrm{poly}(n, 1/\rho)$. Thus, by copying entries from the desymmetrized blocks, into the full tensor we can find $\widehat{T}_{\mathbf{x}}$ such that

$$\|\widehat{T}_{\mathbf{x}} - T_{\mathbf{x}}\| \leq \varepsilon \cdot \mathrm{poly}(n, 1/\rho).$$

□

5.4 Step 3: Tensor Decomposition

Lemma 5.7 (Tensor Decomposition)**.** *Let d, D be constants. Let $T_{\mathbf{x}} = \sum_{i=1}^{m} (a_i(\mathbf{x})^D)^{\otimes 3}$, with all components $\|a_i(\mathbf{y})\| < \mathrm{poly}(n)$, and $m < n^{2dD/(3d-2)-\iota}$ for some $\iota > 0$. Let $\widehat{T}_{\mathbf{x}}$ be an approximation such that $\|\widehat{T}_{\mathbf{x}} - T_{\mathbf{x}}\| \leq \varepsilon \cdot \mathrm{poly}(n, 1/\rho)$. Let A_{base} be the matrix of the $a_i(\mathbf{x})$. Then, we can recover a matrix $\widehat{A}_{\text{base}}$ such that*

$$\|\widehat{A}_{\text{base}} - A_{\text{base}} M_\pi\| < \mathrm{poly}(\varepsilon, n, 1/\rho),$$

for some permutation matrix over the columns $M_\pi : \mathbb{R}^m \to \mathbb{R}^m$ (where each column is correct up to sign when D is even).

Proof. First we use Jennrich's tensor decomposition algorithm to recover estimates of the $a_i(\mathbf{x})^D$. Then we can take their roots to recover the underlying $a_i(\mathbf{x})$.

Tensor decomposition. This follows from the robust analysis of Jennrich's tensor decomposition algorithm from [BCMV14]. Let $A : \mathbb{R}^m \to \mathbb{R}[\mathbf{x}]^{=dD}$ be the factor matrix of the tensor T. That is, it has column i given by $A_i = a_i(\mathbf{x})^D$. By [BCPV19] (Theorem 2.1), since $a_i(\mathbf{x})$ are ρ-perturbed and have magnitude bounded by $\mathrm{poly}(n)$, and since $m < c_1 n^{dD}$ for c_1 a sufficiently small constant (i.e., m is sufficiently smaller than the ambient space of the image $\mathbb{R}[\mathbf{x}]^{=dD}$), we have that $\sigma_m(A) > \mathrm{poly}(1/n, \rho)$. Thus, we meet Condition 2.2 of [BCMV14] and can apply Theorem 2.3 of [BCMV14] to conclude that Jennrich's algorithm applied to $\widehat{T}_{\mathbf{x}}$ will give a decomposition $\widehat{U}, \widehat{V}, \widehat{W}$, where for some permutation matrix of the columns $M_\pi : \mathbb{R}^m \to \mathbb{R}^m$, we have

$$\max_i \left\| \widehat{U}_i \otimes \widehat{V}_i \otimes \widehat{W}_i - (AM_\pi)_i^{\otimes 3} \right\| < \varepsilon \cdot \mathrm{poly}(n, 1/\rho).$$

After relabeling, we now recover the scale of each powered component from its rank-one tensor. Let $e_i = \widehat{U}_i / \|\widehat{U}_i\|$, let $\widehat{R}_i = \widehat{U}_i \otimes \widehat{V}_i \otimes \widehat{W}_i$, and set

$$\widehat{a}_i^{(D)} = \sqrt[3]{\langle \widehat{R}_i, e_i^{\otimes 3} \rangle}\, e_i.$$

This gives $\|\widehat{a}_i^{(D)} - A_i\| \le \varepsilon \mathrm{poly}(n, 1/\rho)$.

Polynomial roots. Finally for each $i \in [m]$, we have an estimate $\widehat{a}_i^{(D)}(\mathbf{x})$ such that

$$\left\| a_i(\mathbf{x})^D - \widehat{a}_i^{(D)}(\mathbf{x}) \right\| \le \mathrm{poly}(\varepsilon, n, 1/\rho).$$

This can be done by the algorithm provided by [BHKX22] Lemma 5.9 for stable computation of Dth roots (upto sign when D is even). □

6 Application: Robust Uniqueness of Tensor Decomposition

Given a homogeneous degree d polynomial $T(\mathbf{x})$, its Waring rank is defined as the smallest integer m for which we can write $T(\mathbf{x}) = \sum_{i=1}^m \ell_i(\mathbf{x})^d$, where each ℓ_i is a linear form (in the variables $\mathbf{x}$). Waring rank is a symmetric variant of the *tensor rank*. The example of Shitov [Shi18] (see also Lovitz [Lov26]) shows that the two notions are actually different in general.

Throughout this section, we associate linear forms in n variables with vectors in $\mathbb{R}^n$; specifically, the form $\langle \ell, \mathbf{x} \rangle$ is associated with the vector $\ell \in \mathbb{R}^n$. For an ordered tuple $\mathbf{a} = (a_1, \ldots, a_m)$ of linear forms, we define

$$\Phi(\mathbf{a}) = \sum_{i=1}^m a_i^3.$$

Here, a_i^3 is the cubic form defined by $\langle a_i, \mathbf{x} \rangle^3$, viewed as a vector in $\mathbb{R}^{N_3}$, where

$$N_3 := \dim \mathbb{R}[\mathbf{x}]^{=3} = \binom{n+2}{3}.$$

Thus, $\Phi(\mathbf{a})$ is a cubic polynomial in the variables $\mathbf{x}$. Throughout, we also write $p = mn$. Our goal will be to reason about the uniqueness of decompositions. For this, we need appropriate notions of distance, that capture permutation among summands. Given two tuples of linear forms $\mathbf{a} = (a_1, a_2, \ldots, a_m)$ and $\mathbf{b} = (b_1, b_2, \ldots, b_m)$ of equal length, define the following:

$$\delta(\mathbf{a}, \mathbf{b}) = \|\Phi(\mathbf{a}) - \Phi(\mathbf{b})\|, \qquad d_{\mathrm{perm}}(\mathbf{a}, \mathbf{b}) = \min_{\pi \in S_m} \Big(\sum_i \|a_i - b_{\pi(i)}\|^2 \Big)^{1/2}.$$

Definition 6.1 (Uniqueness of Decomposition)**.** Given a homogeneous cubic polynomial $T(\mathbf{x})$ and a decomposition $\mathbf{a} = (a_1, a_2, \ldots, a_m)$ such that $\Phi(\mathbf{a}) = T(\mathbf{x})$, we say that $\mathbf{a}$ is the unique rank-m decomposition of T if for all $\mathbf{b} = (b_1, b_2, \ldots, b_m)$, we have that

$$\delta(\mathbf{a}, \mathbf{b}) = 0 \implies d_{\mathrm{perm}}(\mathbf{a}, \mathbf{b}) = 0.$$

In applications to learning and numerical analysis, tensors are usually estimated from data, and are thus known only up to a small error. In many of these settings (e.g., in tensors estimated via empirical moments; see [AGH+14]), the error can be driven drown, at the expense of sample complexity. In particular, using N data samples, we can only estimate tensor entries to an error $\approx N^{-1/2}$. Thus to conclude approximate *identifiability* of the components,[13] we need a more robust notion of uniqueness.

Definition 6.2 (Robust Uniqueness)**.** Given a homogeneous cubic polynomial $T(\mathbf{x})$ and decomposition $\mathbf{a}$ as above, we say that $\mathbf{a}$ is robustly unique if for any radius parameter R, there is a function $\varrho : \mathbb{R}_{\geq 0} \mapsto \mathbb{R}_{\geq 0}$ such that $\varrho(\xi) \to 0$ when $\xi \to 0$, and an $\varepsilon_0 > 0$ such that for all $\varepsilon < \varepsilon_0$, we have:

$$\forall \mathbf{b} = (b_1, b_2, \ldots, b_m) \text{ such that } \max_i \|b_i\| < R, \quad \delta(\mathbf{a}, \mathbf{b}) < \varepsilon \implies d_{\mathrm{perm}}(\mathbf{a}, \mathbf{b}) < \varrho(\varepsilon).$$

Furthermore, we say $\mathbf{a}$ is *polynomially robust* if ε_0 is polynomial in the "parameters" (defined as n, m, R, and $\max_i \|a_i\|$), and $\varrho(\varepsilon)$ is bounded by a polynomial in ε and the parameters.

Unlike matrices, tensors of small rank *typically* satisfy the unique decomposition property. The classic result of Kruskal [Kru77] showed that for *generic* tensors of rank $m < \lfloor 3n/2 \rfloor$, the decomposition is unique, in the sense defined above. The work of [BCV14] extended this result by showing that under a smoothed analysis model, tensors of rank $m < \lfloor 3n/2 \rfloor$ satisfy the robust uniqueness property (defined above), with polynomial robustness. This was shown to have applications to machine learning, specifically to the recoverability of parameters in latent variable models.

It is natural to ask if the bound $m < \lfloor 3n/2 \rfloor$ can be improved. Following the work of Strassen [Str83], Chiantini, Ottaviani, and Vannieuwenhoven [COV17] showed that uniqueness of decomposition extends to a much larger range of m. Using arguments from algebraic geometry, specifically Terracini's lemma about tangent spaces of secant varieties, they showed that the unique decomposition property holds as long as $m < n^2/6$. However, their method of proof does not yield a robust analog. This raises the question: for what range of m do smoothed rank-m tensors satisfy robust uniqueness with polynomial robustness?

In this section, we provide answers to these questions. Before doing so, let us define the smoothed model formally. Smoothing is controlled by a parameter $\rho \in (0, 1]$, which will appear in our bounds.

[13] Identifiability is an information theoretical notion that says that given T, we can uniquely recover the decomposition $\mathbf{a}$. The notion does not require an efficient algorithm for recovery.

Smoothed decomposition. Let $\bar{\mathbf{a}} = (\bar{a}_1, \bar{a}_2, \ldots, \bar{a}_m)$ be an arbitrary m-tuple of linear functions. A *ρ-smoothing* of $\bar{\mathbf{a}}$ is an m-tuple $\mathbf{a} = (a_1, a_2, \ldots, a_m)$ defined as follows:

$$a_i = \bar{a}_i + \rho g_i, \text{ where } g_i \sim \mathcal{N}(0, \mathbf{I}_n).$$

For convenience, we introduce the following notation for the various parameters of interest:

$$R_a = \max_i \|\bar{a}_i\|, \qquad R_b = \max_i \|b_i\|, \qquad Q = n(2 + R_a + R_b + \rho^{-1}). \tag{29}$$

The main result is the following.

Theorem 6.3. *Suppose $m = O(n^{3/2})$, and let $\bar{\mathbf{a}}$ be any m-tuple of linear functions. Let $\mathbf{a}$ be a ρ-smoothing of $\bar{\mathbf{a}}$. There exist constants c, C_0 such that with probability at least $1 - \exp(-cn)$, the decomposition $\mathbf{a}$ is robustly unique with polynomial robustness. Formally, for any $\mathbf{b} = (b_i)_{i=1}^m$ and Q as defined in (29), we have*

$$d_{\text{perm}}(\mathbf{a}, \mathbf{b}) \leq Q^{C_0} \cdot \delta(\mathbf{a}, \mathbf{b}).$$

This result significantly improves on previously known results of this nature [BCV14], from $m < \lfloor 3n/2 \rfloor$ to $m < O(n^{3/2})$. Interestingly, an $m = o(n^{3/2})$ bound for recoverability appears for tensors with purely random components [GM15], although the bound there comes up for reasons very different from ours. It is natural to ask if polynomial robustness can be improved all the way to $m = O(n^{2-\varepsilon})$ rank. Indeed, when prompted, ChatGPT Astra was able to obtain such a result, using a much more intricate version of the compression scheme we present below.[14] However, the two proofs have the same overall structure, and more relevantly for us, the same use of the approximate syzygy theorem, so we present only the proof until rank $n^{3/2}$.

The proof proceeds by first showing a *local* version of the uniqueness theorem, which can be stated as follows.

Theorem 6.4. *Suppose $m = O(n^{2-\varepsilon})$ for some $\varepsilon > 0$, and let $\bar{\mathbf{a}}$ be any m-tuple of linear functions. Let $\mathbf{a}$ be a ρ-smoothing of $\bar{\mathbf{a}}$. There exist constants c, C, C_0 such that with probability at least $1 - \exp(-cn)$, the decomposition $\mathbf{a}$ is robustly* locally *unique with polynomial robustness: formally, for any $\mathbf{b} = (b_i)_{i=1}^m$ and Q as defined in (29), we have*

$$d_{\text{perm}}(\mathbf{a}, \mathbf{b}) < Q^{-C} \implies d_{\text{perm}}(\mathbf{a}, \mathbf{b}) \leq Q^{C_0} \cdot \delta(\mathbf{a}, \mathbf{b}).$$

Remark. Note that the only difference from Theorem 6.3 is that the conclusion holds for $\mathbf{b}$ that are already "close enough" to $\mathbf{a}$. We also note that this statement holds for any $m = O(n^{2-\varepsilon})$. In fact, the proof turns out to be a fairly simple consequence of our result on smoothed syzygies, Theorem 4.1.

6.1 Notation and preliminaries

Throughout this section, we will write $p = mn$ and $N_2 = \binom{n+1}{2}$. We will also use the following maps. The first is $\mu_{\mathbf{a}} : \mathbb{R}^m \mapsto \mathbb{R}^{N_2}$. For a $\alpha \in \mathbb{R}^m$, it is defined by

$$\mu_{\mathbf{a}}(\alpha) = \sum_i \alpha_i a_i^2.$$

[14] In the final version, we will outline details of this improvement in an appendix.

The range of $\mu_{\mathbf{a}}$ will be denoted by the linear space $\mathcal{U} \subset \mathbb{R}^{N_2}$. Next, we define the *tangent* map $T_{\mathbf{a}} : (\mathbb{R}^n)^m \mapsto \mathbb{R}^{N_3}$. It is defined as follows: for a vector of linear forms $\mathbf{h} = (h_1, \ldots, h_m)$,

$$T_{\mathbf{a}}(\mathbf{h}) = \sum_i a_i^2 h_i,$$

where the RHS is viewed as a (vectorized) cubic polynomial. The range of $T_{\mathbf{a}}$ is denoted as $\mathcal{T} \subset \mathbb{R}^{N_3}$. We call this the tangent map because it is essentially the Jacobian of the map Φ introduced earlier, $\Phi(\mathbf{a}) = \sum_{i=1}^m a_i^3$. More generally, we will use the notation D to denote the Jacobian or tangent maps. In this notation, we have $\mathsf{D}\Phi(\mathbf{a}) = 3T_{\mathbf{a}}$. We will also be consider the *joint* map $\Psi(\mathbf{a}, \mathbf{b}) = \Phi(\mathbf{a}) - \Phi(\mathbf{b})$. One of our key claims (Lemma 6.12) will rely on analyzing the map $\mathsf{D}\Psi(\mathbf{a}, \mathbf{b})$.

We will follow the proof outline from Section 2.3. Before formalizing it, we start with some simple properties about the maps $\mu_{\mathbf{a}}$ and $T_{\mathbf{a}}$ that hold with high probability for a smoothed $\mathbf{a}$.

Lemma 6.5 (Properties of smoothed maps; Good event $\mathcal{G}_1$)**.** *Suppose $m = O(n^{3/2})$ and $0 < \rho \le 1$. Let $\mathbf{a}$ be a ρ-smoothing of $\bar{\mathbf{a}}$, and let Q be as in (29). Fix a coordinate subspace $Y \subseteq \mathbb{R}^n$ of dimension $\lfloor n/2 \rfloor$ (before sampling $\mathbf{a}$). Write $a_i = a_i^Y + a_i^\perp$ according to $Y \oplus Y^\perp$, and define*

$$\mu_{\mathbf{a},Y}(\alpha) = \sum_i \alpha_i (a_i^Y)^2, \qquad T_{\mathbf{a},Y}((h_i)_i) = \sum_i (a_i^Y)^2 h_i, \quad h_i \in Y,$$

and the rectangular cross-block map

$$\mu^{\mathrm{cross}}_{\mathbf{a},Y}(\alpha) = \sum_i \alpha_i a_i^\perp (a_i^Y)^\top.$$

There are positive absolute constants c, C, c_0, such that with probability $\ge 1 - e^{-cn}$, for all sufficiently large n, all the conditions below hold:

1. Norm bounds, *i.e.*,
$$\max_i \|a_i\| \le Q.$$

2. Conditioning of the square and tangent maps, *i.e.*,
$$\sigma_{\min}(\mu_{\mathbf{a}}), \quad \sigma_{\min}(T_{\mathbf{a}}), \quad \sigma_{\min}(\mu_{\mathbf{a},Y}), \quad \sigma_{\min}(T_{\mathbf{a},Y}) \ge Q^{-C}.$$

3. Conditioning of small sets of components or robust Kruskal rank. *For every $I \subseteq [m]$ with $1 \le |I| \le n/8$,*
$$\sigma_{\min}([a_i]_{i \in I}) \ge Q^{-C}.$$

4. Rigidity of the square and cross-block maps. *For every $\alpha \ne 0$ and every integer $0 \le k \le c_0 n$,*
$$\mathrm{dist}(\mu_{\mathbf{a}}(\alpha), \mathcal{R}_k) \ge Q^{-C} \frac{\mathrm{dist}(\alpha, \Sigma_k)^2}{\|\alpha\|},$$
$$\mathrm{dist}(\mu^{\mathrm{cross}}_{\mathbf{a},Y}(\alpha), \mathcal{R}_k) \ge Q^{-C} \frac{\mathrm{dist}(\alpha, \Sigma_k)^2}{\|\alpha\|}.$$
Here Σ_k is the set of k-sparse coefficient vectors, and $\mathcal{R}_k$ is the set of matrices of rank at most k in the appropriate ambient matrix space.

The proof is given in Appendix B.2.

In what follows, we will denote by $\mathcal{G}_1$ the event that all the properties in Lemma 6.5 hold.

A key idea in the proof is a powerful geometric lemma, that shows that if a map whose outputs are rational functions of its inputs usually has many "large" singular values for its Jacobian, then it is very unlikely to be close to zero. This is an anticoncentration statement, and is very similar to the result of Kane [Kan17]. It is also along the lines of ideas in [BEG+24], but is more general.

Theorem 6.6 (Anti-concentration of rational maps via Jacobians)**.** *Let $x \in \mathbb{R}^{p_0}$ and $z \in \mathbb{R}^{p_1}$ be two sets of parameters. Suppose x has independent Gaussian coordinates of variance ρ^2, with arbitrary means (which induces a measure γ_ρ), and suppose the $z \in \mathbb{R}^{p_1}$ space is given the Lebesgue measure ν_R on $\mathbb{R}^{p_1}$ restricted to $[-R, R]^{p_1}$. Let*

$$f : \Omega \longrightarrow \mathbb{R}^N, \qquad f_i = \frac{P_i}{S_i},$$

where P_i, S_i are real polynomials of degrees at most $\Delta \geq 1$, and $\Omega \subseteq \mathbb{R}^{p_0+p_1}$ is the open set on which all the relevant denominators are nonzero. Suppose $Q > 2$ is chosen such that

$$p_0 + p_1,\ N,\ \Delta,\ R,\ \rho^{-1} \leq Q.$$

Then there is a constant C such that for every integer $1 \leq r \leq \min\{p_0 + p_1, N\}$, and parameters $\eta \in (0, 1)$ and $J > 0$, the measure

$$(\gamma_\rho \otimes \nu_R)\left\{(x, z) : \|f(x, z)\| \leq \eta, \quad \prod_{i=1}^{r} \sigma_i\big(Df(x, z)\big) \geq J\right\} \leq Q^{C(p_1+r)} \frac{\eta^r}{J}. \tag{30}$$

The key point here is that the RHS does not have a factor of Q^{Cp_0}. It only depends on the number of z variables, and the number of "good" eigenvalues of the Jacobian. The proof closely reminiscent of the argument of Kane [Kan17]. For completeness, we include the proof in Appendix B.4.

6.2 Local uniqueness

As discussed in the outline, we first show Theorem 6.4.

Lemma 6.7. *Let $\mathbf{a} = (a_1, \ldots, a_m)$ and $\mathbf{b} = (b_1, \ldots, b_m)$ satisfy*

$$\max_i\{\|a_i\|, \|b_i\|\} \leq R, \qquad \sigma_{\min}(T_{\mathbf{a}}) \geq \kappa,$$

where $R, \kappa > 0$. If $d_{\text{perm}}(\mathbf{a}, \mathbf{b}) < \kappa/(3R)$, then

$$\|\Phi(\mathbf{b}) - \Phi(\mathbf{a})\| \geq 2\kappa\, d_{\text{perm}}(\mathbf{a}, \mathbf{b}).$$

Proof. After permuting the components of $\mathbf{b}$, write $\mathbf{b} = \mathbf{a}+\mathbf{h}$, where $\|\mathbf{h}\| = d_{\text{perm}}(\mathbf{a}, \mathbf{b})$. Expanding the cubes gives

$$\Phi(\mathbf{b}) - \Phi(\mathbf{a}) = 3T_{\mathbf{a}}(\mathbf{h}) + \sum_{i=1}^{m} h_i^2(2a_i + b_i).$$

By contractivity of polynomial multiplication,

$$\left\|\sum_{i=1}^{m} h_i^2(2a_i + b_i)\right\| \leq \sum_{i=1}^{m} \|h_i\|^2(2\|a_i\| + \|b_i\|) \leq 3R\|\mathbf{h}\|^2.$$

Therefore,

$$\|\Phi(\mathbf{b}) - \Phi(\mathbf{a})\| \geq 3\kappa\|\mathbf{h}\| - 3R\|\mathbf{h}\|^2 \geq 2\kappa\|\mathbf{h}\|,$$

using the assumed bound on $\|\mathbf{h}\|$. □

We can now prove local uniqueness.

Proof of Theorem 6.4. Set $S = \max\{1, R_a, \rho\}$ and write $\widetilde{a}_i = a_i/S$. The normalized components have centers of norm at most one and smoothing parameter $\rho/S \leq 1$.

We can now apply Theorem 4.1 to the generators $\widetilde{a}_i^2$, with linear multipliers. Since these generators have degree two, there are no Koszul syzygies in output degree three. Thus, for $m = O(n^{2-\varepsilon})$, the theorem gives an inverse-polynomial lower bound on $\sigma_{\min}(T_{\widetilde{\mathbf{a}}})$, with probability at least $1 - e^{-cn}$. Re-normalizing, this translates to a bound on $\sigma_{\min}(T_{\mathbf{a}})$.

Also with probability at least $1 - e^{-cn}$, Gaussian norm bounds give

$$\max_i \|a_i\| \leq R_a + 2\rho\sqrt{n} \leq S(1 + 2\sqrt{n}).$$

Both events depend only on $\mathbf{a}$. Under their intersection, fix any candidate tuple $\mathbf{b}$, and put

$$R = \max\left\{R_b, \max_i \|a_i\|\right\}, \qquad \kappa = S^2 Q^{-C_1}.$$

Since $R \leq SQ^{C_2}$ and $S \geq 1$, we have

$$\frac{\kappa}{3R} \geq Q^{-C}$$

for a sufficiently large constant C. Thus, whenever $d_{\mathrm{perm}}(\mathbf{a}, \mathbf{b}) < Q^{-C}$, Lemma 6.7 yields

$$d_{\mathrm{perm}}(\mathbf{a}, \mathbf{b}) \leq \frac{\delta(\mathbf{a}, \mathbf{b})}{2\kappa} \leq Q^{C_0}\delta(\mathbf{a}, \mathbf{b}),$$

as required. □

6.3 Small dimension orthogonal to $\mathcal{U}$

As stated in the outline, we first consider the case where the space $\mathcal{V} = \mathrm{span}\{b_i^2 : i \in [m]\}$ has only a small "number of directions" orthogonal to $\mathcal{U}$, which was defined to be the range of the map $\mu_{\mathbf{a}}$ (i.e., $\mathcal{U} = \mathrm{span}\{a_i^2 : i \in [m]\}$). Formally, let us first define $M_{\mathbf{b}}$ to be the matrix of dimensions $N_2 \times m$ whose ith column is the square b_i^2, expressed as a vector in $\mathbb{R}^{N_2}$. Now the key quantity we study is the number of *large* singular values of the matrix

$$Z_{\mathbf{a}}(\mathbf{b}) := \Pi_{\mathcal{U}^\perp} M_{\mathbf{b}}.$$

We have our first crucial lemma.

Lemma 6.8. *Assume that* $\mathbf{a}$ *satisfies* $\mathcal{G}_1$. *For some small constant* $c_* \in (0, 1)$ *and absolute constants* D_0, D_1, C, *we have the following result: define* $s_0 = \lfloor c_* n \rfloor$; *for any* $\mathbf{b}$ *such that:*

$$\sigma_{s_0+1}(Z_{\mathbf{a}}(\mathbf{b})) < Q^{-D_0} \quad \text{and} \quad \delta(\mathbf{a}, \mathbf{b}) < Q^{-D_1},$$

we have $d_{\mathrm{perm}}(\mathbf{a}, \mathbf{b}) \leq Q^C \delta(\mathbf{a}, \mathbf{b})$.

The lemma states essentially that if the squares of a "candidate decomposition" $\mathbf{b}$ do not add sufficiently many new directions to the squares of $\mathbf{a}$, then $\Phi(\mathbf{a})$ being close to $\Phi(\mathbf{b})$ implies closeness in permutation distance.

This lemma follows from relatively simple linear algebra arguments, but is also quite tricky. We will start by stating a helper lemma, which will essentially "localize" all the components of $\mathbf{b}$ into a either smaller subspace of $\mathbb{R}^n$, or onto a *line* corresponding to one of the components a_i. For a vector $v \in \mathbb{R}^n$, we will write $\mathbb{R}v = \operatorname{span}\{v\}$, for convenience.

Lemma 6.9. *Fix a constant $C_0 > 0$. Write $d \le cn$, for a sufficiently small constant $c > 0$, and suppose $\mathbf{a}$ satisfies the assumptions from Lemma 6.5 (i.e., property $\mathcal{G}_1$) sufficiently large n. Let $V \subset \mathbb{R}^n$ be a given subspace of $\mathbb{R}^n$ with $\dim V = d$. There exists a subspace $V^* \supset V$ with $\dim V^* \le 3d$ that depends only on $\mathbf{a}, V$, and a constant C depending on C_0 and the constants in $\mathcal{G}_1$ such that the following property holds: for any $\eta \in (0,1)$ and for any $b \in \mathbb{R}^n$, if b can be expressed as*

$$b^2 = \mu_{\mathbf{a}}(\alpha) + h + g, \text{ with } h \in \mathsf{Sym}^2(V) \text{ and } \|g\| < Q^{-C}\eta^4,$$

with $\|\alpha\| \le Q^{C_0}$, then either $\mathsf{dist}(b, V^*) \le \eta$ or $\min_i \mathsf{dist}(b, \mathbb{R}a_i) \le \eta$.

The Lemma is a structural one, and it says that for any d-dimensional subspace V, we can come up with a V^* of almost the same dimension, such that all b for which b^2 is a combination of $\{a_i^2\}$ and squares from V either come from V^* or is a scaled version of one of the $\{a_i\}$. Its proof is deferred to Appendix B.5. We now assume this lemma and show how to prove Lemma 6.8.

Proof of Lemma 6.8. Consider any candidate tuple $\mathbf{b}$ such that $\sigma_{s_0+1}(Z_{\mathbf{a}}(\mathbf{b})) < Q^{-D_0}$, for D_0 to be chosen later. By the definition of $Z_{\mathbf{a}}(\mathbf{b})$, this means that the matrix $\Pi_{\mathcal{U}^\perp} M_{\mathbf{b}}$ has at most s_0 "significant" (right) singular value directions. By standard results on column subset selection [DV06], there exists a subset of columns, indexed by $I \subset [m]$ and $|I| \le s_0$, such that

$$\forall j \in [m], \qquad b_j^2 = \mu_{\mathbf{a}}(\alpha) + \sum_{i \in I} \gamma_{ij} b_i^2 + g, \qquad \text{where } \|g\| < m \cdot Q^{-D_0}. \tag{31}$$

Here we use Theorem B.2 on $Z_{\mathbf{a}}(\mathbf{b})$, selecting at most s_0 columns. We set $r = \min\{s_0, \operatorname{rank} Z_{\mathbf{a}}(\mathbf{b})\}$ and apply the theorem when $r > 0$; if $r = 0$, we take $I = \varnothing$. The selected coefficients satisfy $|\gamma_{ij}| \le 1$, so

$$\|\alpha\| \le \sigma_{\min}(\mu_{\mathbf{a}})^{-1} \left((|I|+1) R_b^2 + \|g\| \right) \le Q^{O(1)}.$$

Thus the coefficient bound in Lemma 6.9 holds for a fixed C_0. We apply it with $V = \operatorname{span}\{b_i : i \in I\}$, whose dimension is at most s_0.

The remainder of the proof goes as follows. First we apply Lemma 6.9 to restrict to a space V^*. We then look at $\Phi(\mathbf{a}) - \Phi(\mathbf{b})$ as a polynomial, take derivatives orthogonal to V^*, and then conclude that very few terms remain. Once we have this, we can apply a "classical" uniqueness result (e.g., [BCMV14]) to conclude the desired result.

First, define V^* to be the space given by applying Lemma 6.9, satisfying $\dim V^* \le 3s_0$. Because of (31), we have the guarantee that for a constant C depending on D_0, for each j, either $\mathsf{dist}(b_j, V^*) \le Q^{-C}$ or $\mathsf{dist}(b_j, \mathbb{R}a_i) \le Q^{-C}$, for some index $i \in [m]$. If j satisfies the latter we say that j is "matched" to i, and in this case we can write b_j as a multiple of a_i, up to error Q^{-C}. By doing this for all j, we have that $\Phi(\mathbf{b})$ can be expressed as follows:

$$\Phi(\mathbf{b}) = \sum_i d_i a_i^3 + \mathbf{T} + \mathbf{G}, \qquad \mathbf{T} \in \mathsf{Sym}^3(V^*), \quad \|\mathbf{G}\| < mQ^{-C}. \tag{32}$$

Here $\mathsf{Sym}^3(V^*)$ denotes the span of cubes of linear forms in V^*. The key step is now the following. Consider $\Phi(\mathbf{a}) - \Phi(\mathbf{b})$ as a cubic polynomial (in the variables $\mathbf{x}$). For any direction $\omega \in (V^*)^\perp$,[15] we can take the derivative along ω. This yields

$$\frac{\partial}{\partial \omega}\left(\Phi(\mathbf{a}) - \Phi(\mathbf{b})\right) = \sum_{i \in [m]} 3(1-d_i)a_i^2\langle a_i, \omega\rangle + g,$$

where as before, g is an error term, with $\|g\| < 3mQ^{-C}$. Because the LHS as well as the error g are small, we have that the summation on the RHS is also small. Now, we use the fact that the map $\mu_{\mathbf{a}}$ is well-conditioned to conclude that this can only happen if the coefficients are all small!

More formally, by the hypothesis together with the bound on $\|g\|$ above, we have

$$\left\| \sum_{i \in [m]} 3(1-d_i)\langle a_i, \omega\rangle a_i^2 \right\| < Q^{-D_1} + 3mQ^{-C} \implies \sum_{i \in [m]} (1-d_i)^2 \langle a_i, \omega\rangle^2 < Q^{-D_2},$$

for some constant D_2 depending on the conditioning exponent in $\mathcal{G}_1$ and D_0, D_1, C. We can sum this over ω in an orthonormal basis for $(V^*)^\perp$, to obtain

$$\sum_i (1-d_i)^2 \mathsf{dist}(a_i, V^*)^2 < nQ^{-D_2}.$$

This implies that for every i, either $(1-d_i)^2$ is small or $\mathsf{dist}(a_i, V^*)$ is small. The final trick in the argument is to observe that the number of i for which the latter holds cannot be too large. Formally, define I^* as the set of indices $i \in [m]$ such that $\mathsf{dist}(a_i, V^*) < \theta^*$ for θ^* to be defined shortly. Then, by our robust Kruskal rank assumption on the decomposition $\mathbf{a}$ (Lemma 6.5), we can conclude that if we choose θ^* to be Q^{-C}, then $|I^*| \le \dim V^* \le 3s_0$.

To summarize, we have shown that each b_j is either close to one of the a_i's, in which case its cube contributes to the coefficient d_i in (32), or it is close to the subspace V^*. Following the definition above, let J^* be the set of indices for which the latter holds. Now, let us consider all the terms $i \notin I^*$ as defined in the paragraph above (note that I^* was defined based on a_i being close to V^*). We have shown that $|[m] \setminus I^*| \ge m - 3s_0$. For each such i, we have $(1-d_i)^2$ is small, specifically, $\ll 1$. This means that at least one of the b_j must be contributing to the d_i. This then implies (since each b_j can contribute to at most one d_i) that $|[m] \setminus J^*| \ge |[m] \setminus I^*|$, or $|J^*| \le |I^*|$.

Finally, observe that since $(1-d_i)^2 < nQ^{-D_2}$ for all $i \notin I^*$, we actually have:

$$\Phi(\mathbf{a}) - \Phi(\mathbf{b}) = \sum_{i \in I^*} a_i^3 - \sum_{j \in J^*} b_j^3 + \mathbf{G}',$$

where the error term $\mathbf{G}'$ satisfies $\|\mathbf{G}'\| < R^3 mnQ^{-D_2}$, where R is a bound on the lengths of a_i, b_i as before. This error is also $< Q^{-D_3}$ for an appropriate constant D_3. But now, note that $|I^*|$ and $|J^*|$ are both $\le 3s_0 < n$, which means we are in the *undercomplete* regime of tensor decompositions!

The robust Kruskal guarantee on $\mathbf{a}$ (Lemma 6.5) implies that the components $\{a_i\}_{i \in I^*}$ are well-conditioned, and thus we can use the uniqueness result implicit in [BCMV14] to conclude that (i) we have $|J^*| = |I^*|$, and (ii) $d_{\mathrm{perm}}(\mathbf{a}_{|I^*}, \mathbf{b}_{|J^*}) < Q^{-D_4}$, for an appropriate constant D_4.

[15] The fact that such ω exists is important, and this is why d is chosen to be only a small constant multiple of n.

Now if $|I^*| = |J^*|$, the argument earlier argues that to every d_i, we have *exactly* one b_j contributing, and this implies that we have a permutation of *all indices*, after which we can write $b_i = \beta_i a_i + q_i$, for some q_i orthogonal to a_i, satisfying $\|\mathbf{q}\| < Q^{-D_5}$. Thus, we can write $\Phi(\mathbf{a}) - \Phi(\mathbf{b})$, up to an error of $mR^3Q^{-D_5}$, as $\sum_i (1-\beta_i^3)a_i^3$. Using the fact that the tangent map $T_{\mathbf{a}}$ is well-conditioned, we thus obtain

$$\sum_i (1-\beta_i^3)^2 \|a_i\|^2 \le Q^{O(1)}\big(\delta(\mathbf{a},\mathbf{b}) + mR^3Q^{-D_5}\big)^2.$$

Since $|1-\beta_i^3| \ge |1-\beta_i| \cdot \frac{3}{4}$ (which is easy to see), and $\|a_i\|$ are all lower bounded, the bound on $\|\mathbf{q}\|$ and the triangle inequality give

$$d_{\mathrm{perm}}(\mathbf{a},\mathbf{b}) \le Q^{O(1)}\big(\delta(\mathbf{a},\mathbf{b}) + mR^3Q^{-D_5}\big).$$

We choose D_0, D_1 sufficiently large so that the resulting bound is smaller than Q^{-C}, where C is large enough to ensure $Q^{-C} \le \sigma_{\min}(T_{\mathbf{a}})/(3R)$ on $\mathcal{G}_1$. We can then apply Lemma 6.7 to obtain the conclusion without an additive error. □

6.4 Tangent directions of the joint map

In Section 6.3 above, we saw that if s, defined as the number of sufficiently large singular values of $\Pi_{\mathcal{U}^\perp} M_{\mathbf{b}}$, is $\le s_0$, then the uniqueness result follows from a direct linear algebraic argument combined with *local* uniqueness.

In the remainder of the section, we will thus assume that $s > s_0 = \lfloor c_* n \rfloor$, for the small constant c_* as before. This subsection will focus on the tangent (or Jacobian) maps of $\Phi(\mathbf{a})$, $\Phi(\mathbf{b})$, as well as that of the joint map $\Psi(\mathbf{a},\mathbf{b}) = \Phi(\mathbf{a}) - \Phi(\mathbf{b})$ defined earlier. For a smoothed $\mathbf{a}$, we showed in Lemma 6.5 that $\mathsf{D}\Phi(\mathbf{a})$ is well conditioned, i.e., it has $p = mn$ singular values $\ge Q^{-C}$ for some constant C. We also defined $\mathcal{T}$ to be the range space of $\mathsf{D}\Phi(\mathbf{a})$.

Our first lemma formalizes the following intuition: the tangent space $\mathcal{T}$ consists of the span of polynomials of the form $\{a_i^2 h_i\}_{i\in[m]}$; if $M_{\mathbf{b}}$ (containing the squares of $\mathbf{b}$) has s significant directions orthogonal to $\mathcal{U}$, then the tangent space of $\Phi(\mathbf{b})$ must have roughly $n \cdot s$ directions orthogonal to $\mathcal{T}$. We expect this to hold because for each direction (say b_j^2), $\mathsf{D}\Phi(\mathbf{b})$ has all terms of the form $b_j^2 \ell$, where ℓ is a linear function.

Lemma 6.10. *Let $\mathbf{a}$ be a smoothed decomposition that satisfies the well-conditioned properties $\mathcal{G}_1$, and let $\mathbf{b}$ be a candidate decomposition, and $1 \le s \le m$. Then for $k_s := \lfloor ns/10 \rfloor$, we have*

$$\sigma_{k_s}\big(\Pi_{\mathcal{T}^\perp} D\Phi(\mathbf{b})\big) \ge Q^{-C}\sigma_s\big(Z_{\mathbf{a}}(\mathbf{b})\big). \tag{33}$$

The proof follows easily from a linear algebraic lemma that we first state:

Lemma 6.11. *Assume that $\mathbf{a}$ satisfies $\mathcal{G}_1$, and that n is sufficiently large. Suppose $\mathcal{N} \subset \mathcal{U}^\perp$ is a space of quadratics with $\dim \mathcal{N} = s \ge 1$. Define the map $C_{\mathcal{N}} : \mathcal{N} \otimes \mathbb{R}^n \mapsto \mathcal{T}^\perp$, defined as $C_{\mathcal{N}}(\xi \otimes h) = \Pi_{\mathcal{T}^\perp} \xi h$. We then have*

$$\sigma_{k_s}(C_{\mathcal{N}}) \ge Q^{-C}, \quad \text{where } k_s = \lfloor ns/10 \rfloor.$$

The proof is given in Appendix B.6, and it uses the trick of partitioning coordinates as in Section 4. For now, let us see why Lemma 6.11 implies Lemma 6.10.

Proof of Lemma 6.10. Note that we may assume that $\sigma_s(Z_{\mathbf{a}}(\mathbf{b})) > 0$, as the conclusion is trivial otherwise. Suppose $\mathcal{V}_s \subseteq \mathbb{R}^m$ is the span of the top s right singular vectors of $Z_{\mathbf{a}}(\mathbf{b})$, and let $\mathcal{N}$ denote the image $Z_{\mathbf{a}}(\mathbf{b})\mathcal{V}_s$. Clearly, $\dim \mathcal{N} = s$ and $\mathcal{N} \subset \mathcal{U}^\perp$.

Now take any $v \in \mathcal{V}_s$ and any linear form ℓ; now $\ell \otimes v$, being in $(\mathbb{R}^n)^m$, is a valid input to the tangent map $\mathsf{D}\Phi(\mathbf{b})$. More interestingly, we can see that

$$\mathsf{D}\Phi(\mathbf{b})(\ell \otimes v) = 3(M_{\mathbf{b}} v)\ell.$$

Now note that $M_{\mathbf{b}} v$ and $Z_{\mathbf{a}}(\mathbf{b})v$ only differ from each other by some element in $\mathcal{U}$, and thus, using the definition of $C_{\mathcal{N}}$ from Lemma 6.11,

$$\Pi_{\mathcal{T}^\perp} \mathsf{D}\Phi(\mathbf{b})(\ell \otimes v) = 3\Pi_{\mathcal{T}^\perp}\big((Z_{\mathbf{a}}(\mathbf{b})v)\ell\big) = 3C_{\mathcal{N}}\big((Z_{\mathbf{a}}(\mathbf{b})v) \otimes \ell\big).$$

By the way we defined $\mathcal{N}$ above, we can now apply Lemma 6.11 and conclude that $\Pi_{\mathcal{T}^\perp}\mathsf{D}\Phi(\mathbf{b})$ has k_s singular values at least $Q^{-C}\sigma_s(Z_{\mathbf{a}}(\mathbf{b}))$. (The additional multiplicative term arises because $Z_{\mathbf{a}}(\mathbf{b})v$ is a polynomial $\xi \in \mathcal{N}$ of length $\geq \sigma_s(Z_{\mathbf{a}}(\mathbf{b}))\|v\|$.) □

Lemma 6.12. *Suppose* $\mathbf{a}$ *satisfies the properties* $\mathcal{G}_1$*, and consider any alternate decomposition* $\mathbf{b}$ *with* $\|b_i\| \leq R_b$ *as before, and let* $\sigma_s(Z_{\mathbf{a}}(\mathbf{b})) \geq \tau$ *for some parameters* $s, \tau > 0$*. Define* $k_s = \lfloor ns/10 \rfloor$ *as above. Then for some absolute constant* C*, the Jacobian of the joint function* $\Psi(\mathbf{a}, \mathbf{b}) = \Phi(\mathbf{a}) - \Phi(\mathbf{b})$ *satisfies:*

$$\sigma_p(D\Psi(\mathbf{a}, \mathbf{b})) \geq Q^{-C}, \quad \text{and } \sigma_{p+k_s}(D\Psi(\mathbf{a}, \mathbf{b})) \geq \tau \cdot Q^{-C}.$$

(Recall $p = mn$*.) Furthermore, there exists a constant* C' *such that whenever* $\|\mathbf{b}' - \mathbf{b}\| \leq \tau Q^{-C'}$*, the same bounds above hold up to a constant* $1/2$*, for* $D\Psi(\mathbf{a}, \mathbf{b}')$*.*

The "furthermore" part of the Lemma will be useful for our volume argument.

Proof. The proof follows relatively easily from Lemma 6.10. First consider two perturbation tuples $\mathbf{h}, \mathbf{k} \in (\mathbb{R}^n)^m$. we have

$$\mathsf{D}\Psi(\mathbf{a}, \mathbf{b})(\mathbf{h}, \mathbf{k}) = \mathsf{D}\Phi(\mathbf{a})\mathbf{h} - \mathsf{D}\Phi(\mathbf{b})\mathbf{k}.$$

If we only restrict to $\mathbf{k} = 0$, we obtain the first inequality, just using the well conditioned property $\mathcal{G}_1$. Next, let K be the span of the top k_s right singular vectors of $\Pi_{\mathcal{T}^\perp}\mathsf{D}\Phi(\mathbf{b})$.

We will bound both input perturbations in terms of the output, first recovering $\mathbf{k}$ by projecting away from $\mathcal{T}$, and then recovering $\mathbf{h}$. For any $\mathbf{h} \in (\mathbb{R}^n)^m$ and $\mathbf{k} \in K$, let $w = \mathsf{D}\Psi(\mathbf{a}, \mathbf{b})(\mathbf{h}, \mathbf{k})$. Since $\mathsf{D}\Phi(\mathbf{a})\mathbf{h} \in \mathcal{T}$, we have

$$\Pi_{\mathcal{T}^\perp} w = -\Pi_{\mathcal{T}^\perp}\mathsf{D}\Phi(\mathbf{b})\mathbf{k}, \qquad \|w\| \geq \|\Pi_{\mathcal{T}^\perp} w\| \geq \tau Q^{-C}\|\mathbf{k}\|,$$

where the last inequality follows from Lemma 6.10 and our choice of K. We now use $\mathsf{D}\Phi(\mathbf{a})\mathbf{h} = w + \mathsf{D}\Phi(\mathbf{b})\mathbf{k}$, and the conditioning of $\mathsf{D}\Phi(\mathbf{a})$ to get

$$\|\mathbf{h}\| \leq Q^C\big(\|w\| + \|\mathsf{D}\Phi(\mathbf{b})\|_{\mathrm{op}}\|\mathbf{k}\|\big) \leq Q^{O(1)}\tau^{-1}\|w\|.$$

Here we used the component norm bounds, which give $\|\mathsf{D}\Phi(\mathbf{b})\|_{\mathrm{op}} \leq Q^{O(1)}$ and $\tau \leq \|Z_{\mathbf{a}}(\mathbf{b})\|_{\mathrm{op}} \leq Q^{O(1)}$. Together, we have

$$\|\mathsf{D}\Psi(\mathbf{a}, \mathbf{b})(\mathbf{h}, \mathbf{k})\| \geq \tau Q^{-O(1)}\|(\mathbf{h}, \mathbf{k})\| \qquad \text{on } (\mathbb{R}^n)^m \times K.$$

This input subspace has dimension $p + k_s$, so the min–max principle gives the second inequality.

For the "furthermore" part, note that since the components of $\mathbf{b}$ are bounded in length, the map $\mathbf{b} \mapsto \mathsf{D}\Psi(\mathbf{a}, \mathbf{b})$ is Q^{C_0}-Lipschitz, for a small absolute constant C_0. Choosing C' appropriately, we have that the change in the Jacobian is bounded in Frobenius norm by $\frac{\tau}{2}Q^{-C}$ (in Frobenius norm). Thus by Weyl's inequality, we have the desired conclusion. □

Remark: A weaker robustness bound. The Jacobian bound above already gives a robustness guarantee when combined with Theorem 6.6. We give a rough sketch below.

Fix a bound R_b on the candidate component norms, set $\tau = Q^{-D_0}$, and take $s = s_0 + 1$ and $k_s = \lfloor ns/10 \rfloor = \Theta(n^2)$. At any candidate satisfying $\sigma_s(Z_{\mathbf{a}}(\mathbf{b})) > \tau$, the product of the largest $p + k_s$ singular values of $\mathsf{D}\Psi(\mathbf{a}, \mathbf{b})$ is at least $Q^{-C(p+k_s)}\tau^{k_s}$. The anticoncentration theorem, applied with $\mathbf{a}$ as the Gaussian variables and $\mathbf{b}$ as the Lebesgue variables, therefore bounds the joint measure of pairs with tensor error at most 2η and these Jacobian bounds by

$$Q^{C(p+k_s)}\eta^{p+k_s}\tau^{-k_s}.$$

We can turn this into a probability bound for the existence of a candidate as described in the outline in Section 2.3. Observe that a single candidate $\mathbf{b}$ with tensor error at most η supplies an entire ball of nearby candidates with tensor error at most 2η. For $\eta \le \tau Q^{-C'}$, the neighborhood conclusion of Lemma 6.12 preserves the Jacobian bounds throughout a ball of radius $\eta Q^{-C'}$, whose volume is at least $Q^{-Cp}\eta^p$. Integrating over $\mathbf{a}$ and dividing by this volume gives

$$\mathop{\mathbb{P}}_{\mathbf{a}}\left[\mathbf{a} \in \mathcal{G}_1,\ \exists\, \mathbf{b} : \max_i \|b_i\| \le R_b, \quad \delta(\mathbf{a}, \mathbf{b}) \le \eta, \quad \sigma_s(Z_{\mathbf{a}}(\mathbf{b})) > \tau\right] \le Q^{Cp}\left(\frac{\eta}{\tau}\right)^{k_s}.$$

Since $p = mn$ and $k_s = \Theta(n^2)$, choosing $\eta = Q^{-D(1+m/n)}$ for a sufficiently large constant D makes this probability exponentially small. All remaining candidates fall under Lemma 6.8, which gives the desired linear bound on permutation distance. Thus we obtain robustness at tensor-error scale $Q^{-O(1+m/n)}$. This scale is inverse-polynomial whenever $m = O(n)$, which is already better than [BCV14]; improving it to larger m is the goal of the parameter compression argument below.

6.5 Parameter compression and polynomial robustness

In the computation above, we were able to bound the probability of a smoothed $\mathbf{a}$ having an alternate decomposition $\mathbf{b}$ by $Q^{Cp}\eta^{k_s}$, where $p = mn$ is the number of parameters used to represent $\mathbf{b}$ (using the bound from Lemma 6.6). With the guarantee we have on k_s, i.e., $k_s > cn^2$, we cannot make the bound $\ll 1$, unless η is $< Q^{-Cm/n}$, which is inverse-polynomial only when $m = O(n)$.

We will now show how to improve the bound above, and the key will be to "compress" the number of parameters used to express $\mathbf{b}$. The number of retained parameters (or coordinates) will be controlled by ns and by the number q of "weak" tangent directions of $\mathsf{D}\Phi(\mathbf{b})$. When q is small, this substantially reduces the number of candidate variables. When q is large, the volume argument will instead exploit the larger neighborhoods supplied by those weak directions. We will approximately reconstruct the remaining ones as rational functions of the retained parameters and $\mathbf{a}$.

We will start by describing the reconstruction, and prove it over the next two subsections. Finally in Section 6.8, we will prove our main result of the Section, Theorem 6.3.

Notation and setup. Throughout, we will assume $m = O(n^{3/2})$. Also recall that all the components of $\mathbf{b}$ are bounded in length by R_b and $Q = n(2 + R_a + R_b + \rho^{-1})$. Let $s_0 = \lfloor c_* n \rfloor$ be the threshold from Lemma 6.8. Choose a precision parameter $0 < \gamma \le Q^{-a_0}$, where a_0 is a sufficiently large absolute constant, and set

$$\lambda = \gamma^4, \qquad \tau = \gamma^{16}, \qquad \eta = \gamma^{22}. \tag{34}$$

Here, τ will be the threshold used for singular values of $Z_{\mathbf{a}}(\mathbf{b})$, λ will be the threshold for solving the linear reconstruction systems in Step III of the argument below, and η is the tensor-error scale

used in the volume argument. For a candidate tuple $\mathbf{b}$, define

$$s = \#\{j : \sigma_j(Z_{\mathbf{a}}(\mathbf{b})) > \tau\}, \qquad q = \#\{1 \le j \le p : \sigma_j(\mathsf{D}\Phi(\mathbf{b})) \le \gamma\}. \tag{35}$$

We call (s, q) the *layer* of $\mathbf{b}$. Let $E_\gamma \subseteq \mathbb{R}^p$ be the q-dimensional right singular subspace corresponding to the latter singular values, including the kernel. Thus, $\|\mathsf{D}\Phi(\mathbf{b})|_{E_\gamma}\| \le \gamma$, whereas the singular values on $E_\gamma^\perp$ exceed γ. Since section 6.3 already handled the case $s \le s_0$, assume that $s > s_0$.

Retained coordinates. A *reconstruction index* α specifies a fixed set of scalar coordinates of $\mathbf{b}$, indexed by a subset of $[m] \times [n]$. We write z for the retained coordinates and u for the remaining ones, so that, after permuting the coordinates, $\mathbf{b} = (z, u)$. Let p_z be the number of retained coordinates and $\Pi_z : \mathbb{R}^p \to \mathbb{R}^{p_z}$ the corresponding coordinate projection. The index α also comes with a rational formula $u_\alpha^0(\mathbf{a}, z)$ for predicting u. Once α is fixed, the only numerical inputs to this formula are $\mathbf{a}$ and the retained values z.

Theorem 6.13 (Parameter compression)**.** *There are constants $a_0, C, c > 0$ and a conditioning event $\mathcal{G}_2$, depending only on $\mathbf{a}$, with $\mathbb{P}(\mathcal{G}_2) \ge 1 - e^{-cn}$, such that the following holds: for every $0 < \gamma \le Q^{-a_0}$ and every layer (s, q) with $s_0 < s \le m$, there is a list of at most*

$$Q^{C(ns+q)} \quad \textit{reconstruction indices } \alpha,$$

chosen independently of $\mathbf{a}$ and $\mathbf{b}$. Every index retains at most

$$p_z \le \min\{p, Cns + 2q\} \tag{36}$$

coordinates, and each coordinate of u_α^0 is a rational function with numerator and denominator degrees at most Q^C.

For every $\mathbf{a} \in \mathcal{G}_2$ and every bounded candidate $\mathbf{b}$ in layer (s, q), there is an index α in the list such that, writing $\mathbf{b} = (z_*, u_*)$ according to the coordinates specified by α, the following properties hold:

1. Local reconstruction with the same formula. *The rational formula $u_\alpha^0(\mathbf{a}, \cdot)$ is well-defined on the entire ball*

$$\|z - z_*\| \le Q^{-C}\gamma^{20}, \tag{37}$$

and throughout this ball satisfies

$$\|u_\alpha^0(\mathbf{a}, z) - u_*\| \le Q^C\gamma^{12}. \tag{38}$$

2. Weak tangent directions. *The retained coordinates satisfy*

$$\|\Pi_z v\| \ge Q^{-C}\|v\| \qquad \textit{for all } v \in E_\gamma. \tag{39}$$

The constants in the guarantees are independent of γ.

Suppose $\mathbf{b} = (z_*, u_*)$ is the candidate of interest. Then the theorem actually considers values of the parameters z that are "near" z_*. At $z = z_*$, the reconstruction bound gives an approximation to the missing coordinates of $\mathbf{b}$. The stronger assertion is that the *same formula* remains defined and its predictions remain close to the fixed vector u_* throughout a specified neighborhood. Such a neighborhood guarantee turns out to be crucial for the volume based argument used in the final step of proving Theorem 6.3. Also, we note that the compression statement does not need to assume that $\delta(\mathbf{a}, \mathbf{b})$ is small; its guarantee comes only from the square relations defining the layer.

Finally, we remark that the weak tangent directions need to be preserved for an important technical reason that we discuss in Section 6.8

6.6 Reconstruction from square relations

We begin the proof of Theorem 6.13. We fix the precision parameters from (34), and take a bounded candidate tuple $\mathbf{b}$ in a layer (s, q) with $s > s_0$. In this subsection, we will give the main components of the compression argument: we first select a small subset of the components of $\mathbf{b}$, retain them in full, and use the dependencies among the squares $\{b_j^2\}$ to reconstruct most other components from $\ll n$ of their components. But unfortunately, the reconstruction can fail for some components; these are handled in the next subsection.

Step I: Selecting "generator" components. Since $Z_{\mathbf{a}}(\mathbf{b})$ has exactly s singular values above τ, we can use the column-selection guarantee used earlier (from [DV06]), to obtain a set $I \subseteq [m]$ with $|I| = s$ and coefficients $|\Gamma_{ij}| \le 1$ such that every column of $Z_{\mathbf{a}}(\mathbf{b})$ is expressible as a combination of the columns in I, up to an error $(m+1)\tau$. Consequently, for every $j \in [m]$,

$$b_j^2 = \mu_{\mathbf{a}}(c_j) + \sum_{i \in I} \Gamma_{ij} b_i^2 + e_j, \qquad \|c_j\| \le Q^C, \quad \|e_j\| \le (m+1)\tau. \tag{40}$$

We apply Theorem B.2 with $M = Z_{\mathbf{a}}(\mathbf{b})$ and $r = s$.

Note that the bound on $\|c_j\|$ follows because $\mu_{\mathbf{a}}$ is well-conditioned, and the norms of the components are bounded. We retain the vectors b_i, $i \in I$, in full, at a cost of ns scalar coordinates. These will be referred to as the *generator* columns of $\mathbf{b}$. The coefficients c_j and Γ_{ij} are only used to describe the reconstruction; they are not retained as additional variables (since there are too many of them).

Next, the idea is to partition the coordinates $[n]$ into blocks, and for each $j \notin I$, the goal is to retain coordinates corresponding to only one of the blocks. For each $s > s_0$, choose

$$\ell = \left\lceil K\sqrt{s+1} \right\rceil, \qquad \nu = \left\lceil 2\sqrt{s+1} \right\rceil, \qquad t_s = \lfloor n/\ell \rfloor, \tag{41}$$

where K is a sufficiently large constant. We fix disjoint coordinate blocks $S_1, \dots, S_{t_s} \subseteq [n]$, each of size ℓ; fewer than ℓ coordinates remain outside these blocks. This partition is chosen independently of $\mathbf{a}$. By our choice of parameters (specifically $c_* n \le s \le m$), we have $\ell = o(n)$.

For a block S, define the rectangular square map that maps $\mathbb{R}^m$ to $\mathbb{R}^{(n-\ell)\times\ell}$

$$\mu_{\mathbf{a},S}(\alpha) = \sum_i \alpha_i (a_i)_{S^c} (a_i)_S^\top,$$

and write A_S for the matrix whose columns are the vectorized $(a_i)_{S^c}(a_i)_S^\top$. We define a block S to be *good* if the following rigidity condition holds:

$$\operatorname{dist}(\mu_{\mathbf{a},S}(\alpha), \mathcal{R}_k) \ge Q^{-C_{\mathrm{g}}} \frac{\operatorname{dist}(\alpha, \Sigma_k)^2}{\|\alpha\|} \qquad (\text{for } \alpha \ne 0,\ 0 \le k \le c_{\mathrm{rank}}\ell). \tag{42}$$

As before, $\Sigma_k \subseteq \mathbb{R}^m$ is the set of vectors supported on at most k coordinates, and $\mathcal{R}_k$ is the set of rectangular matrices (of appropriate dimensions) of rank at most k. In particular, taking $k = 0$ shows that $\sigma_{\min}(A_S) \ge Q^{-C}$.

Recall that $c_0 n$ is the range of ranks for which the map $\mu_{\mathbf{a}}$ stays rigid (condition $\mathcal{G}_1$ defined earlier). Now, we define a stronger conditioning property, using the quantities $c_{\mathrm{load}}, c_{\mathrm{rank}} > 0$, and allowing at most a $c_0/4$ fraction of bad blocks. Choose K large enough that

$$\nu \le c_{\mathrm{rank}}\ell, \qquad m \le c_{\mathrm{load}}\ell(n-\ell), \qquad \frac{\nu - 1}{\ell} \le \frac{c_0}{4}. \tag{43}$$

We can meet the load condition by increasing K, since $m = O(n^{3/2})$ and $s = \Omega(n)$ give $m = O(n\sqrt{s})$, while $n - \ell = \Theta(n)$.

For sufficiently large n, we also have $\ell \le c_0 n/4$.

Definition 6.14 (Block-based conditioning)**.** We say that $\mathbf{a}$ satisfies the property (or event) $\mathcal{G}_2$ if *in addition to* $\mathcal{G}_1$ defined earlier, for some absolute constants $c_{\text{rank}}, c_{\text{load}}$ satisfying the relations above, for every integer $s_0 < s \le m$, at least $(1 - c_0/4)$ fraction of the corresponding fixed blocks $S_1, \dots, S_{t_s}$ are good as defined in (42).

Note that this event depends only on $\mathbf{a}$, not on the precision parameter or the candidate. We prove in Lemma B.1 in Appendix B that the condition $\mathcal{G}_2$ holds with probability at least $1 - \exp(-cn)$ over the randomness in $\mathbf{a}$.

For now, we will work with some $\mathbf{a}$ that satisfies $\mathcal{G}_2$.

Step II: Column selection for good blocks. Let $\mathcal{U}_S = \operatorname{range} A_S$, and form the matrix of generator cross-products

$$B_S = \left[\operatorname{vec}\left((b_i)_{S^c}(b_i)_S^\top\right)\right]_{i \in I}.$$

Both A_S and B_S are known from $\mathbf{a}$ and the fully retained set of generators I. Now, given the threshold τ, suppose r_S is the number of singular values of $\Pi_{\mathcal{U}_S^\perp} B_S$ above τ. We claim that there exists a subset $I_S \subseteq I$ such that $|I_S| = r_S$, and further, for D_S and the associated projector defined as

$$D_S = [A_S, (B_S)_{|I_S}], \qquad P_S = I - D_S D_S^\dagger, \tag{44}$$

we have two properties (i) $\sigma_{\min}(D_S) \ge Q^{-C}\tau$, and (ii) every column of B_S is a combination of columns of D_S with coefficients bounded by Q^C, up to error $Q^C\tau$. Both properties follow from Corollary B.3 with $A = A_S$ and $B = B_S$.

The claim follows from a straightforward argument based on column subset selection, along with the property $\mathcal{G}_2$ which implies $\sigma_{\min}(A_S) \ge Q^{-C}$. We will refer to D_S as the "dictionary" corresponding to the good block S.

Step III: Recovery and regular components Next, we give conditions under which a component b_j can be recovered only using a portion of its coordinates corresponding to a single good block. Suppose S is a good block, and consider some component b_j such that $j \notin I$. Write $x = (b_j)_S$ and $y = (b_j)_{S^c}$. The idea will be to recover y using x. Let us define the vectorization operator $\mathcal{L}_x(y) = \operatorname{vec}(yx^\top)$. Taking the $S^c \times S$ block of (40), and replacing the generator columns by their representations in the dictionary D_S above, we have that $\mathcal{L}_x(y)$ can be expressed as a combination of the vectors in D_S, up to an error $Q^C\tau$. Or alternatively,

$$\|P_S \mathcal{L}_x(y)\| \le Q^C \tau. \tag{45}$$

Now suppose we retain all the coordinates in x. The above yields an approximate homogeneous linear system for y. For intuition about what follows, consider the case $\tau = 0$; here we have the equation $P_S\mathcal{L}_x(y) = 0$. This is a system of $\ell(n-\ell)$ equations in $(n-\ell)$ variables. If it happens to be full rank, the system has only the zero solution. But if the rank is $n - \ell - k$, we can hope to retain k of the variables in y, and now we obtain an invertible affine system, that we can solve for the other variables!

We will now generalize the intuition above. Say that b_j is *regular on* S if S is good and

$$\sigma_{n-\ell-\nu+1}(P_S\mathcal{L}_x) > \lambda, \tag{46}$$

for the parameter ν defined earlier. We say that component b_j is regular if this holds on at least one good block, and *exceptional* otherwise. For each regular component b_j outside I, choose such a block $S = S(j)$ for which b_j is regular on S. Now we can once again apply column subset selection, this time to the operator $P_S\mathcal{L}_x$. It yields a set $J_j \subseteq S^c$ of size $n - \ell - \nu + 1$ for which

$$\sigma_{\min}((P_S\mathcal{L}_x)_{|J_j}) \geq \lambda/n. \tag{47}$$

We use Theorem B.2 with $M = P_S\mathcal{L}_x$ and $r = n - \ell - \nu + 1$. Let $F_j = S^c \setminus J_j$, so that $|F_j| = \nu - 1$. Retain the coordinates of b_j in F_j, in addition to those in x, and then we can predict the remaining coordinates, indexed by J_j. Each regular component therefore costs only $\ell + \nu - 1$ retained coordinates.

Writing $y = (y_{J_j}, y_{F_j})$, the prediction is the least-squares solution

$$\widehat{y}_{J_j} = -((P_S\mathcal{L}_x)_{|J_j})^\dagger (P_S\mathcal{L}_x)_{|F_j} y_{F_j}. \tag{48}$$

Equations (45) and (47) give

$$\|\widehat{y}_{J_j} - y_{J_j}\| \leq Q^C \frac{\tau}{\lambda} = Q^C \gamma^{12}. \tag{49}$$

Rational functions. The key observation now is that once the discrete choices $I, I_S, S(j), F_j$ are made, we can actually compute the predictions without explicitly choosing bases for any of the projected spaces. Define

$$M_j = [D_S, -\mathcal{L}_{x,J_j}], \text{ and } v_j = \mathcal{L}_{x,F_j} y_{F_j},$$

and solve the full least-squares system

$$\begin{pmatrix} \widehat{\alpha} \\ \widehat{y}_{J_j} \end{pmatrix} = (M_j^\top M_j)^{-1} M_j^\top v_j. \tag{50}$$

This is exactly the same as the earlier formulation (48), as can be seen by eliminating the α variables. The matrix M_j has full column rank, and in fact,

$$\sigma_{\min}(M_j) \geq Q^{-C}\tau\lambda. \tag{51}$$

This can be seen as follows: applying P_S to $O = D_S\alpha - \mathcal{L}_{x,J_j} w$ first gives $\|w\| \leq n\lambda^{-1}\|O\|$. Using $D_S^\dagger$ to recover α (and using its least singular value bound from above) then gives $\|\alpha\| \leq Q^C(\tau\lambda)^{-1}\|O\|$.

Once we have fixed all the discrete choices I, I_S, $S(j)$ and F_j, all entries of M_j and v_j are polynomials of degree at most two in $\mathbf{a}$ and the retained coordinates. Their dimensions are polynomial in Q. The adjugate formula for the inverse of a Gram matrix in (50) therefore expresses each predicted coordinate as a rational function with numerator and denominator degrees at most Q^C. This completes the argument for this step.

Remark: same formula for neighborhoods. It is now straightforward to prove that there is a uniform reconstruction guarantee for the regular components in a small ball around the original component. Fix $\mathbf{a}$ and all the discrete choices above. Let z_* denote the actual values of the coordinates retained so far (including the generators I) and perturb them to z that satisfies

$$\|z - z_*\| \leq r_{\text{nb}}, \text{ where } \qquad r_{\text{nb}} = Q^{-C}\tau\lambda = Q^{-C}\gamma^{20}. \tag{52}$$

The dictionary D_S changes by at most $Q^C r_{\text{nb}}$. Because of our earlier bound $\sigma_{\min}(D_S) \geq Q^{-C}\tau$, its column-space projector changes by at most $Q^C r_{\text{nb}}/\tau$. The maps $\mathcal{L}_{x,J_j}$ and $\mathcal{L}_{x,F_j}$ change by at

most r_{nb}. Choosing the constant in (52) large enough, the $\sigma_{\min}$ bounds on D_S and (47) persist up to a factor two. Hence the same Gram-matrix inverse remains defined throughout this ball.

For the error bound, we can use the *same coefficients* as before to express $\mathcal{L}_x(y)$ in terms of the columns D_S (denote these coefficients by $c_{j,S}$ for now), and also keep the original missing coordinates $y_{J_j,*}$ fixed. As the retained values vary, their residual in the perturbed system remains bounded by

$$\|D_S(z)c_{j,S} - \mathcal{L}_{x(z),J_j}y_{J_j,*} - \mathcal{L}_{x(z),F_j}y_{F_j}(z)\| \le Q^C(\tau + r_{\text{nb}}) \le Q^C\tau.$$

Projecting off $D_S(z)$ and using the persisting column gap gives

$$\|\widehat{y}_{J_j}(\mathbf{a}, z) - y_{J_j,*}\| \le Q^C\tau/\lambda = Q^C\gamma^{12} \tag{53}$$

throughout the neighborhood. Summing the squared errors over all regular components only changes the polynomial factor in Q.

This completes the step of retaining some generators and the regular components in $\mathbf{b}$ that are not generators.

6.7 Exceptional components and proof of Theorem 6.13

We complete the proof of Theorem 6.13. So far we have seen how every "regular" component can be recovered from a small set of its coordinates. We will now focus on the exceptional components, and we in fact retain them fully. The key will be to prove that their total coordinate cost is at most $Cns + q$. They key trick will be the rigidity property $\mathcal{G}_2$, which we assume $\mathbf{a}$ satisfies.

The following auxiliary lemma is crucial to the proof. Suppose we have a map $\mu : \mathbb{R}^m \mapsto \mathbb{R}^{a\times b}$ that satisfies the rigidity property (a μ with at least k non-zero entries results in a matrix of rank at least k, until a certain bound on k). Now suppose we have a linear space of matrices W with the property that each of its elements is a rank k matrix. (But since we are viewing it as a linear space of matrices, its dimension could be as large as $\frac{k(k+1)}{2}$.) But the claim is that $\dim(W \cap \text{range}(\mu) \le k$. This is because any element in the intersection is a linear combination of some of the $\{\mu_{e_j}\}$ for indices j (call this these indices the *support* of the element). The key point is that if W_1, W_2 in the intersection (both rank k) have unequal support, then a generic combination of them would have a coefficient vector with $> k$ entries, and thus the result will be a matrix of rank $> k$, by rigidity! This contradicts the assumption on W. Thus the supports must all be the same, which implies the lemma. This has a nice consequence, that we will need in our arguments for exceptional matrices.

Lemma 6.15 (Intersections with Rigid Subspaces)**.** *Let $Q \ge 2$, and let $\mu : \mathbb{R}^m \to \mathbb{R}^{a\times b}$ be an injective linear map; define $U = \operatorname{range}\mu$. Let $m \ge k \ge 0$ be an integer, and assume $\|\mu\|_{\text{op}}, k \le Q^{C_0}$, for some constant C_0. Suppose that for some $\beta \ge Q^{-C_0}$, the map μ satisfies the following rigidity property:*

$$\operatorname{dist}(\mu(c), \mathcal{R}_k) \ge \beta\frac{\operatorname{dist}(c, \Sigma_k)^2}{\|c\|} \qquad (c \ne 0), \tag{54}$$

where $\Sigma_k \subseteq \mathbb{R}^m$ is the set of vectors with at most k nonzero coordinates, and $\mathcal{R}_k$ is the set of matrices of rank at most k. Let $W \subseteq \mathbb{R}^{a\times b}$ be another linear space, with the property that every element has rank at most k. There is a constant C, depending only on C_0, such that:

1. *There exists a subspace $W_0 \subseteq W$, with $\dim W_0 \le k$, satisfying*

$$\operatorname{dist}(X, W_0) \le Q^C \operatorname{dist}(X, U) \qquad \text{for every } X \in W. \tag{55}$$

In particular, $W \cap U \subseteq W_0$.

2. *For any linear map* $F : \mathbb{R}^d \to \mathbb{R}^{a\times b}$, *integer* $s \geq 0$, *and* $e \geq 0$, *if*

$$\|F - \Pi_W F\|_{\text{op}} \leq e, \qquad \sigma_{s+1}(\Pi_{U^\perp} F) \leq e,$$

then

$$\sigma_{s+k+1}(F) \leq Q^C e. \tag{56}$$

The proof is deferred to Appendix B.7.

The next lemma shows that if for a good block S, we look at all the components of $\mathbf{b}$ that are not regular with respect to S, they effectively lie in a space of low dimension.

Lemma 6.16. *Following the notations above, fix a good block* S *of size* ℓ. *Let* $\mathcal{E}_S \subseteq [m]$ *be the set of all components that fail the regularity test on* S. *Consider the matrix:*

$$X_S = [(b_j)_S]_{j\in\mathcal{E}_S} \in \mathbb{R}^{\ell\times|\mathcal{E}_S|}.$$

Then, for an absolute constant C *independent of* γ,

$$\sigma_\nu(X_S) \leq Q^C \lambda. \tag{57}$$

Note that if the conclusion holds for $\mathcal{E}_S$ as defined in the lemma, it also holds for a subset; so it also holds when we look at the set of exceptional components (that are not regular for any block).

Proof. If $\operatorname{rank} X_S < \nu$, the conclusion is immediate. Otherwise, by column subset selection, we have ν indices $j_1, \ldots, j_\nu \in \mathcal{E}_S$ satisfying the following: write $x_i = (b_{j_i})_S$ and $X_* = [x_1, \ldots, x_\nu]$, then

$$\beta_X := \sigma_{\min}(X_*) \geq \frac{\sigma_\nu(X_S)}{|\mathcal{E}_S|} \geq \frac{\sigma_\nu(X_S)}{m} > 0. \tag{58}$$

We use Theorem B.2 with $M = X_S$ and $r = \nu$. We will show that $\beta_X \leq Q^C\lambda$. First, note that because the component j_i is not regular,

$$\sigma_{n-\ell-\nu+1}(P_S\mathcal{L}_{x_i}) \leq \lambda.$$

Recall that the map P_S was defined as the projector onto $(\operatorname{range} D_S)^\perp$, the orthogonal complement of the space spanned by the dictionary vectors. Also, $P_S\mathcal{L}_{x_i}$ has $(n-\ell)$ input coordinates, so the above implies that the span of the bottom ν right singular values span a space $V_i \subseteq \mathbb{R}^{S^c}$ of dimension ν satisfying

$$\forall w_i \in V_i, \quad \|P_S\mathcal{L}_{x_i}(w_i)\| \leq \lambda\|w_i\|. \tag{59}$$

Next, define the following map that takes one vector from each V_i and outputs an $(n-\ell)\times\ell$ matrix:

$$\mathcal{M} : \bigoplus_{i=1}^{\nu} V_i \longrightarrow \mathbb{R}^{(n-\ell)\times\ell}, \qquad \mathcal{M}((w_i)_i) = \sum_{i=1}^{\nu} w_i x_i^\top.$$

The domain has dimension ν^2. We can also bound the least singular value of the map easily. Since the matrix X_* above is least singular value β_X, we can simply use this coordinate-wise, and obtain

$$\|\mathcal{M}((w_i)_i)\|_F^2 \geq \beta_X^2 \sum_{i=1}^{\nu} \|w_i\|^2. \tag{60}$$

Thus $\mathcal{M}$ is injective, has least singular value at least β_X, and hence its range W has dimension exactly ν^2. Furthermore, every matrix in W has rank at most ν, since it is a sum of ν rank-one matrices. On the other hand, because $w_i \in V_i$, by definition, the entire image of $\mathcal{M}$ is almost annihilated by P_S. Formally,

$$\|P_S \operatorname{vec}(\mathcal{M}((w_i)_i))\| \le \sum_{i=1}^{\nu} \|P_S \mathcal{L}_{x_i}(w_i)\| \le \lambda \sum_{i=1}^{\nu} \|w_i\| \le \sqrt{\nu}\lambda \left(\sum_{i=1}^{\nu} \|w_i\|^2\right)^{1/2}. \tag{61}$$

This means that if β_X is large, most of the mass of $\mathcal{M}((w_i)_i)$ must lie in $\ker P_S$, which is the space spanned by the chosen dictionary for the block (and is contained in $\mathcal{U}_S + \operatorname{range} B_S$). We can now use the rigidity of $\mu_{\mathbf{a},S}$ which bounds the "number of directions" of overlap of W (range of $\mathcal{M}$) with $\mathcal{U}_S$, and then a simple bound of s for the B_S portion.

Formally, apply Lemma 6.15 to $\mu_{\mathbf{a},S}$ and W, with $k = \nu$. The block is good and $\nu \le c_{\text{rank}}\ell$, so the hypotheses of the Lemma hold. Identifying rectangular matrices with their vectorizations, the lemma gives a $W_0 \subseteq W$, $\dim W_0 \le \nu$, such that

$$\forall X \in W, \qquad \operatorname{dist}(X, W_0) \le Q^C \operatorname{dist}(X, \mathcal{U}_S). \tag{62}$$

(Recall that $\mathcal{U}_S$ was defined as the range of $\mu_{\mathbf{a},S}$.) The dictionary also contains selected generator columns B_S. Their new directions outside $\mathcal{U}_S$ lie in

$$\mathcal{H}_S = \Pi_{\mathcal{U}_S^\perp}(\operatorname{range} B_S), \qquad \dim \mathcal{H}_S \le s,$$

and $\mathcal{U}_S + \operatorname{range} B_S = \mathcal{U}_S \oplus \mathcal{H}_S$.

Orthogonality to W_0 imposes at most ν linear conditions on W, and orthogonality to $\mathcal{H}_S$ imposes at most s more. Hence

$$\dim(W \cap W_0^\perp \cap \mathcal{H}_S^\perp) \ge \nu^2 - \nu - s.$$

By our choice $\nu = \lceil 2\sqrt{s+1} \rceil$, the RHS above is strictly > 0. Thus, choose a nonzero matrix X in this intersection. Since $X \perp W_0$, $\operatorname{dist}(X, W_0) = \|X\|_F$. Also $\Pi_{\mathcal{U}_S^\perp} X$ is orthogonal to $\mathcal{H}_S$: for $H \in \mathcal{H}_S \subseteq \mathcal{U}_S^\perp$,

$$\langle \Pi_{\mathcal{U}_S^\perp} X, H \rangle = \langle X, H \rangle = 0.$$

Thus adding the generator directions does not reduce the distance of this particular X from $\mathcal{U}_S$. Equation (62) gives

$$\operatorname{dist}(X, \mathcal{U}_S + \operatorname{range} B_S) = \operatorname{dist}(X, \mathcal{U}_S) \ge Q^{-C} \|X\|_F. \tag{63}$$

We can now compare the upper and lower bounds to complete the proof. Write $X = \mathcal{M}(w)$, where $w = (w_1, \ldots, w_\nu) \ne 0$. Since $\operatorname{range} D_S \subseteq \mathcal{U}_S + \operatorname{range} B_S$ and P_S projects onto $(\operatorname{range} D_S)^\perp$,

$$\|P_S \operatorname{vec}(X)\| \ge \operatorname{dist}(X, \mathcal{U}_S + \operatorname{range} B_S) \ge Q^{-C} \|X\|_{\mathrm{F}} \ge Q^{-C} \beta_X \|w\|.$$

Combining this with (61) and cancelling $\|w\| > 0$ yields $\beta_X \le Q^C \sqrt{\nu}\lambda$. Finally, (58) implies

$$\sigma_\nu(X_S) \le m\beta_X \le Q^C m \sqrt{\nu}\lambda \le Q^{C'}\lambda,$$

because m and ν are polynomially bounded in Q. This completes the proof of the lemma. □

We write $\mathcal{E}$ for the set of exceptional components. Since these components fail the regularity test on every good block, we have $\mathcal{E} \subseteq \mathcal{E}_S$ for each such block. As a corollary, we now obtain that *all* the exceptional components (in full) can be placed within a subspace V of dimension $< c_0 n$, for a small constant $c_0 < 1/2$, up to a small error. This is purely a structural statement.

To see this, for every good block S we can pick $V_S \subseteq \mathbb{R}^S$ to be the (left) top singular value subspace of X_S of dimension at most $\nu - 1$ (by the previous Lemma, this space contains most of the mass of the corresponding block of the exceptional vectors). Define $V \subseteq \mathbb{R}^n$ to be the orthogonal sum of these spaces, along with all the coordinate spaces of bad blocks, and the coordinates outside the fixed blocks (if any). Then we get

$$d := \dim V \leq \frac{c_0}{4} \cdot \frac{n}{\ell} \cdot \ell + (\nu - 1)\frac{n}{\ell} + \ell \leq c_0 n. \tag{64}$$

Moreover, (57) gives

$$\sum_{j \in \mathcal{E}} \operatorname{dist}(b_j, V)^2 \leq \sum_{S \text{ good}} \ell\, \sigma_\nu(X_S)^2 \leq Q^C \lambda^2.$$

Thus all exceptional components approximately lie in one small subspace. This subspace is used only in the proof; it is not retained as part of the reconstruction index. Also, note that the gain we get in $\dim V$ is only the constant c_0 – all the exceptions (indeed all the b_i!) lie in $\mathbb{R}^n$. But we will see now why this helps.

Bounding exceptional components. Recall that we defined q to be the number of singular values of the tangent map $\mathsf{D}\Phi(\mathbf{b})$ that are "small" ($\leq \gamma$). We may also refer to these directions as *weak* tangent space directions. We will now prove that the total number of coordinates in all the exceptional candidates, i.e., $n|\mathcal{E}|$ is at most $Cns + q$, for a constant C.

Let $\mu_{\text{exc}} : \mathbb{R}^{|\mathcal{E}|} \to \mathbb{R}^{N_2}$ be the square map with columns b_j^2, $j \in \mathcal{E}$. Bounded component norms and the preceding approximation imply

$$\|\mu_{\text{exc}} - \Pi_{\mathsf{Sym}^2(V)} \mu_{\text{exc}}\| \leq Q^C \lambda.$$

Also, $\Pi_{\mathcal{U}^\perp} \mu_{\text{exc}}$ is a column submatrix of $Z_{\mathbf{a}}(\mathbf{b})$, so

$$\sigma_{s+1}(\Pi_{\mathcal{U}^\perp} \mu_{\text{exc}}) \leq \sigma_{s+1}(Z_{\mathbf{a}}(\mathbf{b})) \leq \tau \leq \lambda.$$

Every symmetric matrix in $\mathsf{Sym}^2(V)$ has rank at most d. We can thus apply the second part of Lemma 6.15 with $\mu = \mu_{\mathbf{a}}$ and $k = d$ to obtain

$$\sigma_{s+d+1}(\mu_{\text{exc}}) \leq Q^C \lambda.$$

Geometrically, the squares b_j^2 with $j \in \mathcal{E}$ have at most s substantial square directions outside $\mathcal{U}$, and the small approximate intersection with $\mathcal{U}$ (which comes from Lemma 6.15) accounts for at most d more. Hence the entire exceptional square map has at most $s + d$ singular directions above the error scale $Q^C \lambda$. Hence there is a coefficient space W_{exc} of dimension at least $(|\mathcal{E}| - s - d)_+$ on which μ_{exc} has norm at most $Q^C \lambda$. Tensoring with $\mathbb{R}^n$ gives a space of component perturbations, supported on the exceptional components, of dimension at least $n(|\mathcal{E}| - s - d)_+$. On this space, $\mathsf{D}\Phi(\mathbf{b})$ has norm at most $Q^C \lambda = Q^C \gamma^4 \leq \gamma$, after choosing a_0 sufficiently large. By the definition of q (and the min–max principle for eigenvalues),

$$q \geq n(|\mathcal{E}| - s - d)_+.$$

Since $d \le c_0 n$ and since $s > s_0 = \lfloor c_* n \rfloor$ implies $s = \Omega(n)$, this gives

$$n|\mathcal{E}| \le n(s+d) + q \le Cns + q. \tag{65}$$

We can thus retain the exceptional components not already in I in full, by adding at most $Cns + q$ coordinates.

Retaining coordinates for weak tangent directions. The final set of coordinates we retain will be to ensure that the weak tangent directions (of the map $\mathsf{D}\Phi(\mathbf{b})$) are "maintained" by our retained vectors. Let $U_\gamma \in \mathbb{R}^{p\times q}$ be a matrix with orthonormal columns spanning E_γ. If $q > 0$, we can again use column subset selection to $U_\gamma^\top$, and select q coordinates $I_{\mathrm{wk}} \subseteq [m] \times [n]$ such that

$$\forall v \in E_\gamma, \quad \|\Pi_{I_{\mathrm{wk}}} v\| \ge p^{-1}\|v\|.$$

We apply Theorem B.2 with $M = U_\gamma^\top$ and $r = q$. For $q = 0$, take $I_{\mathrm{wk}} = \varnothing$. We will retain these coordinates as well (noting that some of the coordinates may already have been retained). The final retained projection therefore satisfies $\|\Pi_z v\| \ge p^{-1}\|v\| \ge Q^{-C}\|v\|$ for all $v \in E_\gamma$), which is (39), the second part of our recovery Theorem. Values retained at this stage replace their earlier predictions, and every still-missing coordinate uses its original regular-component formula.

Summary and counts of retained coordinates and reconstruction indices. Let $I_{\mathrm{exc}} = \mathcal{E} \setminus I$. The retained coordinates consist of the generators, the components indexed by I_{exc}, the base coordinates of regular non-generators, and the additional coordinates I_{wk}. By (65),

$$p_z \le ns + n|I_{\mathrm{exc}}| + m(\ell+\nu-1) + q \le Cns + m(\ell+\nu) + 2q.$$

Since $\ell + \nu = O(\sqrt{s})$ and $m = O(n\sqrt{s})$, we obtain

$$p_z \le Cns + 2q,$$

and trivially $p_z \le p$. We can now complete the proof of the main compression statement, Theorem 6.13.

Proof of Theorem 6.13. The above argument bounds the number of coordinates z that have been retained. We only need to specify the details of a "reconstruction index" (denoted α in the Theorem): it records I, I_{exc}, the dictionary selections I_S, the block $S(j)$ and the set F_j for each reconstructed component, and I_{wk}. Enumerating all choices with the coordinate bounds above gives us that there are at most

$$(m+1)^s\, 2^m\, 2^{st_s}\, (t_s n^{\nu-1})^m\, (p+1)^q \le Q^{C(ns+q)} \tag{66}$$

indices. In particular, the factor t_s^m for guessing a block for each regular component is included. The bound follows from $st_s \le ns$, $m\nu \le Cns$, and the polynomial bounds on m, n, p.

Crucially, note that the enumeration is independent of $\mathbf{a}$ and the candidate $\mathbf{b}$; our arguments so far show that *there exists* a good choice of the indices for which the recovery guarantees hold (the formulas themselves only use the retained coordinates in a fixed way).

For a valid index, the regular-coordinate formulas are the rational Gram-matrix inverse formulas from (50); all other coordinates are retained. Their degrees are at most Q^C. Summing (53) over the reconstructed components proves the uniform local reconstruction property of Theorem 6.13. □

6.8 Volume argument with Compressed Coordinates

We now combine Theorem 6.13 with the volume estimate from 6.6 to finish the proof of polynomial robustness. The reconstruction formulas that we have developed reduce the number of candidate variables, but introduce a considerable error in approximating the components. The idea is to first perform one Newton step on the function $\Psi(\mathbf{a}, \mathbf{b})$ (as we describe below). This reduces the error by a quadratic factor. Then, for every candidate $\mathbf{b}$, we construct an ellipsoid of a sufficiently large volume (which gets larger as the number of "weak" tangent directions q increases). Together, these imply our main result.

We use the same notation of R_b, Q as before, and as described earlier, we use the scales

$$\tau = \gamma^{16}, \qquad \eta = \gamma^{22}.$$

The main technical lemma is the following.

Lemma 6.17 (Compressed volume bound)**.** *For a layer (s, q) with $s > s_0$, let $E_{s,q}$ be the event that there exists a candidate $\mathbf{b}$ in that layer with $\max_i \|b_i\| \leq R_b$ and $\delta(\mathbf{a}, \mathbf{b}) \leq \eta$. Recall that $k_s = \lfloor ns/10 \rfloor$. There is a constant C such that*

$$\mathbb{P}(\mathcal{G}_2 \cap E_{s,q}) \leq Q^{C(ns+q)} \left(\frac{\eta}{\tau}\right)^{k_s} \gamma^q. \tag{67}$$

We will outline the key ideas in the proof and defer the proof to Appendix B.8.

Recall that the main idea of compressing coordinates is the following: for a given $\mathbf{b}$, we retain a subset z of its coordinates, and express the remaining coordinates u as rational functions of z; up to a permutation of coordinates determined by α, $\mathbf{b}$ can be expressed as (z, u). Now the main challenge in the proof is that the reconstruction $u = u^0(\mathbf{a}, z)$ introduces a non-zero error (say ε), that depends on condition number parameters of $\mathbf{a}$. Thus, if we wish to use Theorem 6.6 to bound the probability of $\delta(\mathbf{a}, \mathbf{b}) < \eta$ for a very small η, in terms of the recovered vector $\mathbf{b}' = (z, u)$, $\delta(\mathbf{a}, \mathbf{b}')$ can range in a much larger interval (roughly $\eta + \varepsilon \gg \eta$), which makes the volume argument fail. The novel idea here is to use one Newton iteration to obtain a *different* reconstruction formula (still a rational function), that reduces the error to roughly $\eta + \varepsilon^2$.

Let us illustrate the idea in a very simple one dimensional example: suppose we have a function $\phi(z, u)$ in two variables that we wish to minimize. Suppose we are able to approximate u using $u = r(z)$. Now consider the partial derivative $d(z) := \frac{\partial \phi(z,u)}{\partial u}$ (evaluated at the point $(z, r(z))$). Now the Newton correction is the function $r_1(z) := r(z) - \frac{1}{d(z)}\phi(z, u)$. When we are at a point $(z, r(z))$ such that $\phi(z, u)$ is already much smaller than $d(z)$ (or if we have nice lower bounds on $d(z)$), then we can evaluate ϕ at the point $(z, r_1(z))$ using a Taylor expansion.

$$\phi(z, r_1(z)) \approx \phi\left(z, r(z) - \frac{\phi(z, r(z))}{d(z)}\right) = \phi(z, r(z)) - d(z)\frac{\phi(z, r(z))}{d(z)} + \frac{\partial^2 \phi(z, u)}{\partial^2 u}\left(\frac{\phi(z, r(z))}{d(z)}\right)^2.$$

The key point in Newton's method is that the first two terms on the RHS cancel, and as long as the second derivative term and the derivative $d(z)^{-1}$ are bounded, the value $\phi(z, r_1(z))$ is quadratically smaller than $\phi(z, r(z))$.

The challenge in our (high dimensional) setting is to reason about the derivatives and Hessians of the associated functions, which makes the argument technical. A second novel aspect of the proof is the way in which the q retained coordinates corresponding to the "weak" tangent directions

are handled. As such, since q can be as large as $p = mn$, this can make the number of retained coordinates, and hence the resulting Q^{p_1} term arising in Theorem 6.6) impossible to cancel out using the η^r term. The trick, as it turns out, is to consider ellipsoids around candidate $\mathbf{b}$, that yields the additional γ^q term on the RHS of (67).

We are now ready to state the main result (re-formulation of Theorem 6.3).

Theorem 6.18 (Polynomial robustness)**.** *Suppose $m = O(n^{3/2})$ and let $\mathbf{a}$ be a ρ-smoothing of any tuple $\bar{\mathbf{a}}$. Let $R_a = \max_i \|\bar{a}_i\|$, and fix a radius $R_b \geq 0$. Define Q as before. There are constants $c, C_0, C_1 > 0$ such that, for all sufficiently large n, with probability at least $1 - e^{-cn}$, simultaneously for every $\mathbf{b}$ with $\max_i \|b_i\| \leq R_b$,*

$$\delta(\mathbf{a}, \mathbf{b}) \leq Q^{-C_1} \quad \Longrightarrow \quad d_{\mathrm{perm}}(\mathbf{a}, \mathbf{b}) \leq Q^{C_0} \delta(\mathbf{a}, \mathbf{b}).$$

Here we fix the candidate radius and assume that the tensor error is sufficiently small. After proving this statement, we obtain Theorem 6.3 by taking a union bound over dyadic radii and using the norm bounds for larger tensor errors.

Proof. We choose a sufficiently large constant $a \geq a_0$, and set

$$\gamma = Q^{-a}, \qquad \tau = \gamma^{16}, \qquad \eta = \gamma^{22}.$$

Take a large enough that $\tau < Q^{-D_0}$ and $\eta < Q^{-D_1}$, where D_0, D_1 are the square-residual and tensor-error exponents in Lemma 6.8. If $m \leq s_0$, Lemma 6.8 already applies to every candidate with tensor error at most η on $\mathcal{G}_2$. Otherwise, consider all layers with $s > s_0$.

For such layers, $k_s \geq cns$. Lemma 6.17 therefore bounds each layer's probability by

$$\begin{aligned} Q^{C(ns+q)} (\eta/\tau)^{k_s} \gamma^q &= Q^{Cns - 6ak_s + (C-a)q} \\ &\leq Q^{-cns-2q}, \end{aligned}$$

after choosing a sufficiently large. Summing over $s_0 < s \leq m$ and $0 \leq q \leq p$, and adding $\mathbb{P}(\mathcal{G}_2^c) \leq e^{-cn}$, leaves failure probability at most e^{-cn}, with a change of the constant c.

Outside this failure event, every candidate with tensor error at most η has at most s_0 singular values of $Z_{\mathbf{a}}(\mathbf{b})$ above τ. Thus

$$\sigma_{s_0+1}(Z_{\mathbf{a}}(\mathbf{b})) \leq \tau < Q^{-D_0}, \qquad \delta(\mathbf{a}, \mathbf{b}) \leq \eta < Q^{-D_1}.$$

Lemma 6.8 gives $d_{\mathrm{perm}}(\mathbf{a}, \mathbf{b}) \leq Q^{C_0} \delta(\mathbf{a}, \mathbf{b})$, with no additive reconstruction error. Since $\eta = Q^{-22a}$, take $C_1 = 22a$. □

Proof of Theorem 6.3. Theorem 6.18 gives the desired bound for a fixed candidate radius and sufficiently small tensor error. We will first make this bound hold for all radii R_b at once, and then use the component norm bounds to handle larger tensor errors.

We define a common event $\mathcal{G}_2$ by evaluating its conditions at $R_b = 0$, using the size parameter $Q = n(2 + R_a + \rho^{-1})$. We keep this event fixed throughout. Increasing the candidate radius only weakens the conditioning requirements, so this event is sufficient at every radius.

Allowing all candidate radii. To cover candidates of arbitrary size, we apply the proof of Theorem 6.18 with norm bounds $R_b = 1, 2, 4, \ldots$. Every candidate is covered by one of these bounds, and we will sum the failure probabilities so that the conclusions hold simultaneously. For each integer $j \geq 0$, we write Q_j for the value of Q corresponding to the bound $R_b = 2^j$:

$$Q_j = n(2 + R_a + 2^j + \rho^{-1}).$$

We apply the Theorem 6.18 with Q replaced by Q_j in the precision parameters and probability bounds, while the common event $\mathcal{G}_2$ is unchanged. Here we use $\gamma = Q_j^{-a}$ and $\tau = \gamma^{16}$, as in the proof of Theorem 6.18. We group the candidate tuples according to two counts: the number s of singular values of $Z_{\mathbf{a}}(\mathbf{b})$ above τ, and the number q of singular values of $\mathsf{D}\Phi(\mathbf{b})$ at most γ. On this event, Lemma 6.8 handles every candidate with $\delta(\mathbf{a}, \mathbf{b}) \leq Q_j^{-C_1}$ and $s \leq s_0$. We only need to exclude candidates with the same tensor error but $s > s_0$.

For each fixed pair of counts (s, q), Theorem 6.18 bounds the probability that $\mathcal{G}_2$ holds and such a candidate exists by $Q_j^{-cns-2q}$. Summing over these pairs gives a bound of

$$\sum_{s=s_0+1}^{m} \sum_{q=0}^{p} Q_j^{-cns-2q} \leq m(p+1) Q_j^{-cn(s_0+1)} \leq Q_j^{-\Omega(n^2)},$$

where we used $s_0 + 1 = \Theta(n)$ and $m(p+1) \leq Q_j^{O(1)}$.

We now take a union bound over the radii. Since $Q_j \geq n 2^j$, these bounds decrease geometrically in j. We count the failure of the common event $\mathcal{G}_2$ only once, giving total failure probability at most

$$\mathbb{P}(\mathcal{G}_2^c) + \sum_{j \geq 0} Q_j^{-\Omega(n^2)} \leq \exp(-\Omega(n)).$$

Outside this failure event, the small-error conclusion (Theorem 6.18) holds for every radius, with the same constants C_0, C_1.

Removing the small-error restriction. We fix any candidate $\mathbf{b}$ and choose the smallest radius $2^j \geq \max\{1, \max_i \|b_i\|\}$. If $\delta(\mathbf{a}, \mathbf{b}) \leq Q_j^{-C_1}$, the bound we just obtained gives $d_{\mathrm{perm}}(\mathbf{a}, \mathbf{b}) \leq Q_j^{C_0} \delta(\mathbf{a}, \mathbf{b})$.

For larger tensor errors, a crude triangle-inequality bound on the permutation distance suffices. Since $\mathcal{G}_2$ includes $\mathcal{G}_1$, on this event we have

$$d_{\mathrm{perm}}(\mathbf{a}, \mathbf{b}) \leq \sqrt{m}\big(\max_i \|a_i\| + 2^j\big) \leq Q_j^{O(1)} \leq Q_j^{C_1 + O(1)} \delta(\mathbf{a}, \mathbf{b}).$$

The last inequality uses $\delta(\mathbf{a}, \mathbf{b}) > Q_j^{-C_1}$. Thus both cases give $d_{\mathrm{perm}}(\mathbf{a}, \mathbf{b}) \leq Q_j^{O(1)} \delta(\mathbf{a}, \mathbf{b})$. Rounding the candidate radius up to 2^j changes the size parameter by at most a factor two: $Q_j \leq 2Q$, where Q is evaluated at $R_b = \max_i \|b_i\|$. We can absorb this factor by increasing the final exponent. This completes the proof. □

7 Application: Robust Ideal Membership

In this section, we provide an algorithm for the robust ideal membership problem. The ideal membership problem asks, given a set of generating polynomials $b_1, \ldots, b_m$ and a target polynomial

p, whether p lies in the ideal generated by $b_1, \dots, b_m$. We consider a robust version of this problem, where we are given the polynomials $b_1, \dots, b_m, p$ upto some noise, and we want to find if p lies *close* to the ideal generated by them.

Formally, we are given $\widehat{b}_1, \dots, \widehat{b}_m, \widehat{p}$ such that $\sum_{i\in[m]} \left\|\widehat{b}_i - b_i\right\| + \|p - \widehat{p}\| \le \delta$ for $\delta > 0$. We want an algorithm that outputs YES if p lies in the ideal generated by $b_1, \dots, b_m$, and NO if p lies far from the ideal. In the case when p lies in the ideal, we also want to find multipliers $h_1, \dots, h_m$ so that $p \approx \sum_{i\in[m]} b_i h_i$.

We provide an algorithm for the above problem. Our algorithm essentially attempts to solve the linear system $\mathsf{M}^{(t)}_{\widehat{\mathbf{b}}}(\mathbf{h}) = \widehat{p}$, where $\mathsf{M}^{(t)}_{\widehat{\mathbf{b}}}$ denotes the Macaulay matrix of the noisy generators $\widehat{\mathbf{b}} = (\widehat{b}_1, \dots, \widehat{b}_m)$.

Theorem 7.1. *Under the setting in Theorem 4.1, there is a constant C depending only on ε, $d_{\max}$, $D_{\max}$, and T, and a polynomial-time algorithm $\mathcal{A}$ such that the following holds with $\eta = n^{-C}\rho^{C}$. Simultaneously for all $\delta > 0$, all $t \in [T]$, all polynomials $p \in \mathbb{R}[\mathbf{x}]^{=t}$ with $\|p\| \le 1$, and all noisy estimates $\hat{b}_i \in \mathbb{R}[\mathbf{x}]^{=\deg b_i}$ and $\hat{p} \in \mathbb{R}[\mathbf{x}]^{=t}$ with*

$$\sum_{i=1}^{m} \|\hat{b}_i - b_i\| + \|\hat{p} - p\| \le \delta,$$

$\mathcal{A}$ takes $\{\hat{b}_i\}_{i=1}^m$, $\hat{p}$, η, and δ as input and has the following guarantee:

1. *If $p \in \langle b_1, \dots, b_m\rangle$, then $\mathcal{A}$ outputs YES, and outputs $\mathbf{h} \in \bigoplus_{i=1}^m \mathbb{R}[\mathbf{x}]^{=t-\deg b_i}$, where*

$$\left\|p - \sum_{i=1}^{m} b_i h_i\right\| \le \frac{3\delta}{\eta} \quad \text{and} \quad \|\mathbf{h}\| \le \frac{1}{\eta}$$

   with probability $1 - \exp\big(-\Omega(n^{\min d_i})\big)$.

2. *If $\mathsf{dist}\Big(p, \operatorname{colspan}\big(\mathsf{M}^{(t)}_{\mathbf{b}}\big)\Big) \ge 4\delta/\eta$, then $\mathcal{A}$ outputs NO with probability $1 - \exp\big(-\Omega(n^{\min d_i})\big)$.*

Proof. We will condition on the good event in Theorem 4.1 throughout. On this event, every nonzero singular value of the transposed Macaulay matrix $\mathsf{M} = \mathsf{M}^{(t)}_{\mathbf{b}}$ is at least $\eta = \rho^C n^{-C}$, simultaneously for all $t \in [T]$. Since the generators $\{b_i\}$ are homogeneous, the degree-t components of the ideal is just

$$\langle b_1, \dots, b_m\rangle \cap \mathbb{R}[\mathbf{x}]^{=t} = \operatorname{im} \mathsf{M}.$$

Thus every p in this space with $\|p\| \le 1$ has a representation $p = \mathsf{M}\mathbf{h}_0$ with $\|\mathbf{h}_0\| \le 1/\eta$, by the pseudoinverse bound.

Let $\widehat{\mathsf{M}}$ be the Macaulay matrix of the noisy generators, and let $e_b = \sum_i \|\hat{b}_i - b_i\|$ and $e_p = \|\hat{p} - p\|$, so that $e_b + e_p \le \delta$. Since polynomial multiplication is contractive, we have $\|\widehat{\mathsf{M}} - \mathsf{M}\|_{\mathrm{op}} \le e_b$.

Since $\eta \le 1$, for every $\|\mathbf{h}\| \le 1/\eta$, the noisy and true residuals differ by at most

$$\begin{aligned}
\Big|\|\widehat{\mathsf{M}}\mathbf{h} - \hat{p}\| - \|\mathsf{M}\mathbf{h} - p\|\Big| &\le \|(\widehat{\mathsf{M}} - \mathsf{M})\mathbf{h} - (\hat{p} - p)\| \\
&\le \|\widehat{\mathsf{M}} - \mathsf{M}\|_{\mathrm{op}}\|\mathbf{h}\| + \|\hat{p} - p\| \\
&\le e_b\|\mathbf{h}\| + e_p \le \frac{\delta}{\eta}.
\end{aligned}$$

Let $\mathcal{A}$ be the algorithm that computes a solution $\mathbf{h}$ to the least-squares problem

$$\min_{\|\mathbf{h}\| \le 1/\eta} \|\widehat{\mathsf{M}}\mathbf{h} - \hat{p}\|,$$

up to additive accuracy δ/η, and returns $\mathbf{h}$ if its objective value is at most $2\delta/\eta$; otherwise it returns $\perp$.

This takes polynomial time because the optimization problem is convex and has polynomially many variables, since $t \le T$ is bounded.

If $p \in \langle b_1, \ldots, b_m \rangle$, the feasible point $\mathbf{h}_0$ above has objective value at most

$$\|\widehat{\mathsf{M}}\mathbf{h}_0 - \hat{p}\| \le \|\mathsf{M}\mathbf{h}_0 - p\| + \frac{\delta}{\eta} = \frac{\delta}{\eta},$$

so the algorithm returns a coefficient vector. Whenever it returns $\mathbf{h}$, we have $\|\mathbf{h}\| \le 1/\eta$ and

$$\begin{aligned} \|p - \mathsf{M}\mathbf{h}\| &\le \|p - \hat{p}\| + \|\hat{p} - \widehat{\mathsf{M}}\mathbf{h}\| + \|(\widehat{\mathsf{M}} - \mathsf{M})\mathbf{h}\| \\ &\le \|\hat{p} - \widehat{\mathsf{M}}\mathbf{h}\| + \frac{\delta}{\eta} \le \frac{3\delta}{\eta}. \end{aligned}$$

On the other hand, if $\mathsf{dist}(p, \operatorname{im}\mathsf{M}) \ge 4\delta/\eta$, then every feasible $\mathbf{h}$ satisfies

$$\|\widehat{\mathsf{M}}\mathbf{h} - \hat{p}\| \ge \|\mathsf{M}\mathbf{h} - p\| - \frac{\delta}{\eta} > \frac{2\delta}{\eta},$$

so the algorithm returns $\perp$. The argument is deterministic on the single event from Theorem 4.1, so all guarantees hold uniformly with its stated probability. □

8 AI Methodology

AI Disclosure. The original problem of approximate syzygies in the smoothed analysis setting was formulated by the authors over a year ago, while trying to develop a broader framework for proving smoothed analysis guarantees for learning arithmetic circuits. In the terminology of [Fei26], we view the AI contribution as being in the category of "AI as a Co-author". The proof for the main result (Theorem 4.1 presented in Section 4) was developed over a period of several months through extensive iterative interactions, using GPT 5.5, GPT 5.6 Sol and GPT 6 Astra. We realized the connection of the syzygy result to local uniqueness of tensor decompositions. Subsequently, the proof for robust uniqueness was obtained (using the proven syzygy claims) through the use of GPT 6 Astra, and forms part of Section 6. The authors simplified, organized and presented the ideas and proofs in the paper. We verified the correctness and originality of all proofs presented in the paper, and take responsibility for any errors, including references.

The authors also note that the breadth of results we were able to obtain would not be possible in a reasonable time-frame without the help of AI. For example, earlier studies on smoothed analysis on syzygies were limited to cases such as powers of linear forms and quadratics... [BEG+24]. Moreover, the proofs involve many moving parts, with a careful combination of deep technical ideas from many different areas (e.g., from the Alexander-Hirschowitz theorem in algebraic geometry). Likewise the result on robust uniqueness of tensor decompositions builds on ideas from across fields, including numerical linear algebra, probability theory, optimization and the Newton method, and the algebra of polynomials.

We also note that some of the key technical ideas in the AI proofs were not attributed correctly initially. For example, on further investigation, we traced back the ideas in the global uniqueness proof involving the Bezout-based volume argument to a remarkable paper by Kane [Kan17], and the triangular elimination idea to the celebrated theorem of Alexander and Hirschowitz [AH95, BO08].

References

[ADM+18] Nima Anari, Constantinos Daskalakis, Wolfgang Maass, Christos H. Papadimitriou, Amin Saberi, and Santosh Vempala. Smoothed analysis of discrete tensor decomposition and assemblies of neurons. In *Advances in Neural Information Processing Systems*, volume 31, 2018.

[AGH+14] Animashree Anandkumar, Rong Ge, Daniel Hsu, Sham M. Kakade, and Matus Telgarsky. Tensor decompositions for learning latent variable models. *Journal of Machine Learning Research*, 15(80):2773–2832, 2014.

[AH95] J. Alexander and A. Hirschowitz. Polynomial interpolation in several variables. *Journal of Algebraic Geometry*, 4(2):201–222, 1995.

[AMP16] Kwangjun Ahn, Dhruv Medarametla, and Aaron Potechin. Graph matrices: Norm bounds and applications. *arXiv: Combinatorics*, 2016.

[BCMV14] Aditya Bhaskara, Moses Charikar, Ankur Moitra, and Aravindan Vijayaraghavan. Smoothed analysis of tensor decompositions. In *Proceedings of the 46th Annual ACM Symposium on Theory of Computing (STOC)*, 2014.

[BCPV19] Aditya Bhaskara, Aidao Chen, Aidan Perreault, and Aravindan Vijayaraghavan. Smoothed analysis in unsupervised learning via decoupling. In *Proceedings of the 60th Annual IEEE Symposium on Foundations of Computer Science (FOCS)*, 2019.

[BCV14] Aditya Bhaskara, Moses Charikar, and Aravindan Vijayaraghavan. Uniqueness of tensor decompositions with applications to polynomial identifiability. In *Proceedings of the 27th Conference on Learning Theory*, volume 35 of *Proceedings of Machine Learning Research*, pages 742–778. PMLR, 2014.

[BDD14] Kim Batselier, Philippe Dreesen, and Bart De Moor. The canonical decomposition of $\mathcal{C}_d^n$ and numerical Gröbner and border bases. *SIAM Journal on Matrix Analysis and Applications*, 35(4):1242–1264, 2014.

[BDL26] Mats Boij, Eric Dannetun, and Samuel Lundqvist. Independence of generic forms and the Fröberg conjecture, 2026. arXiv:2605.03872.

[BEG+24] Aditya Bhaskara, Eric Evert, Anxin Guo, Tanmay Sinha, Vaidehi Srinivas, and Aravindan Vijayaraghavan. New tools for smoothed analysis: Least singular value bounds for random matrices with dependent entries, 2024. arXiv:2405.01517. https://arxiv.org/abs/2405.01517.

[Bha97] R. Bhatia. *Matrix analysis*. Graduate texts in mathematics. Springer, 1997.

[BHKX22] Mitali Bafna, Jun-Ting Hsieh, Pravesh K. Kothari, and Jeff Xu. Polynomial-time power-sum decomposition of polynomials. In *2022 IEEE 63rd Annual Symposium on Foundations of Computer Science (FOCS)*, pages 956–967, 2022.

[BL98] Bernhard Beckermann and George Labahn. When are two numerical polynomials relatively prime? *Journal of Symbolic Computation*, 26(6):677–689, 1998.

[BO08] Maria Chiara Brambilla and Giorgio Ottaviani. On the alexander–hirschowitz theorem. *Journal of Pure and Applied Algebra*, 212(5):1229–1251, 2008.

[Buc65] Bruno Buchberger. *Ein Algorithmus zum Auffinden der Basiselemente des Restklassenringes nach einem nulldimensionalen Polynomideal*. PhD thesis, University of Innsbruck, 1965.

[BV18] Paul Breiding and Nick Vannieuwenhoven. The condition number of join decompositions. *SIAM Journal on Matrix Analysis and Applications*, 39(1):287–309, January 2018.

[CGK+24] Pritam Chandra, Ankit Garg, Neeraj Kayal, Kunal Mittal, and Tanmay Sinha. Learning arithmetic formulas in the presence of noise: A general framework and applications to unsupervised learning. In *15th Innovations in Theoretical Computer Science Conference (ITCS 2024)*, volume 287 of *Leibniz International Proceedings in Informatics (LIPIcs)*, pages 25:1–25:19. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2024.

[CLO05] David A. Cox, John Little, and Donal O'Shea. *Using Algebraic Geometry*, volume 185 of *Graduate Texts in Mathematics*. Springer, 2 edition, 2005.

[CLO15] David A. Cox, John Little, and Donal O'Shea. *Ideals, Varieties, and Algorithms: An Introduction to Computational Algebraic Geometry and Commutative Algebra*. Undergraduate Texts in Mathematics. Springer, 4 edition, 2015.

[CO12] Luca Chiantini and Giorgio Ottaviani. On generic identifiability of 3-tensors of small rank. *SIAM Journal on Matrix Analysis and Applications*, 33, 2012.

[COV17] Luca Chiantini, Giorgio Ottaviani, and Nick Vannieuwenhoven. On generic identifiability of symmetric tensors of subgeneric rank. *Transactions of the American Mathematical Society*, 369(6):4021–4042, 2017.

[DV06] Amit Deshpande and Santosh Vempala. Adaptive sampling and fast low-rank matrix approximation. In *Approximation, Randomization, and Combinatorial Optimization. Algorithms and Techniques*, volume 4110 of *Lecture Notes in Computer Science*, pages 292–303. Springer, 2006.

[Eis95] David Eisenbud. *Commutative Algebra: With a View Toward Algebraic Geometry*, volume 150 of *Graduate Texts in Mathematics*. Springer, 1995.

[Fau99] Jean-Charles Faugère. A new efficient algorithm for computing Gröbner bases (F4). *Journal of Pure and Applied Algebra*, 139(1–3):61–88, 1999.

[Fau02] Jean-Charles Faugère. A new efficient algorithm for computing Gröbner bases without reduction to zero (F5). In *Proceedings of the 2002 International Symposium on Symbolic and Algebraic Computation (ISSAC)*, pages 75–83. ACM, 2002.

[Fei26] Uriel Feige. A track system for handling submissions to competitive TCS conferences, August 2026. Manuscript dated August 5, 2026.

[FH94] Ralf Fröberg and Joachim Hollman. Hilbert series for ideals generated by generic forms. *Journal of Symbolic Computation*, 17(2):149–157, 1994.

[Frö85] Ralf Fröberg. An inequality for Hilbert series of graded algebras. *Mathematica Scandinavica*, 56:117–144, 1985.

[Ful98] William Fulton. *Intersection Theory*. Springer, 2 edition, 1998.

[GE96] Ming Gu and Stanley C. Eisenstat. Efficient algorithms for computing a strong rank-revealing QR factorization. *SIAM Journal on Scientific Computing*, 17(4):848–869, 1996.

[GHK15] Rong Ge, Qingqing Huang, and Sham M. Kakade. Learning mixtures of Gaussians in high dimensions. In *Symposium on Theory of Computing*, 2015.

[GKS20] Ankit Garg, Neeraj Kayal, and Chandan Saha. Learning sums of powers of low-degree polynomials in the non-degenerate case. In *2020 IEEE 61st Annual Symposium on Foundations of Computer Science (FOCS)*, pages 889–899, 2020.

[GM15] Rong Ge and Tengyu Ma. Decomposing overcomplete 3rd order tensors using sum-of-squares algorithms. In *Proceedings of the 19th International Conference on Randomization and Computation (RANDOM 2015)*, pages 829–849, 2015.

[Har72] Richard A. Harshman. Determination and proof of minimum uniqueness conditions for PARAFAC1. *UCLA Working Papers in Phonetics*, 22:111–117, 1972.

[Hås90] Johan Håstad. Tensor rank is NP-complete. *Journal of Algorithms*, 11(4), 1990.

[HL87] Melvin Hochster and Dan Laksov. The linear syzygies of generic forms. *Communications in Algebra*, 15(1–2):227–239, 1987.

[HL13] Christopher J. Hillar and Lek-Heng Lim. Most tensor problems are NP-hard. *Journal of the American Mathematical Society*, 60, 2013.

[JPR+22] Chris Jones, Aaron Potechin, Goutham Rajendran, Madhur Tulsiani, and Jeff Xu. Sum-of-Squares Lower Bounds for Sparse Independent Set . In *2021 IEEE 62nd Annual Symposium on Foundations of Computer Science (FOCS)*, pages 406–416, Los Alamitos, CA, USA, February 2022. IEEE Computer Society.

[Kan17] Daniel M. Kane. A structure theorem for poorly anticoncentrated polynomials of Gaussians and applications to the study of polynomial threshold functions. *The Annals of Probability*, 45(3):1612–1679, 2017.

[Kru77] Joseph B. Kruskal. Three-way arrays: rank and uniqueness of trilinear decompositions, with application to arithmetic complexity and statistics. *Linear Algebra and its Applications*, 18(2):95–138, 1977.

[Lic19] Daniel Lichtblau. Approximate polynomial GCD by approximate syzygies. *Mathematics in Computer Science*, 13:517–532, 2019.

[Lov26] Benjamin Lovitz. A $27 \times 27 \times 27$ counterexample to Comon's conjecture, 2026. arXiv:2609.28292.

[MRX20] Sidhanth Mohanty, Prasad Raghavendra, and Jeff Xu. Lifting sum-of-squares lower bounds: degree-2 to degree-4. In *Proceedings of the 52nd Annual ACM SIGACT Symposium on Theory of Computing*, STOC 2020, page 840–853, New York, NY, USA, 2020. Association for Computing Machinery.

[MSS16] Tengyu Ma, Jonathan Shi, and David Steurer. Polynomial-time tensor decompositions with sum-of-squares. In *Proceedings of the 57th Annual IEEE Symposium on Foundations of Computer Science (FOCS)*, pages 438–446. IEEE, 2016.

[Nen17] Gleb Nenashev. A note on Fröberg's conjecture for forms of equal degrees. *Comptes Rendus. Mathématique*, 355(3):272–276, 2017.

[Nic17] Lisa Nicklasson. On the Hilbert series of ideals generated by generic forms. *Communications in Algebra*, 45(8):3390–3395, 2017.

[Ott13] Giorgio Ottaviani. Identifiability and weak defectivity. Lecture notes for the Lukecin School, 2013. September 2–6, 2013.

[Shi18] Yaroslav Shitov. A counterexample to Comon's conjecture. *SIAM Journal on Applied Algebra and Geometry*, 2(3):428–443, 2018.

[SST06] Arvind Sankar, Daniel A. Spielman, and Shang-Hua Teng. Smoothed analysis of the condition numbers and growth factors of matrices. *SIAM Journal on Matrix Analysis and Applications*, 28(2):446–476, 2006.

[ST04] Daniel A. Spielman and Shang-Hua Teng. Smoothed analysis of algorithms: Why the simplex algorithm usually takes polynomial time. *Journal of the ACM*, 51(3):385–463, 2004.

[Str83] Volker Strassen. Rank and optimal computation of generic tensors. *Linear Algebra and its Applications*, 52–53:645–685, 1983.

[Vij20] Aravindan Vijayaraghavan. *Beyond the Worst-Case Analysis of Algorithms*, chapter 19 Efficient Tensor Decomposition. Cambridge University Press, 2020.

[Wed73] Per-Åke Wedin. Perturbation theory for pseudo-inverses. *BIT Numerical Mathematics*, 13(2):217–232, 1973.

A Useful Lemmas

Here we record some lemmas that we use in our proofs.

A.1 Bombieri inner product bounds for polynomial

We will use the following properties of the Bombieri inner product on polynomials.

Proposition A.1 (Properties of Bombieri Norm)**.** *The Bombieri norm on homogenous polynomials satisfies the following properties.*

- *For any* $p \in \mathbb{R}[\mathbf{x}]^{=d}$*,*
$$\sum_{\ell=1}^{n} \left\| \frac{\partial p}{\partial x_\ell} \right\|^2 = d^2 \|p\|^2$$
- *For any* $p_1, \ldots, p_n \in \mathbb{R}[\mathbf{x}]^{=d}$*,*
$$\left\| \sum_{\ell=1}^{n} x_\ell p_\ell \right\|^2 \leq \sum_{\ell=1}^{n} \|p_\ell\|^2$$

- *For any homogenous polynomials $p(\mathbf{x}), q(\mathbf{x})$,*

$$\|p(\mathbf{x}) \cdot q(\mathbf{x})\| \leq \|p(\mathbf{x})\| \cdot \|q(\mathbf{x})\|.$$

Proof. For the first identity, write p in the scaled monomial basis:

$$p(\mathbf{x}) = \sum_{\alpha \in \mathbb{Z}^n_{\geq 0} : |\alpha| = d} p_\alpha \sqrt{\frac{d!}{\alpha!}} \mathbf{x}^\alpha.$$

Then,

$$\begin{aligned}
\sum_{\ell=1}^n \left\| \frac{\partial p}{\partial x_\ell} \right\|^2 &= \sum_{\ell=1}^n \left\langle \frac{\partial p}{\partial x_\ell}, \frac{\partial p}{\partial x_\ell} \right\rangle \\
&= \sum_{\ell=1}^n \sum_{\alpha \in \mathbb{Z}^n_{\geq 0} : |\alpha| = d} p_\alpha^2 \frac{d!}{\alpha!} \alpha_\ell^2 \frac{\alpha!}{\alpha_\ell (d-1)!} \\
&= \sum_{\alpha \in \mathbb{Z}^n_{\geq 0} : |\alpha| = d} p_\alpha^2 d \sum_{\ell=1}^n \alpha_\ell = d^2 \sum_{\alpha \in \mathbb{Z}^n_{\geq 0} : |\alpha| = d} p_\alpha^2 \\
&= d^2 \|p\|^2.
\end{aligned}$$

For the second part, if P_ℓ is the symmetric tensor corresponding to p_ℓ and e_ℓ is the ℓ-th standard unit vector, we have

$$\begin{aligned}
\left\| \sum_{\ell=1}^n x_\ell p_\ell \right\|^2 &= \left\| \sum_{\ell=1}^n \mathsf{Sym}_{d+1}(e_\ell \otimes P_\ell) \right\|^2 \\
&\leq \left\| \sum_{\ell=1}^n e_\ell \otimes P_\ell \right\|^2 \\
&= \sum_{\ell=1}^n \|P_\ell\|^2 = \sum_{\ell=1}^n \|p_\ell\|^2
\end{aligned}$$

where in the inequality we used the fact that Sym_{d+1} is a projection, and can only decrease the norm.

We can use a similar argument for the third part. If P, Q are the symmetric tensor representations of $p(\mathbf{x}), q(\mathbf{x})$, then the symmetric tensor representation of $p(\mathbf{x}) \cdot q(\mathbf{x})$ is $\mathsf{Sym}_{d_1+d_2}(P \otimes Q)$ (here d_1 and d_2 are the degrees of p, q respectively. We have that

$$\begin{aligned}
\|p(\mathbf{x}) \cdot q(\mathbf{x})\| &= \|\mathsf{Sym}_{d_1+d_2}(P \otimes Q)\| \\
&\leq \|P \otimes Q\| = \|P\| \cdot \|Q\| \\
&= \|p(\mathbf{x})\| \cdot \|q(\mathbf{x})\|.
\end{aligned}$$

□

A.2 Probability bounds for Gaussian random variables

Lemma A.2 (Gaussian norm bound)**.** *Let $N \geq 1$, $\rho > 0$, and $X \sim \mathcal{N}(\mu, \rho^2 I_N)$, where $\mu \in \mathbb{R}^N$. For every $t \geq 0$,*

$$\mathbb{P}\left[\|X - \mu\| > 2\rho\sqrt{N + t}\right] \leq e^{-N/2-t}.$$

In particular, taking $t = 0$, we have

$$\mathbb{P}\left[\|X - \mu\| > 2\rho\sqrt{N}\right] \leq e^{-N/2}.$$

Proof. We write $G = (X - \mu)/\rho \sim \mathcal{N}(0, I_N)$. By the independence of its coordinates, we have $\mathbb{E}\exp(\|G\|^2/4) = 2^{N/2}$. Applying Markov's inequality, we get

$$\mathbb{P}\left[\|G\| > 2\sqrt{N + t}\right] \leq e^{-N-t}\mathbb{E}\exp(\|G\|^2/4) = e^{-N-t}2^{N/2} \leq e^{-N/2-t}.$$

□

Lemma A.3 (Gaussian small-ball bound)**.** *Let $X \sim \mathcal{N}(\mu, \Sigma)$ in $\mathbb{R}^N$, with arbitrary mean μ. Suppose at least r eigenvalues of Σ are at least σ^2, where $1 \leq r \leq N$ and $\sigma > 0$. Then, for every $t \geq 0$,*

$$\mathbb{P}\left[\|X\| \leq t\right] \leq \left(\frac{O(1)t}{\sigma}\right)^r.$$

Proof. We project X onto the span of r orthonormal eigenvectors whose eigenvalues are at least σ^2. The projected Gaussian has density at most $(2\pi\sigma^2)^{-r/2}$ everywhere, regardless of its mean. If $\|X\| \leq t$, its projection lies in the ball of radius t, whose volume is at most $(2t)^r$. Multiplying these bounds, we get

$$\mathbb{P}\left[\|X\| \leq t\right] \leq (2\pi\sigma^2)^{-r/2}(2t)^r \leq \left(\frac{O(1)t}{\sigma}\right)^r.$$

□

B Deferred Proofs of Section 6

B.1 Probability of the good-block event.

We verify the probability bound for $\mathcal{G}_2$, the event where Definition 6.14 holds.

Lemma B.1 (Block-based conditioning)**.** *Suppose $m = O(n^{3/2})$, $0 < \rho \leq 1$, and let $\mathbf{a}$ be a ρ-smoothing of $\bar{\mathbf{a}}$. Choose c_{load} and c_{rank} sufficiently small, and then K sufficiently large, in (41) and (43). For a sufficiently large absolute constant C_{g} and all sufficiently large n, the following holds with probability at least $1 - \exp(-\Omega(n))$, simultaneously for every integer $s_0 < s \leq m$. Among the fixed disjoint blocks $S_1, \ldots, S_{t_s}$ of size ℓ in the corresponding layer, at least a $(1 - c_0/4)$ fraction have the following property: for every nonzero $\alpha \in \mathbb{R}^m$ and every integer $0 \leq k \leq \lfloor c_{\mathrm{rank}}\ell \rfloor$,*

$$\operatorname{dist}\left(\sum_{i=1}^{m} \alpha_i (a_i)_{S^c} (a_i)_S^\top, \mathcal{R}_k\right) \geq Q^{-C_{\mathrm{g}}} \frac{\operatorname{dist}(\alpha, \Sigma_k)^2}{\|\alpha\|},$$

where S denotes such a block, Σ_k is the set of vectors supported on at most k coordinates, and $\mathcal{R}_k$ is the set of matrices of rank at most k.

Proof. We first fix a block S of size ℓ. We use three bad events, with the following intuition:

- E_{short}: a small set of vectors is ill-conditioned after restriction to S.
- E_{long}: the same failure on the longer coordinate set S^c.
- E_{cancel}: small column subsets on both sides are well-conditioned, but many significant terms still cancel to give an approximately low-rank matrix.

To define these events formally, we define $X = [(a_i)_S]_{i\in[m]}$ and $Y = [(a_i)_{S^c}]_{i\in[m]}$, so that $\mu_{\mathbf{a},S}(\alpha) = Y\,\mathrm{diag}(\alpha)X^\top$. We use X_i and Y_i to denote the ith columns of these matrices.

We set $\ell' = \lfloor \ell/2 \rfloor < n$ for large n; this is half the short-side dimension ℓ, and to keep the column submatrices tall (which makes E_{short} possible), while retaining $\Omega(\ell)$ directions of effective randomness showing when E_{cancel}. We also choose a sufficiently small inverse-polynomial threshold $\kappa = Q^{-O(1)}$.

Now we formally define E_{short} to be the event that we either have $\max_i \|X_i\| > Q^3$ or $\sigma_{\min}(X_J) < \kappa$, for some $1 \le |J| \le \ell'$. E_{long} is defined in the same way using Y in place of X.

We write $|\alpha|_{(\ell')}$ for the ℓ'-th largest coordinate magnitude, and define E_{cancel} to be the event that neither of these events occurs, but some $\alpha \neq 0$ and integer $0 \le k \le c_{\text{rank}}\ell$ satisfy

$$\mathrm{dist}(\mu_{\mathbf{a},S}(\alpha), \mathcal{R}_k) < \kappa^2 \frac{|\alpha|^2_{(\ell')}}{\|\alpha\|}.$$

We will show that these events cover every bad block, and that

$$\mathbb{P}(E_{\text{short}}) \le Q^{-\Omega(\ell)}, \qquad \mathbb{P}(E_{\text{long}}) \le Q^{-\Omega(n)}, \qquad \mathbb{P}(E_{\text{cancel}}) \le Q^{-\Omega(n\ell)}.$$

The first event depends only on the coordinates in S, so its occurrences on disjoint blocks are independent since the blocks are disjoint.

Bounding E_{short} and E_{long}. We bound these two probabilities by the same argument, taking $Z = X$ or Y.

For upper bounding the norm, we apply Lemma A.2 with $t = Q^2$ and take a union bound over the m columns. The column means have norm at most $R_a \le Q$, and since $\rho \le 1$ and each column has at most $n \le Q$ coordinates, the resulting norm bound is at most Q^3. Thus, the norm of the columns can be bounded by

$$\mathbb{P}(\max_i \|Z_i\| > Q^3) \le m \exp(-Q^2) \le Q^{-\Omega(n)}.$$

When $\max_i \|Z_i\| \le Q^3$, we also have, for every column subset J,

$$\|Z_J\|_{\text{op}} \le \|Z_J\|_F \le \sqrt{|J|}\, Q^3 \le Q^{O(1)}.$$

Now we show that Z is unlikely to be near-singular. Let m_Z be the number of rows in Z. For fixed J of size $j \le \ell'$ and unit $v \in \mathbb{R}^j$, the vector $Z_J v$ is Gaussian with covariance $\rho^2 I_{m_Z}$. Applying Lemma A.3, we get $\mathbb{P}(\|Z_J v\| \le 2\kappa) \le (C\kappa/\rho)^{m_Z}$ for an absolute constant C.

Consider a $\kappa/Q^{O(1)}$-net of the unit sphere of v's which detects every failure of $\sigma_{\min}(Z_J) \ge \kappa$ when $\max_i \|Z_i\| \le Q^3$. Specifically, we choose the denominator in the net mesh large enough that $\|Z_J\|_{\text{op}}$ times the mesh is at most κ. If $\sigma_{\min}(Z_J) < \kappa$, we choose a unit v with $\|Z_J v\| < \kappa$ and a nearest net point v_0 to v, such that

$$\|Z_J v_0\| \le \|Z_J v\| + \|Z_J\|_{\text{op}} \|v_0 - v\| \le 2\kappa.$$

The net has size at most $(Q^{O(1)}/\kappa)^j$, and there are at most m^j choices of J of size j. Union bounding over the net and all column subsets, we bound its probability by

$$\sum_{j=1}^{\ell'} m^j (Q^{O(1)}/\kappa)^j (C\kappa/\rho)^{m_Z} \le Q^{O(m_Z)} \kappa^{m_Z - \ell'} \le Q^{-\Omega(m_Z)},$$

since $\ell' \le m_Z/2$, $\rho^{-1} \le Q$, and κ is sufficiently small. In particular, the net costs at most ℓ' powers of κ^{-1}, whereas the small-ball bound gives m_Z powers of κ. At least $m_Z/2$ powers remain, so choosing κ to be a sufficiently small inverse polynomial absorbs the factor $Q^{O(m_Z)}$.

Taking $m_Z = \ell$ for X and $m_Z = n - \ell = \Theta(n)$ for Y, THIS prove the first two probability bounds.

Parameter regime for bounding E_{cancel}. We will condition on X outside E_{short}, leaving the independent Gaussian perturbations in Y unexposed, and show that the resulting matrix is unlikely to be approaimtely low-rank. We will fix a rank $0 \le k \le \lfloor c_{\text{rank}} \ell \rfloor$, and apply a union bound at the end.

We balance the parameters in the following way: we would net over the α's (which are m-dimensional), as well k-dimsnional subspaces $L \subseteq \mathbb{R}^{n-\ell}$ that the matrix can possibly be close to. The exponent in this net size must not exceed the dimension of the effective randomness that survives a projection onto $L^\perp$, which is characterized by $\ell' \cdot (n - \ell - k)$.

Now we pick c_{load} and c_{rank} to be small enough so that, by the load condition $m \le c_{\text{load}} \ell(n-\ell)$,

$$\ell'(n - \ell - k) \ge 4\big(m + (n-\ell)k\big).$$

The left side is $\Theta(n\ell)$. We will use this gap to take a union bound over the coefficients and low-rank subspaces.

We fix $\zeta = 2^{-j}$ for an integer $j \ge 0$ as the sparsity threshold, and we will sum over j at the end. At this scale, we first fix a unit α with $|\alpha|_{(\ell')} \ge \zeta/2$ and a k-dimensional subspace $L \subseteq \mathbb{R}^{n-\ell}$. Let J be the index of the ℓ' largest coefficients in α, then $X \operatorname{diag}(\alpha_i^2) X^\top \succeq (\zeta^2/4) X_J X_J^\top$.

We represent L by an orthonormal basis matrix $U \in \mathbb{R}^{(n-\ell)\times k}$ and project the output onto $L^\perp$:

$$\Pi_L^\perp \mu_{\mathbf{a},S}(\alpha) = (I - UU^\top) Y \operatorname{diag}(\alpha) X^\top.$$

The Frobenius norm of this projected matrix measures the residual outside L.

We now count the Gaussian directions after this projection. Choose any orthonormal coordinates on $L^\perp$, then the $n - \ell - k$ rows of the projected matrix have independent Gaussian perturbations, each with covariance

$$\rho^2 X \operatorname{diag}(\alpha_i^2) X^\top \succeq (\rho^2 \zeta^2/4) X_J X_J^\top.$$

Indeed, taking orthonormal combinations of the rows of Y preserves the independent Gaussian perturbations with variance ρ^2; multiplying each row by $\operatorname{diag}(\alpha) X^\top$ gives the displayed covariance. Since $\sigma_{\min}(X_J) \ge \kappa$, this covariance has at least ℓ' eigenvalues at least $\rho^2 \kappa^2 \zeta^2/4$. We get these ℓ' directions independently in each of the $n - \ell - k$ rows.

We therefore conclude that its vectorization is Gaussian with at least $\ell'(n - \ell - k)$ covariance eigenvalues at least $\Omega(\rho^2 \kappa^2 \zeta^2)$. Applying Lemma A.3 to this vectorization, we get

$$\begin{aligned}\mathbb{P}\big(\|\Pi_L^\perp \mu_{\mathbf{a},S}(\alpha)\|_F \le 8\kappa^2\zeta^2 \mid X\big) &\le (C\kappa\zeta/\rho)^{\ell'(n-\ell-k)} \\ &\le (Q^{O(1)}\kappa\zeta)^{\ell'(n-\ell-k)},\end{aligned}$$

since $\rho^{-1} \le Q$.

Union bounding. Now we apply union bound to make this uniform. We will cover the unit coefficient sphere and all orthonormal frames U for L by nets. We need to check that replacing both α and U by nearby net points changes the projected matrix by only a small amount. For a fixed L, projection does not increase the approximation error in the matrix. However, our net also replaces L by a nearby subspace, so we must bound the error from changing the projection as well. We do this in two steps.

We work on the event $\max_i \|Y_i\| \le Q^3$. Note that we already assume $X \notin E_{\text{short}}$, so $\max_i \|X_i\| \le Q^3$.

First, consider changing the coefficients from α to another unit vector β. By linearity and Cauchy–Schwarz, we have

$$\|\mu_{\mathbf{a},S}(\alpha) - \mu_{\mathbf{a},S}(\beta)\|_F \le \sum_i |\alpha_i - \beta_i| \|Y_i\| \|X_i\| \le \sqrt{m} Q^6 \|\alpha - \beta\|.$$

The same estimate gives $\|\mu_{\mathbf{a},S}(\beta)\|_F \le \sqrt{m} Q^6$. Thus, keeping the projection operator fixed, we change the projected matrix by at most $Q^{O(1)}\|\alpha - \beta\|$.

Next, we keep β fixed and replace the orthonormal basis matrix U by a nearby orthonormal basis matrix V. The projection changes from $I - UU^\top$ to $I - VV^\top$. We use

$$UU^\top - VV^\top = (U - V)U^\top + V(U - V)^\top, \qquad \|UU^\top - VV^\top\|_{\text{op}} \le 2\|U - V\|_F.$$

Multiplying this projection error by $\|\mu_{\mathbf{a},S}(\beta)\|_F$, and combining the two steps, we have

$$\begin{aligned}
&\|(I - UU^\top)\mu_{\mathbf{a},S}(\alpha) - (I - VV^\top)\mu_{\mathbf{a},S}(\beta)\|_F \\
&\quad \le \|\mu_{\mathbf{a},S}(\alpha) - \mu_{\mathbf{a},S}(\beta)\|_F + 2\|U - V\|_F \|\mu_{\mathbf{a},S}(\beta)\|_F \\
&\quad \le Q^{O(1)}\big(\|\alpha - \beta\| + \|U - V\|_F\big).
\end{aligned}$$

By the triangle inequality,

$$\Big| \|(I - UU^\top)\mu_{\mathbf{a},S}(\alpha)\|_F - \|(I - VV^\top)\mu_{\mathbf{a},S}(\beta)\|_F \Big| \le Q^{O(1)}\big(\|\alpha - \beta\| + \|U - V\|_F\big).$$

This is the Lipschitz bound we need for the nets.

We use nets of mesh $\kappa^2\zeta^2/Q^{O(1)}$, in Euclidean norm for the coefficients and Frobenius norm for the basis matrices. We take a unit coefficient vector α with $|\alpha|_{(\ell')} \ge \zeta$ and an orthonormal basis matrix U for which $\|(I - UU^\top)\mu_{\mathbf{a},S}(\alpha)\|_F \le 4\kappa^2\zeta^2$. We choose their nearest net points β and V. Our mesh is at most $\zeta/2$, so each coordinate changes by at most $\zeta/2$. The ℓ' coordinates of α with magnitude at least ζ therefore still have magnitude at least $\zeta/2$ in β, giving $|\beta|_{(\ell')} \ge \zeta/2$.

The mesh also ensures that the projected matrix changes by at most $\kappa^2\zeta^2$, by the Lipschitz bound. Thus, by triangle inequality,

$$\|(I - VV^\top)\mu_{\mathbf{a},S}(\beta)\|_F \le (4 + 1)\kappa^2\zeta^2 \le 8\kappa^2\zeta^2.$$

We can therefore apply this small-ball estimate to each net pair (β, V). Thus, if any coefficients and subspace give a projected norm at most $4\kappa^2\zeta^2$, some net pair gives a projected norm at most $8\kappa^2\zeta^2$. We will take a union bound over these net pairs.

We choose net points within the coefficient sphere and the set of orthonormal basis matrices. These sets have ambient dimensions m and $(n-\ell)k$, and radii 1 and $\sqrt{k}$, respectively. The volume bound with our mesh gives total net size at most $(Q^{O(1)}/(\kappa\zeta)^2)^{m+(n-\ell)k}$. We use the column norm restriction only to pass to net points; each small-ball estimate conditions only on X.

Taking a union bound, conditional on X, the probability that (a) the norm bound on Y holds, (b) some unit α has $|\alpha|_{(\ell')} \geq \zeta$, and (c) $\operatorname{dist}(\mu_{\mathbf{a},S}(\alpha), \mathcal{R}_k) \leq 4\kappa^2\zeta^2$ is at most

$$\left(\frac{Q^{O(1)}}{(\kappa\zeta)^2}\right)^{m+(n-\ell)k} (Q^{O(1)}\kappa\zeta)^{\ell'(n-\ell-k)} \leq Q^{O(n\ell)}(\kappa\zeta)^{\Omega(n\ell)}.$$

Here we used $\operatorname{dist}(M, \mathcal{R}_k) = \min_{\dim L = k} \|\Pi_L^{\perp} M\|_F$ and the dimension gap above.

For any unit α satisfying the inequality defining E_{cancel}, we choose $\zeta = 2^{-j} \leq |\alpha|_{(\ell')} < 2\zeta$. Its residual is less than $4\kappa^2\zeta^2$, so the preceding bound applies. For sufficiently small κ, this bound is at most $Q^{-\Omega(n\ell)}2^{-\Omega(jn\ell)}$. We sum over $j \geq 0$ and the $O(\ell)$ ranks k, and average over X, to obtain $\mathbb{P}(E_{\text{cancel}}) \leq Q^{-\Omega(n\ell)}$.

Obtaining rigidity outside the bad events. We assume that none of the three events occurs, take $\|\alpha\| = 1$, and fix an integer $0 \leq k \leq c_{\text{rank}}\ell$; we will show this leads to the desired $\mathsf{dist}(\alpha, \Sigma_k)$ bound.

Let $e = \operatorname{dist}(\mu_{\mathbf{a},S}(\alpha), \mathcal{R}_k)$, and it suffices to consider $e < 1$. We first truncate α to the ℓ' largest coefficients indexed by J, and set the others to zero, to get sparse vector α^J. Since E_{cancel} does not occur, the discarded tail satisfies

$$\|\alpha - \alpha^J\|^2 \leq m|\alpha|_{(\ell')}^2 \leq m\kappa^{-2}e \leq Q^{O(1)}e.$$

By the column norm bounds for X and Y, removing this tail changes the output by at most $Q^{O(1)}\|\alpha - \alpha^J\|$. Adding this to the original error e gives $\operatorname{dist}(\mu_{\mathbf{a},S}(\alpha^J), \mathcal{R}_k) \leq Q^{O(1)}\sqrt{e}$.

We next reduce the support of α from ℓ' to k as needed for the statement. Since neither E_{short} nor E_{long} occurs, X, Y have left pseudo-inverses $X_J^{\dagger}$, $Y_J^{\dagger}$ of norm at most κ^{-1}. For any rank-at-most-k approximation B to $\mu_{\mathbf{a},S}(\alpha^J) = Y_J \operatorname{diag}(\alpha_J) X_J^{\top}$, the matrix $Y_J^{\dagger} B (X_J^{\dagger})^{\top}$ still has rank at most k, and

$$\|\operatorname{diag}(\alpha_J) - Y_J^{\dagger} B (X_J^{\dagger})^{\top}\|_F \leq \kappa^{-2}\|\mu_{\mathbf{a},S}(\alpha^J) - B\|_F.$$

The singular values of $\operatorname{diag}(\alpha_J)$ are the coefficient magnitudes, so its distance to rank k equals $\operatorname{dist}(\alpha^J, \Sigma_k)$. Adding back the discarded tail gives

$$\operatorname{dist}(\alpha, \Sigma_k) \leq \|\alpha - \alpha^J\| + \kappa^{-2} \operatorname{dist}(\mu_{\mathbf{a},S}(\alpha^J), \mathcal{R}_k) \leq Q^{O(1)}\sqrt{e}.$$

Finally, we square this bound and rescale to obtain (42).

Putting the blocks together. We take a union bound over the at most mn predetermined blocks in all layers to rule out E_{long} and E_{cancel} on every block, with failure probability at most $\exp(-\Omega(n))$. For a fixed layer s, the events E_{short} for its blocks $S_1, \ldots, S_{t_s}$ are independent, since these blocks use disjoint coordinates of the Gaussian perturbations.

If E_{short} occurs on more than $c_0 t_s/4$ blocks, it occurs on every block in some subset of $\lceil c_0 t_s/4 \rceil$ of these fixed blocks. We take a union bound over the at most 2^{t_s} choices of this subset, multiplying

the individual block probabilities by independence, to get

$$2^{t_s}(Q^{-\Omega(\ell)})^{c_0 t_s/4} \le \exp(-\Omega(n)),$$

where we used $\ell t_s = \Theta(n)$ and $Q \ge n$. We now take a union bound over s to prove the lemma. Finally, we intersect with $\mathcal{G}_1$ to get the stronger event used in Theorem 6.13, with the same probability bound. □

B.2 Properties of smoothed maps

Proof of Lemma 6.5. Let $\ell = \lfloor n/2 \rfloor = \dim Y$. We prove the four properties and then take a union bound over their failure events.

Norm bounds. For each i, we have $a_i \sim \mathcal{N}(\bar{a}_i, \rho^2 I_n)$. Applying Lemma A.2 with $N = n$ and $t = 0$, we have $\mathbb{P}(\|a_i - \bar{a}_i\| > 2\rho\sqrt{n}) \le e^{-n/2}$.

By triangle inequality, we have $\|a_i\| \le R_a + 2\rho\sqrt{n} \le R_a + 2n \le Q$ outside this failure event. Applying a union bound over the m components, we have with probability at least $1 - me^{-n/2}$, this holds simultaneously for every i.

Conditioning of the square and tangent maps. We first rescale the linear forms so that we can apply Theorem 4.1, and we apply it on $\widetilde{a}_i = a_i/R$ where $R = \max\{1, R_a\}$. The centers now have norm at most one, and the smoothing parameter is $\rho/R \le 1$.

We apply the theorem with root degrees $d_i = 1$, powers $D_i = 2$, maximum output degree $T = 3$, and slack $\varepsilon = 1/4$. We will do this both in the full space and after restricting to Y. The restrictions are still independent smoothed linear forms with the same bounds on the centers and the same smoothing parameter. In either case, the number of components is $m \ll n^{2-\varepsilon}$ so the conditions apply.

For quadratic generators, there are no Koszul syzygies in output degrees two and three. Thus, the distance to the Koszul space in the theorem is just the norm of the input. For $t \in \{2, 3\}$, we obtain

$$\|(h_i)_i\| \le n^{C_0}(R/\rho)^{C_0} \left\| \sum_i \widetilde{a}_i^2 h_i \right\|, \qquad h_i \in \mathbb{R}[\mathbf{x}]^{=t-2}.$$

For $t = 2$ the h_i are scalars, so this is the square map; for $t = 3$ they are linear forms, so this is the tangent map. Since this holds for every input, the least singular value of each map is at least $n^{-C_0}(\rho/R)^{C_0}$.

Now we scale back to the original linear forms. Both maps are multiplied by R^2, so their least singular values are at least $R^2 n^{-C_0}(\rho/R)^{C_0} \ge Q^{-C}$, since all of R, ρ^{-1} and n are at most Q. This gives

$$\sigma_{\min}(\mu_{\mathbf{a}}), \quad \sigma_{\min}(T_{\mathbf{a}}), \quad \sigma_{\min}(\mu_{\mathbf{a},Y}), \quad \sigma_{\min}(T_{\mathbf{a},Y}) \ge Q^{-C}$$

Since the dimensions n and ℓ in the two applications are both $\Theta(n)$, a union bound gives total failure probability $\exp(-\Omega(n))$.

Conditioning of small component sets. Let $X = [a_i^Y]_{i\in[m]}$ and $Z = [a_i^\perp]_{i\in[m]}$, in orthonormal coordinates on Y and $Y^\perp$, and let $\ell' = \lfloor \ell/2 \rfloor$. We will show that every small set of columns is well-conditioned in both X and Z.

We will reuse the bounds on E_{short} and E_{long} from the proof of Lemma B.1, taking the block S to be the coordinates of Y. It applies here because m is polynomial in Q and $\ell' \le \frac{1}{2}\min\{\ell, n-\ell\}$. Thus, we can choose $\kappa = Q^{-L}$ for a sufficiently large absolute constant L, so that except with probability $Q^{-\Omega(\ell)} + Q^{-\Omega(n-\ell)} = \exp(-\Omega(n))$, both matrices have column norms at most Q^3 and

$$\sigma_{\min}(X_J),\ \sigma_{\min}(Z_J) \ge \kappa \qquad \big(1 \le |J| \le \min\{m, \ell'\}\big).$$

For sufficiently large n, we have $n/8 \le \ell'$. For any set J of at most $n/8$ components, projecting onto Y can only decrease the norm of a linear combination. Hence, $\sigma_{\min}([a_i]_{i\in J}) \ge \sigma_{\min}(X_J) \ge \kappa$. This proves the robust Kruskal rank bound.

Rigidity of the cross-block and square maps. Since $\ell = \Theta(n)$ and $m = O(n^{3/2})$, then the load condition $m \le c_{\text{load}}\ell(n-\ell)$ holds for sufficiently large n. Choose $c_0 > 0$ small enough that $c_0 n \le c_{\text{rank}}\ell$.

For $m > \ell'$, we can apply the single-block argument in the proof of Lemma B.1, using coordinates of Y in place of S. The exact same argument gives, with probability $1 - \exp(-\Omega(n))$, for all $\alpha \ne 0$ and $0 \le k \le c_0 n$,

$$\operatorname{dist}(\mu^{\text{cross}}_{\mathbf{a},Y}(\alpha), \mathcal{R}_k) \ge Q^{-C}\frac{\operatorname{dist}(\alpha, \Sigma_k)^2}{\|\alpha\|}.$$

If $m \le \ell'$, applying the left pseudoinverses of X and Z to a rank-at-most-k approximation gives the stronger lower bound $\kappa^2 \operatorname{dist}(\alpha, \Sigma_k)$, so the same conclusion holds.

We now use the rectangular bound above to prove the square-map bound. View $\mu_{\mathbf{a}}(\alpha)$ as the symmetric matrix $M = \sum_i \alpha_i a_i a_i^\top$. Its $Y^\perp \times Y$ block is exactly $\mu^{\text{cross}}_{\mathbf{a},Y}(\alpha)$. For any matrix A of rank at most k, its corresponding block $A_{Y^\perp,Y}$ also has rank at most k, and

$$\|\mu^{\text{cross}}_{\mathbf{a},Y}(\alpha) - A_{Y^\perp,Y}\|_F = \|(M-A)_{Y^\perp,Y}\|_F \le \|M - A\|_F.$$

Thus, a rank-k approximation to the whole matrix gives a rank-k approximation to the rectangular block with no larger error. Taking the infimum over such A, we obtain

$$\operatorname{dist}(\mu_{\mathbf{a}}(\alpha), \mathcal{R}_k) \ge \operatorname{dist}(\mu^{\text{cross}}_{\mathbf{a},Y}(\alpha), \mathcal{R}_k).$$

The proven bound for the right-hand side therefore gives the required square-map bound. Finally, we apply a union bound over the four parts of the claim, and the proof is complete by adjusting c, C if necessary. □

B.3 Column subset selection.

We use the following column-selection formulation of the strong rank-revealing QR result of Gu and Eisenstat [GE96, Section 3, Theorem 3.2], with parameter $f = 1$. Matrices with fewer rows than columns can be padded with zero rows.

Theorem B.2 (Column subset selection)**.** *Let $M \in \mathbb{R}^{d\times N}$, where $d, N \ge 1$, and let r be an integer with $1 \le r \le \operatorname{rank} M$. Write M_I for the submatrix with column indices in I, and set $D = \sqrt{1 + r(N-r)}$. There exist a set $I \subseteq [N]$ of size r and a matrix $\Gamma \in \mathbb{R}^{r\times N}$ such that*

1. $\Gamma_{:,I} = I_r$ *and* $|\Gamma_{ij}| \le 1$ *for every entry;*
2. $\sigma_{\min}(M_I) \ge \sigma_r(M)/D$*;*
3. $\|M - M_I\Gamma\|_{\mathrm{op}} \le D\sigma_{r+1}(M)$*.*

Here, singular values are ordered nonincreasingly and taken to be zero beyond the rank. In particular, since $D \le N$, the selected columns have minimum singular value at least $\sigma_r(M)/N$, and every column of M is within distance $N\sigma_{r+1}(M)$ of a linear combination of them with coefficients of absolute value at most one.

Corollary B.3 (Column selection modulo a subspace)**.** *Let $Q \ge 2$ and $C_0 > 0$. Let $A \in \mathbb{R}^{d\times t}$ have full column rank, and let $B \in \mathbb{R}^{d\times N}$, where $d, t, N \ge 1$. Suppose that*

$$\sigma_{\min}(A) \ge Q^{-C_0}, \qquad \|B\|_{\mathrm{op}}, N \le Q^{C_0}.$$

Set $P = \Pi_{\mathrm{range}(A)^\perp}$, fix $0 < \tau \le 1$, and let r be the number of singular values of PB that exceed τ. There is a constant C_1 depending only on C_0, a set $I \subseteq [N]$ of size r, and matrices $C \in \mathbb{R}^{t\times N}$, $\Gamma \in \mathbb{R}^{r\times N}$, and $E \in \mathbb{R}^{d\times N}$ such that

$$B = AC + B_I\Gamma + E, \qquad |\Gamma_{ij}| \le 1,$$

and

$$\|C\|_{\mathrm{op}} \le Q^{C_1}, \qquad \|E\|_{\mathrm{op}} \le N\tau, \qquad \sigma_{\min}([A, B_I]) \ge Q^{-C_1}\tau.$$

When $r = 0$, the set I and matrix Γ are empty, and $[A, B_I] = A$.

Proof. If $r = 0$, take $C = A^\dagger B$ and $E = PB$. Then $\|E\|_{\mathrm{op}} \le \tau$, and all the bounds follow from the assumptions. Suppose now that $r > 0$. Apply Theorem B.2 to PB to obtain I and Γ with $|\Gamma_{ij}| \le 1$ and

$$\sigma_{\min}(PB_I) \ge \tau/N, \qquad \|P(B - B_I\Gamma)\|_{\mathrm{op}} \le N\tau.$$

Set $C = A^\dagger(B - B_I\Gamma)$ and $E = P(B - B_I\Gamma)$. Since $AA^\dagger + P = I_d$, these choices give the claimed representation. Moreover, $r \le N$ and $\|\Gamma\|_{\mathrm{op}} \le \|\Gamma\|_F \le N$, so

$$\|C\|_{\mathrm{op}} \le \|A^\dagger\|_{\mathrm{op}}\|B\|_{\mathrm{op}}(1 + N) \le Q^{C_1}$$

for a sufficiently large C_1 depending only on C_0.

To bound the conditioning of the concatenated matrix, write $h = Au + B_I v$. Projection by P gives $\|v\| \le N\tau^{-1}\|h\|$, and hence

$$\|u\| \le \|A^\dagger\|_{\mathrm{op}}(\|h\| + \|B\|_{\mathrm{op}}\|v\|) \le Q^{C_1}\tau^{-1}\|h\|.$$

Thus $\|(u, v)\| \le Q^{C_1}\tau^{-1}\|h\|$, after increasing C_1 if necessary, which proves the last bound. □

B.4 Anti-concentration of rational maps via Jacobians

Proof of Theorem 6.6. Write $d = p_0 + p_1$, and let F be the set whose measure appears on the LHS of (30).

The square case. First consider a rational map $g : O \subseteq \mathbb{R}^r \to \mathbb{R}^r$, where O is an open set, and all numerator and denominator degrees are at most Δ. Write $g_i = \widetilde{P}_i/\widetilde{S}_i$, with nonzero denominators on O. For a fixed output $w \in \mathbb{R}^r$, the equations $g(u) = w$ imply

$$\widetilde{P}_i(u) - w_i\widetilde{S}_i(u) = 0, \qquad i = 1, \dots, r.$$

At any such solution,

$$\mathsf{D}(\widetilde{P}_i - w_i \widetilde{S}_i)(u) = \widetilde{S}_i(u)\, \mathsf{D}g_i(u).$$

Thus, if $\det \mathsf{D}g(u) \neq 0$, the polynomial system also has an invertible Jacobian at u, and thus the solution is *isolated*. For such solutions, Bézout's theorem bounds the number by Δ^r. This is true even if the system also has other non-isolated solutions that form a higher dimension variety. (We refer to, e.g., [Ful98] for details.)

Consequently, for any $A \subseteq O$ that satisfies:

$$g(A) \subseteq [-\eta, \eta]^r, \qquad |\det \mathsf{D}g(u)| \geq \lambda > 0 \quad (u \in A),$$

then change of variables formula (with multiplicity) gives

$$\lambda \cdot \mathsf{vol}_r(A) \leq \int_A |\det \mathsf{D}g(u)|\, du = \int_{\mathbb{R}^r} \#\{u \in A : g(u) = w\}\, dw \leq \Delta^r (2\eta)^r. \tag{68}$$

This essentially implies our required bound for the case of square maps.

General maps. As it turns out, we can reduce the general case to the square case above by selecting an appropriate minor of the map. Consider the original map $f : \mathbb{R}^d \mapsto \mathbb{R}^N$, where $d = p_0 + p_1$. There are

$$L = \binom{N}{r}\binom{d}{r} \leq Q^{C_1 r}$$

minors of size $r \times r$ in $\mathsf{D}f$, for some constant C_1. By the Cauchy–Binet formula,

$$\sum_{\substack{|I|=r \\ |K|=r}} |\det(\mathsf{D}f)_{I,K}|^2 \geq \prod_{i=1}^{r} \sigma_i(\mathsf{D}f)^2.$$

Every point of F (the set whose measure is being bounded in (68)) therefore belongs to the set associated with its I, K minor satisfying

$$|\det(\mathsf{D}f)_{I,K}| \geq J/\sqrt{L}. \tag{69}$$

We bound these sets separately and sum over the L choices to get a bound on the measure of F.

Fix a minor (I, K), and fix the values of all input coordinates outside K. The selected output coordinates define a rational map g in the r selected inputs, still with numerator and denominator degrees at most Δ. If any denominator vanishes identically, then there are no points in F with this fixing of non-K variables (because F only counts points where the denominators do not vanish). Otherwise, its Jacobian determinant is the selected minor. Moreover, $\|f\| \leq \eta$ ensures that g takes values in $[-\eta, \eta]^r$ on the counted section. We can thus use the square case result (68) along with (69), to bound its Lebesgue volume by

$$\frac{\sqrt{L}}{J}\, \Delta^r (2\eta)^r \leq Q^{C_2 r} \frac{\eta^r}{J}. \tag{70}$$

Now some of the selected inputs may be Gaussian coordinates, and thus we need to convert the measure from Lesbegue. Because the Gaussian coordinates are independent, their joint density conditional on the frozen coordinates is at most

$$\max\{1, (\sqrt{2\pi}\rho)^{-1}\}^r \leq Q^{C_3 r},$$

for some constant C_3 (irrespective of the means). Multiplying (70) by this density bound gives an estimate of the same form, even for the mixed Gaussian–Lebesgue measure of the section. Finally we need to integrate over all the choices of freezings, and then sum over all L. Note that the Gaussian coordinates have total mass one. The non-frozen Lebesgue coordinates range over a box of volume at most $(2R)^{p_1}$. Summing over the $L \le Q^{C_1 r}$ minors gives

$$(\gamma_\rho \otimes \nu_R)(F) \le Q^{C(p_1+r)} \frac{\eta^r}{J},$$

This completes the proof. □

B.5 Proof of Lemma 6.9

We restate Lemma 6.9. Fix a constant $C_0 > 0$, and assume that $\mathbf{a}$ satisfies $\mathcal{G}_1$. Let $V \subseteq \mathbb{R}^n$ have dimension $d \le cn$, for a sufficiently small constant $c > 0$ and sufficiently large n. There exists a subspace $V^* \supseteq V$ with $\dim V^* \le 3d$, depending only on $\mathbf{a}, V$, and a constant C depending on C_0 and the constants in $\mathcal{G}_1$, such that the following holds. For every $\eta \in (0,1)$ and $b \in \mathbb{R}^n$, if

$$b^2 = \sum_i \alpha_i a_i^2 + h + g, \qquad h \in \mathsf{Sym}^2(V), \quad \|\alpha\| \le Q^{C_0}, \quad \|g\| < Q^{-C}\eta^4,$$

then either $\operatorname{dist}(b, V^*) \le \eta$ or $\min_i \operatorname{dist}(b, \mathbb{R}a_i) \le \eta$.

Proof of Lemma 6.9. We can pick c small enough that $d+1 \le c_0 n$ and $2d+2 \le n/8$. The first condition is so we can apply square rigidity from Lemma 6.5 at rank $d+1$: since $b^2 - h$ has matrix rank at most $d+1$, we will be able to discard all but $d+1$ coefficients of α, up to a small error. The second lets us use conditioning from Lemma 6.5 on $2d+2$ original components to show that the difference vectors from $d+1$ disjoint pairs cannot all lie close to the d-dimensional space V. The same conditioning will also give dual vectors to isolate the remaining terms.

By Lemma 6.5, we can fix $\kappa = Q^{-O(1)} \le 1$ as a common lower bound for $\sigma_{\min}([a_i]_{i\in S})$ over $1 \le |S| \le 2d+2$. We also have $\max_i \|a_i\| \le Q$.

The proof has two parts. We first construct V^* using only $\mathbf{a}$ and V, so that whenever two nonzero projected components are nearly parallel, at least one of their original vectors lies in V^*. We then reduce the expression for b^2 to a small number of terms and project away from V. If two or more of the remaining terms have non-negligible size after this projection, we will use the property of V^* to put b close to V^*. If at most one has non-negligible size, b will be close to a line $\mathbb{R}a_i$, or to zero. We specify the size threshold below.

Constructing V^*. Write $\hat{a}_i = \Pi_{V^\perp} a_i$. For nonzero vectors u, v, we measure the angle between their lines by $\|\Pi_{\mathbb{R}u} - \Pi_{\mathbb{R}v}\|_{\mathrm{op}}$. This quantity satisfies the triangle inequality, and equals $\operatorname{dist}(u, \mathbb{R}v)/\|u\|$.

Choose a sufficiently small inverse polynomial $\vartheta = Q^{-O(1)} \le 1$ such that $\sqrt{n}\, Q\vartheta < \kappa/2$. We greedily select pairs of distinct indices (i,j) with nonzero $\hat{a}_i, \hat{a}_j$ and

$$\|\Pi_{\mathbb{R}\hat{a}_i} - \Pi_{\mathbb{R}\hat{a}_j}\|_{\mathrm{op}} \le \vartheta.$$

Each index can be used in at most one pair, and we continue until no further pair can be added. The reason for taking disjoint pairs is that each pair gives a difference vector close to V, while

conditioning prevents too many such vectors from fitting near a d-dimensional space. Indeed, for a selected pair (i, j), set $\phi_{ij} = \langle \hat{a}_i, \hat{a}_j \rangle / \|\hat{a}_j\|^2$. Then

$$\operatorname{dist}(a_i - \phi_{ij} a_j, V) = \|\hat{a}_i - \phi_{ij} \hat{a}_j\| \le \vartheta \|\hat{a}_i\| \le \vartheta Q. \tag{71}$$

We claim that we select at most d pairs. Otherwise, take $d+1$ selected pairs and form the matrix X with their difference vectors $a_i - \phi_{ij} a_j$ as columns. Since the pairs are disjoint, a linear combination with coefficient vector u has a coefficient vector on the original a_i's of norm at least $\|u\|$. Small-set conditioning applied to these $2d+2$ components therefore gives $\sigma_{\min}(X) \ge \kappa$. On the other hand, by (71),

$$\|X - \Pi_V X\|_F \le \sqrt{d+1}\, \vartheta Q < \kappa/2.$$

This is impossible: $\Pi_V X$ has rank at most d, whereas X has $d+1$ singular values at least κ.

Let J be the set of indices used in the selected pairs, and define

$$V^* = V + \operatorname{span}\{a_i : i \in J\}.$$

We have $|J| \le 2d$, so $\dim V^* \le 3d$. Every pair satisfying the angle bound has at least one index in J: if both indices were unused, we could have added that pair. Thus V^* has the property we wanted. If $\hat{a}_i = 0$, then a_i already lies in V. The construction uses only $\mathbf{a}, V$, not b, α, or η.

Reducing to a small set of components. Now take b, α, h, g as in the statement. We identify quadratics with symmetric matrices, so the given identity becomes $bb^\top = \sum_i \alpha_i a_i a_i^\top + H + G$, where the rows and columns of H lie in V and $\|G\|_F = \|g\|$. Since $bb^\top - H$ has rank at most $d+1$, we apply square rigidity from Lemma 6.5 at rank $d+1$. For $\alpha \neq 0$,

$$\operatorname{dist}(\alpha, \Sigma_{d+1})^2 \le Q^{O(1)} \|\alpha\|\, \|g\| \le Q^{O(1)} \|g\|.$$

We keep the $d+1$ largest coefficients of α (or all of them if $m < d+1$), with index set I. Since $\|\mu_{\mathbf{a}}\|_{\mathrm{op}} \le \sqrt{m}\, Q^2 = Q^{O(1)}$, the discarded terms contribute at most $Q^{O(1)} \|g\|^{1/2}$ to the matrix norm. We thus obtain

$$bb^\top = \sum_{i \in I} \alpha_i a_i a_i^\top + H + G', \qquad |I| \le d+1, \qquad \varepsilon := \|G'\|_F \le Q^{O(1)} \|g\|^{1/2}. \tag{72}$$

If $\alpha = 0$, we take $I = \varnothing$ and $G' = G$, with the same bound.

We will only need

$$\varepsilon \le \frac{\kappa \vartheta}{4nQ} \eta^2.$$

This follows from $\|g\| < Q^{-C} \eta^4$ by taking C sufficiently large, since all other factors are fixed powers of Q. In particular, this also ensures $\varepsilon \le \eta^2/2$.

Isolating each remaining term. By small-set conditioning, for each $i \in I$ we can choose a unit vector w_i orthogonal to all a_j with $j \in I \setminus \{i\}$, but with $\langle a_i, w_i \rangle \ge \kappa$. We obtain it by normalizing the transpose of the ith row of $[a_j]_{j \in I}^\dagger$, whose norm is at most κ^{-1}.

Multiplying (72) by w_i removes every other term:

$$\langle b, w_i \rangle b = \alpha_i \langle a_i, w_i \rangle a_i + H w_i + G' w_i. \tag{73}$$

We write $\hat{b} = \Pi_{V^\perp} b$ and project onto $V^\perp$. Since $Hw_i \in V$, this gives

$$\|\langle b, w_i\rangle \hat{b} - \alpha_i \langle a_i, w_i\rangle \hat{a}_i\| \le \varepsilon. \tag{74}$$

Thus, if $|\alpha_i|\|\hat{a}_i\|$ is large enough compared with the error, $\hat{a}_i$ must be nearly parallel to $\hat{b}$. We now make this comparison quantitative.

If $\|\hat{b}\| \le \eta$, then $\operatorname{dist}(b, V^*) \le \operatorname{dist}(b, V) \le \eta$. Otherwise, call $i \in I$ significant if

$$|\alpha_i|\|\hat{a}_i\| \ge \frac{\eta\|\hat{b}\|}{2nQ}.$$

After projecting (72) on the right onto $V^\perp$, the ith term becomes $\alpha_i a_i \hat{a}_i^\top$. Thus significance depends on $|\alpha_i|\|\hat{a}_i\|$, not just on the coefficient $|\alpha_i|$; if $a_i \in V$, the term vanishes regardless of its coefficient. We choose the threshold so that all non-significant terms together are small: there are at most n such terms, and each has norm $\|\alpha_i a_i \hat{a}_i^\top\|_F \le Q|\alpha_i|\|\hat{a}_i\|$. Their total norm is therefore at most $\eta\|\hat{b}\|/2$.

Two or more significant terms. For any significant i, project (74) orthogonally to $\hat{b}$ and divide by $|\alpha_i\langle a_i, w_i\rangle|\|\hat{a}_i\|$. Using $\|\hat{b}\| > \eta$, we obtain

$$\|\Pi_{\mathbb{R}\hat{a}_i} - \Pi_{\mathbb{R}\hat{b}}\|_{\mathrm{op}} \le \frac{\varepsilon}{\kappa|\alpha_i|\|\hat{a}_i\|} \le \frac{2nQ\varepsilon}{\kappa\eta\|\hat{b}\|} \le \frac{\vartheta}{2}.$$

By triangle inequality, any two significant indices satisfy the angle bound used to select pairs. If there are at least two, at least one must belong to J; otherwise we could have added another disjoint pair. Choose such an index i, so $a_i \in V^*$. The triangle inequality in (74) also gives

$$|\langle b, w_i\rangle|\|\hat{b}\| \ge \kappa|\alpha_i|\|\hat{a}_i\| - \varepsilon \ge \frac{\kappa\eta\|\hat{b}\|}{4nQ},$$

where we used $\varepsilon \le \kappa\eta^2/(4nQ)$. Thus $|\langle b, w_i\rangle| \ge \kappa\eta/(4nQ)$. We can now return to (73) and project onto $(V^*)^\perp$. Both a_i and Hw_i disappear, leaving

$$\operatorname{dist}(b, V^*) \le \frac{\varepsilon}{|\langle b, w_i\rangle|} \le \frac{4nQ\varepsilon}{\kappa\eta} \le \vartheta\eta \le \eta.$$

At most one significant term. Multiplying (72) on the right by $\Pi_{V^\perp}$ removes H and gives

$$b\hat{b}^\top = \sum_{i\in I} \alpha_i a_i \hat{a}_i^\top + G'\Pi_{V^\perp}.$$

If there is exactly one significant index i^*, the bound on the non-significant terms gives

$$\|b\hat{b}^\top - \alpha_{i^*} a_{i^*} \hat{a}_{i^*}^\top\|_F \le \varepsilon + \frac{\eta\|\hat{b}\|}{2}.$$

To recover b from the left-hand side, we multiply on the right by $\hat{b}/\|\hat{b}\|^2$. The other term becomes a multiple of a_{i^*}, so

$$\operatorname{dist}(b, \mathbb{R}a_{i^*}) \le \frac{\varepsilon}{\|\hat{b}\|} + \frac{\eta}{2} \le \eta.$$

If no index is significant, the same calculation without the i^* term gives $\|b\| \le \eta$, and hence $\operatorname{dist}(b, V^*) \le \eta$. This proves the lemma. □

B.6 Multiplication outside the tangent space

We restate Lemma 6.11: Recall that $\mathcal{U} = \operatorname{span}\{a_i^2 : i \in [m]\}$ is the span of the original squares. Assume that $\mathbf{a}$ satisfies $\mathcal{G}_1$, and let $\mathcal{N} \subseteq \mathcal{U}^\perp$ be a space of quadratics of dimension $s \geq 1$. Let $\mathcal{T} = \{\sum_{i=1}^m a_i^2 u_i : u_i \in \mathbb{R}^n\}$ be the tangent space. For

$$C_{\mathcal{N}} : \mathcal{N} \otimes \mathbb{R}^n \longrightarrow \mathcal{T}^\perp, \qquad C_{\mathcal{N}}(\xi \otimes h) = \Pi_{\mathcal{T}^\perp}(\xi h),$$

we claim that $\sigma_{\lfloor ns/10 \rfloor}(C_{\mathcal{N}}) \geq Q^{-C}$ for sufficiently large n.

Proof of Lemma 6.11. We use the fixed coordinate split $\mathbb{R}^n = Y \oplus Y^\perp$ from $\mathcal{G}_1$, where $\dim Y = \lfloor n/2 \rfloor$. Write $\mathbf{x} = (\mathbf{y}, \mathbf{z})$ for the coordinates on $Y \oplus Y^\perp$. As in Section 4, we write $f[r]$ for the part of f of $\mathbf{z}$-degree r, and use the same notation componentwise for tuples of polynomials. Our goal is to find $k_s = \lfloor ns/10 \rfloor$ input directions, whose products remain large after projection onto $\mathcal{T}^\perp$. We will find these directions on the restricted subspace $h \in Y^\perp$.

Adjusting the quadratics. We can subtract any element of $\mathcal{U}$ from ξ without changing $\Pi_{\mathcal{T}^\perp}(\xi h)$. Indeed, for any scalar coefficients α_i,

$$\left(\sum_i \alpha_i a_i^2\right) h = \sum_i a_i^2 (\alpha_i h) \in \mathcal{T},$$

since each $\alpha_i h$ is a linear form. We will use this freedom to remove the part of $\xi[0]$ in the span of the squares $(a_i^Y)^2$. Specifically, define the linear map

$$L\xi = \widetilde{\xi} := \xi - \mu_{\mathbf{a}}\big(\mu_{\mathbf{a},Y}^\dagger \xi[0]\big), \qquad \widetilde{\mathcal{N}} = L\mathcal{N}.$$

Restricting this identity to Y, we have

$$\widetilde{\xi}[0] = (I - \mu_{\mathbf{a},Y}\mu_{\mathbf{a},Y}^\dagger)\xi[0] \perp \operatorname{range}(\mu_{\mathbf{a},Y}).$$

Also, since $\xi \in \mathcal{U}^\perp$ and $L\xi - \xi \in \mathcal{U}$, we have $\Pi_{\mathcal{U}^\perp} L\xi = \xi$. Hence $\|L\xi\| \geq \|\xi\|$, so $L : \mathcal{N} \to \widetilde{\mathcal{N}}$ is invertible and $\dim \widetilde{\mathcal{N}} = s$.

Separating the products from $\mathcal{T}$. Consider the multiplication map

$$\mathsf{M} : \widetilde{\mathcal{N}} \otimes Y^\perp \longrightarrow \mathsf{Sym}^3(\mathbb{R}^n), \qquad \mathsf{M}(\widetilde{\xi} \otimes h) = \widetilde{\xi} h.$$

We first show that every f in its range satisfies

$$\operatorname{dist}(f, \mathcal{T}) \geq Q^{-O(1)} \|f\|. \tag{75}$$

The reason for our choice of $\widetilde{\xi}$ is that the first two parts of f are now easy to distinguish from an element of $\mathcal{T}$: we have $f[0] = 0$, and $f[1]$ is orthogonal to $\operatorname{range}(\mu_{\mathbf{a},Y}) \cdot Y^\perp$.

To prove the bound, take any tuple $\mathbf{h} = (h_i)_{i \in [m]}$ of linear forms and write $e = f - T_{\mathbf{a}}(\mathbf{h})$ for the error. We will show that a small error forces $\mathbf{h}$ to be small. Then $T_{\mathbf{a}}(\mathbf{h})$ is small as well, so $f = e + T_{\mathbf{a}}(\mathbf{h})$ cannot be large while remaining close to $\mathcal{T}$. We first recover $\mathbf{h}[0]$ using $T_{\mathbf{a},Y}^\dagger$, then

recover $\mathbf{h}[1]$ using the first substitution row in the proof of Lemma 4.3, with $\mathbf{p} = (a_i^2)_{i\in[m]}$, output degree $t = 3$, and $R = 1$. There are no Koszul terms in this degree, so we get

$$\mathbf{h}[0] = -\mathsf{W}_{\mathbf{p}}^{(3)}[0]\, e[0],$$
$$\mathbf{h}[1] = -\widehat{\mathsf{W}}_{\mathbf{p}}^{(2)}\left(e[1] + \sum_i p_i[1]h_i[0]\right).$$

Here $\mathsf{W}_{\mathbf{p}}^{(3)}[0] = T_{\mathbf{a},Y}^{\dagger}$, and $\widehat{\mathsf{W}}_{\mathbf{p}}^{(2)}$ applies $\mu_{\mathbf{a},Y}^{\dagger}$ to each quadratic coefficient of the $\mathbf{z}$ variables. There is no contribution from f in the first formula because $f[0] = 0$. In the second, each quadratic coefficient of $f[1]$ is orthogonal to range$(\mu_{\mathbf{a},Y})$, so $\mu_{\mathbf{a},Y}^{\dagger}$ sends it to zero. By $\mathcal{G}_1$ and the extended-pseudoinverse bound in the proof of Lemma 4.3, both recovery operators have norm $Q^{O(1)}$. Since $\|\mathbf{p}\| \le Q^{O(1)}$, the first formula bounds $\mathbf{h}[0]$ by $Q^{O(1)}\|e\|$, and the second gives the same bound for $\mathbf{h}[1]$. Thus $\|\mathbf{h}\| \le Q^{O(1)}\|e\|$, and by triangle inequality, we have

$$\|f\| \le \|e\| + \|T_{\mathbf{a}}(\mathbf{h})\| \le Q^{O(1)}\|e\|.$$

Every element of $\mathcal{T}$ has the form $T_{\mathbf{a}}(\mathbf{h})$, so taking the infimum over $\mathbf{h}$, this proves (75).

Counting large singular values. We now show that M has many large singular values before projection. Writing its input in a coordinate basis of $Y^{\perp}$, contractivity of multiplication gives $\|\mathsf{M}\|_{\mathrm{op}} \le 1$.

We note that for any unit quadratic $\widetilde{\xi}$ and unit linear form h, we also have $\|\widetilde{\xi}h\|^2 \ge 1/3$. To see this, rotate coordinates so that $h = x_1$. Multiplication of a degree-two monomial x^{α} by x_1 multiplies its squared Bombieri norm by $(\alpha_1 + 1)/3 \ge 1/3$. Distinct monomials remain orthogonal after multiplication, so the same lower bound holds for every quadratic.

Different products can cancel, so this does not give a lower bound on the least singular value of M. Instead, we sum the squared output norms over tensor products of orthonormal bases of $\widetilde{\mathcal{N}}$ and $Y^{\perp}$. If r singular values are at least $1/\sqrt{6}$, the bounds above give

$$\frac{s \dim Y^{\perp}}{3} \le \|\mathsf{M}\|_F^2 \le r + \frac{s \dim Y^{\perp} - r}{6}.$$

Rearranging, this gives $r \ge s \dim Y^{\perp}/5 \ge ns/10$. In particular, $\sigma_{k_s}(\mathsf{M}) \ge 1/\sqrt{6}$.

By (75), projection onto $\mathcal{T}^{\perp}$ decreases these singular values by at most a polynomial factor. To return to the original quadratics, we use

$$C_{\mathcal{N}}\big|_{\mathcal{N}\otimes Y^{\perp}} = \Pi_{\mathcal{T}^{\perp}}\mathsf{M}(L \otimes I_{Y^{\perp}}),$$

since $L\xi - \xi \in \mathcal{U}$. Since L is invertible and does not decrease norms, we have k_s singular values at least $Q^{-O(1)}$ even after restricting $C_{\mathcal{N}}$ to $\mathcal{N} \otimes Y^{\perp}$. Allowing all multipliers in $\mathbb{R}^n$ can only increase them. Hence $\sigma_{k_s}(C_{\mathcal{N}}) \ge Q^{-C}$, as required. □

B.7 Intersections with rigid subspaces

Proof of Lemma 6.15. We start with a simple claim: suppose that $k < m$. Then every $(k+1)$-dimensional linear subspace $V \subseteq \mathbb{R}^m$ contains a unit vector c such that

$$\operatorname{dist}(c, \Sigma_k)^2 \ge \frac{1}{k+1}. \tag{76}$$

To see this, maximize $\|x\|$ over the compact set $V \cap [-1,1]^m$. The coordinate functionals corresponding to saturated coordinates, those with $|x_i| = 1$, must span V^*. Otherwise there is a nonzero $w \in V$ vanishing on all saturated coordinates. For sufficiently small $t > 0$, both $x + tw$ and $x - tw$ remain in the cube, and at least one has larger squared norm than x, a contradiction. Thus at least $k+1$ coordinates are saturated, so $\|x\|^2 \geq k+1$. For $c = x/\|x\|$, any k coordinates carry squared mass at most $k/\|x\|^2 \leq k/(k+1)$, proving (76).

Constructing W_0. Set $P = \Pi_{U^\perp}$ and

$$\theta = \min\left\{\frac{1}{2}, \frac{\beta}{4(k+1)\max\{1, \|\mu\|_{\mathrm{op}}\}}\right\}.$$

Consider the linear map $T : W \to U^\perp$ defined by $T(X) = PX$. Let W_0 be the span of its right singular vectors with singular values below θ. The remaining right singular directions have singular values at least θ, and their images are orthogonal to $T(W_0)$. Consequently,

$$\forall X \in W, \quad \operatorname{dist}(X, U) = \|PX\| \geq \theta \operatorname{dist}(X, W_0). \tag{77}$$

We claim $\dim W_0 \leq k$. If not, take a $(k+1)$-dimensional subspace $L \subseteq W_0$. On L we have $\|PX\| < \theta\|X\|$ for nonzero X. Since $\theta \leq 1/2$, projection onto U is injective on L. Its image $L_U = \Pi_U L$ therefore has dimension $k+1$. For $Y = \Pi_U X$ with $X \in L$,

$$\|X\| \leq (1-\theta^2)^{-1/2}\|Y\| \leq 2\|Y\|, \qquad \operatorname{dist}(Y, W) \leq \|Y - X\| < 2\theta\|Y\|$$

when $Y \neq 0$. Apply (76) to the $(k+1)$-dimensional coefficient space $\mu^{-1}(L_U)$, obtaining a unit vector c with the stated sparse-tail bound. Since $W \subseteq \mathcal{R}_k$,

$$\operatorname{dist}(\mu(c), \mathcal{R}_k) \leq \operatorname{dist}(\mu(c), W) < 2\theta\|\mu(c)\| \leq \frac{\beta}{2(k+1)},$$

whereas (54) and (76) give the lower bound $\beta/(k+1)$. This contradiction proves the claim. Since $\theta \geq Q^{-C}$, equation (77) proves (55).

If $k \geq m$, instead take $W_0 = \Pi_W U$, whose dimension is at most $m \leq k$. Every $X \in W \cap W_0^\perp$ is orthogonal to U, so $PX = X$. Moreover, its image is orthogonal to $P(W_0)$. It follows that $\|PX\| \geq \operatorname{dist}(X, W_0)$ for every $X \in W$. Thus (55) holds in this case with constant one. In either case, setting $X \in W \cap U$ in the distance bound shows $W \cap U \subseteq W_0$.

Transferring the rank bound. Let $F' = \Pi_W F$, and write the bound just proved as

$$\operatorname{dist}(X, W_0) \leq A\|PX\| \qquad (X \in W), \qquad A \leq Q^C.$$

Since $\|F - F'\|_{\mathrm{op}} \leq e$, Weyl's inequality gives $\sigma_{s+1}(PF') \leq 2e$. Hence there is a map G of rank at most s such that $\|PF' - G\|_{\mathrm{op}} \leq 2e$. Put $K = \ker G$, so $\dim K^\perp \leq s$. For $v \in K$, the distance bound gives

$$\|(I - \Pi_{W_0})F'v\| \leq A\|PF'v\| \leq 2Ae\|v\|.$$

Define

$$F_0 = F'\Pi_{K^\perp} + \Pi_{W_0}F'\Pi_K.$$

The two summands have ranks at most s and k, respectively. Also,

$$F' - F_0 = (I - \Pi_{W_0})F'\Pi_K, \qquad \|F' - F_0\|_{\mathrm{op}} \leq 2Ae.$$

Thus F is within $e + 2Ae \leq Q^C e$ of a map of rank at most $s + k$, proving (56). □

B.8 Compressed volume bound

Proof of Lemma 6.17. We fix a layer (s, q) and a reconstruction index α from Theorem 6.13. Let E_α be the event that $\mathbf{a} \in \mathcal{G}_2$ and there is a candidate $\mathbf{b}$ in this layer with $\max_i \|b_i\| \le R_b$ and $\delta(\mathbf{a}, \mathbf{b}) \le \eta$, for which α satisfies the two reconstruction guarantees of that theorem. We will bound $\mathbb{P}(E_\alpha)$ and then sum over the indices. The number p_z of retained coordinates may depend on α. We write

$$u^0(\mathbf{a}, z) = u^0_\alpha(\mathbf{a}, z), \qquad R(\mathbf{a}, z, u) = \Phi(\mathbf{a}) - \Phi(z, u).$$

Here $\Phi(z, u)$ uses the coordinate ordering specified by α.

The reconstruction error is at most $Q^C\gamma^{12}$, which is too large to guarantee tensor error $O(\eta)$, since $\eta = \gamma^{22}$. We use one Newton step to reduce the residual to this smaller scale, while keeping the retained coordinates fixed. If $p_z < p$, we define

$$\begin{aligned} J_0(\mathbf{a}, z) &= D_u R(\mathbf{a}, z, u^0(\mathbf{a}, z)), \\ u^1(\mathbf{a}, z) &= u^0(\mathbf{a}, z) - (J_0^\top J_0)^{-1} J_0^\top R(\mathbf{a}, z, u^0(\mathbf{a}, z)), \\ f_\alpha(\mathbf{a}, z) &= R(\mathbf{a}, z, u^1(\mathbf{a}, z)). \end{aligned} \tag{78}$$

We discard any index for which a denominator is identically zero. For the remaining indices, we use these formulas on the open set where all denominators, including the Gram determinants, are nonzero. The correction changes only the missing coordinates; all retained values, including those added to capture weak directions, stay fixed. If $p_z = p$, there are no missing coordinates, and we simply set $f_\alpha(\mathbf{a}, z) = \Psi(\mathbf{a}, z)$, with no correction. Every coordinate of f_α is rational with numerator and denominator degrees at most Q^C: the base formulas have this property, and the correction uses a fixed number of matrix operations on polynomial-size matrices.

Ellipsoid in the retained coordinates. To turn the joint volume bound into a probability bound, we need a whole neighborhood of retained values on which the corrected residual is small. We can use longer axes along the weak tangent directions, since the tensor changes slowly in those directions. These q longer axes will give an extra factor γ^{-q} in the volume.

We fix $\mathbf{a} \in E_\alpha$ and a corresponding candidate $\mathbf{b} = (z_*, u_*)$. Let $E_\gamma \subseteq \mathbb{R}^p$ be the weak right singular space at $\mathbf{b}$, and set

$$V_z = \Pi_z E_\gamma \subseteq \mathbb{R}^{p_z}.$$

The bound (39) from Theorem 6.13 implies that $\dim V_z = q \le p_z$, and that the coordinate projection from E_γ onto V_z has an inverse $L_\gamma : V_z \to E_\gamma$ with $\|L_\gamma\| \le Q^C$.

For a retained-coordinate displacement $v = v_\parallel + v_\perp \in V_z \oplus V_z^\perp$, we lift $v_\parallel$ to a weak tangent direction using L_γ. For $v_\perp$, we change only the retained coordinates. This gives the following *reference tuple*:

$$\mathbf{b}_v = \mathbf{b} + L_\gamma v_\parallel + (v_\perp, 0) = (z_* + v, u_v). \tag{79}$$

We choose a sufficiently large absolute constant B, and let $\mathcal{Z} \subseteq \mathbb{R}^{p_z}$ be the ellipsoid centered at zero with semiaxes of length

$$Q^{-B}\eta/\gamma \quad \text{on } V_z, \qquad Q^{-B}\eta \quad \text{on } V_z^\perp.$$

Thus $\mathcal{Z}$ has q long axes and $p_z - q$ short axes. Since $\|R(\mathbf{a}, z_*, u_*)\| \le \eta$, Taylor expansion at $\mathbf{b}$ gives, for every $v \in \mathcal{Z}$,

$$\|\mathbf{b}_v - \mathbf{b}\| \le Q^C\gamma^{21}, \qquad \|R(\mathbf{a}, z_* + v, u_v)\| \le C\eta. \tag{80}$$

Indeed, the derivative has norm at most γ on E_γ, so the long axes contribute at most $Q^{C-B}\eta$ to the linear term. The short axes contribute at most the same amount, using the polynomial bound on $\|\mathsf{D}\Phi(\mathbf{b})\|$. The quadratic remainder is at most $Q^C(\eta/\gamma)^2 = Q^C\gamma^{42} \ll \eta$. The reference tuples need not stay in layer (s, q); below we use their distance from the original candidate $\mathbf{b}$.

Newton correction throughout the ellipsoid. We next check that the correction is defined throughout $\mathcal{Z}$ and reduces the residual to $O(\eta)$.

Suppose first that $p_z < p$. Since $\|v\| \le Q^{-B}\gamma^{21}$, the whole ellipsoid lies in the neighborhood supplied by Theorem 6.13, for B sufficiently large. Uniform local reconstruction, together with (80), gives

$$e_0 := \|u^0(\mathbf{a}, z_* + v) - u_v\| \le \|u^0(\mathbf{a}, z_* + v) - u_*\| + \|u_* - u_v\| \le Q^C\gamma^{12}.$$

The predicted tuple $(z_* + v, u^0(\mathbf{a}, z_* + v))$ is also within $Q^C\gamma^{12}$ of the original candidate $\mathbf{b}$.

We first explain why its missing-coordinate Jacobian is well conditioned. Every weak tangent direction has a nonnegligible projection onto the retained coordinates, by (39). Thus a perturbation that leaves all retained coordinates fixed must have a sufficiently large component outside the weak tangent space. To quantify this, let $w \in \mathbb{R}^p$ be supported on the missing coordinates, so that $\Pi_z w = 0$, and decompose $w = w_{\mathrm{wk}} + w_{\mathrm{st}}$ according to $E_\gamma \oplus E_\gamma^\perp$. The same projection bound gives

$$Q^{-C}\|w_{\mathrm{wk}}\| \le \|\Pi_z w_{\mathrm{wk}}\| = \|\Pi_z w_{\mathrm{st}}\| \le \|w_{\mathrm{st}}\|.$$

Thus $\|w_{\mathrm{st}}\| \ge Q^{-C}\|w\|$. The two singular subspaces have orthogonal images under $\mathsf{D}\Phi(\mathbf{b})$, and all singular values on $E_\gamma^\perp$ exceed γ. Consequently,

$$\|\mathsf{D}\Phi(\mathbf{b})w\| \ge \gamma\|w_{\mathrm{st}}\| \ge Q^{-C}\gamma\|w\|.$$

Equivalently,

$$\sigma_{\min}(D_u\Phi(\mathbf{b})) \ge Q^{-C}\gamma. \tag{81}$$

Since the predicted tuple is within $Q^C\gamma^{12}$ of $\mathbf{b}$, the polynomial Lipschitz bound on the derivative shows that its missing-coordinate Jacobian changes by at most $Q^C\gamma^{12}$. For sufficiently large a_0, this is less than half the lower bound in (81). Weyl's inequality therefore gives

$$\sigma_{\min}(J_0(\mathbf{a}, z_* + v)) \ge Q^{-C}\gamma.$$

In particular, the Newton correction is defined throughout the ellipsoid, and $\|J_0^\dagger\| \le Q^C\gamma^{-1}$ there. This also shows that any index with $E_\alpha \neq \varnothing$ has nonzero denominators somewhere, so none of these indices was discarded.

We first bound the distance from the corrected tuple to the reference tuple. This will let us use the joint Jacobian bounds near $\mathbf{b}$. We suppress the fixed arguments $(\mathbf{a}, z_* + v)$ and write $r_v = R(\mathbf{a}, z_* + v, u_v)$. Taylor expansion about u^0 gives

$$R(u^0) = r_v + J_0(u^0 - u_v) + \mathcal{E}_0, \qquad \|\mathcal{E}_0\| \le Q^C e_0^2.$$

Since $J_0^\dagger J_0 = I$, the Newton formula cancels the linear reconstruction error:

$$\begin{aligned} u^1 - u_v &= -J_0^\dagger(r_v + \mathcal{E}_0), \\ \|u^1 - u_v\| &\le Q^C\gamma^{-1}(\eta + \gamma^{24}) \le Q^C\gamma^{21}. \end{aligned} \tag{82}$$

It follows that $\|u^1 - u^0\| \le \|u^1 - u_v\| + e_0 \le Q^C \gamma^{12}$.

The bound on the coordinate error is still larger than η. To bound the tensor residual at this smaller scale, we expand again about u^0:

$$\begin{aligned} R(u^1) &= R(u^0) + J_0(u^1 - u^0) + \mathcal{E}_1 \\ &= (I - J_0 J_0^\dagger)(r_v + \mathcal{E}_0) + \mathcal{E}_1, \end{aligned}$$

where

$$\|\mathcal{E}_1\| \le Q^C \|u^1 - u^0\|^2 \le Q^C \gamma^{24}.$$

The matrix $I - J_0 J_0^\dagger$ is an orthogonal projection. Using $\|r_v\| \le C\eta$, we therefore obtain

$$\|f_\alpha(\mathbf{a}, z_* + v)\| \le C\eta + Q^C \gamma^{24} \le C'\eta. \tag{83}$$

The projection cancellation is essential here: applying a Lipschitz bound directly to (82) would only give the larger error scale $Q^C \gamma^{21}$. If $p_z = p$, there are no missing coordinates and no correction is needed; (83) follows directly from (80).

Jacobian bounds also hold for the corrected functions. In either case, the corrected tuple, denoted by $\widehat{\mathbf{b}}^1(\mathbf{a}, z)$, satisfies throughout the ellipsoid

$$\|\widehat{\mathbf{b}}^1(\mathbf{a}, z_* + v) - \mathbf{b}\| \le Q^C \gamma^{21}.$$

This is smaller than the perturbation radius $Q^{-C}\tau = Q^{-C}\gamma^{16}$ in Lemma 6.12, after increasing a_0. The corrected components also have norms at most $R_b + 1$. Consequently, the joint Jacobian of Ψ at the corrected tuple has

$$\sigma_p(\mathsf{D}\Psi) \ge Q^{-C}, \qquad \sigma_{p+k_s}(\mathsf{D}\Psi) \ge Q^{-C}\tau.$$

The rational graph parametrization

$$\iota_\alpha(\mathbf{a}, z) = (\mathbf{a}, z, u^1(\mathbf{a}, z))$$

retains $(\mathbf{a}, z)$ among its coordinates, and therefore $\|\mathsf{D}\iota_\alpha(w)\| \ge \|w\|$. Its tangent image has dimension $p + p_z$ inside the original $2p$-dimensional input space. Intersecting the span of the top p or $p + k_s$ right singular vectors of $\mathsf{D}\Psi$ with this tangent image loses at most $p - p_z$ dimensions. Thus the p directions with singular value at least Q^{-C} leave at least p_z such directions, and the $p + k_s$ directions with singular value at least $Q^{-C}\tau$ leave at least $p_z + k_s$. Since $\mathsf{D}\iota_\alpha$ does not shorten input vectors, these bounds also hold after parametrizing the tangent image by $(\mathbf{a}, z)$. Using $f_\alpha = \Psi \circ \iota_\alpha$ and the min–max principle, we obtain

$$\sigma_{p_z}(\mathsf{D}f_\alpha) \ge Q^{-C}, \qquad \sigma_{p_z+k_s}(\mathsf{D}f_\alpha) \ge Q^{-C}\tau \tag{84}$$

throughout the ellipsoid. If $p_z = p$, the graph is simply $\iota_\alpha(\mathbf{a}, z) = (\mathbf{a}, z)$ and the same argument applies. The first inequality is omitted when $p_z = 0$. In particular, any index with $E_\alpha \neq \varnothing$ has $p_z + k_s \le N_3$. No upper bound on the derivatives of the reconstruction formulas is needed.

Joint volume and section volume. Let $\mathcal{J}_\alpha$ be the set of pairs $(\mathbf{a}, z)$ with z in the fixed box $[-R_b - 1, R_b + 1]^{p_z}$, where the formulas are defined and

$$\|f_\alpha(\mathbf{a}, z)\| \le C'\eta, \qquad \prod_{i=1}^{p_z+k_s} \sigma_i(\mathsf{D}f_\alpha(\mathbf{a}, z)) \ge Q^{-C(p_z+k_s)}\tau^{k_s}.$$

The constants can be fixed uniformly over the indices. By (84), only the last k_s singular values need to pay a factor τ. We apply Theorem 6.6 with $p_0 = p$, $p_1 = p_z$, $r = p_z + k_s$, and output threshold $C'\eta$. We take its size parameter to be a sufficiently large fixed power of Q, since the dimensions and rational degrees here are at most $Q^{O(1)}$. The Gaussian variables are all coordinates of $\mathbf{a}$, while only the retained candidate coordinates receive Lebesgue measure. Writing γ_ρ for the original Gaussian law, we get

$$(\gamma_\rho \otimes \mathrm{Leb}_{p_z})(\mathcal{J}_\alpha) \le Q^{C(p_z+k_s)} \eta^{p_z+k_s} \tau^{-k_s}. \tag{85}$$

For fixed $\mathbf{a}$, we write $(\mathcal{J}_\alpha)_{\mathbf{a}} = \{z : (\mathbf{a}, z) \in \mathcal{J}_\alpha\}$. Whenever $\mathbf{a} \in E_\alpha$, the preceding bounds show that this section contains the ellipsoid $z_* + \mathcal{Z}$. The ellipsoid stays in the chosen box because $\|z_*\|_\infty \le R_b$ and $\|v\| \le 1$. Its volume in the retained coordinates satisfies

$$\mathrm{vol}_{p_z}(\mathcal{Z}) \ge Q^{-Cp_z} \eta^{p_z} \gamma^{-q}. \tag{86}$$

Indeed, q of its axes have the additional factor γ^{-1}, and the unit-ball volume is absorbed into Q^{-Cp_z} since $p_z \le p \le Q^C$.

We now integrate the section volumes over $\mathbf{a}$. Every $\mathbf{a} \in E_\alpha$ contributes at least the volume in (86), so

$$\begin{aligned}(\gamma_\rho \otimes \mathrm{Leb}_{p_z})(\mathcal{J}_\alpha) &= \int \mathrm{vol}_{p_z}((\mathcal{J}_\alpha)_{\mathbf{a}})\, d\gamma_\rho(\mathbf{a}) \\ &\ge \mathbb{P}(E_\alpha)\, Q^{-Cp_z} \eta^{p_z} \gamma^{-q}.\end{aligned}$$

Dividing by this lower bound on each section volume and using (85), we obtain

$$\mathbb{P}(E_\alpha) \le Q^{C(p_z+k_s)} \left(\frac{\eta}{\tau}\right)^{k_s} \gamma^q.$$

No disjointness of the ellipsoids is required: for each fixed $\mathbf{a}$, we only use a lower bound on the volume of its section. By Theorem 6.13, every $\mathbf{a} \in \mathcal{G}_2 \cap E_{s,q}$ belongs to E_α for at least one of the indices. We therefore take a union bound over the at most $Q^{C(ns+q)}$ indices. Since $p_z \le Cns + 2q$ and $k_s \le ns$, this proves (67). □